Class 7

Physics

(Solutions)

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1. Units and Measurements

Solutions

LEVEL-I

1. Correct option: (A)

To know and understand a physical situation completely, one is likely to measure the quantities such as distance, speed, time, mass, acceleration, force etc. which are called the physical quantities. Physical quantity is that in terms of which laws of physics can be expressed and which can be measured directly or indirectly.

2. Correct options: (A, B, C)

Fundamental quantities are the quantities whose measurement is done with the independent units. The quantities in option (D) Length, time and mass are measured by independent units such as kilogram, meter and second respectively. In all the other options velocity is there which is expressed in terms of derived units which is meters per second. So, all the other sets cannot enter into the list of fundamental quantities in any system of units.

3. Correct option: (B)

Mass is a physical quantity as it is used to measure a physical situation. Rest all are the emotional situations

4. Correct option: (A)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. For example, quantities Length, time and mass are measured independently and hence they are fundamental physical quantities.

5. Correct option: (B)

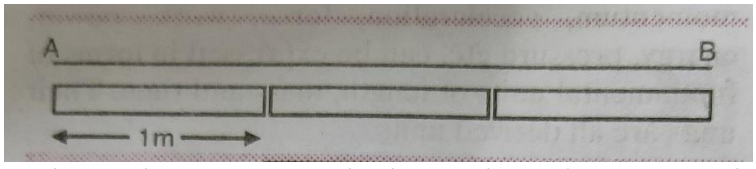
Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. For example, velocity is a derived physical quantity which depends on fundamental quantities distance(length) and time.

6. Correct option: (B)

Measurement is basically a comparison process. It involves the selection of a unit of measurement and comparing the quantity with standard unit. The standard chosen should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod, we have to take length as the standard quantity.

7. Correct option: (A)

In comparing the quantity with the standard unit, we have to find the number of times this unit is contained in the quantity. For example, suppose we have to measure the length of a wire AB and we select meter as the unit



of measurement. We place the meter rod along the wire AB and find that it is contained 3 times in AB. Thus 3 is the numerical value of the length AB, when m is the unit of measurement. We write AB= 3 meter

8. Correct option: (B)

The SI unit of temperature as per the International System of Units is Kelvin which is represented by the symbol K.

9. Correct option: (B)

In general magnitude of a physical quantity (P)= numerical value × size of its unit.

For example. If length of a rod = 3 m

Then 3 = numerical value

1m = size of the unit

10. Correct option: (A)

In general magnitude of a physical quantity (P)= numerical value(N) × size of its unit(U) Or P=NU

For example. If length of a rod = 3 m

Then 3 = numerical value

1m = size of the unit

11. Correct option: (D)

As we know

physical quantity (P)= numerical value(N) × size of its unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of its measurement. Hence, we may write

$$P = n_1U_1 = n_2U_2$$

Where

U_1, U_2 are the two unts of measurement of the same quantity and n_1 and n_2 are their respective numerical values .

12. Correct option:(C)

Derived quantity is a quantity whose measurement is done with the help of other fundamental physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. For example, velocity is a derived physical quantity which depends on fundamental quantities distance(length) and time.

13.Correct option:(B)

Fundamental quantities are the quantities whose measurement is done with the independent units. The quantities Length, time and mass are measured by independent units such as kilogram, meter and second respectively. These units can neither be changed nor be related to any other unit.

14. Correct option: (B)

The SI unit of temperature as per the International System of Units is Kelvin which is represented by the symbol (K).

15.Correct option: (A)

Strength of Electric current is measured in Amperes, Hence SI unit of strength of electric current is Ampere.

Its symbol is (A).

16. Correct option:(D)

None of these unit is written by putting a dot between the individual units.

- Units like seconds, meters are fundamental units/independent units.
- units like meter /second are an example of a derived unit/dependent unit.
- Compound units are those composed of more than one unit. For example, miles/hour

17. Correct option: (A)

The normal force (thrust)(F) exerted by an object per unit area(A) of the surface in contact with it is called pressure (P).

$$P = \frac{F}{A}$$

SI unit of pressure is Pascal (Pa)

1 Pascal = $1Nm^{-2}$

18. Correct option:(B), (D)

Full name of the unit named in the honor of scientist is written with small letters/lower initial letter.

For example, SI unit of force is written as **newton** and not **Newton or NEWTON**.

19.Correct option:(C)

Measurement is basically a comparison process. It involves the selection of a unit of measurement and comparing the quantity with standard quantity/unit. The standard chosen should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod we have to take (m) as standard unit.

20. Correct option:(A), (B)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence,

- Symbol for C.G.S unit of volume is cm^3
- Symbol for SI unit of volume is m^3

21. Correct option:(B)

The meter is the base unit of length in the International System of Units (SI). Its unit symbol is (m).

22. Correct option:(C)

The second is the base unit of time in the International System of Units (SI). Its unit symbol is (s).

23. Correct option:(C)

Density $= \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kilogram(kg).

SI unit of volume is m^3

So, SI unit of density is $\frac{kg}{m^3}$

24. Correct option:(C)

velocity $= \frac{\text{displacement}}{\text{time taken}}$

SI unit of displacement is m.

SI unit of time is s.

SI unit of velocity is m/s.

25. Correct option:(C)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence,

C.G.S unit of mass is gram.

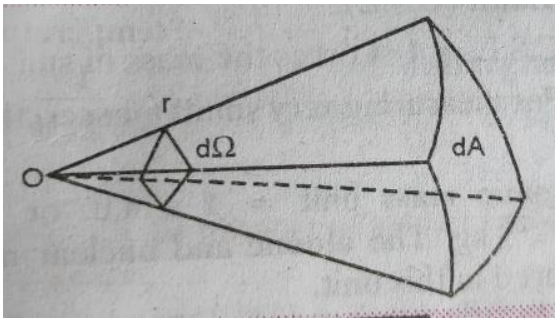
26.Correct option:(A)

Steradian is the unit of solid angle (angle in space)

1 steradian is the solid angle subtended at the center of a sphere, by that surface of the sphere, which is equal in area, to the square of radius of the sphere.

If an area dA of a spherical surface subtends a solid angle dΩ at the center of the sphere of radius r, then,

$$d\Omega = \frac{dA}{r^2} \text{ steradian}$$



27. Correct option:(B)
M.K.S is the system of units based on measuring lengths in meters, mass in kilograms, time in seconds.
The M.K.S unit of mass is kilogram.
28. Correct option:(D)
M.K.S is the system of units based on measuring lengths in meters, mass in kilograms, time in seconds.
The M.K.S unit of length is meter.
29. Correct option:(C)
F.P.S system is the British Engineering system of units, which uses foot as the unit of length, pound as the unit of mass and second as the unit of time.
Symbol for unit of mass pound is (lb)
30. Correct option:(B)
M.K.S is the system of units based on measuring lengths in meters, mass in kilograms, time in seconds.
The M.K.S unit of time is second having symbol(s).
31. Correct option:(B)
F.P.S system is the British Engineering system of units, which uses foot as the unit of length, **pound as the unit of mass** and second as the unit of time.
32. Correct option:(A)
For a unit named after a scientist, the symbol is a capital letter. For example, symbol for SI unit of force (newton)is (N).
33. Correct option:(C)
C.G.S is centimeter(cm) –gram(g)- second(s) system of units
Hence,
C.G.S unit of length is centimeter.
34. Correct option:(C)
SI unit of work is joule having symbol(J).
35. Correct option:(C)
C.G.S system of units was set up in France and it is based on centimeter, gram and second as the fundamental units of length, mass and time respectively.
F.P.S system is the British Engineering system of units, which uses foot as the unit of length, **pound as the unit of mass** and second as the unit of time.
M.K.S system was set up in France. It makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.
However, these three fundamental quantities were not sufficient to obtain the units of quantities in other branches of physics. So, a new system of units named **SI system** was Introduced in which there are seven fundamental quantities: Mass, length, Time,

Temperature, Electric current, Luminous Intensity, Quantity of matter and two Supplementary physical quantities.

36. Correct option:(A)

SI system is the internationally accepted system of units for measurements. The SI, with standard scheme of symbols, units and abbreviations, was developed and recommended by General conference on weights and measures in 1971 for international usage in scientific, technical, industrial and commercial work.

There are seven fundamental quantities in SI system: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter and two Supplementary physical quantities.

LEVEL-II

1. Correct option:(C)

Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. Volume depends on other physical quantities for its measurement.

2. Correct option:(B)

Area is the amount of surface covered by a two-dimensional shape, measured in square units.

SI unit of area is m^2.

3. Correct option:(A)

SI unit of luminous intensity is Candela.

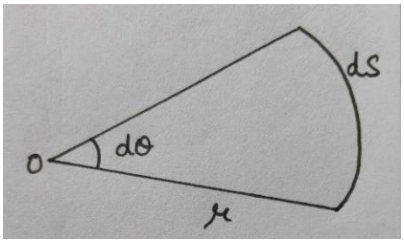
4.Correct option:(B)

Plane angle is measured in radians.

1 radian is the angle subtended at the center of a circle by an arc equal in length to the radius of the circle.

If an arc of length ds subtends an angle dθ at the centre O of a circle of radius r, then

$d\theta = \frac{ds}{r}$ radian



5. Correct option:(A)

Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities. Units of derived quantities are called derived units.

6. Correct option:(D)

Strength of Electric current is measured in Amperes, Hence SI unit of strength of electric current is Ampere.

Its symbol is (A).

7. Correct option:(C)

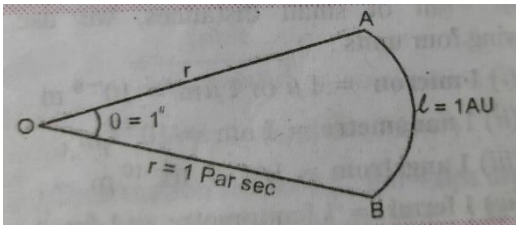
Volume is the amount of space that something fills.

SI unit of area is m^3.

8. Correct option:(D)

Parallactic second (parsec) is a unit of long distances. One parsec is the radius of a circle at the center of which an arc of the circle,1 astronomical unit (1 AU= 1.496×10^{11}m) long subtends an angle of 1 second (1")

1 parsec = 3.1×10^{16}m



9. Correct option: (A), (B), (C)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter. These quantities can be measured independently.

Length can't be a derived quantity in any system as it is a fundamental quantity.

10. Correct options: (A, B, D)

M.K.S system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

However, these three fundamental quantities were not sufficient to obtain the units of quantities in other branches of physics. So, a new system of units named SI system was Introduced in which there are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter and two Supplementary physical quantities.

11. Correct options: (A), (B), (C)

Fundamental quantities are the quantities whose measurement is done with the independent units. The quantities Length, time and mass are measured by independent/fundamental units such as kilogram, meter and second respectively.

Newton is a derived unit for derived physical quantity (force).

12. Correct options: (A), (D)

Mass of a body is the quantity of matter it contains.

SI unit of mass is Kilogram.

1 kilogram = 1000000 milligrams

Hence, kilogram and milligram are units used to measure mass. Kilometer and millimeter are used to measure distance.

13. Correct options: (A), (B), (D)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter/amount of substance. These quantities can be measured independently.

14. Correct options: (A), (B), (C), (D)

Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities.

- Density $= \frac{\text{mass}}{\text{volume}}$

density depends on mass and volume so it is a derived quantity.

- velocity $=\frac{\text{displacement}}{\text{time taken}}$

Velocity depends on displacement and time so it is a derived quantity.

- Volume and acceleration are also derived quantities as their measurement depends on other physical quantities.

15.Correct options:(A), (D)

- Mole is the fundamental unit for quantity of matter.
- Radian is one of the two supplementary fundamental units in SI system.
- Coulomb is the derived SI unit for charge. It is constructed from fundamental units' ampere and seconds.
- Pascal is a derived SI unit for pressure.it is constructed from fundamental units' kilogram, meter, seconds.

16. Correct options: (A, B, C)

Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities.

- Newton is a derived unit for derived physical quantity (force).
- Pascal is a derived unit for derived physical quantity (pressure).
- Joule is a derived unit for derived physical quantity (work/energy).

17. Correct options: (A, D)

velocity $=\frac{\text{displacement}}{\text{time taken}}$

SI unit of displacement is m.

SI unit of time is s.

SI unit of velocity is m/s or ms^{-1}

18. Correct option:(B, D)

Density $=\frac{\text{mass}}{\text{volume}}$

SI unit of mass is kilogram(kg).

SI unit of volume is m^3

So, SI unit of density is $\frac{kg}{m^3}$ or kgm^{-3}

19. Correct options: (A, B, D)

M.K.S system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

However, these three fundamental quantities were not sufficient to obtain the units of quantities in other branches of physics. So, a new system of units named SI system was Introduced in which there are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter and two Supplementary physical quantities.

20. Correct options: (A)

SI unit of length is meter.

As distance between two points is the path length between two points so its unit is same as the length.

21. Correct options: (C)

Unit of time is **second** in all the system of units (F.P.S, C.G.S, M.K.S, SI).

22.Correct option:(B)

Steradian is the SI unit of solid angle (angle in space)

23. Correct option: (A)

As we know

physical quantity (P)= numerical value(N) × size of its unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of its measurement. Hence, we may write

$$P = N_1U_1 = N_2U_2$$

Where

U_1, U_2 are the two unts of measurement of the same quantity and N_1 and N_2 are their respective numerical values .

If $U_1 > U_2$, then $N_1 < N_2$ which means larger is the unit , lesser is the numerical value and vice – versa. For example,

Length of rod = 3 m = 300 cm

Here, unit(m) is larger and numerical value 3 is lesser

Unit (cm) is lesser and numerical value 300 is larger

24. Correct option: (A)

As we know

physical quantity (P)= numerical value(N) × size of its unit(U)

Further we know that magnitude of a quantity remains the same, whatever may be the units of its measurement. Hence, we may write

$$P = N_1U_1 = N_2U_2$$

Where

U_1, U_2 are the two unts of measurement of the same quantity and N_1 and N_2 are their respective numerical values .

If $U_1 > U_2$, then $N_1 < N_2$ which means larger is the unit , lesser is the numerical value and vice – versa. For example,

Length of rod = 3 m = 300 cm

Here, unit(m) is larger and numerical value 3 is lesser

Unit (cm) is lesser and numerical value 300 is larger

Hence numerical value and size of unit are inversely proportional to each other.

25. Correct option: (B)

1 kilogram = 1000 grams or 10^3 grams

now , 1 gram = 1000 mg

so , 1 kilogram = 1000000 mg or 10^6mg

Hence 1kilogram can't be expressed as 10000 grams.

26. Correct Matching is: a-PQ; b-S; c-T; d-R

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities.

Time is a fundamental quantity having fundamental unit (second).

Length is a fundamental quantity having fundamental unit (meter).

Luminous intensity is a fundamental quantity having fundamental unit (candela).

Derived quantity is a quantity whose measurement is done with the help of other physical quantities and it can be obtained by writing the defining equation in terms of fundamental physical quantities.

Energy is a derived quantity.

Force is a derived quantity having derived unit (newton).

27. Correct Matching is: a-T; b-R; c-Q; d-P

Unit of time is **second** in all the system of units (F.P.S, C.G.S, M.K.S, SI).

SI unit of length is meter.

F.P.S: It uses foot as the unit of length, pound as the unit of mass and second as the unit of time.

C.G.S: It is based on centimeter, **gram** and second as the fundamental units of length, **mass** and time respectively.

Multiple And Sub Multiples
LEVEL-I

1. Correct option: (C)
1 millisecond $= 10^{-3}$seconds
1microsecond $= 10^{-6}$seconds
1 nanosecond $= 10^{-9}$seconds
1shake $= 10^{-8}$seconds

2. Correct option: (A)
1Decimeter = 10 centimeter

3.Correct option:(D)
The average of the varying solar days, when earth completes one revolution around the sun is called the mean solar day.

4. Correct option: (C)
Light year is an important unit of long distances. One light year is the distance travelled by light in vacuum in one year.
as velocity of light in vacuum $= 3 \times 10^8$ ms^{-1}
1year $= 365 \times 24 \times 60 \times$ 60 seconds
so, 1 light year $= 3 \times 10^8 \times 365 \times 24 \times 60 \times 60$ meter
1 light year $= 9.46 \times 10^{15}$m

5. Correct option:(C)
1 kg $= 10^6$mg
500kg $= 500 \times 10^6$ mg
500kg $= 5 \times 10^8$ mg

6. Correct option:(B)
1g= 1000mg
1 mg= 1 / 1000 g
1mg= 0.001 g

7. Correct option: (B)
1 quintal = 100 kg

8. Correct option: (C)
The standard unit chosen for measurement should be of the same nature as that of the quantity to be measured. For example, if we have to measure the length of a rod, we have to take length as the standard quantity and meter as the standard unit.

9. Correct option: (C)
1 hectometer (hm) = 100 m
Meter (m) is a smaller unit of length as compared to hectometer

10. Correct option: (D)
1 meter = 100 cm
1 cm $= \frac{1}{100}$m
$= 0.01$ m
Hence, 1 centimeter = 0.01 m

11. Correct option: (D)
1 picometer $= 10^{-12}$m
1m $= 10^2$cm
so, 1 picometer $= 10^{-12} \times 10^2$ cm
1 picometer $= 10^{-10}$ cm

12.Correct option:(C)

As the gap between two points is measured in units of metres.so, it can measure as length because SI unit of length is meter.

13. Correct option: (A)

$1\text{cm} = \frac{1}{100}\text{m}$

$300\text{ cm} = \frac{300}{100}\text{m}$

300 cm = 3 m

14. Correct option: (A)

1 gram = 10^6 micrograms

Or

1 microgram = 10^{-6}grams

15. Correct option: (B)

(nm) is the symbol for nanometer

1 cm = 10^7nm

$3\text{cm} = 3 \times 10^7$ nm

$3\text{cm} = 0.3 \times 10^8$ nm

16. Correct option: (A)

1J = 10^7ergs

Joule is the SI(MKS) unit of energy and ergs is the cgs unit of energy.

17. Correct option: (B)

1 min = 60 s

40 min = 40 ×60 s

= 2400 s

18. Correct option: (A)

The diameter of the sun = 1.39□10^9 m

19. Correct option: (B)

1 quintal = 100 kg

1quintal = 10^2kg

20. Correct option: (D)

Diameter of hydrogen atom is approximately 1.06□10^{-10} m. It is defined by the diameter of the spherical

electron cloud around the nucleus.

21. Correct option: (B)

A century is a period of 100 years

So, 1 century = 100 years

22. Correct option: (D)

1 day = 24 hours

1 hour = 60 minutes

1minute = 60 seconds

So, 1 day =24×60 minutes

=1440 min

Hence, 1 day = 24×60×60 seconds

1 day = 86,400 seconds

23.Correct option:(A)

The exponential part of a particular measurement is called the order of magnitude of a quantity. An order of magnitude is an exponential change of plus or minus 1 in the value of a quantity or unit.

24. Correct option: (D)
Both option(A) and (B) are the advantages of using scale. It can be easily taken from one place to another and it does not change during measurements.
25. Correct option: (D)
1km=100000cm
1cm = 1/100000km
1centimeter=1/100000 kilometer
26. Correct option: (D)
1 centimeter = 10 millimeters
Or
1 millimeter = 1/10 centimeter
27. Correct option: (D)
The most appropriate unit to measure the length of a book is centimeter. meter would be too large and millimeter or micrometer would be too small to measure the length of a book.
28. Correct option: (C)
1 meter = 1000 millimeters
Or
1 millimeter = 1/1000 meter
29. Correct option: (A)
1 decade = 10 years
30. Correct option: (A)
1 centimeter = 10 millimeters

LEVEL-II

1. Correct option: (C)
(Mm) is the symbol of Mega meter,
(pm) is the symbol for Pico meter.
$1Mm=10^{6}m$
$1m=10^{12}pm$
So, $1Mm=10^{6}\times 10^{12}pm$
$1Mm=10^{18}pm$
2. Correct option: (D)
$1nm=10^{-9}m$
$1m=10^{10}A°$
So, $1nm=10^{-9}\times 10^{10}A°$
1 nm =10 A°
3. Correct option: (A)
1 fermi or femtometer (FM) = $10^{-15}m$
These units of length are commonly used for measuring very small distances in scientific work.
4. Correct option: (B)
1 ton = 1000 kg
5. Correct option: (A)
$1\ m = 10^{9}\ nm$
Or
$1\ nm = 10^{-9}m$
(nm) is the symbol for nanometer. It is a smaller unit than meter.

6. Correct option: (B)

1 km = 1000 m

Meter (m) is smaller unit of length as compared to km

7.Correct option:(B)

(ng) is the symbol for nanogram.

1 metric ton is also known as 1 ton

1 metric ton = 1000 kg

1Kg = 1000g

And

1g= 10^9 ng

So,

1 metric ton = 1000 x 1000 x 10^9 ng

$=10^{15}$ ng

8. Correct option: (D)

$1\mu m=10^{-6}m$

$1m=10^{10}A°$

$1\mu m=10^{-6} \times 10^{10}A°$

$1\mu m=10^4 A°$

9. Correct options:(C, D)

$1mm=10^{-3}m$

$1m=10^2cm$

So, $1mm=10^{-3} \times 10^2cm$

$1mm=10^{-1}cm$

$=1/10$ cm

1mm =0.1cm

10.Correct options:(A), (B), (D)

$1mg=10^{-3}g$--------(1)

$1g=10^6\mu g$

Putting in equation 1

$1mg=10^{-3} \times 10^6\mu g$

$1mg=10^3 \mu g$

Also

1g= 10^9 ng

Putting in equation 1

$1mg=10^{-3}$ x 10^9 ng

1mg =10^6 ng

11. Correct options:(A, B)

1 metric ton is also known as 1 ton

1 metric ton= 10 quintal

And we know

1 quintal = 100 kg

So,

1 metric ton =10×100 kg

= 1000 kg

12.Correct options:(B, C)

1 decagram= 10 g -------(1)

$1g = 10^{-3}kg$

Putting in equation 1

1 decagram= 10 x 10^{-3}kg

1 decagram =10^{-2}kg

13. Correct options: (A), (B), (C), (D)

1μ g =10^{-6} g

5μ g =5×10^{-6} g----(1)

as, $1g = 10^{-3}kg$

So, from equation 1

5μ g =$5\times10^{-6}\times10^{-3}$kg

=5×10^{-9} kg - - - - (2)

Also,

$1g = 10^{3}mg$

So, from equation 1

5μ g =$5\times10^{-6}\times10^{3}$mg

=5×10^{-3} mg

Also, $1kg = 10^{-3}$tonne

So, from equation 2

5μ g = $5\times10^{-9}\times10^{-3}$tonne

= 5×10^{-12}tonne

14.Correct option:(A)

Shake is an informal metric unit of time equal to 10^{-8} seconds.

15. Correct options: (A), (B), (D)

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.

1 day = 24 hours

1 hour = 60 minutes

1minute = 60 seconds

So,

1 day =24×60 minutes

=1440 min

Hence, 1 day = 24×60×60 seconds

= 86,400 seconds

16.Correct option:(D)

1 fermi or femtometer (FM) = 10^{-15}m

These units of length are commonly used for measuring very small distances in scientific work.

17.Correct option:(B)

Diameter of hydrogen atom in scientific notation is 1.06×10^{-10} m

Diameter of hydrogen atom in microcosm is 0.000,000,000,106 m

18.Correct option:(A)

Diameter of sun in scientific notation is 1.39×10^{9} m

Diameter of in macrocosm is 1,390,000,000m

19. Correct options: (B)

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.

1 mean solar day = 24 hours

$\frac{1}{24}$th part of the mean solar day = 1 hour

20. Correct Matching is: a-T; b-P; c-Q; d-R

larger measurements, we use Some prefixes with the units known as multiples. For example $1\ km = 10^3 m$

Submultiples: For expressing smaller measurements, we use Some prefixes with the units known as submultiples. For example

1 picometre $= 10^{-12} m$

1 femtometre $= 10^{-15} m$

1 nanometre $= 10^{-9} m$

Length and Time
LEVEL-I

1.Correct option:(B)

Mass of a body is the quantity of matter it contains.

2.Correct option:(A)

The kilogram is the base unit of mass in the International System of Units (SI). Its unit symbol is (kg).

3. Correct option:(A)

Scalar quantities: These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km. similarly mass is a scalar quantity as it has magnitude only and it does not need any other information to specify it.

4. Correct option:(A)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence,

C.G.S unit of length is centimeter.

5. Correct option:(A)

Scalar quantities: These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.

Similarly, mass is a scalar quantity as it has magnitude only and it does not need any other information to specify it.

Vector quantities are those quantities which have both magnitude(size) and direction.

Velocity, displacement and force are **vector quantities** because they have both magnitude and they act in a particular direction.

6. Correct option:(B)

As the gap between two points is measured in units of metres.so, it can measure as length because SI unit of length is meter.

7. Correct option:(B)

The meter is the base unit of length in the International System of Units (SI). Its unit symbol is (m).

8. Correct option:(A)

Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. There are seven fundamental quantities: Mass, length, Time, Temperature, Electric current, Luminous Intensity, Quantity of matter. These quantities can be measured independently.

9. Correct option:(D)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence, C.G.S unit of length is centimeter.

10.Correct option:(C)

1 km = 1000 m

Meter (m) is smaller unit of length as compared to km

1m=100cm

Centimeter (cm) is smaller unit of length as compared to km and m

1cm=10 mm

millimeter(mm) is smaller unit of length as compared to km, m and cm

Meter (m) is smaller unit of length as compared to km

11. Correct option:(D)

1 km = 1000 m

Meter (m) is smaller unit of length as compared to km

12. Correct option:(C)

Average distance of the center of the sun from the center of the earth is

1.496×10^{11}m(1 Asronomical unit)

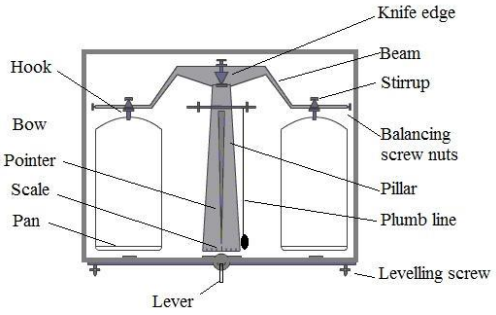
Which is approximately = 149 million km

13.Correct option:(A)

The gravitational mass of a body can be measured in particular with respect to a standard known physical unit by using a physical balance.

(When a body is under the effect of gravity in absence of motion, the mass of the body measured is called gravitational mass.)

The body to be measured is placed on the left pan and the standard mass on the right pan. The mass is adjusted till the beam becomes horizontal, when it is so, the gravitational mass



of the body will be equal to the standard known mass.

14. Correct option:(A)

A spring balance is used to measure the weight of a body.

(Weight is defined as the force exerted on a body by gravity).



Its working is based on the **Hooke's law** in elasticity which states that the extension produced in a spring is directly proportional to the force applied, provided the spring is not loaded beyond the elastic limit.

15. Correct option:(B)

- Mass is the quantity of matter contained by a body. It is a constant quantity and hence cannot be zero for any material body.
- Weight is defined as the force exerted on a body by gravity.

Weight is often expressed as

W= mg

Where, w= weight of the body

m = mass of the body

g= acceleration due to gravity

weight can be zero where there is no gravity(g=0) like in space.

16.Correct option:(B)

The mole is SI unit of amount of substance/quantity of matter.

It has the symbol(mol).

Amount of substance (n) = $\frac{\text{Given weight (g)}}{\text{molecular weight (gmol}^{-1})}$

17. Correct option:(C), (D)

Time can be measured in seconds, minutes, hours, days, weeks, months, and years with clocks and calendars.

18. Correct option:(A)

Scalar quantities: These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.

Similarly, time is a scalar quantity as it has magnitude only and it does not need any other information to specify it.

19. Correct option:(B)

1shake = 10^{-8}seconds

20. Correct option:(D)

Time = $\frac{\text{distance}}{\text{speed}}$

Average distance of the center of the sun from the center of the earth is 1.496×10^{11}m(1 Asronomical unit) or approximately 1.5×10^{11}m

Speed of light = 3×10^{8}m/s

Time = $\frac{1.5 \times 10^{11}\text{m}}{3 \times 10^{8}\text{m/s}}$

Time = $\frac{15 \times 10^{11}}{3x\ 10 \times 10^{8}}$ s

Time = 500s

So, the time taken by the light rays to reach the ground from the sun= 500 sec

21. Correct option:(C)

1 picometre = 10^{-12}m

1Megametre=10^{6}m

1 micrometre = 10^{-6}m

22. Correct option:(B)

1milligram=10^{-3}gram

1 centigram = 10^{-2}gram

1Decagram = 10 gram

1 hectogram = 100 gram

23. Correct option:(A)

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.

1 mean solar day = 24 hours= 86400 seconds

$\frac{1}{86400}$th part of the mean solar day = 1 second

24. Correct option:(A)

1g= 10^3mg

1 mg =10^{-3} g

25. Correct option:(D)

Earth revolves around the sun and completes 1 revolution in 365 days, 5 hours, 59 minutes and 16 seconds.

This time is taken as equal to 1 year.

26. Correct option: (B)

1 millisecond = 10^{-3}seconds

1microsecond = 10^{-6}seconds

1 nanosecond = 10^{-9}seconds

1shake = 10^{-8}seconds

27. Correct option:(C)

There are **7** months in a year having 31 days. These are January, March, May, July, August, October and December.

28. Correct option:(B)

There are 28 days in the month of February (in normal year).

29. Correct option:(A)

The time between two successive passages of mean sun across the meridian at noon is called mean solar day. Mean solar day is equal to 24 hours.

1 mean solar day = 24 hours= 24x60 minutes= 1440 minutes

$\frac{1}{1440}$th part of the mean solar day = 1 minute

30. Correct option:(C)

Interval between two events is called Duration/Time.

LEVEL-II

1. Correct option:(A)

The wall clock has a least count (minimum measurable value) of 1 second. It cannot measure a time below one second.

2. Correct option:(D)

length of meter scale = 1 m

as the scale is divided into 500 equal parts

so,

500 parts = 1m

1 part = $\frac{1}{500}$m

1 part = 0.002 m

1m = 1000 mm

So, 1part = 0.002 x 1000mm

= $\frac{2}{1000}$x 1000 mm

1 part = 2mm

Hence, smallest measurement that can be measured by scale is 2 mm.

3. Correct option:(B)
Given:
On atlas
10cm= 15000km
Or
1 cm = $\frac{15000}{10}$km
1cm = 1500km----------(1)
Actual distance between Bhopal and Kochi = 1500km
From equation (1)
Distance between Bhopal and Kochi on atlas = 1cm

4. Correct option:(A)
A lunar month is the duration between successive new moons, it has a mean period of 29.53059 days (29 days 12 hours and 44 minutes).

5.Correct options:(A, B, C)
Mass can be measured in kilograms, grams, milligrams etc.

6. Correct options:(C, D)
Length can be measured in meter, kilometer etc.
Acre and hectare are units of area.

7. Correct option:(A, B, C, D)
Time can be measured in seconds, minutes, hours, days, weeks, months, and years with clocks and calendars.

8. Correct option:(A, B, C)
Scalar quantities: These have a size or magnitude only and need no other information to specify them.
For example, distance/Length between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.
Similarly, time and mass are also scalar quantities as they have magnitude only and they do not need any other information to specify them.

9.Correct option:(C)
Mass of earth = 5.972×10^{24} kg

10. Correct option:(A)
Distance between mars to earth is approximately 54.6 million km

11. Correct option:(A)
1 calendar year=365 days
1 day = 24 hours
1 hour = 60 minutes
1 minute=60 seconds
So,
1 calendar year = 365 x 24 x 60 x 60 seconds
1 calendar year = 365 x 24 x 60 x 60 seconds
1 calendar year =31,536,000 seconds

12. Correct option is: a-RT; b-ST; c-QT; d-P
Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities. For example, quantities Length, time and mass are measured independently and hence they are fundamental physical quantities.

- **SI** unit of mass is kilogram.
- **SI** unit of length is meter.

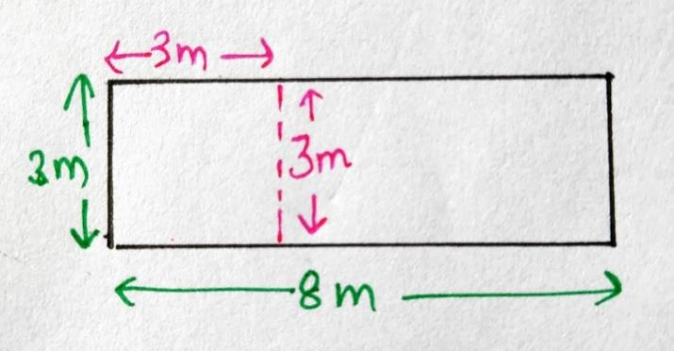
- **SI** unit of time is second.
- Weight is defined as the force exerted on a body by gravity.
  Weight is often expressed as:
  W= mg
  Where, w= weight of the body
  m = mass of the body
  g= acceleration due to gravity

Area

LEVEL-I

1. Correct option:(A)

The Maximum length of the side of a square sheet that can be cut off from a rectangular sheet of size 8m x 3m
is 3m because square has all its four sides equal and for that the maximum possible side length which can be cut is 3m.



2.Correct option:(B)

Given: edge/side of cube =5m
Total surface area of cube = 6 x $(side)^2$
= 6 x $(5m)^2$
= 6 x 25 m^2
Total surface area of cube = 150m^2

3. Correct option:(B)

Area is the amount of surface covered by a two-dimensional shape, measured in square units.
SI unit of area is m^2(square meter)

4. Correct option:(C)

1 km = 10 hectometer
1km x 1km = 10 hectometer x 10 hectometer
1km^2 = 100 hectare
1 sq.km= 100 hectare

5. Correct option:(C)

Area is the amount of surface covered by a two-dimensional shape, measured in square units.
SI unit of area is m^2(square meter).

6.Correct option:(D)

Given: area of the square = 256m^2
To find: perimeter of square
We know

Area of square = $(side)^2$

$256m^2=(side)^2$

$\sqrt{256}$ m = side

16m= side

Or side of square = 16m

Now,

Perimeter of square = 4 x side

= 4 x 16 m

Perimeter of square = 64m

7. Correct option:(D)

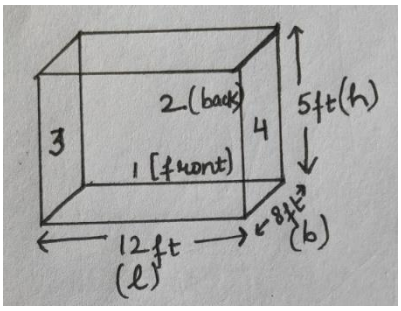
1 km = 1000 m

1km x 1km = 1000 m x 1000 m

$1km^2 = 10^6 m^2$

1 sq.km= $10^6 m^2$

8.Correct option:(A)



Given:

Length of the shop(l) = 12ft

Breadth of the shop(b) = 8ft

Height of the shop(h) = 5ft

Area of the four walls= curved surface area of the cuboid

=area of (wall 1 + 2 +3 +4)

=lh + lh + bh + bh

= 2x (lh + bh)

= 2x (12 x 5 + 8 x 5) ft^2

=2x (60 + 40)ft^2

= 2 x 100 ft^2

=200 ft^2

Hence, area of the four walls is 200 ft^2

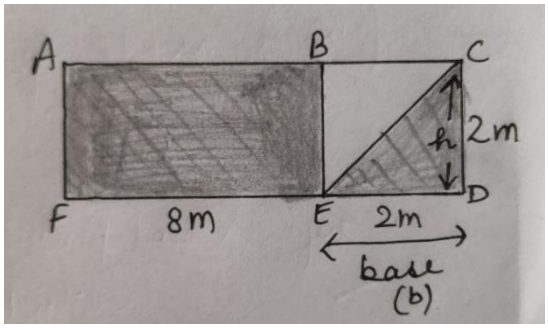
9.Correct option:(C)

In order to find the area of a rose leaf the best option is a **millimeter graph paper.**

A millimeter graph paper has ten squares per centimeter.

It would be suitable to find the area enclosed by the smaller portions of the rose leaf.

10. Correct option:(B)



Area of rectangle ABEF = Length × breadth
= 8m × 2m
= $16m^2$

Area of triangle CDE = $\frac{1}{2}$ base × height
= $\frac{1}{2} \times 2 \times 2$
= $2\ m^2$

Area of shaded region = area of (rectangle ABEF + Triangle CDE)
= (16 + 2) m^2
= 18 m^2

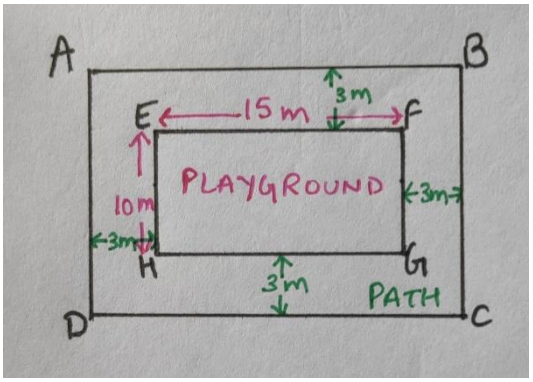
11. Correct option: (B)

Given: radius = 21 cm
And π= 22/7
Area of circle = πr^2
$=\frac{22}{7} \times 21 \times 21$
$=1386\ cm^2$

12. Correct option:(A)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units
Area is the amount of surface covered by a two-dimensional shape, measured in square units.
C.G.S unit of area is cm^2(square centimeter).

13.Correct option:(D)



Let EFGH is the rectangular ground.
ABCD is the boundary of the path around the rectangular field.
Length of rectangular ground =15m
Breadth of the rectangular ground=10m
Area of rectangular ground (EFGH)= length x breadth
= 15m x 10m
= $150m^2$
Width of the path = 3m
Length of rectangle ABCD = 15m+3m+3m
= 21m
Breadth of rectangle ABCD = 10m+3m+3m
= 16m
Area of rectangle ABCD = length x breadth
= 21m x 16m
= $336m^2$
Area of the path= Area of rectangle ABCD- Area of rectangular ground (EFGH)

$=336m^2 - 150m^2$

$=186m^2$

Hence, area of the path = $186m^2$

14.Correct option:(C)

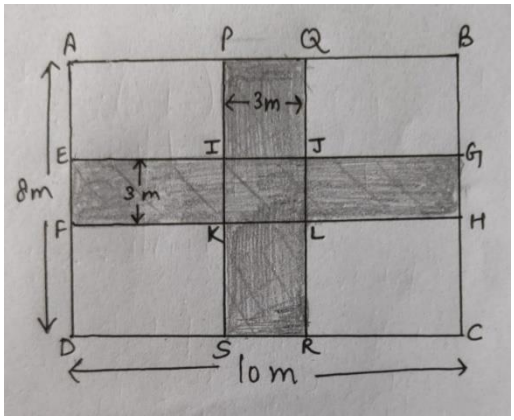
Hectare is a bigger unit of area.

1 hectare = $10000m^2$

The area of a circular cricket ground is approximately 24.64 hectares.

So, hectare is an appropriate unit to measure the area of a cricket ground.

15. Correct option: (D)



Given: length of rectangular field = 10m

Breadth = 8 m

Width of the paths (PQRS and EGHF) = 3 m

Length of Path PQRS =8m

Breadth = 3m

Area of PQRS= length x breadth

= 8m x 3m

= 24 m^2

Length of Path EGHF =10m

Breadth = 3m

Area of EGFH= length x breadth

= 10m x 3m

= $30m^2$

Side of square IJLK= 3m

Area of square IJLK= side x side

= 3m x 3m

= 9 m^2

Total area occupied by the paths = area of (rectangle PQRS + rectangle EGHF – square IJLK)

= (24+30-9) m^2

=$45m^2$

16.Correct option:(A)

The appropriate unit to measure the area of a textbook is cm^2,because the dimensions of a normal textbook are in cm.

km^2 and m^2 would be too large to measure the area of a textbook.

mm^2 would be too small to measure the area of a textbook.

17.Correct option:(B)

Area of rectangle FEBA = Length × breadth

= 12m × 4m

= 48 m^2

Area of triangle EBC = $\frac{1}{2}$ base $\times$ height

$= \frac{1}{2} \times 4 \times 4$

$= 8\ m^2$

Area of shaded region = area of (rectangle FEBA + Triangle EBC)

$= (48 + 8)\ m^2$

$= 56m^2$

18.Correct option:(A)

1 barn=10^{-28} square meters

19.Correct option:(B)

Given:

Radius of the wire = 3.5m

Ratio of length to breadth =6:5

Let the length = 6x

Breadth = 5x

To find: area of the rectangle

As the circular wire is bent to form a rectangle, perimeter remains the same

Perimeter of rectangle = perimeter of circle

2× (length + breadth) = 2πr

2× (6x + 5x) = 2 ×$\frac{22}{7}$ × 3.5m

2×(11x) = 22m

22x = 22m

x = $\frac{22m}{22}$

x = 1m

length = 6x = 6 ×1m=6m

Breadth = 5x =5×1m=5m

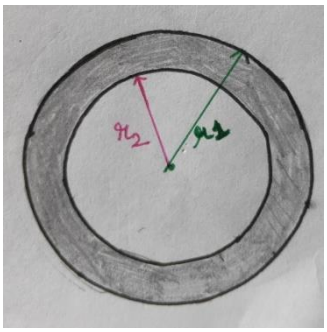
Area of rectangle= length ×breadth

=6m ×5m

=$30m^2$

Hence, area of the rectangle is $30m^2$

20.Correct option:(A)



Given: outer radius of the ring (r_1) =4cm

Inner radius of the ring (r_2)=3cm

Area of ring = $\pi(r_1^2 - r_2^2)$

$=\frac{22}{7}(4^2 - 3^2)cm^2$

$=\frac{22}{7}(16 - 9)cm^2$

$= \frac{22}{7} x\ 7cm^2$

Area of the ring =$22cm^2$

21.Correct option:(A)

Given: ratio of volume of two cubes = 729:1331=$\frac{729}{1331}$----------(1)

To find: ratio of total surface area of two cubes

Let the side of cube 1 = x

The side of cube 2 = y

Volume of cube =$(side)^3$

Volume of 1st cube =$(x)^3$

Volume of 2nd cube =$(y)^3$

ratio of volume of 1st cube to 2nd cube = $\frac{(x)^3}{(y)^3}$ ----------(2)

Equating (1) and (2)

$$\frac{(x)^3}{(y)^3} = \frac{729}{1331}$$

$$\frac{(x)^3}{(y)^3} = \frac{(9)^3}{(11)^3}$$

Cancelling the cubes both side

$\frac{x}{y} = \frac{9}{11}$ ----------(3)

Total surface area of cube =$6 \times (side)^2$

Total surface area of 1st cube =$6 \times (x)^2$

Total surface area of 2nd cube =$6 \times (y)^2$

ratio of total surface area of 1st cube to 2nd cube = $\frac{6(x)^2}{6(y)^2}$

$=\left(\frac{x}{y}\right)^2$----------(4)

Putting value of $\frac{x}{y}$ from equation (3) in equation (4)

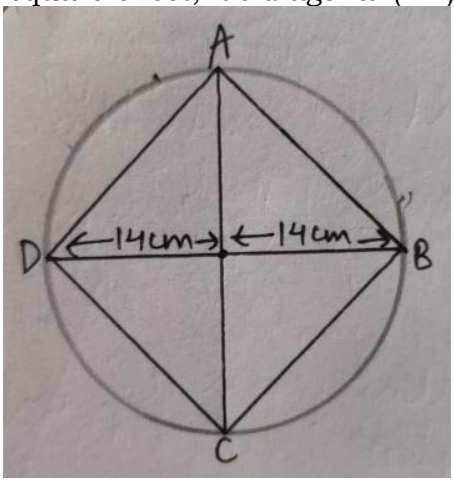
ratio of total surface area of 1st cube to 2nd cube =$\left(\frac{9}{11}\right)^2$

ratio of total surface area of 1st cube to 2nd cube =$\frac{81}{121}$

ratio of total surface area of 1st cube to 2nd cube =81:121

22. Correct option: (A)

To get maximum area of the square sheet, it's diagonal (DB) is equal to the diameter of the



circle

Given: radius of the circle= 14cm

Diameter (DB) = 2r

= 2 x 14 cm

= 28 cm

Diagonal of square (DB)= diameter of the circle

$\sqrt{2}$ x side = 28 cm

Side = $\frac{28}{\sqrt{2}}$ cm

Area of the square ABCD = side x side

$= \frac{28}{\sqrt{2}}$ cm x $\frac{28}{\sqrt{2}}$ cm

$= \frac{28\times28}{2}$ cm^2

$= 392$ cm^2

Area of the circle = πr^2

$= \frac{22}{7} \times 14 \times 14$

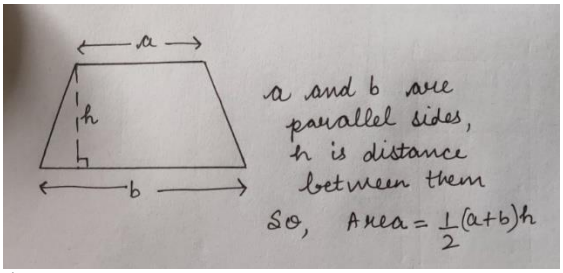
$= 616$ cm^2

Area of remaining cardboard = Area of circle – area of square ABCD

$= 616\text{-}392$ cm^2

$= 224$ cm^2

23. Correct option: (A)



Area of trapezium = $\frac{1}{2}$ × (sum of parallel sides) × (distance between them)

$= \frac{1}{2} \times (a + b) \times (h)$

$= \frac{1}{2} \times (24 + 20) \times (7)$

$= \frac{1}{2} \times (44) \times (7)$

=154 cm^2

Area of trapezium =**154** cm^2

According to question

Area of circle = area of trapezium

$\pi r^2 = 154 cm^2$

$\frac{22}{7} r^2 = 154\ cm^2$

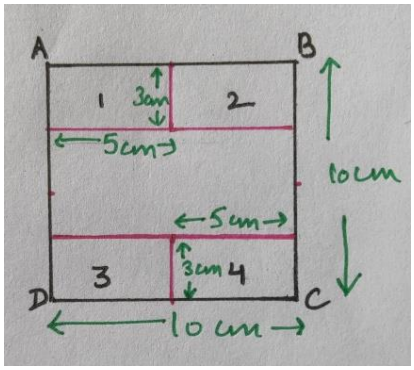
$r^2 = \frac{7}{22} \text{x } 154 cm^2$

$r^2 = 49 cm^2$

$r = \sqrt{49} cm$

r = 7 cm

radius of the circle = 7cm



24. Correct option: (B)

Given: side of the square sheet =10cm

Length of rectangular sheet =5cm

Breadth of rectangular sheet= 3cm

Area of the square sheet (ABCD)= side x side

= 10cm x 10cm

=100cm^2

Area of 1 rectangular piece =length x breadth

=5cm x3cm

=15cm^2

Area of 4 rectangular pieces =4 x15cm^2

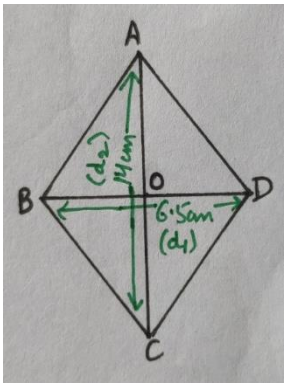
=60 cm^2

Area of aluminum sheet leftover = Area of the square sheet (ABCD)- Area of 4 rectangular pieces

= 100 cm^2-60 cm^2

=40 cm^2

25.Correct option:(A)



Given: diagonal (d_1) = 6.5cm

diagonal (d_2) = 14cm

Area of rhombus = $\frac{1}{2}$ x d_1 x d_2

Area of rhombus = $\frac{1}{2}$ x 6.5cm x 14cm

Area of rhombus = $\frac{1}{2}$ x91 cm^2

Area of rhombus = 45.5 cm^2

As rhombus is divided into 4 equal triangles

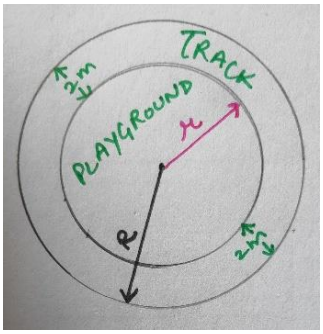
So,

Area of one the 4 triangles (AOD) = $\frac{1}{4}$x Area of rhombus

Area of one the 4 triangles (AOD) = $\frac{1}{4}$x 45.5 cm^2

Area of one the 4 triangles (AOD) = 11.375cm^2

26.Correct option:(C)



Given: circumference of the playground = 132m
Width of the circular track = 2m
To find: area of the circular track

- Calculation for Radius and area of the playground

We know
Circumference of circular track = $2\pi r$

$$132m = 2\pi r \ (\pi = \frac{22}{7})$$

$$132m = 2x\frac{22}{7}x\,r$$

$$132 = \frac{44}{7}r$$

$$\frac{132x7}{44} = r$$

$$21m = r$$

So, radius(r) of the playground = 21m
Area of the playground = πr^2

$$= \pi \times 21 \times 21\ m^2$$

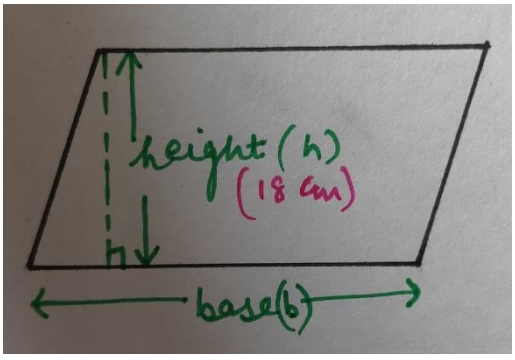
Area of the playground $= 441\pi\ m^2$

- Calculation for area of the circular track

Radius (R)of the playground including track=21m + 2m
= 23m
Area of the playground including track = πR^2

$$= \pi \times 23 \times 23\ m^2$$

Area of the playground including track $= 529\pi m^2$
Area of the circular track = Area of the playground including track- Area of the playground
$= 529\pi m^2 - 441\pi\ m^2$
$= 88\ \pi m^2$
Hence, area of the circular track = $88\pi m^2$



27.Correct option:(D)
Given:
Area of parallelogram $= 144cm^2$
Height=18 cm
To find: length of base
Area of parallelogram = base x height
$144cm^2$ = base x 18cm
$\frac{144}{18}$ cm = base
8cm = base
Hence, length of base is 8 cm.

28. Correct option: (B)

Given: radius = 2.1 m

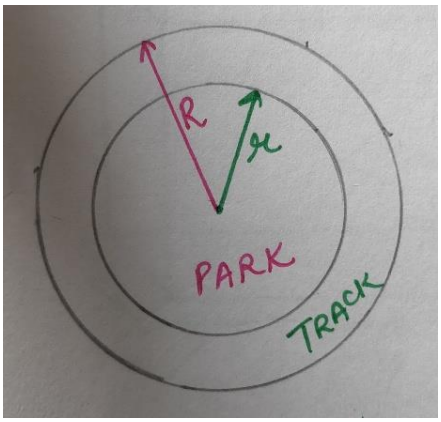
And π= 22/7

Area of circle = πr^2

$$=\frac{22}{7} \times 2.1 \times 2.1$$

$$=13.86 \text{ m}^2$$

29.Correct option:(A)



Given: difference between the circumference of track and the park =132m

Let the radius of the park = r

The radius of the park including the track = R

Width of the track= R-r

Circumference of the park = 2πr

Circumference of the park including the track = 2πR

Difference between circumference of track and park = 2πR-2πr

$$=2\pi(R-r)$$

Now, given difference between the circumference of track and park=132m

So, we get

$$2\pi(R-r) = 132\text{m}$$

$$2 \text{ x} \frac{22}{7}(R-r) = 132\text{m}$$

$$\frac{44}{7}(R-r) = 132\text{m}$$

$$(R-r) = 132 \text{ x } \frac{7}{44}\text{m}$$

$$(R-r) = 132 \text{ x } \frac{7}{44}\text{m}$$

$$(R-r) = 21 \text{ m}$$

So, the width of the track is 21m

30.Correct option:(B)

Given: radius = 14cm

And π= 22/7

Area of semicircle = $\frac{\pi r^2}{2}$

$$=\frac{1}{2} \times \frac{22}{7} \times 14 \times 14$$

$$= 308\text{cm}^2$$

LEVEL-II

1. Correct option: (A)

1hectare = 10000 m^2

50hectare = $50 \times 10000 \text{ m}^2$

$$= 5 \times 10^5 \text{ m}^2$$

2. Correct option: (A)

Area of Triangle = $\frac{1}{2}$ base $\times$ height

$= \frac{1}{2} \times 5 \times 8$

$= 20\ m^2$

3. Correct option: (A)

Given: radius of circle (r) = 200m

Area of circle = $\pi\ r^2$

$= \pi \times 200^2$

$= \pi \times 200 \times 200$

$= 40000\ \pi$

4. Correct option: (C)

Given: side of square = 10 cm

Perimeter of square = 4×side

= 4×10 cm

= 40 cm

5. Correct option: (A)

Given:

radius of circle (r) =7 cm

Area of circle = πr^2

$= \pi \times 7^2$

$= \pi \times 7 \times 7$

$= 49\ \pi cm^2$

6. Correct option: (A)

Given: area = $120 cm^2$

breadth (b) = 6 cm

To find: length (L)

Area of rectangle =length ×breadth

120 = L×6

120/6 = L

20 = L

Or length = 20 cm

7. Correct option: (B)

Given: length (L) = 9 cm

breadth (b) = 6cm

Perimeter = 2(length + breadth)

= 2(9+6)

=2(15)

= 30 cm

8. Correct option: (B)

Given: area = $120\ m^2$

length (L) = 12 m

To find: breadth(b)

Area of rectangle =length ×breadth

120 = 12×b

120/12 = b

10 = b

Or breadth = 10m

9.Correct option:(B)

Given: edge/side of cube =8m

Total surface area of cube = 6 x $(side)^2$

= 6 x $(8m)^2$

= 6 x 64 m^2

Total surface area of cube = 384m^2

10.Correct option:(D)

Given:

Total surface area of the cube= 726m^2

We know,

Total surface area of cube = 6 x $(side)^2$

726m^2 = 6 x $(side)^2$

$\frac{726}{6}$ m^2 = $(side)^2$

121m^2 = $(side)^2$

$\sqrt{121}$ m = side

11m= side

Edge/side of cube = 11m

11.Correct option:(A)

Given:

Perimeter of square = 16cm

To find: area of square

We know

Perimeter of square = 4 x side

16 cm = 4 x side

$\frac{16}{4}$cm = side

4cm=side

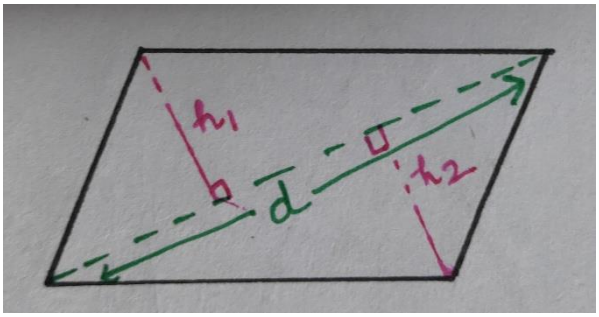
Area of square = $(side)^2$

=$(4cm)^2$

= 16cm^2

Hence, area of square = 16cm^2

12.Correct option:(D)



Given: area of quadrilateral= 420cm^2

Altitude (h_1) = 9cm

Altitude (h_2) = 12cm

Let the diagonal=d

Area of quadrilateral= $\frac{1}{2}$x d x($h_1 + h_2$)

420 = $\frac{1}{2}$x d x (9 + 12)

420 x 2 = d x 21

$840 = d \times 21$

$\frac{840}{21} = d$

$40\ cm = d$

Hence, Length of diagonal= 40 cm

13.Correct options:(A), (D)

In Standard International system (SI), the unit of area is square meter (m^2)

In short form square meter is written as sq m.

Hectare is a bigger unit of area.

1 hectare =$10000m^2$

14. Correct options:(A, B, C)

$1km = 10^3m$

$1km \times 1\ km = 10^3\ m \times 10^3m$

$1km^2 = 10^6m^2$--------(1)

$1m = 10^2cm$

$1m \times 1\ m = 10^2cm \times 10^2cm$

$1m^2 = 10^4cm^2$

Putting in equation 1

$1km^2 = 10^6 \times 10^4\ cm^2$

$1km^2 = 10^{10}cm^2$----------(2)

Also, $1cm = 10\ mm$

$1cm \times 1\ cm = 10mm \times 10\ mm$

$1cm^2 = 10^2mm^2$

Putting in equation 2

$1km^2 = 10^{10} \times 10^2mm^2$

$1km^2 = 10^{12}mm^2$

15. Correct options:(A, D)

$1cm = 10^{-2}m$

$1cm \times 1\ cm = 10^{-2}m \times 10^{-2}m$

$1cm^2 = 10^{-4}m^2$

Or $1cm^2 = 1/10000\ m^2$

16. Correct options:(A, B)

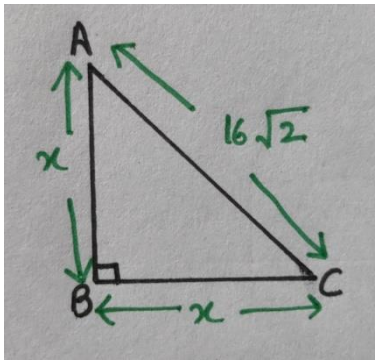
$1m = 10^2cm$

$1m \times 1\ m = 10^2cm \times 10^2cm$

$1m^2 = 10^4cm^2$

Or $1m^2 = 10000\ cm^2$

17.Correct option:(A)



- **To find the base and height of triangle**

Triangle ABC is a right-angled isosceles triangle

Given: Hypotenuse (AC)=$16\sqrt{2}$cm

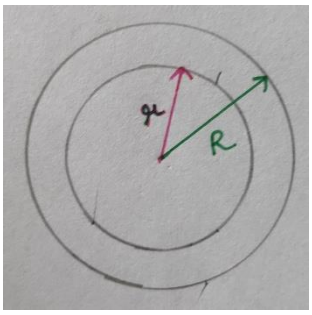
Let Base (BC) = Perpendicular (AB)= x cm
Using Pythagoras theorem in triangle ABC
$(\text{hypotenuse})^2 = (\text{perpendicular})^2 + (\text{base})^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$\left(16\sqrt{2}\right)^2 = x^2 + x^2$
$16 \times 16 \times \sqrt{2} \times \sqrt{2} = 2x^2$
$512 = 2x^2$
$\frac{512}{2} = x^2$
$256 = x^2$
$\sqrt{256} = x$
$16 = x$

Hence,
Base =height =16 cm

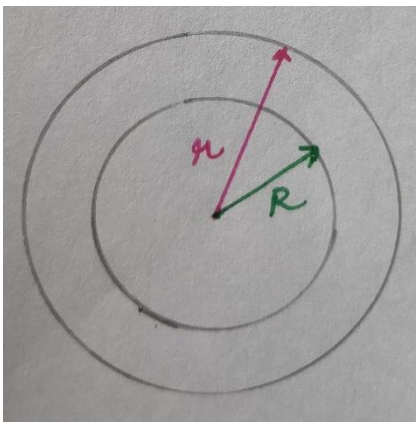
- **To find the area of the triangle**

Area of triangle = $\frac{1}{2}$x base x height
= $\frac{1}{2}$x 16cm x 16cm
Area of triangle =$128cm^2$

18.Correct option:(A), (D)

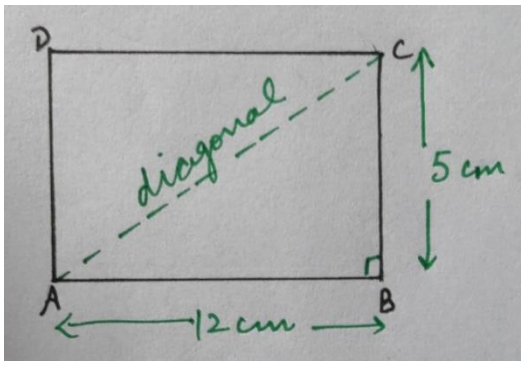


If,
outer radius of the ring= (R)
Inner radius of the ring = (r)
Area of ring = $\pi(R^2 - r^2)$



If,
outer radius of the ring= (r)
Inner radius of the ring = (R)
Area of ring = $\pi(r^2 - R^2)$

19.Correct option:(D)



Triangle ABC is a right-angled triangle
AB = base = 12cm
BC=perpendicular=5cm
AC= hypotenuse/diagonal
Using Pythagoras theorem in triangle ABC
$(\text{hypotenuse})^2 = (\text{base})^2 + (\text{perpendicular})^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$(AC)^2 = (12)^2 + (5)^2$
$(AC)^2 = 144 + 25$
$(AC)^2 = 169$
$AC = \sqrt{169}$
AC= 13cm
So, length of diagonal = 13cm

20.Correct option:(B)

Given:
Breadth(b) = 3cm
length (l) = 8cm
To find: area of rectangle
Area of rectangular sheet =length ×breadth
= 8cm x 3cm
Area of rectangular sheet = 24 cm^2

21.Correct option:(A)

Given: area = $15m^2$
breadth(b)=3m
To find: length (l)
Area of rectangle =length ×breadth
15 = l × 3
15/3 = l
5m = l
Or
length of rectangular plot = 5m

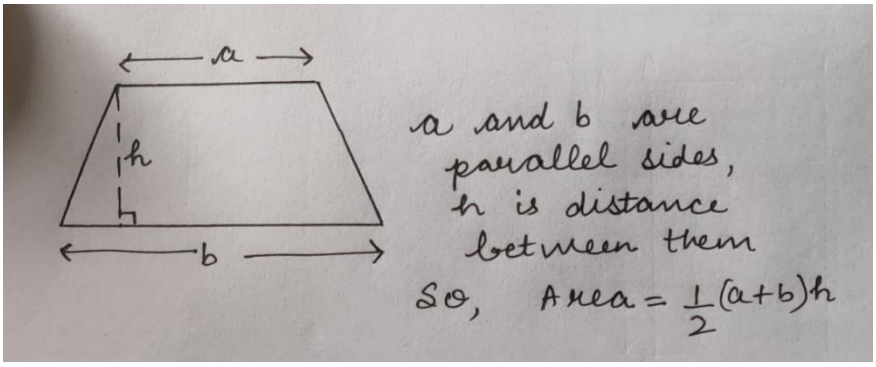
22. Correct option:(B)

Given: area = 120 m^2
length (L) = 12 m
To find: breadth(b)
Area of rectangle =length ×breadth
120 = 12×b
120/12 = b
10m = b

Or
breadth = 10m
1m=100cm
10m= 1000cm
So, breadth of rectangular table = 1000cm

23. Correct Matching is: a-Q; b-S; c-R; d-P

(a) Area of trapezium $=\frac{1}{2}\times$ (sum of parallel sides) $\times$ (distance between them)



(b) Given: diagonal of square = $\sqrt{8}$m

$= 2\sqrt{2}$m

$= \sqrt{2}\times 2$ m-------(1)

Also,

diagonal of square = $\sqrt{2}\times$ side ----------(2)

comparing equation 1 and 2

we get

$\sqrt{2}\times 2 = \sqrt{2}\times$ side

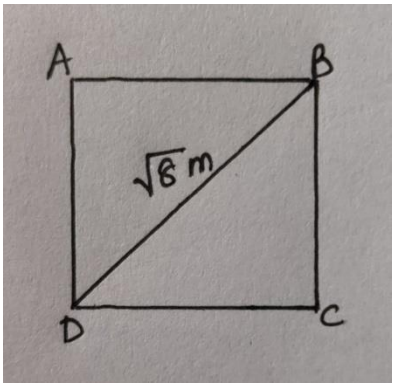
Or

Side =2 m

Area of square = side x side

= 2 m x 2 m

= 4 m^2



(c) Total surface area of cube = 6 x $side^2$

=6 x $\sqrt{10^2}$

= 6 x $\sqrt{10}$cm x $\sqrt{10}$cm

= 60 cm^2

(d) we know

diagonal of square = $\sqrt{2}\times$ side

= $\sqrt{2}\times$ 10 cm

=$10\sqrt{2}$ cm

24. Correct Matching is: a-R; b-T; c-Q; d-P; e-S

(a) area of circle = πr^2

(b) Total surface area of sphere = $4\pi r^2$

(c) Area of a ring= $\pi(R^2 - r^2)$

(d) Total surface area of cube= $6r^2$ (r= side of the cube)

(e) Total surface area of cylinder= $2\pi r(r + h)$ (r= radius of the cylinder

(h= height of the cylinder)

Volume

LEVEL-I

1.Correct option:(A)

Volume is the amount of space occupied by a three-dimensional shape, measured in cubic units.

SI unit of volume is m^3(cubic meter)

2. Correct option:(A)

Volume is the amount of space occupied by a three-dimensional shape, measured in cubic units.

SI unit of volume is m^3(cubic meter)

3. Correct option:(B)

$1m = 10^2cm$

$1m \times 1 m \times 1m = 10^2cm \times 10^2cm \times 10^2cm$

$1m^3 = 10^6cm^3$

4. Correct option:(B)

$1cm = 10mm$

$1cm \times 1 cm \times 1cm = 10mm \times 10mm \times 10mm$

$1cm^3 = 10^3mm^3$

5. Correct option:(C)

Volume of cuboid = length x width x height

6. Correct option:(C)

Volume of block of wood/cuboid = length x breadth x height

=20 cm x 10 cm x 8 cm

$= 1600cm^3$

7. Correct option:(D)

Volume of block of wood/cuboid = length x breadth x height

=30 cm x 15cm x 10 cm

$= 4500cm^3$

8. Correct option:(B)

Tank is in the shape of cuboid

Given:

Length = 1.4m

Width =1m

Height/depth=0.7m

Volume of the tank= length x width x height

=1.4 m x 1m x 0.7m

$= 0.98m^3$

$1\ m^3 = 1000$ litres

$0.98\ m^3 = 0.98 \times 1000$ litres

$= \frac{98}{100}$x1000 litres

0.98 m^3 = 980 litres

Volume of tank = 980 liters

9. Correct option:(A)

Given: radius = 3 cm

Height= 4cm

And $\pi = \frac{22}{7}$

Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 4$

$= 37.7 cm^3$

10. Correct option:(B)

1 liter= 1000 ml

We know 1ml=1 cm^3

So, 1litre = 1000 cm^3

(1 cm^3=10^{-6} m^3)

1litre= 1000 x 10^{-6} m^3

1 liter =10^{-3} m^3

0r

1 liter = $\frac{1}{1000}$ m^3

1 m^3 = 1000 litres

11.Correct option:(A)

Let the length of the cuboid = x units

According to question

Breadth= $\frac{x}{2}$ units

Height = 2x units

Volume of cuboid = length ×breadth × height

= x units × $\frac{x}{2}$ units× 2x units

Volume of cuboid = **x^3 cubic units**

12.Correct option:(B)

Given:

Length = 2m= 200cm (1m=100cm)

breadth =10cm

Height=1mm=0.1cm (1mm=0.1cm)

Volume of the cuboid= length x breadth x height

=200cm x 10cm x 0.1cm

= $200 cm^3$

Volume of the cuboid=$200 cm^3$

13. Correct option:(C)

Given:

Length = 1.8m= 1.8 X 100cm = 180cm (1m=100cm)

Breadth/width =9cm

Height=60cm

Volume of the box/cuboid= length x breadth x height

=180cm x 9cm x 60cm

= $97200 cm^3$

Volume of the cuboid=$97200 cm^3$

14. Correct option:(D)

Given:

Length = 6cm

breadth =4.5cm

Height=40mm= 40x0.1cm =4cm (1mm=0.1cm)

Volume of the soap box/cuboid= length x breadth x height

=6cm x 4.5cm x 4cm

= $108cm^3$

Volume of the soap box=$108cm^3$

15.Correct option:(B)

Given: radius = 4cm

Height= 21mm = 21 x0.1 cm = 2.1 cm (1mm=0.1cm)

And $\pi = \frac{22}{7}$

Volume of cone =$\frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 4 \times 4 \times 2.1$

= $35.2cm^3$

Volume of cone = $35.2cm^3$

16.Correct option:(B)

Length of the cuboid(L)= 21cm

Breadth of the cuboid(b)=9cm

Height of the cuboid(h)=8cm

Volume of the wooden block/cuboid= length x breadth x height

=21cm x 9cm x 8cm

= 1512 cm^3

Volume of the wooden block =1512 cm^3

Number of cubes that can be cut along the length = $\frac{21}{3} = 7$

Number of cubes that can be cut along the breadth = $\frac{9}{3} = 3$

Number of cubes that can be cut along the height = $\frac{8}{3} = 2.6$

As the number of cubes cut along height is not a natural number so we can just cut 2 cubes along the height and rest of the volume is wasted.

Total number of cubes cut = 7 x 3 x 2

= 42

Volume of each cube = $side^3$

= $(3cm)^3$

Volume of each cube =$27cm^3$

Volume of all 42 cubes =27 x 42 cm^3

=$1134cm^3$

Volume of the waste wood = volume of wooden block – volume of all the 42 cut cubes

= 1512 cm^3-$1134cm^3$

=$378cm^3$

17.Correct option:(D)

Given: ratio of the sides of cubes= 2:5

To find: ratio of volumes

let the side of first cube= 2x

the side of second cube =5x

Volume of cube = $side^3$

Volume of first cube = $(2x)^3$
Volume of first cube = $2\times2\times2\ x^3$
Volume of first cube = $8x^3$
Volume of second cube = $(5x)^3$
Volume of second cube = $5\times5\times5\ x^3$
Volume of second cube = $125x^3$

Ratio of volume of first cube to second cube = $\frac{\text{Volume of first cube}}{\text{volume of second cube}}$

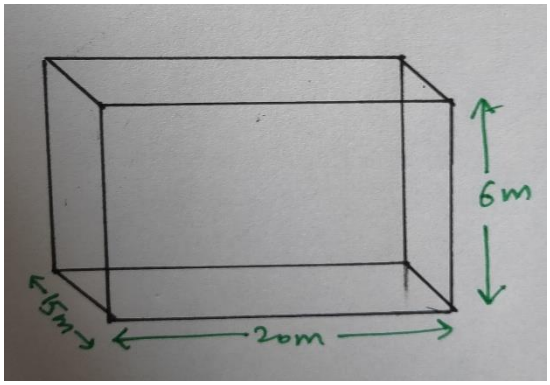
$= \frac{8x^3}{125x^3}$

$= \frac{8}{125}$

= 8:125

Hence, Ratio of volume of first cube to second cube=8:125

18.Correct option:(A)



Given:
Total population of village =4000
water required by one person=150 liters

Dimensions of the tank:

Length =20m
Breadth=15m
Height=6m
To find: number of days for which water is sufficient

Number of days for which the water will be sufficient = $\frac{\text{volume of the tank}}{\text{total water required per day}}$----------(1)

- Calculation for volume of the tank

Volume of the tank/cuboid = length x breadth x height
=20m x 15m x 6m
= $1800m^3$

Volume of the tank=$1800m^3$

- Calculation for water required per day

water required by 1 person=150 liters
water required by 4000 persons per day=150x4000 liters
=600000litres

1 liter = $\frac{1}{1000}$ m^3

600000 liters = $600000x\frac{1}{1000}$ m^3
=$600m^3$

water required by 4000 persons per day=$600m^3$

Total water required by 4000 persons per day=$600m^3$

From equation (1)

Number of days for which the water will be sufficient = $\frac{\text{volume of the tank}}{\text{total water required per day}}$

Number of days for which the water will be sufficient = $\frac{1800m^3}{600m^3}$

Number of days for which the water will be sufficient = 3 days

19.Correct option:(D)

Given:

Length = 8.2m

breadth =5.3m

Height=2.6m

Volume of cuboid= length x breadth x height

=8.2m x 5.3m x 2.6m

= $112.996m^3$

Volume of the cuboid=$112.996m^3$

20.Correct option:(C)

Given:

Length of cuboid = 12m

breadth of cuboid=6m

Height of cuboid=3m

To find: side of the cube

When cuboid is recast into cube its volume remains same

Volume of cube= volume of cuboid

$Side^3$ = length x breadth x height

$Side^3 = 12m \times 6m \times 3m$

$Side^3 = 216m^3$

$side = \sqrt[3]{216}\ m$

side= 6m

so,

side of the cube is 6m

21. Correct option:(B)

Volume of cuboid = length x breadth x height

=24 cm x 12cm x 5 cm

= $1440cm^3$

22. Correct option: (B)

Given: radius of the cylinder = 5.5 cm

Height of the cylinder = 14 cm

Side of cube = l cm

Also π= 22/7

When cylinder is recast into cube its volume remains same

Volume of cube = volume of cylinder

$Side^3 = \pi r^2 h$

$l^3 = \frac{22}{7} \times 5.5^2 \times 14$

$l^3 = \frac{22}{7} \times 5.5 \times 5.5 \times 14$

$l^3 = 1331\ cm^3$

$l = 11cm$

23.Correct option:(C)

Given:

Area of the square =$100\ cm^2$

$(side)^2 = 100\ cm^2$

$(side)^2 = (10cm)^2$
Side of square = 10cm
According to the question
height of cuboid = side of the square = 10cm
length of cuboid = height + 2cm= 10cm+2cm = 12cm
breadth of cuboid = height -2cm= 10cm-2cm = 8cm
Volume of cuboid= length x breadth x height
=12cm x 8cm x 10cm
$= 960cm^3$
Volume of the cuboid=$960cm^3$
24.Correct option:(A)
Given:
Length = 6m
Breadth/width =3m
Height/depth=2m
Capacity of tank means volume of the tank
Volume of water tank/cuboid= length x breadth x height
=6m x 3m x 2m
$= 36m^3$
Volume of the water tank=$36m^3$
So,
capacity of the water tank =$36m^3$
25.Correct option:(A)
Given: radius of the cone= 22m
Height of the cone = 21m
Side of cube = a
Also π= 22/7
When cone is recast into cube its volume remains same
Volume of cube = volume of cone

$$Side^3 = \frac{1}{3}\pi r^2 h$$

$$a^3 = \frac{1}{3} x \frac{22}{7} \times 22^2 \times 21$$

$$a^3 = \frac{1}{3} x \frac{22}{7} \times 22 \times 22 \times 21 \quad m^3$$

$$a^3 = 10648\ m^3$$

$$a = \sqrt[3]{10648}\ m$$

a= 22m
26. Correct option:(B)
Given: volume of cube= $512\ cm^3$
To find: length of edge/side of cube
Also Volume of cube = $side^3$
$512cm^3 = (side)^3$
$\sqrt[3]{512}\ cm$ = side
8cm=side/edge
512= 2x2x2x2x2x2x2x2x2
$512 = 2^3 \times 2^3 \times 2^3$
Hence, Length of edge of cube=8cm
$512 = (2 \times 2 \times 2)^3$
$\sqrt[3]{512} = 8$

27. Correct option:(B)

Given: radius = 3 cm

Height= 7cm

And $\pi = \frac{22}{7}$

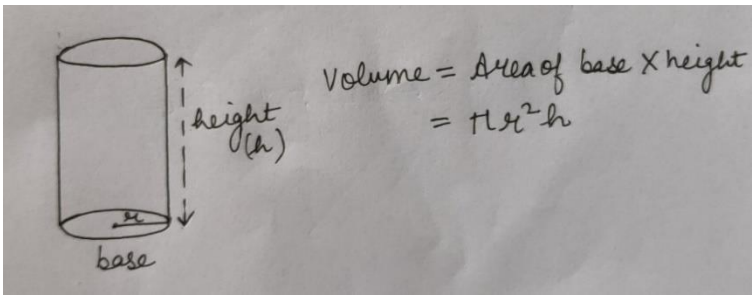
Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 7$

$= 66 cm^3$

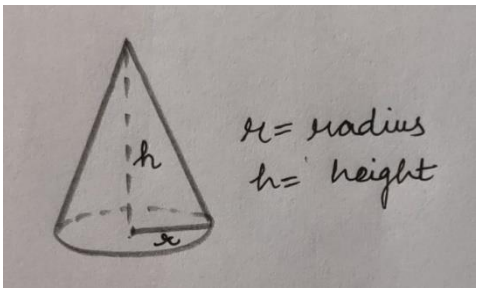
28. Correct option:(C)

Volume of cylinder $= \pi r^2 h$



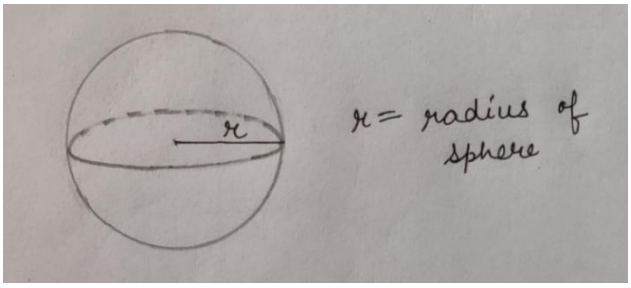
29. Correct option:(A)

Volume of cone = $\frac{1}{3}\pi r^2 h$



30. Correct option:(C)

Volume of sphere = $\frac{4}{3}\pi r^3$



LEVEL-II

1. Correct option: (C)

(A) A meter scale is a one-meter-long scale used to measure length.

(B) Graph paper or coordinate paper is a writing paper printed with fine lines making up a regular grid. These lines help in plotting graphs.

(C) A measuring jar is used primarily to measure the volume of liquids. A measuring jar has scale markings at varying heights used to measure fixed volume of liquids.

(D) A physical balance is used to measure mass of a body.

2. Correct option: (B)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

Hence,

C.G.S unit of volume is cm^3

3. Correct option: (B)

1 ml = 1 cm^3

The reason for this is, volume of liquid that is present in 1 ml of a liquid is equal to the volume of water that fits in a unit cube of dimensions (1 cm length x 1 cm breadth x 1 cm height)

4. Correct option: (B)

Volume is the amount of space occupied by any liquid.

5. Correct option: (B)

Length (l)=10m

Breadth(b)=7m

Height(h)=5m

Total surface area of cuboid = 2(lb+bh+hl)

=2(10x7+7x5+5x10)

=2(70+35+50)

=2(155)

= 310 m^2

6. Correct option: (D)

(A) A measuring jar is used primarily to measure the volume of liquids. A measuring jar has scale markings at varying heights used to measure fixed volume of liquids.

(B) A pipette is a laboratory instrument. It is used to measure out or transfer small quantities of liquid in volumes of milliliters or microliters.

(C) A burette is a graduated glass tube with tap at one end. It is used to deliver known volumes of a liquid. It is usually used in titrations.

7. Correct option:(B)

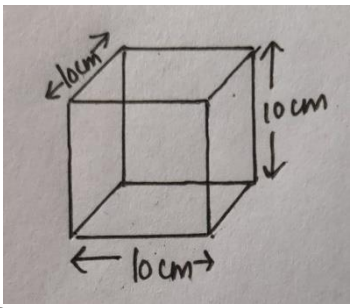
Given: side of the cube = 10 cm

We know

Volume of cube = $side^3$

= $(10cm)^3$

= 10 x 10 x 10 cm^3



=1000 cm^3

8.Correct option:(C)

Volume of cube= $(side)^3$

As same cube is cut into x cubes so

Volume of cube of edge 9cm = x × volume of cube of edge 3cm

$(9cm)^3 = x \times (3cm)^3$

9cm x 9cm x 9cm = x × 3cm x 3cm x 3cm

$729cm^3 = x \times 27cm^3$

$\frac{729}{27}$ **= x**

27 = x

Or x=27

9.Correct option:(A)

Given: volume of cube= 343 cm^3

To find: total surface area of cube

Also

Volume of cube = $side^3$

$343cm^3 = (side)^3$

$\sqrt[3]{343}$ cm = side

7cm=side

$343 = 7 \times 7 \times 7 = (7)^3$

$\sqrt[3]{343} = 7$

Or side of cube = 7cm

Total surface area of cube = 6 x $side^2$

$= 6 \times (7cm)^2$

$= 6 \times 7cm \times 7cm$

$= 294cm^2$

Total surface area of cube =$294cm^2$

10.Correct options:(A), (B), (D)

1 liter= 1000 ml

We know 1ml=1 cm^3

1ml=1cc

So, 1litre = 1000cc

Now,

1litre = 1000 cm^3

(1 $cm^3=10^{-6}$ m^3)

1litre= 1000 x 10^{-6} m^3

1 liter =10^{-3} m^3----------(2)

(1 $m^3=10^9$ mm^3)

1litre=10^{-3}x 10^9 mm^3

1litre= 10^6 mm^3

11. Correct options:(A), (B), (D)

1 liter= 1000 ml

We know 1ml=1 cm^3

Now,

1litre = 1000 cm^3

(1 $cm^3=10^{-6}$ m^3)

1litre= 1000 x 10^{-6} m^3

1 liter =10^{-3} m^3

Or, 1 liter =1/1000 m^3

(1 $m^3=10^9$ mm^3)

1litre=10^{-3}x 10^9 mm^3

1litre= 10^6 mm^3

12. Correct options:(A), (D)

- A measuring jar and measuring flask are kind of vessels used to measure the volume of liquids. They have scale markings at varying heights used to measure fixed volume of liquids.

- A pipette is a type of glass tube. It is used to measure out or transfer small quantities of liquid in volumes of milliliters or microliters.
- A burette is a graduated glass tube with tap at one end. It is used to deliver known volumes of a liquid. It is usually used in titrations.

13.Correctoptions(A), (B), (C), (D)

Volume of a given liquid can be measured in any of the given units.

SI unit of volume is m^3.

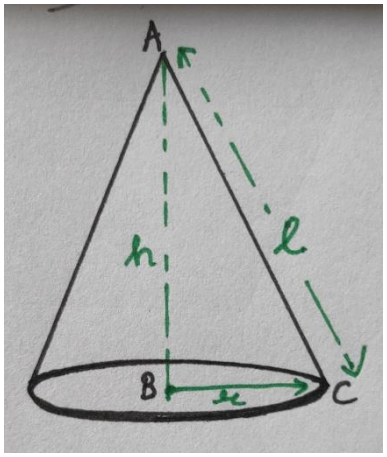
1 m^3 = 1000 litres

1 liter= 1000 ml

1litre = 1000 cm^3

1litre = 1000 cc

14.Correct option:(A)



Given:

Slant height of the cone(l)=17cm

Radius of the cone(r)= 15cm

To find: volume of the cone

- Calculation of height(h) of the cone

Triangle ABC is a right-angled triangle

BC= radius of the base(r) = 15cm

AC= hypotenuse/slant height(l)=17cm

To find: AB=perpendicular/height(h)

Using Pythagoras theorem in triangle ABC

$(\text{hypotenuse})^2 = (\text{perpendicular})^2 + (\text{base})^2$

$(AC)^2 = (AB)^2 + (BC)^2$

$(17)^2 = (AB)^2 + (15)^2$

$(AB)^2 = (17)^2 - (15)^2$

$(AB)^2 = 289 - 225$

$(AB)^2 = 64$

$(AB)^2 = (8)^2$

$AB = 8\ cm$

So, height of the cone = 8cm

- Calculation of volume of the cone

Given: radius = 15cm

Height= 8cm (calculated above)

And $\pi = 3.14$

Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times 3.14 \times 15 \times 15 \times 8$

$= 1884 cm^3$

Hence, volume of the cone = $1884 cm^3$

15. Correct option:(C)

Volume of cuboid = length x breadth x height

=10 cm x 8 cm x 6 cm

= 480 cm^3

16. Correct option:(B)

Given: side of the cube = 10 cm

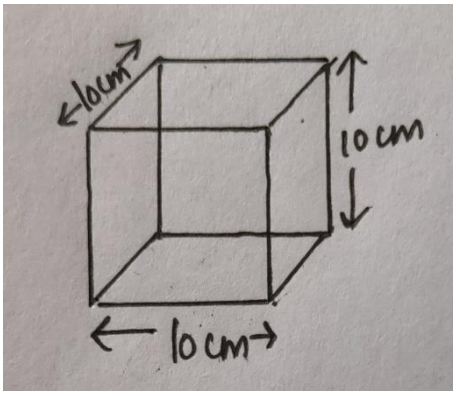
We know

Volume of cube = $side^3$

$= (10cm)^3$

= 10 x 10 x 10 cm^3

=1000 cm^3



17. Correct option:(B)

Volume of cuboid = length x breadth x height

=20cm x 18 cm x 15 cm

= 5400 cm^3

18. Correct option:(D)

Volume of cuboid = length x breadth x height

=30m x 28 m x 12 m

= $10080 m^3$

19. Correct option:(D)

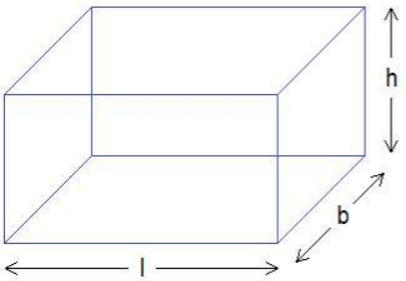
Volume of cuboid = length x breadth x height

=14cm x 10cm x 5 cm

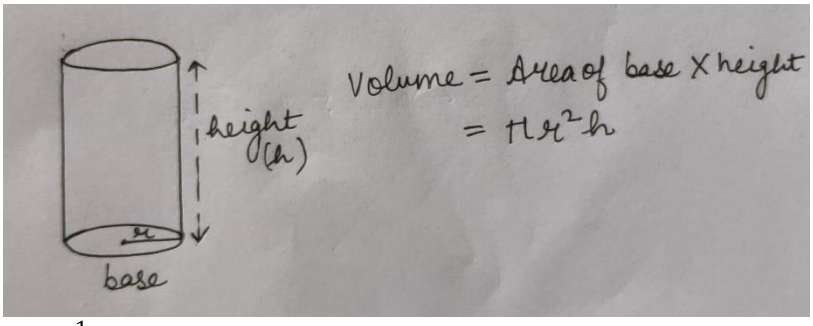
= $700 cm^3$

20. Correct Matching is: a-S; b-R; c-Q; d-P

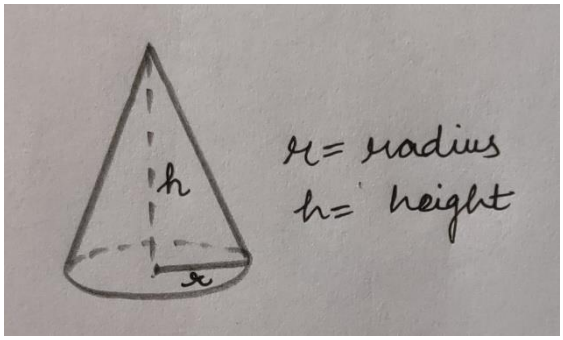
(a) Volume of cuboid = length x breadth x height



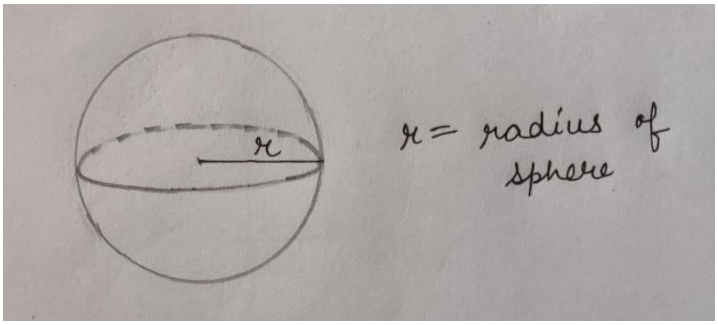
(b) Volume of cylinder $=\pi r^2 h$



(c) Volume of cone $= \frac{1}{3}\pi r^2 h$



(d) Volume of sphere $= \frac{4}{3}\pi r^3$



21. Correct Matching is: a-Q; b-P; c-ST; d-R

(a) Given: side/edge of the cube = 10 cm

We know

Volume of cube = side^3

$= (10\text{cm})^3$

$= 10 \times 10 \times 10\ \text{cm}^3$

$= 1000\ \text{cm}^3$

(b) Volume of cuboid = length x breadth x height

$= 10\text{cm} \times 5\text{cm} \times 2\ \text{cm}$

$= 100\text{cm}^3$

(c) Given: radius of the sphere(r) = 7cm

Volume of sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times 7 \times 7 \times 7\ \text{cm}^3$

Volume of sphere $= 1437.3\text{cm}^3$

$1\ \text{cm}^3 = \frac{1}{1000000}\text{m}^3$

$1437\ \text{cm}^3 = \frac{1437.3}{1000000}\text{m}^3$

$1437\ cm^3 = 0.0014373m^3$

Hence, volume of sphere $=0.0014373m^3$

(d) Given: radius of the hemisphere(r) = 7cm

Volume of hemisphere = $\frac{2}{3}\pi r^3$

$= \frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7\ cm^3$

Volume of sphere = $718.6cm^3$

Density and Relative Density

LEVEL-I

Density:

1.Correct option:(A)

Density of a substance is defined as the mass per unit volume of the substance, i.e.,

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

2. Correct option:(C)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kg

SI unit of volume is m^3

So,

SI unit of density is kg/m^3

3. Correct option:(A)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

C.G.S unit of mass is g

C.G.S unit of volume is cm^3

So,

C.G.S unit of density is g/cm^3

4. Correct option:(A), (B)

$1kg = 1000g$

$1kgm^{-3} = 1000gm^{-3}$

Also,

$1m = 100cm$

$1m^3 = 1000000cm^3$

$1kgm^{-3} = \frac{1000}{1000000} gcm^{-3}$

$1kgm^{-3} = \frac{1}{1000} gcm^{-3}$

5. Correct option:(B)

Density of water at 4°C(277K) is $1.0 \times 10^3 kg/m^3$

6. Correct option:(A)

Density of ice is $9 \times 10^2 kg/m^3$

7. Correct option:(C)

$1kg = 1000g$

Also,

$1m = 100cm$

$1m^3 = 1000000cm^3$

$1\text{kgm}^{-3} = \frac{1000}{1000000}\text{gcm}^{-3}$

$1\text{kgm}^{-3} = \frac{1}{1000}\text{gcm}^{-3}$

Or

$1\text{gcm}^{-3} = 1000\text{kgm}^{-3}$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

8. Correct option:(B)

Given:

Volume of liquid = 500cc

mass of liquid= 1kg=1000g

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{1000}{500}\text{g/cc}$

Density =2 g/cc

9. Correct option:(C)

Given:

Volume of liquid = 18cc

mass of liquid= 7.2kg=7.2 X 1000g= 7200 g (1kg=1000g)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{7200}{18}\text{g/cc}$

Density =400g/cc

10. Correct option:(D)

M.K.S system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

Given: Density =0.75g/cc

We know,

$1\text{cc} = 1\text{cm}^3$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

$1\text{g/cc} = 10^3\text{kg/m}^3$

$0.75\text{g/cc} = 0.75 \text{ x } 10^3\text{kg/m}^3$

$= \frac{75}{100} \text{ x } 1000\text{kg/m}^3$

$= 750\text{kg/m}^3$

So, Density in M.K.S system is 750kg/m^3

11. Correct option:(B)

C.G.S system is based on centimeter, **gram** and second as the fundamental units of length, **mass** and time respectively.

Given: Density =450kg/m^3

$1\text{kg/m}^3 = \frac{1}{1000}\text{g/cc}$

$450\text{kg/m}^3 = 450 \text{ x} \frac{1}{1000}\text{g/cc}$

$450\text{kg/m}^3 = \frac{45}{100}\text{g/cc}$

$450\text{kg/m}^3 = 0.45 \text{ g/cc}$

Density of substance in C.G.S system is 0.45 g/cc

12. Correct option:(C)

Given: mass of substance= 0.6kg=600g (1kg=1000g)

Density =1.2g/cc

To find: Volume of substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Volume} = \frac{\text{mass}}{\text{density}}$

$\text{Volume} = \frac{600}{1.2}\text{cc}$

$= \frac{600 \text{ x } 10}{12}\text{cc}$

volume =500cc

13. Correct option:(A)

Given: Volume of liquid= 2litre=2000cc (1litre = 1000cc)

Density =2g/cc

To find: mass of liquid

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

2g/cc x 2000cc = mass

4000g = mass

Mass = 4000g

14. Correct option:(D)

SI system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

Given: Density $=4\text{gcm}^{-3}$

We know,

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

$4\text{gcm}^{-3} = 4 \text{ x} 10^3\text{kgm}^{-3}$

$4\text{gcm}^{-3} = 4000\text{kgm}^{-3}$

So, Density in SI system is 4000kgm^{-3}

15. Correct option:(B)

Given:

Volume of liquid = 20m^3

mass of liquid= 5kg

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{5}{20}\text{kgm}^{-3}$

Density $=0.25\text{kgm}^{-3}$

16. Correct option:(D)

Given: Volume of liquid= 5litre=5000cc (1litre = 1000cc)

Mass of the liquid= 2 kg=2000g (1kg=1000g)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{2000}{5000}\text{g/cc}$

Density =0.4g/cc

17. Correct option:(D)

Given: Volume of body= 2cm^3

Density = $0.52\ \text{gcm}^{-3}$

To find: mass of body

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

$0.52\ \text{gcm}^{-3}$ x 2cm^3= mass

1.04g= mass

Mass= 1.04 g

$1\text{g} = 10^{-3}\text{kg}$

$1.04\text{g} = 1.04\text{x}10^{-3}\text{kg}$

So, mass of body in SI system is $1.04\text{x}10^{-3}\text{kg}$

Relative density:

18. Correct option:(A)

$\text{Relative density} = \frac{\text{density of substance}}{\text{density of water at 4°C}}$

19. Correct option:(B)

Relative density has no units because it is equal to the ratio of same physical quantity (density).

20. Correct option:(A)

Given: density of substance = 0.45g/cc

$1\text{g/cc} = 10^3\text{kg/m}^3$

$0.45\text{g/cc} = 0.45 \text{ x } 10^3\text{kg/m}^3$

We know,

Density of water at 4°C(277K) is $1.0 \text{ x } 10^3\text{kg/m}^3$

$\text{Relative density} = \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$\text{Relative density} = \frac{0.45 \text{ x } 10^3\text{kg/m}^3}{10^3\text{kg/m}^3}$

Relative density =0.45

21. Correct option:(B)

Given: Relative density =0.8

We know,

Density of water at 4°C(277K) is $1.0 \text{ x } 10^3\text{kg/m}^3$ or 10^3kg/m^3

$1\text{kg/m}^3 = \frac{1}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = \frac{10^3}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = \frac{1000}{1000}\text{g/cc}$

$10^3\text{kg/m}^3 = 1\text{g/cc}$

$\text{Relative density} = \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.8 = \frac{\text{density of substance}}{1\text{g/cc}}$

0.8 x 1g/cc= density of substance

0.8g/cc= density of substance

Density of substance =0.8g/cc

22. Correct option:(C)

Specific gravity/Relative density= $\frac{\text{density of substance}}{\text{density of water at 4°C}}$

- **To find the density of substance**

Given: specific gravity=0.75

We know,

Density of water at 4°C(277K) is $10^3 kg/m^3$ or $1g/cc$

$1kg/m^3 = \frac{1}{1000} g/cc$

$10^3 kg/m^3 = \frac{10^3}{1000} g/cc$

$10^3 kg/m^3 = \frac{1000}{1000} g/cc$

$10^3 kg/m^3 = 1g/cc$

Specific gravity $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.75 = \frac{\text{density of substance}}{1g/cc}$

0.75 x 1g/cc= density of substance

Density of substance = 0.75 g/cc

- **To find the mass of the substance**

Given: Volume of substance= 800cc

Density of substance = 0.75 g/cc (calculated above)

To find: mass of substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

0.75g/cc x 800cc = mass

$\frac{75}{100}$ x 800 g = mass

600g = mass

600g = mass

Mass = 600g

23. Correct option:(A)

- **To find the density of substance**

Given: Relative density =0.2

We know,

Density of water at 4°C(277K) is $1.0 \times 10^3 kg/m^3$ or $10^3 kg/m^3$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.2 = \frac{\text{density of substance}}{10^3 kg/m^3}$

0.2 x $10^3 kg/m^3$= density of substance

0.2 x$10^3 kg/m^3$= density of substance

Density of substance =0.2 x$10^3 kg/m^3$

- **To find the mass of the substance**

Given: Volume of substance= $20m^3$

Density of substance =0.2 x$10^3 kg/m^3$ (calculated above)

To find: mass of substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

0.2 x$10^3 kg/m^3$ x $20m^3$ = mass

$\frac{2}{10}$ x10^3 x 20 kg = mass

2 x 10^3 kg = mass

2000kg = mass

Mass = 2000kg

24. Correct option:(B)

- **To find the density of substance**

Given: Relative density =0.2

We know,

Density of water at 4°C(277K) is $1.0 \times 10^3 kg/m^3$ or $10^3 kg/m^3$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.2 = \frac{\text{density of substance}}{10^3 kg/m^3}$

$0.2 \times 10^3 kg/m^3$= density of substance

$0.2 \times 10^3 kg/m^3$= density of substance

Density of substance $=0.2 \times 10^3 kg/m^3$

- **To find the volume of the substance**

Given: mass of substance= 500kg

Density of substance $=0.2 \times 10^3 kg/m^3$ (calculated above)

To find: volume of substance

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{Volume} = \frac{\text{mass}}{\text{density}}$$

$$\text{Volume} = \frac{500kg}{0.2 \times 10^3 kg/m^3}$$

$$\text{Volume} = \frac{500 \times 10}{2 \times 1000} m^3$$

$$\text{Volume} = 2.5\ m^3$$

25. Correct option:(B)

Given: density of iron= $7.85 gcm^{-3}$

$1 gcm^{-3} = 10^3 kgm^{-3}$

$7.85 gcm^{-3} = 7.85 \times 10^3 kgm^{-3}$

Density of water at 4°C(277K) is $10^3 kgm^{-3}$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density $= \frac{7.85 \times 10^3 kgm^{-3}}{10^3 kgm^{-3}}$

Relative density =7.85

LEVEL-II

1. Correct option:(C)

Density of a substance is defined as the mass per unit volume of the substance, i.e.,

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

2. Correct option:(D)

Specific gravity= $\frac{\text{density of substance}}{\text{density of water at 4°C}}$

3. Correct option:(B)

Specific gravity= $\frac{\text{density of substance}}{\text{density of water at 4°C}}$

- **To find the density of substance**

Given: specific gravity=0.35

We know,

Density of water at 4°C(277K) is 1g/cc

Specific gravity $=\frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.35=\frac{\text{density of substance}}{1\text{g/cc}}$

0.35 x 1g/cc= density of substance

Density of substance = 0.35 g/cc

4. Correct option:(C)

Given:

Volume of liquid = 49m^3

mass of liquid= 7kg

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{7}{49}\text{kgm}^{-3}$

$\text{Density} = 0.14\frac{\text{kg}}{\text{m}^3}$

5. Correct option:(D)

M.K.S system makes use of meter, kilogram and second as the fundamental units of length, mass and time respectively.

Given: Density =0.85g/cc

We know,

$1\text{cc} = 1\text{cm}^3$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

$1\text{g/cc} = 10^3\text{kg/m}^3$

$0.85\text{g/cc} = 0.85 \text{ x } 10^3\text{kg/m}^3$

$= \frac{85}{100} \text{ x } 1000\text{kg/m}^3$

$= 850\text{kg/m}^3$

So, Density in M.K.S system is $850\ \frac{\text{kg}}{\text{m}^3}$

6. Correct option:(A)

Given: mass of iron piece= 7020g

Length = 3cm

Breadth=15cm

Height=20cm

To find: density of the iron piece

Volume of iron piece/cuboid = length x breadth x height

=3cm x 15cm x 20cm

$= 900\text{cm}^3$

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{7020}{900}\text{gcm}^{-3}$

$\text{Density} = 7.8\ \text{gcm}^{-3}$

$1\text{gcm}^{-3} = 10^3\text{kgm}^{-3}$

$7.8\ \text{gcm}^{-3} = 7.8 \text{ x } 10^3\text{kgm}^{-3}$

7. Correct option:(B)

Given: mass of box= 30kg=30x1000g= 30000g (1kg=1000g)

Length = 6cm

Breadth=3cm

Height=12cm

To find: density of the box

Volume of cubical box = length x breadth x height

=6cm x 3cm x 12cm

= 216cm^3

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{30000}{216}\frac{\text{g}}{\text{cm}^3}$

Density =138.8 $\frac{\text{g}}{\text{cm}^3}$

8.Correct options:(A, B, C, D)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kg

SI unit of volume is m^3

So,

SI unit of density is kgm^{-3} or $\frac{\text{kg}}{\text{m}^3}$

C.G.S unit of mass is g

C.G.S unit of volume is cm^3

So,

C.G.S unit of density is gcm^{-3} or $\frac{\text{g}}{\text{cm}^3}$

9. Correct options:(A, B, D)

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kg

SI unit of volume is m^3

So,

SI unit of density is kgm^{-3}

C.G.S unit of mass is g

C.G.S unit of volume is cm^3

So,

C.G.S unit of density is gcm^{-3}

10. Correct options:(A, B)

1kg = 1000g

Also,

1m = 100cm

1m^3 = 1000000cm^3

$1\text{kgm}^{-3} = \frac{1000}{1000000}\text{gcm}^{-3}$

$1\text{kgm}^{-3} = \frac{1}{1000}\text{gcm}^{-3}$

Or

$1\text{kgm}^{-3} = 10^{-3}\text{gcm}^{-3}$

11. Correct options: (A)

Given: mass of body=100g

Density = 1gcm^{-3}

To find: Volume of body

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Volume} = \frac{\text{mass}}{\text{density}}$

Volume $= \frac{100g}{1gcm^{-3}}$

$= 100cm^3$

volume $= 100cm^3$

12. Correct options: (A, B)

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density has no units because it is equal to the ratio of same physical quantity (density).

13.Correct option: (C)

Given: Volume of substance= 200cc

Density of substance = 0.7 g/cc

To find: mass of substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

Density x Volume =mass

0.7g/cc x 200cc = mass

$\frac{7}{10}$ x 200 g = mass

140g = mass

140g = mass

Mass = 140g

14.Correct option: (A)

Given:

Volume of liquid = 1000cc

mass of liquid= 1kg=1000g

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Density} = \frac{1000}{1000} g/cc$

Density $= 1g/cc$

15.Correct option: (C)

Given: mass of substance= 100kg

Density $= 10kg/m^3$

To find: Volume occupied by the substance

$\text{Density} = \frac{\text{mass}}{\text{volume}}$

$\text{Volume} = \frac{\text{mass}}{\text{density}}$

$\text{Volume} = \frac{100kg}{10kg/m^3}$

$= 10m^3$

volume $= 10m^3$

16.Correct option: (A)

Given: density of substance = 0.5g/cc

We know,

Density of water at 4°C(277K) is 1g/cc

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density $= \frac{0.5g/cc}{1g/cc}$

Relative density =0.5

17.Correct option: (B)

Given: Relative density =0.4

We know,

Density of water at 4°C(277K) is $10^3 kg/m^3$ or 1g/cc

$1kg/m^3 = \frac{1}{1000} g/cc$

$10^3 kg/m^3 = \frac{10^3}{1000} g/cc$

$10^3 kg/m^3 = \frac{1000}{1000} g/cc$

$10^3 kg/m^3 = 1g/cc$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

$0.4 = \frac{\text{density of substance}}{1g/cc}$

0.4 x 1g/cc= density of substance

0.4 g/cc= density of substance

Density of substance =0.4g/cc

18.Correct option: (A)

Given: density of substance = $5000kg/m^3$

We know,

Density of water at 4°C(277K) is $10^3 kg/m^3$

Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density $= \frac{5000kg/m^3}{10^3 kg/m^3}$

$= \frac{5000kg/m^3}{1000kg/m^3}$

Relative density = 5

19. Correct Matching is: a-PQ; b-R; c-P; d-T

(a) Density $= \frac{\text{mass}}{\text{volume}}$

C.G.S unit of mass is g

C.G.S unit of volume is cm^3

So,

C.G.S unit of density is g/cm^3

(b) Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

(c) $1kg = 1000g$

Also,

$1m = 100cm$

$1m^3 = 1000000cm^3$

$1kgm^{-3} = \frac{1000}{1000000} gcm^{-3}$

$1kgm^{-3} = \frac{1}{1000} gcm^{-3}$

Or

$1g/cm^3 = 1000kg/m^3$

(d) Weight is defined as the force exerted on a body by gravity.

Weight is often expressed as

W= mg

Where, w= weight of the body

m = mass of the body

g= acceleration due to gravity

SI unit of mass is kg

SI unit of acceleration due to gravity is m/s^{-2}

SI unit of weight is same as force that is newton (N).

1N = $1kgm/s^{-2}$

20. Correct Matching is: a-Q; b-R; c-P; d-ST

(a)Density $= \frac{\text{mass}}{\text{volume}}$

SI unit of mass is kg

SI unit of volume is m^3

So,

SI unit of density is kg/m^3

(b)

Density $= \frac{\text{mass}}{\text{volume}}$

C.G.S unit of mass is g

C.G.S unit of volume is cm^3

So,

C.G.S unit of density is g/cm^3

(c) Relative density $= \frac{\text{density of substance}}{\text{density of water at 4°C}}$

Relative density has no units because it is equal to the ratio of same physical quantity (density).

(d) **Density** of water at 4°C(277K) is $1000kgm^{-3}$ or $1gcm^{-3}$

$(1000kgm^{-3} = 1gcm^{-3})$

Dimensional Formula

LEVEL-I

1.Correct option:(B)

The seven fundamental or the base quantities chosen in SI are called the seven dimensions of the physical world, they are denoted with square brackets [].

Length =[L]

Mass =[M]

Time= [T]

Electric current =[A]

Thermodynamic temperature =[K] or [θ]

Luminous intensity=[Cd]

Amount of substance=[mol]

The expression which shows how and which of the base quantities represent the dimensions of a physical quantity is called the dimensional formula of the given physical quantity.

The dimensions of a physical quantity are the powers to which the base quantities are raised to represent a derived unit of that quantity.

For example,

Velocity $= \frac{\text{displacement}}{\text{time}} = \frac{[L]}{[T]}$

Velocity = $[L^1\ T^{-1}] = [M^0L^1\ T^{-1}]$

So, $[M^0L^1\ T^{-1}]$ is the dimensional formula of velocity.

It tells that unit of velocity depends on [L] and [T] and does not depend on [M].

2. Correct option:(A)

The dimensions of a physical quantity are the powers to which the base quantities are raised to represent a derived unit of that quantity.

For example,

Volume = length x breadth x height

Length is represented as [L]

Breadth and height also have the units of length.

So,

Volume =[L] x [L] x [L]

= $[L^3]$

Thus, to represent volume we have to raise the [L] to the power 3. Therefore, volume is said to have 3 dimensions in length.

3. Correct option:(B)

Velocity =$\frac{\text{displacement}}{\text{time}} = \frac{[L]}{[T]}$ Length/displacement is represented as [L]

Velocity = $[L^1\ T^{-1}] = [M^0L^1\ T^{-1}]$

So, $[M^0L^1\ T^{-1}]$ is the dimensional formula of velocity.

or

$[M^0L\ T^{-1}]$ is the dimensional formula of velocity.

4. Correct option:(C)

Area = length x breadth

Length is represented as [L]

Breadth also has the units of length.

So,

Area =[L] x [L]

Area = $[L^2]$= $[M^0L^2\ T^0]$

$[M^0L^2\ T^0]$ is the dimensional formula of area.

5. Correct option:(B)

Volume = length x breadth x height

Length is represented as [L]

Breadth and height also have the units of length.

So,

Volume =[L] x [L] x [L]

Volume = $[L^3]$ =$[M^0L^3\ T^0]$

$[M^0L^3\ T^0]$ is the dimensional formula of volume.

6. Correct option:(D)

The general formula for acceleration is:

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

7. Correct option:(D)

The general formula for momentum is:

momentum= mass x velocity

8. Correct option:(A)

The general formula for force is:

force= mass x acceleration

9. Correct option:(D)

The general formula for impulse is:

impulse= force x time

10. Correct option:(C)

Density $= \frac{\text{mass}}{\text{volume}}$

Mass is represented as [M]

$[M^0L^3\ T^0]$ is the dimensional formula of volume.

Density $= \frac{[M]}{[M^0L^3\ T^0]}$

Density $= [M^1L^{-3}\ T^0]$

$[M^1L^{-3}\ T^0]$ is the dimensional formula of density.

0r $[ML^{-3}\ T^0]$ is the dimensional formula of density.

11. Correct option:(D)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Time is represented as [T]

$[M^0L^1T^{-1}]$ is the dimensional formula of velocity.

Acceleration = $\frac{[M^0L^1T^{-1}]}{[T]}$

Acceleration =$[M^0L^1T^{-1-1}]$

Acceleration =$[M^0L^1T^{-2}]$

$[M^0L^1T^{-2}]$ is the dimensional formula of acceleration.

or $[M^0LT^{-2}]$ is the dimensional formula of acceleration.

12. Correct option:(D)

Length is represented as [L] =$[M^0LT^0]$

So, length has 0 dimensions in time.

Dimensional formula of velocity = $[M^0L^1T^{-1}]$ (-1 dimensions in time)

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$ (-2 dimensions in time)

Dimensional formula of force = $[M^1L^1T^{-2}]$ (-2 dimensions in time)

13. Correct option:(C)

Power = $\frac{\text{work}}{\text{time}}$

Now,

work =force x displacement

Power = $\frac{\text{force x displacement}}{\text{time}}$

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Time is represented as [T]

So,

Power =$\frac{[MLT^{-2}]x[L]}{[T]}$

Power =$[ML^{1+1}T^{-2-1}]$

Power =$[ML^2T^{-3}]$

dimensions of Time in power are -3

14. Correct option:(B)

force= mass x acceleration

Mass is represented as [M]

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$

Force = [M]x $[M^0L^1T^{-2}]$

Force = $[M^{1+0}L^1T^{-2}]$

Force = $[M^1L^1T^{-2}]$

Or Force = $[MLT^{-2}]$

Dimensional formula of force = $[MLT^{-2}]$

15. Correct option:(B)

Potential energy = mgh

Where m=mass

g = acceleration due to gravity

h= height

Mass is represented as [M]

Length/height is represented as [L]

Acceleration due to gravity has the same dimensional formula as that of acceleration which is $[M^0L^1T^{-2}]$

So,

Potential energy = [M] x $[M^0L^1T^{-2}]$ x [L]

Potential energy = $[M^{0+1}L^{1+1}T^{-2}]$

Potential energy = $[M^1L^2T^{-2}]$

Or

Potential energy = $[ML^2T^{-2}]$

16. Correct option:(D)

Torque = force x distance

Length/distance is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

So,

Torque = $[MLT^{-2}]$ x [L]

Torque = $[ML^{1+1}T^{-2}]$

Torque = $[ML^2T^{-2}]$

Dimensional formula of torque = $[ML^2T^{-2}]$

17. Correct option:(B)

Length is represented as [L]

$[M^0L^1T^0]$ is the dimensional formula of length.

18. Correct option:(D)

Frequency = number of vibrations/second

Second is the unit of time

And time is represented as [T]

Number of vibrations have no dimensions

So,

Frequency = $\frac{1}{[T]}$

Frequency = $[T^{-1}]$

So, dimensions of time in frequency are -1

19. Correct option:(A)

Impulse = force x time

Time is represented as [T]

Dimensional formula of force = $[MLT^{-2}]$

So,

Impulse = $[MLT^{-2}]$ x [T]

Impulse = $[MLT^{-2+1}]$

Impulse = $[MLT^{-1}]$

Dimensional formula of impulse = $[MLT^{-1}]$

Dimensions of mass in impulse is 1

20. Correct option:(A)

Acceleration due to gravity(g)= $\frac{\text{change in velocity}}{\text{time taken}}$

Time is represented as [T]

$[M^0L^1T^{-1}]$ is the dimensional formula of velocity.

Acceleration due to gravity= $\frac{[M^0L^1T^{-1}]}{[T]}$

Acceleration due to gravity =$[M^0L^1T^{-1-1}]$

Acceleration due to gravity=$[M^0L^1T^{-2}]$

$[M^0L^1T^{-2}]$ is the dimensional formula of acceleration due to gravity.

or $[M^0LT^{-2}]$ is the dimensional formula of acceleration due to gravity.

23. Correct option:(C)

(A)Dimensional formula of Electric charge = $[M^0L^0T^1A^1]$

Dimensional formula of Electrical conductivity = $[M^{-1}L^{-3}T^3A^2]$

(B)Dimensional formula of Impulse = $[M^1L^1T^{-1}]$

Dimensional formula of specific heat = $[M^0L^2T^{-2}K^{-1}]$

(C)Dimensional formula of work = $[ML^2T^{-2}]$

Dimensional formula of Heat = $[ML^2T^{-2}]$

(D)Relative density has no dimensions, as it has no units.

Dimensional formula of Temperature = [K] or [θ]

25. Correct option:(A)

(A)**Velocity of light** in vacuum $=3\times10^8$ m/s which is dimensional constant having dimensional formula $[M^0LT^{-1}]$ and it is invariable.

(B)Resistance of wire depends on various factors like length of wire, thickness of wire, temperature.

(C)Young's modulus of copper also varies with temperature.

(D)Acceleration due to gravity depends on the distance from the earth.

27. Correct option:(C)

- Work and energy both are scalars means they have magnitude only and no direction.
- Impulse and linear momentum both are vectors means they have both magnitude and direction.
- Speed is a scalar quantity, it has magnitude only whereas velocity is a vector quantity, it has both magnitude and direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

Dimensional formula of speed and velocity is same that is = $[M^0L^1T^{-1}]$

- Force is a vector quantity and energy are a scalar quantity.

28. Correct option:(D)

A force, whose source(origin) cannot be identified, is called a fictitious or pseudo force. For example, centrifugal force is not a real force. When the bus takes a turn, the passengers inside the bus simply participate in rotation and experience centrifugal force whose origin cannot be identified.

Dimensions of pseudo force is same as force which is = $[MLT^{-2}]$

29. Correct option:(B)

Given: Force, length and time are fundamental quantities and

Mass depends on force, length and time

Mass =$F^aL^bT^c$----------(1)

$[M] = [M^1L^1T^{-2}]^a[L]^b[T]^C$ (Dimensional formula of force=$[MLT^{-2}]$)

$[M] =[M]^a[L]^{a+b}[T]^{-2a+c}$

Comparing the dimensions of fundamental quantities on both sides
1=a -------(2)
0= a + b -------(3)
0= -2a + c----------(4)
put **a=1** in equation 3
0=1 + b
b = -1
put a =1 in equation 4
0=-2x1 + c
0=-2 + c
c = 2
putting values of a, b, c in equation (1)
we get
Mass $=F^1L^{-1}T^2$
Hence, dimensional formula of Mass = $[FL^{-1}T^2]$

30. Correct option:(A)
Given: Force, acceleration and time are fundamental quantities and
Energy depends on force, acceleration and time
Energy $=F^aA^bT^c$----------(1)
$[M^1L^2T^{-2}]=[M^1L^1T^{-2}]^a[M^0L^1T^{-2}]^b[T]^C$ (Dimensional formula of force=$[M^1LT^{-2}]$)
$[M^1L^2T^{-2}]=[M]^{a+0}[L]^{a+b}[T]^{-2a-2b+c}$ (Dimensional formula of energy=$[M^1L^2T^{-2}]$)
(Dimensional formula of acceleration=$[M^0L^1T^{-2}]$
Comparing the dimensions of fundamental quantities on both sides
1=a -------(2)
2= a + b -------(3)
-2= -2a -2b + c----------(4)
put **a=1** in equation 3
2=1 + b
2-1= b
b =1
put a =1 in equation 4
-2=-2x1 -2x1 + c
-2=-2 -2+ c
-2+2+2=c
c = 2
putting values of a, b, c in equation (1)
we get
Energy $=F^1A^1T^2$
Hence, dimensional formula of energy = $[FAT^2]$

LEVEL-II

1.Correct option:(A)
Weight is defined as the force exerted on a body by gravity.
Weight is often expressed as
W= mg
Where, w= weight of the body
m = mass of the body
g= acceleration due to gravity
SI unit of weight is same as force that is newton (N).

so, its dimensional formula is also same as force
Dimensional formula of force = $[M^1L^1T^{-2}]$
So, Dimensional formula of weight= $[M^1L^1T^{-2}]$

2. Correct option:(A)

Power = $\frac{\text{work}}{\text{time}}$
Now,
work =force x displacement
Power = $\frac{\text{force x displacement}}{\text{time}}$
Length/displacement is represented as [L]
Dimensional formula of force = $[MLT^{-2}]$
Time is represented as [T]
So,
Power =$\frac{[MLT^{-2}]x[L]}{[T]}$
Power =$[ML^{1+1}T^{-2-1}]$
Power =$[ML^2T^{-3}]$
dimensions of mass in power are 1

3. Correct option:(D)

Frequency = number of vibrations/second
Second is the unit of time
And time is represented as [T]
Number of vibrations have no dimensions
So,
Frequency = $\frac{1}{[T]}$
Frequency = $[T^{-1}] = [M^0L^0T^{-1}]$
Dimensional formula of frequency is $[M^0L^0T^{-1}]$

4. Correct option:(C)

Dimensional formula of intensity = $[ML^0T^{-3}]$

5. Correct option:(B)

- Dimensional formula of angular momentum and Planck's constant = $[ML^2T^{-1}]$
- Dimensional formula of moment of inertia = $[ML^2T^0]$ which is same as Dimensional formula of moment of force= $[ML^2T^{-2}]$
- Dimensional formula of work and torque = $[ML^2T^{-2}]$
- Dimensional formula of impulse and momentum = $[ML^1T^{-1}]$

6. Correct option:(D)

Gravitational constant(G) =$\frac{Fr^2}{m_1m_2}$
Where, F is the force between masses m_1 and m_2 at a distance r
Dimensional formula of force(F) = $[MLT^{-2}]$
Length/distance is represented as [L]
Dimensions of r^2 = [L] x [L] =$[L^2]$
Mass is represented as [M]
So,
Gravitational constant = $\frac{[MLT^{-2}]x[L^2]}{[M]x[M]}$
Gravitational constant = $[M^{1-1-1}L^{1+2}T^{-2}]$
Gravitational constant = $[M^{-1}L^3T^{-2}]$
So; Dimensional formula of gravitational constant =$[M^{-1}L^3T^{-2}]$

7. Correct option:(B)

Kinetic energy = $\frac{1}{2}mv^2$

Mass(m) is represented as [M]

Dimensional formula of velocity (v)= $[M^0L^1T^{-1}]$

Dimensions of v^2 = $[M^0L^1T^{-1}]$ x $[M^0L^1T^{-1}]$

Dimensions of v^2 = $[M^0L^{1+1}T^{-1-1}]$

Dimensions of v^2 = $[M^0L^2T^{-2}]$

Kinetic energy = [M] x $[M^0L^2T^{-2}]$

Kinetic energy = $[M^{0+1}L^2T^{-2}]$

Kinetic energy = $[M^1L^2T^{-2}]$

Or Kinetic energy =$[ML^2T^{-2}]$

Dimensional formula of kinetic energy =$[ML^2T^{-2}]$

8. Correct option:(C)

Light year is an important unit of long distances. One light year is the distance travelled by light in vacuum in one year.

1 light year = 9.46×10^{15}m

as light year has the units of length

so, dimensions of light year are [L]

9.Correct option:(A)

Dimensional analysis gives no information about the dimensional constants in the formula. For example, 1, 2, 3, π etc.

10.Correct options:(A, B, C, D)

Dimensional formula of speed = $[M^0L^1T^{-1}]$ (0 dimensions in mass)

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$ (0 dimensions in mass)

Dimensional formula of volume = $[M^0L^3\ T^0]$ (0 dimensions in mass)

Dimensional formula of area = $[M^0L^2\ T^0]$ (0 dimensions in mass)

11.Correct options:(A, B)

Speed = $\frac{\text{distance}}{\text{time}} = \frac{[L]}{[T]}$

Length/displacement/distance is represented as [L] Velocity =$\frac{\text{displacement}}{\text{time}} = \frac{[L]}{[T]}$

Velocity/speed = $[L^1\ T^{-1}] = [M^0L^1\ T^{-1}]$

So, $[M^0L^1\ T^{-1}]$ is the dimensional formula of velocity/speed.

or $[M^0L\ T^{-1}]$ is the dimensional formula of velocity/speed.

Hence, velocity and speed have same dimensional formula.

12.Correct options: (A, B, D)

Dimensional formula of acceleration = $[M^0L^1T^{-2}]$ (-2 dimensions in time)

Dimensional formula of force = $[MLT^{-2}]$ (-2 dimensions in time)

Dimensional formula of Work=$[ML^2T^{-2}]$ (-2 dimensions in time)

Dimensional formula of velocity=$[M^0L\ T^{-1}]$ (-1 dimensions in time)

Hence, acceleration, force, work have same dimensions in time

13.Correct option: (A)

work =force x displacement

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Work=$[MLT^{-2}]$x [L]

Work =$[ML^{1+1}T^{-2}]$

Work=$[ML^2T^{-2}]$

Dimensions of length in work is 2

14.Correct option:(A)

Dimensional formula of Work=$[ML^2T^{-2}]$

Dimensions of mass in work is 1

15.Correct option:(D)

Dimensional formula of Work=$[ML^2T^{-2}]$

Dimensions of time in work is -2

16.Correct option:(B)

Power = $\frac{\text{work}}{\text{time}}$

Now,

work =force x displacement

Power = $\frac{\text{force x displacement}}{\text{time}}$

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Time is represented as [T]

So,

Power =$\frac{[MLT^{-2}]x[L]}{[T]}$

Power =$[ML^{1+1}T^{-2-1}]$

Power =$[ML^2T^{-3}]$

dimensions of length in power are 2

17. Correct option:(A)

Dimensional formula of Power =$[ML^2T^{-3}]$

dimensions of mass in power are 1

18. Correct option:(D)

Dimensional formula of Power =$[ML^2T^{-3}]$

dimensions of time in power are -3

19. Correct Matching is: a-T; b-S; c-QR; d-P

- Mass is represented by [M].
- Length is represented by [L].
- Strength of electric current is represented by [A] or [I].
- Thermodynamic temperature is represented by [θ].

20. Correct Matching is: a-R; b-Q; c-S; d-T

(a)

Density = $\frac{\text{mass}}{\text{volume}}$

Mass is represented as [M]

$[M^0L^3\ T^0]$ is the dimensional formula of volume.

Density = $\frac{[M]}{[M^0L^3\ T^0]}$

Density $= [M^1L^{-3}\ T^0]$

$[M^1L^{-3}\ T^0]$ is the dimensional formula of density.

0r $[ML^{-3}\ T^0]$ is the dimensional formula of density.

(b) work =force x displacement

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Work=$[MLT^{-2}]$x [L]

Work =$[ML^{1+1}T^{-2}]$

Work=$[ML^2T^{-2}]$

Dimensional formula of Work=$[ML^2T^{-2}]$

(c) Power = $\frac{\text{work}}{\text{time}}$

Now,

work =force x displacement

Power = $\frac{\text{force x displacement}}{\text{time}}$

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Time is represented as [T]

So,

Power =$\frac{[MLT^{-2}]x[L]}{[T]}$

Power =$[ML^{1+1}T^{-2-1}]$

Power =$[ML^{2}T^{-3}]$

Dimensional formula of Power =$[ML^{2}T^{-3}]$

(d) Linear momentum= mass x velocity

Mass is represented as [M]

Dimensional formula of velocity=$[M^{0}L\,T^{-1}]$

So,

Linear momentum = [M] x $[M^{0}LT^{-1}]$

Linear momentum = $[M^{1+0}LT^{-1}]$

Linear momentum = $[MLT^{-1}]$

Dimensional formula of Linear momentum = $[MLT^{-1}]$

2. Kinematics

Solutions

LEVEL-I

1. Correct option: (A)

Periodic motion: A motion that repeats itself after equal/regular intervals of time is known as periodic motion. The movement of moon around the earth is repeated after 30 days and hence it is a periodic motion.

2. Correct option: (B)

Rest and motion are relative: It means an object in one situation can be at rest but in another situation the same object can be in motion. For illustration, a person sitting in a moving train is at rest with respect to its fellow passengers but is in motion with respect to the objects outside the train.

3. Correct option: (B)

Rectilinear motion: It is also known as linear motion. It is that motion in which a particle or an object /body moves along a straight line or well-defined straight path. For example, the motion of train along a straight railway track, an object dropped from a height above the ground etc.

4. Correct option: (B)

Periodic motion: A motion that repeats itself after equal/regular intervals of time is known as periodic motion. The oscillations (To and fro motion) of a simple pendulum of a wall- clock repeats itself after a definite interval of time that's why, it is called periodic motion. The time interval of periodic motion is different for different simple pendulums.

5. Correct option: (A)

Periodic motion: A motion that repeats itself after equal intervals of time is known as periodic motion. The oscillations (To and fro motion) of a simple pendulum of a wall- clock repeats itself after a definite interval of time. That time interval is different for different simple pendulums. Hence, motion of a simple of a wall clock is periodic motion.

6. Correct option: (B)

Motion: An object is said to be in motion if it changes its position with time, with respect to its surroundings or some reference point.

For example, A bird flying in air is in motion because it changes its position continuously with respect to a person standing on the ground.

7. Correct option: (D)

Oscillatory motion: It is that motion in which a body moves/fluctuates to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

8. Correct option: (D)

A moving object is said to have a random motion if it does not move in any particular direction and its motion has no specific path.

A butterfly flying in a garden is an example of Random motion.

9. Correct option: (A)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

If in the oscillatory motion, the amplitude is very small, the motion of body is said to be **vibratory motion.**

Hence, the To and fro motion of an object can be vibratory motion.

10. Correct option: (B)

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

A butterfly flying in a garden is an example of Random motion.

11.Correct option: (A)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time. Hence motion of a potter's wheel is an example of rotatory motion.

12.Correct option: (D)

Rectilinear motion: It is also known as linear motion. It is that motion in which a particle or an object /body moves along a straight line or well-defined straight path. For example, the motion of train along a straight railway track, an object dropped from a height above the ground etc.

13.Correct option: (B)

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

Hence javelin thrown by an athlete is an example of curvilinear motion as it travels along a curved path.

14.Correct option: (B), (C)

Non periodic motion: A motion that does not repeat itself after equal intervals of time is known as non-periodic motion. For example, a car running on a curved does not repeat its motion after regular intervals of time hence, it is also a non-periodic motion.

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

When a car runs on a curved road, it follows a curved path. Hence, a car running on a curved road is an example of curvilinear motion.

15.Correct option: (C)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time.

A windmill converts wind power into rotational energy using its blades. Hence, windmill is an example of rotatory motion.

16.Correct option: (A)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite

interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.
Hence the needle of the sewing machine is in oscillatory motion
Wheel of the sewing machine is in rotatory motion.

17.Correct option: (C)

Rest and motion are relative: It means an object in one situation can be at rest but in another situation the same object can be in motion. For illustration, a person sitting in a moving train is at rest with respect to its fellow passengers or inner walls of the train but is in motion with respect to the objects outside the train.

18.Correct option: (A)

Periodic motion: A motion that repeats itself after equal intervals of time is known as periodic motion.

The motion of the earth around the sun is a periodic motion as earth takes $365\frac{1}{4}$ days to complete one revolution.

19.Correct option: (B)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles/ circular path lie on a line, called the axis of rotation and shift through an angle in a given time.

20.Correct option: (D)

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

A kite flying in air is an example of Random motion.

21.Correct option: (B)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles/ circular path lie on a line, called the axis of rotation and shift through an angle in a given time.

Hence, wheel of a sewing machine possesses rotatory motion.

22.Correct option: (A)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles/ circular path lie on a line, called the axis of rotation and shift through an angle in a given time.

Hence, pedals of bicycle possess rotatory motion.

23.Correct option: (D)

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

The motion fishes in water are an example of Random motion.

24.Correct option: (D)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits

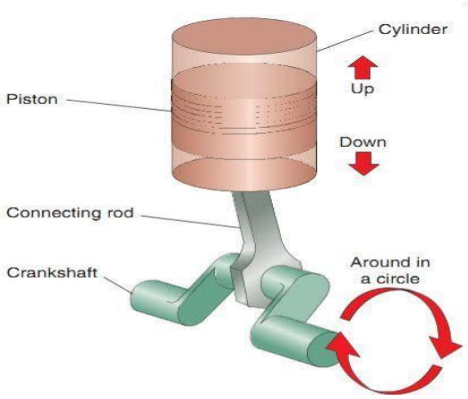
(called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.
Hence, motion of pendulum of wall clock is an example of oscillatory motion as it moves To and fro between two fixed points about a mean position.

25.Correct option: (C)

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.
The motion fishes in water are an example of Random motion.

26.Correct option: (A)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.
Hence, motion of piston of an engine is an example of oscillatory motion as it moves up and down between two fixed points about a mean position.



27.Correct option: (B), (D)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.
If in the oscillatory motion, the amplitude is very small, the motion of body is said to be vibratory motion.
Hence, the motion of strings of a violin is an example of vibratory/ oscillatory motion.

28.Correct option: (B)

Translatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.
hence, marching of soldiers on a straight road is an example of translator motion as the soldiers along parallel straight lines.

29.Correct option: (C)

Translatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.

Hence, the free fall of fruits from the tree under the gravity is an example of translatory motion because the fruit moves downwards along a straight path.

30. Correct option: (A)

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

When a car takes a turn, it follows a curved path. Hence, a car taking a turn executes curvilinear motion.

LEVEL-II

1. Correct option: (A)

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

The motion houseflies are an example of Random motion.

2. Correct option: (B)

Circular motion is that motion in which a particle or a point mass is moving in a circle.

Circular motion is special case of rotatory motion as in this motion the distance from the axis of rotation is always fixed. A spinning top around its own axis executes circular motion.

3. Correct option: (B)

An object is said to be in motion if it changes its position with time, with respect to its surroundings. For example, a bird flying in air, a man walking on the road are in motion.

4. Correct option: (A)

An object is said be at rest if it does not change its position with time, with respect to its surroundings. For example, a book lying on a table, a person sitting on a chair are at rest.

5. Correct option: (D)

An object is said be at rest if it does not change its position with time, with respect to its surroundings. For example, a chair lying in a room.

6. Correct option: (C)

An object is said to be in motion if it changes its position with time, with respect to its surroundings. For example, a bird flying in air, a man walking on the road are in motion.

7. Correct option: (C)

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

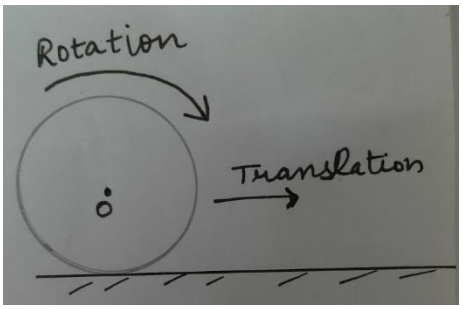
A bus moving on a flyover bridge is an example of curvilinear motion.

8. Correct option: (A), (B)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time.

Translatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.

Hence, a rolling ball executes both rotatory and translatory motion.



9. Correct option: (A), (B), (C), (D)

Periodic motion: A motion that repeats itself after equal intervals of time is known as periodic motion.

For example:

(A) Hands of a wrist watch come to their starting position after a fixed interval of time and then repeat the same motion with same interval of time. Hence, it is a periodic motion.

(B) The rotation of earth around its own axis repeats itself after 24 hours and hence it is a periodic motion.

(C) The strings of a musical instrument vibrate when plucked. vibration repeats itself after certain period of time. so, it represents periodic motion.

(D) Rotation of earth around the sun repeats itself after 365 days and 6 hours. hence, it is a periodic motion.

10.Correct option: (A), (B), (D)

- A running vehicle has both translatory motion and rotatory motion(wheels).
- A spinning billiards ball has both translatory and rotatory motion.
- A playing marble have both translator and rotatory motion.

11. Correct option: (A), (B), (C), (D)

Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles/ circular path lie on a line, called the axis of rotation and shift through an angle in a given time.

For example: spinning top, wind mill, fan, wheels of a car all execute rotatory motion

12.Correct option: (A), (B), (C), (D)

Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

Hence the pendulum of wall clock, see saw, needle of the sewing machine, compressed spring are in oscillatory motion as they move To and Fro between two fixed points.

13. Correct option: (B)

Translatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along parallel straight lines and shift through equal distance in a given interval of time.

Hence, a car moving on a straight level road possess translatory motion.

14.Correct option: (A)

Rest and motion are relative: It means an object in one situation can be at rest but in another situation the same object can be in motion. For illustration, a person sitting in a moving train is at rest with respect to its fellow passengers or inner walls of the train but is in motion with respect to the objects outside the train.

15.Correct option: (A)

Rest and motion are relative: It means an object in one situation can be at rest but in another situation the same object can be in motion. For illustration, a person sitting in a moving train is at rest with respect to its fellow passengers, train driver or inner walls of the train but is in motion with respect to the objects outside the train.

16.Correct option: (B)

Rest and motion are relative: It means an object in one situation can be at rest but in another situation the same object can be in motion. For illustration, a person sitting in a moving train is at rest with respect to its fellow passengers or inner walls of the train but is in motion with respect to the objects outside the train for example, for an observer on the platform, the train and the person sitting inside it are in motion.

17.Correct option: (C)

Rectilinear motion: It is also known as linear/straight line motion. It is that motion in which a particle or an object /body moves along a straight line or well-defined straight path. For example, the motion of train along a straight railway track, an object dropped from a height above the ground etc.

18.Correct option: (A)

Curvilinear motion: The motion of an object moving in a curved path is called curvilinear motion. For example, a stone thrown into the air at an angle.

19.Correct option: (D)

A moving object is said to have a **random motion** if it does not move in any particular direction and its motion has no specific path. The object changes its direction in an irregular manner.

The motion houseflies are an example of Random motion.

20. Correct Matching is: a-R; b-Q; c-S; d-P

(a) Oscillatory motion: It is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well-defined limits (called extreme positions) on either side of mean position. It means a periodic and bounded motion of a body about a fixed point is called oscillatory motion.

Swinging cradle is an example of oscillatory motion because it moves between two fixed

point about a mean position.

(b) Rotatory motion: It is that motion in which a body, which is not a point mass body, is moving such that all its constituent particles move simultaneously along concentric circles lie on a line, called the axis of rotation and shift through an angle in a given time.

Spinning top around its own axis is an example of rotatory motion.

(c) A moving object is said to have a **random motion** if it does not move in any particular direction and its motion

has no specific path. The object changes its direction in an irregular manner.

The motion fishes in water are an example of Random motion.

(d) Rectilinear motion: It is also known as linear motion. It is that motion in which a

particle or an object /body moves along a straight line or well defined straight

path. So, a car moving in a straight line is an example of rectilinear motion.

DISTANCE AND DISPLACEMENT

LEVEL-I

1.Correct option: (A)

Distance: It is the actual path traversed by the object in the given time.

Distance travelled along a circular path is calculated with the help of circumference of the circle.

Distance travelled by the particle when it completes full circle = circumference of the circle

Distance travelled by the particle when it completes half circle (ACB)

$$= \frac{(circumference\ of\ the\ circle)}{2}$$

$$= \frac{2\pi r}{2}$$

$=\pi r$ (r= radius of the semicircle=7m)

$(\pi = 22/7)$

Distance travelled by the particle when it completes half circle (ACB) = $\frac{22}{7}$ x 7 m

= 22m

2. Correct option: (C)

Distance is the actual path traversed/covered by the object in the given time.

3. Correct option: (A)

Scalar quantities: These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km.

4. Correct option: (D)

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

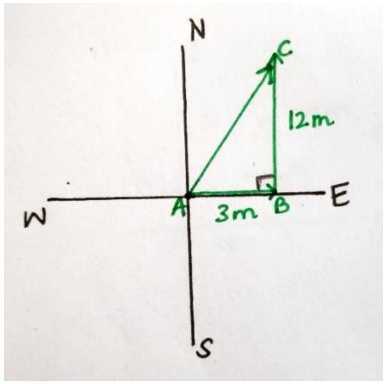
Hence,

C.G.S unit of distance is centimeter.

5. Correct option: (C)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

In the question, let the body is displaced from point A to B to C.



Here, initial position is A and final position is C.
When we join A to C, we get the shortest distance AC= displacement
Triangle ABC is a right-angled triangle
AB = base = $5m$
BC=perpendicular=12m
AC= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (perpendicular)^2 + (base)^2$
$(\mathrm{AC})^2 = (\mathrm{BC})^2 + (\mathrm{AB})^2$
$(\mathrm{AC})^2 = (12)^2 + (5)^2$
$(\mathrm{AC})^2 = 144 + 25$
$(\mathrm{AC})^2 = 169$
$\mathrm{AC} = \sqrt{169}m$
$\mathrm{AC} = 13m$
Hence, displacement (AC)= $13m$

6.Correct option: (B)

SI system is an internationally accepted system of units for measurements.
According to this system unit of length is meter.
As distance is also the length of path.so, SI unit of distance is meter.

7. Correct option: (A)

Displacement: It is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.
Here, initial position is A and final position is C. So, displacement is the line joining Initial and final position of the body.

8. Correct option: (A)

Scalar quantities: These have a size or magnitude only and need no other information to specify them. For example, distance between two towns (say 20km) is a scalar quantity whose magnitude is 20 km, it has no direction.

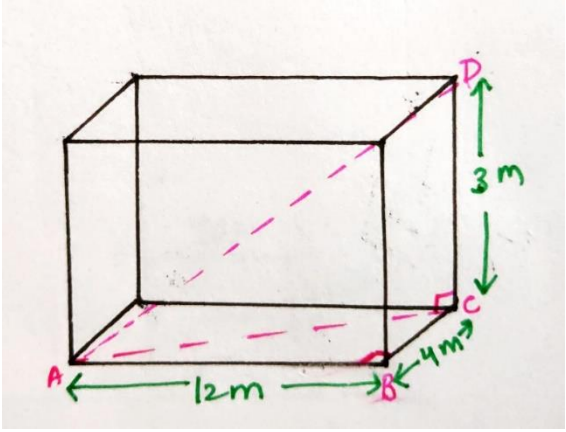
9. Correct option: (C)

Displacement: It is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.
In the given figure, the person is starting from P and returning to P.
As displacement is the difference between the final and initial positions of the object in a given time.
Here, initial position is P and final position is also P.
So, displacement is zero.

10.Correct option: (B)

Vector quantities are those quantities which have both magnitude(size) and direction.
Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. Hence it is distance measured in a particular direction.so it is a vector quantity.

11.Correct option: (B)



• Displacement is the difference between the final and initial positions of the object in a given time.
It is the shortest distance between two positions of the object.
As shown in the figure
Shortest distance /displacement= AD

Step 1: calculation of AC

In right angled triangle ABC
AB = base =12m
BC=perpendicular=4m
AC= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$(AC)^2 = (12)^2 + (4)^2$
$(AC)^2 = 144 + 16$
$(AC)^2 = 160$
AC = $\sqrt{160}$m

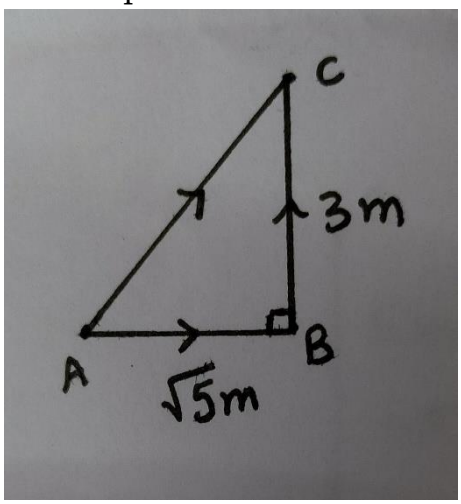
Step 2: Calculation of AD

In right angled triangle ACD
AC = base =$\sqrt{160}$m
DC=perpendicular=3m
AD= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AD)^2 = (AC)^2 + (DC)^2$
$(AD)^2 = (\sqrt{160})^2 + (3)^2$
$(AD)^2 = 160 + 9$
$(AD)^2 = 169$
AD= $\sqrt{169}$m
AD=13m
Displacement AD= 13m

12. Correct option: (A)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.
In the given figure, the object is moving from point A to B to C.

Here, initial position is A and final position is C.



When we join A to C, we get the shortest distance AC= displacement
Triangle ABC is a right-angled triangle
AB = base = $\sqrt{5}m$
BC=perpendicular=3m
AC= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$(AC)^2 = \left(\sqrt{5}\right)^2 + (3)^2$
$(AC)^2 = 5 + 9$
$(AC)^2 = 14$
$AC = \sqrt{14}\, m$
Hence, displacement (AC)= $\sqrt{14}\, m$

13.Correct option: (C)

Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. Hence it is distance measured in a particular direction. If the distance travelled by the body is 0 then displacement is also 0.

14.Correct option: (A)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.
In the given figure, the particle is moving from point A to B through P and another particle through Q.
Here, initial position is A and final position is B.

- **Calculation of displacement for first particle travelling from A TO B through P**

shortest distance AB= displacement
Triangle ABP is a right-angled triangle
AP = base = 8m
BP=perpendicular=6m
AB= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AB)^2 = (AP)^2 + (BP)^2$

$(AB)^2 = (8)^2 + (6)^2$
$(AB)^2 = 64 + 36$
$(AB)^2 = 100$
AB = $\sqrt{100}$ m
AB= 10m
Hence, displacement (AB)= 10m

• **Calculation of displacement for first particle travelling from A TO B through Q**

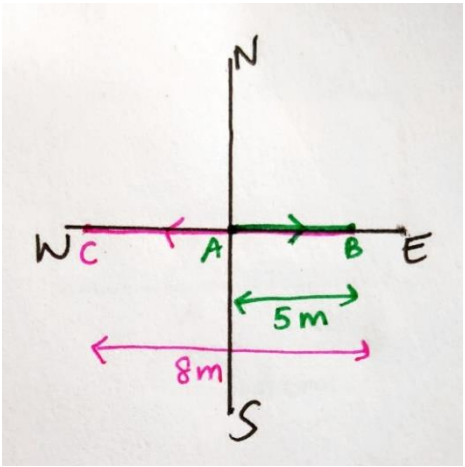
shortest distance AB= displacement
Triangle ABQ is a right-angled triangle
BQ= base = 8m
AQ=perpendicular=6m
AB= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AB)^2 = (BQ)^2 + (AQ)^2$
$(AB)^2 = (8)^2 + (6)^2$
$(AB)^2 = 64 + 36$
$(AB)^2 = 100$
AB = $\sqrt{100}$ m
AB= 10m
Hence, displacement (AB)= 10m

15.Correct option: (C)

Vector quantities are those quantities which have both magnitude(size) and direction.
Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. Hence it is distance(magnitude) measured in a particular direction.

16. Correct option: (B)

Distance is the actual path traversed by the object in the given time.

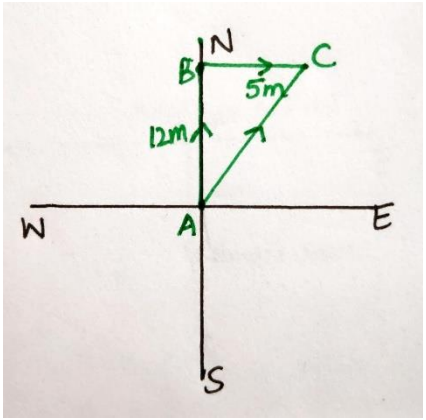


Distance travelled towards east (AB) =5m
Distance travelled towards west (BC)= 8m
Total distance travelled = 5m +8m = 13m

17. Correct option: (B)

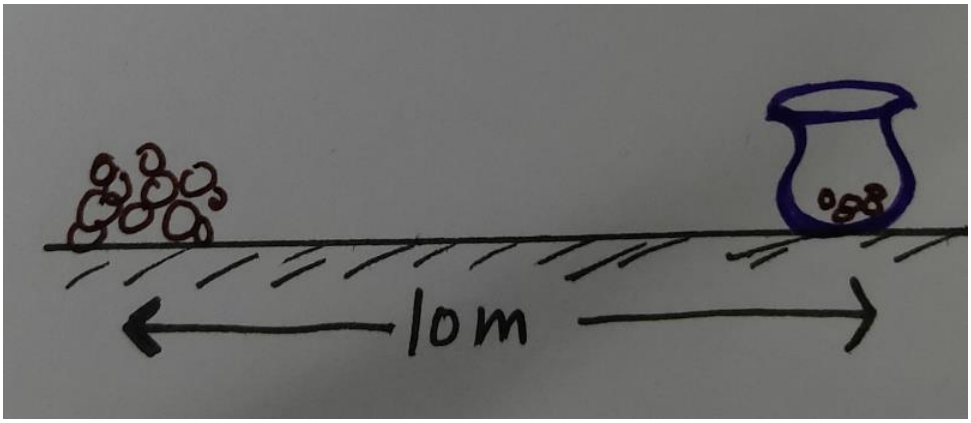
Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

In the question, let the horse is running from point A to B to C.



Here, initial position is A and final position is C.
When we join A to C, we get the shortest distance AC= displacement
Triangle ABC is a right-angled triangle
AB = base = $12m$
BC=perpendicular=5m
AC= hypotenuse
Using Pythagoras theorem in triangle ABC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AC)^2 = (AB)^2 + (BC)^2$
$(AC)^2 = (12)^2 + (5)^2$
$(AC)^2 = 144 + 25$
$(AC)^2 = 169$
AC = $\sqrt{169}m$
AC = $13m$
Hence, displacement (AC)= $13m$

18. Correct option: (A)



Distance travelled by the crow in one trip= distance from pebbles to pot +distance from pot to
pebbles
= 10m+10m
=20m
Distance travelled by the crow in 5 trips = 5 x Distance travelled by the crow in one trip
= 5 x 20m
= 100m

19.Correct option: (B)

• Calculation of displacement of ant with respect to ground

When the ant travels from point A to B on the rod

shortest distance AC= displacement

Triangle ACB is a right-angled triangle

AB = base = 2m

CB=perpendicular=10m

AC= hypotenuse

Using Pythagoras theorem in triangle ABC

$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$

$(\mathrm{AC})^2 = (\mathrm{AB})^2 + (\mathrm{CB})^2$

$(\mathrm{AC})^2 = (2)^2 + (10)^2$

$(\mathrm{AC})^2 = 4 + 100$

$(\mathrm{AC})^2 = 104$

$\mathrm{AC} = \sqrt{104}\mathrm{m}$

Hence, displacement (AC)=$\sqrt{104}$ m

20.Correct option: (C)

Displacement is the difference between the final and initial positions of the object in a

given time.

In the given question, the bird makes 5 trips up and down, starting from the tree and returning back to it.

As displacement is the difference between the final and initial positions of the object in a given time.

Here, initial position is tree and final position is also tree.

So, displacement is 0m.

21. Correct option: (B)

An odometer is an instrument used for measuring the distance travelled by a vehicle such as car.

22. Correct option: (B)

Distance: It is the actual path traversed by the object in the given time.

Distance travelled along a circular path is calculated with the help of circumference of the circle.

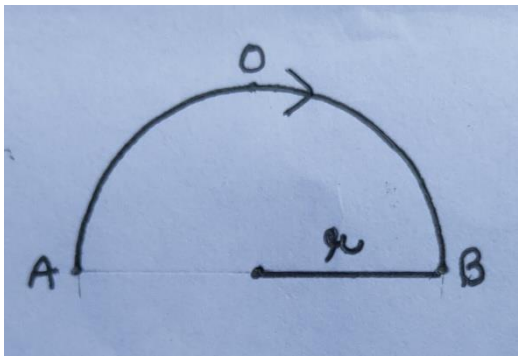
Distance travelled by the particle when it completes full circle = circumference of the circle

Distance travelled by the particle when it completes half circle (AOB)

$$= \frac{(circumference\ of\ the\ circle)}{2}$$

$$= \frac{2\pi r}{2}$$

$$= \pi r$$



23. Correct option: (D)

Distance is the actual path traversed by the object in the given time. Distance travelled by an object can never be negative.

Whereas displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. It can be positive, negative or zero

The displacement of an object in a time can be equal to or less than distance travelled but never greater than the distance travelled.

Displacement ≤ distance

Numerical ratio($\frac{\text{displacement}}{\text{distance}}$)

Can be equal to 1 or less than 1. Since, the denominator(distance) is either greater than or equal to the numerator(displacement).

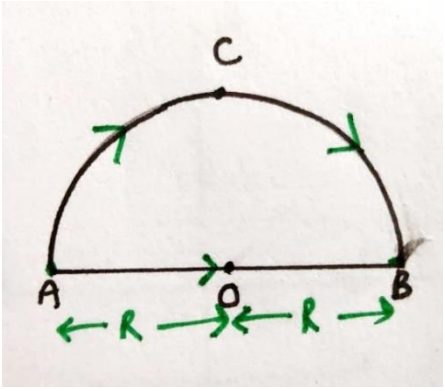
24.Correct option: (A)

Distance and magnitude of displacement are equal when an object is moving in a straight line because in this case the shortest distance/displacement is equal to the actual length covered.

Hence, distance moved and magnitude of displacement will be equal for a car moving on a straight road.

25.Correct option: (A)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.



Consider that while travelling the circular path, the body completes half revolution along the path ACB.

Here, A is the initial position and B is the final position of the body.

So, AB =shortest distance joining initial and final position of the body= displacement of the body

Given: radius of the circle = R

AB= diameter of the circle = 2R

Hence, displacement of the body = 2R

26. Correct option: (B)

Displacement: It is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.

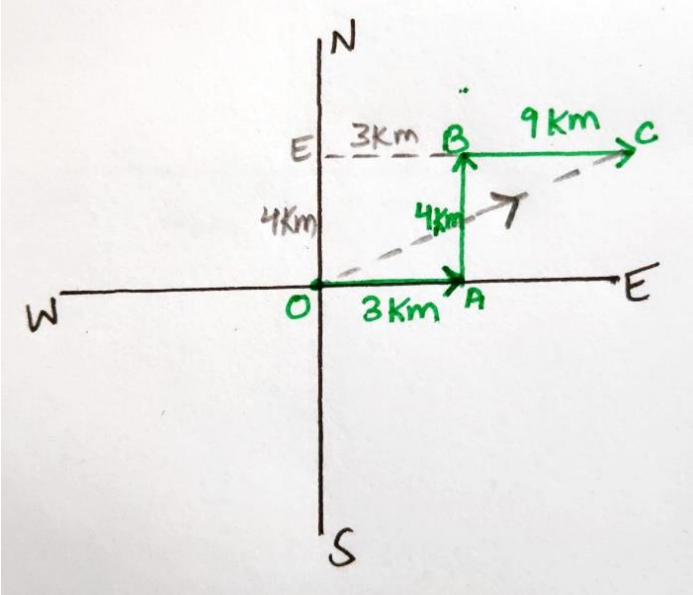
In the given question, we are starting from home and returning to home.

As displacement is the difference between the final and initial positions of the object in a given time.

Here, initial position is home and final position is also home.

So, displacement is 0.

27.Correct option: (D)



• **Calculation of the distance travelled**

Distance is the actual path traversed by the object in the given time.
Distance travelled towards east (OA) = 3km
Distance travelled towards north (AB) = 4km
Distance travelled towards east (BC) = 9km
Total distance travelled = 3km +4km+9km
= 16km

• **Calculation of the displacement**

Displacement is the difference between the final and initial positions of the object in a given time.
It is the shortest distance between two positions of the object.
In the above figure
We have produced CB to E
EB= OA = 3km
OE=AB=4km
EC= EB+BC = 3km + 9km =12km
In the given figure
shortest distance OC= displacement
Triangle OEC is a right-angled triangle
OE = base = 4km
EC=perpendicular=12km
OC= hypotenuse
Using Pythagoras theorem in triangle OEC
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(OC)^2 = (OE)^2 + (EC)^2$
$(OC)^2 = (4)^2 + (12)^2$
$(OC)^2 = 16 + 144$
$(OC)^2 = 160$
$OC = \sqrt{160}$km
$OC = 4\sqrt{10}$km
Hence, displacement (OC)=$4\sqrt{10}$km

28.Correct option: (C)

Distance is the actual path traversed by the object in the given time.
Distance travelled in going from classroom to book shop =20m
Distance travelled in returning from book shop to classroom= 20m
Total distance travelled = 20m +20m = 40m

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.

In the given question, boy is starting from classroom and returning to returning.
As displacement is the difference between the final and initial positions of the object in a given time.
Here, initial position is classroom and final position is also classroom.
So, displacement is 0.

29.Correct option: (A)

Distance is the actual path traversed by the object in the given time.
Distance (AB) =4m
Distance (BC)= 6m (given: C is the midpoint of BD so BC=CD=6m)
Total distance travelled = 4m +6m = 10m

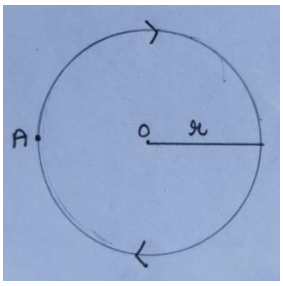
30. Correct option: (B)

- Distance: It is the actual path traversed by the object in the given time.

Distance travelled by the particle when it completes 1 revolution (A to A)
= circumference of the circle
$= 2\pi r$ (r= radius of the circle)

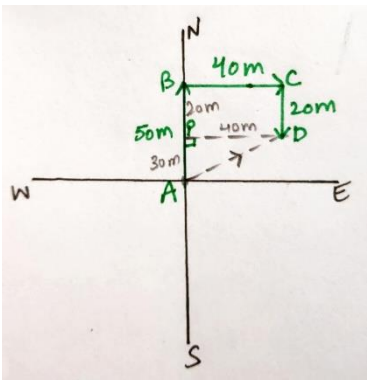
- Displacement is the difference between the final and initial positions of the object in a given time.

In a circular motion when the particle completes 1 revolution it returns to its initial position(A). So, initial and final position are same.
Hence, displacement of the particle is 0.



LEVEL-II

1.Correct option: (B)



Calculation of distance

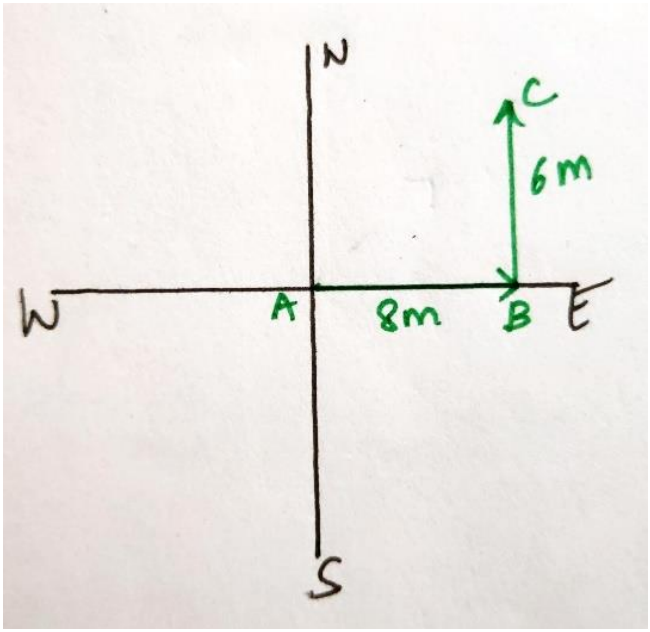
Distance moved towards north (AB)=50m
Distance moved towards east (BC)=40m
Distance moved towards south (CD)=20m
total distance moved = 50m + 40m + 20m
= 110m
total distance moved = 110 m

- **Calculation of displacement**

Displacement is the difference between the final and initial positions of the object in a given time.
It is the shortest distance between two positions of the object.
As shown in the figure
Shortest distance /displacement= AD
Drop a perpendicular DP on AB
In right angled triangle APD
AP = base = 30m
DP=perpendicular=40m
AD= hypotenuse
Using Pythagoras theorem in triangle APD
$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$
$(AD)^2 = (AP)^2 + (DP)^2$
$(AD)^2 = (30)^2 + (40)^2$
$(AD)^2 = 900 + 1600$
$(AD)^2 = 2500$
$(AD)^2 = (50)^2$
AD = 50m
Hence, displacement (AD)= 50m

2.Correct option: (B)

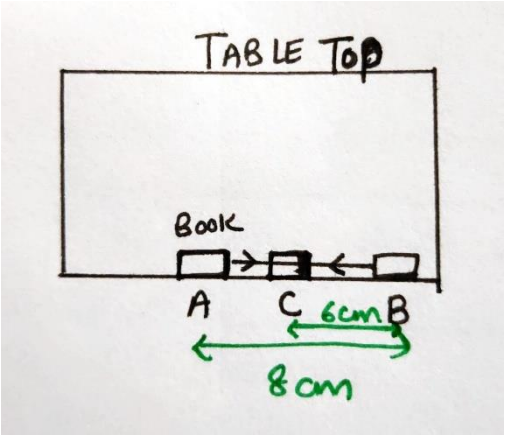
Distance is the actual path traversed by the object in the given time.



Distance travelled towards east (AB) =8m
Distance travelled towards north (BC)= 6m
Total distance travelled = 8m +6m = 14m

3.Correct option: (A)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.



According to the question
Let the initial position of the book = A
After that it is moved through a distance of 8cm
Let the book reaches at point B
Now table is moved in the opposite direction through 6cm
Let the final position of the book =C
Displacement = shortest path joining the initial and the final position AC= AB-CB
=8cm -6cm
=2cm
Hence, displacement of the book = 2cm

4.Correct option: (A)

Given: 1 unit = 1km

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.

In the given figure
shortest distance AC= displacement
Triangle ADC is a right-angled triangle
DC = base = 5km
AD=perpendicular=3km
AC= hypotenuse
Using Pythagoras theorem in triangle ABC

$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$

$(AC)^2 = (DC)^2 + (AD)^2$

$(AC)^2 = (5)^2 + (3)^2$

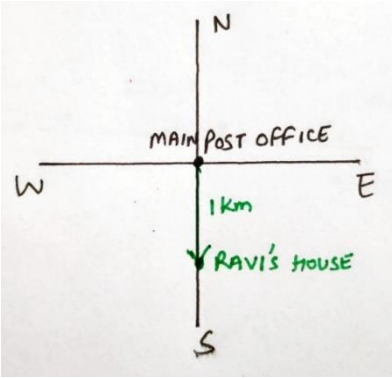
$(AC)^2 = 25 + 9$

$(AC)^2 = 34$

AC = $\sqrt{34}$km

Hence, displacement (AC)=$\sqrt{34}$ km

5.Correct option: (A)



According to the given question
Reference point is the main post office
Distance moved by ravi's friend towards his house = 1 km
Displacement of ravi's friend from post office to ravi's house = 1km towards south
Magnitude of displacement = distance moved =1km

6.Correct option: (B)

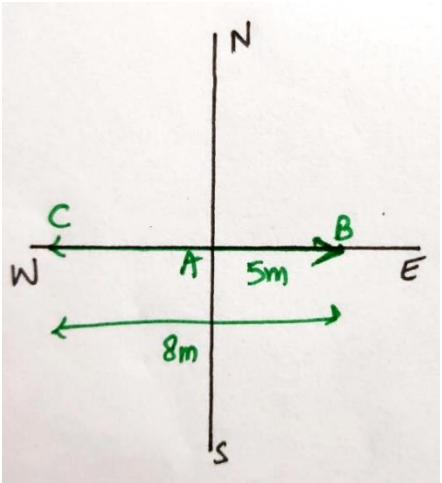
Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.
The particle is travelling along the path ACB.
Here, A is the initial position and B is the final position of the body.
So, AB =shortest distance joining initial and final position of the body= displacement of the body
Given: radius of the circle = 14m
AB= diameter of the circle = 2R= 2x14m= 28m
Hence, displacement of the body = 28m

7.Correct option: (A)

Distance is the actual path traversed by the object in the given time.
Distance (AB) =5m
Distance (BC)= 5m
Distance (CD)= 2m
Total distance travelled by the particle = 5m +5m+2m = 12m

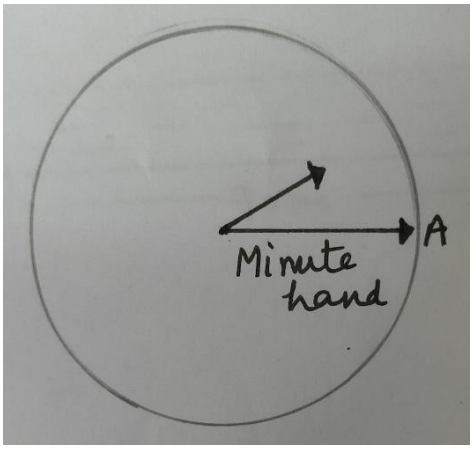
8.Correct option: (A)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.



Taking A as the reference point.
distance travelled towards east (AB) =5m
distance travelled towards west (BC)=8m
Here, A is the initial position and C is the final position of the body.
So, AC =shortest distance joining initial and final position of the body= displacement of the body
AC= BC-BA
AC = 8m-5m
AC= 3m

9. Correct option: (A), (D)



In one hour, minute's hand of the clock moves in a circle and completes one revolution.
• Displacement is the difference between the final and initial positions of the object in a given time.
In a circular motion, when the minute's hand completes 1 revolution it returns to its initial position(A). So, initial and final position are same.
Hence, displacement of the minute's hand is 0.
• Distance is the actual path traversed by the object in the given time.
Distance travelled along a circular path is calculated with the help of circumference of the circle.
Distance travelled by the minute's hand after one hour when it completes 1 revolution (A to A)
= circumference of the circle
= $2\pi R$ (R= radius of the circle)

10.Correct option: (A), (B), (C), (D)
SI system is an internationally accepted system of units for measurements.
According to this system unit of length is meter.
As distance is also the length of path.so, SI unit of distance is meter.
C.G.S unit of distance is centimeter.
Millimeter and kilometer are other units of distance.

11.Correct option: (A), (D)
Distance is the actual path traversed by the object in the given time. It has just the magnitude and no direction so it is a scalar quantity.
Fundamental physical quantities are the quantities whose measurement does not depend on other physical quantities.
As distance is also the length of path and we know that length is a fundamental physical quantity so, distance is also a fundamental physical quantity.

12. Correct option: (A), (B), (C)

Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. It can be positive, negative or zero

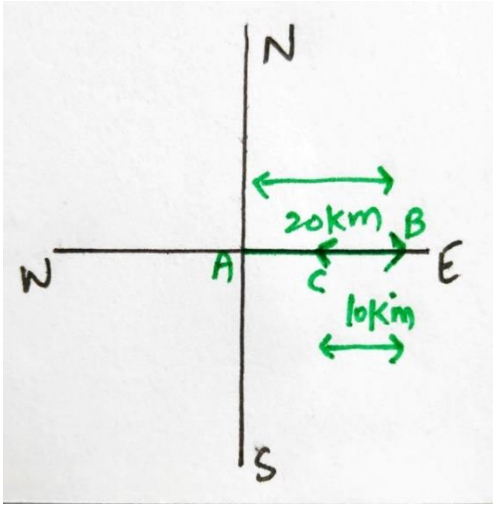
Displacement is positive when the final position is at a greater distance than the initial position.

Displacement is negative when the final position is at a smaller distance than the initial position.

Displacement is zero when the final position is same as the initial position.

13.Correct option: (A)

Distance is the actual path traversed by the object in the given time.

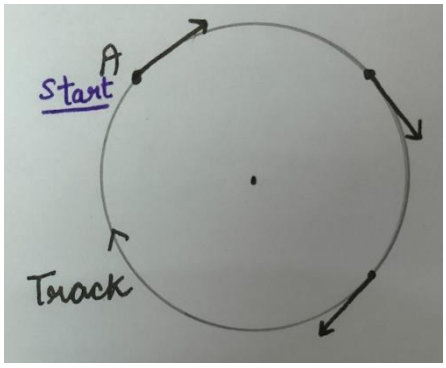


Distance travelled by the car towards east (AB) =20km

Distance travelled by the car towards west (BC)= 10km

Total distance travelled = 20km +10km = 30km

14.Correct option: (C)



Distance is the actual path traversed by the object in the given time.

Distance covered in 1 round (A to A) = 200m

Distance covered in 4 rounds A to A) = 4 x 200m

= 800m

15.Correct option: (C)

Displacement is the difference between the final and initial positions of the object in a given time.

In the given question after taking 4 rounds of the track the athlete returns back to its initial position that is starting point(A). So, initial and final position are same.

Hence, displacement of the athlete is 0.

16.Correct option: (A)

Distance is the actual path traversed/covered by the object in the given time.

17.Correct option: (A)

Distance travelled from A to C = Distance travelled along A to B + Distance travelled along semicircle B to C

Distance travelled along A to B=10m

Distance travelled by the particle along semicircle (B to C)

$=\frac{(circumference\ of\ the\ circle)}{2}$

$=\frac{2\pi r}{2}$

$=\pi r$ (r= radius of the semicircle=7m)

($\pi = 22/7$)

Distance travelled by the particle along semicircle (B to C) = $\frac{22}{7}$x 7 m

= 22m

Distance travelled from A to C= 10m + 22m

=32m

18.Correct option: (A)

Distance travelled from B to D = Distance travelled along semicircle B to C + Distance travelled along C to D

Distance travelled along C to D=10m

Distance travelled by the particle along semicircle (B to C)

$=\frac{(circumference\ of\ the\ circle)}{2}$

$=\frac{2\pi r}{2}$

$=\pi r$ (r= radius of the semicircle=7m)

($\pi = 22/7$)

Distance travelled by the particle along semicircle (B to C) = $\frac{22}{7}$x 7 m

= 22m

Distance travelled from B to D= 22m +10 m

=32m

19.Correct option: (B)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.

The particle is travelling along the path ABC.

Here, A is the initial position and C is the final position of the body.

So, AC =shortest distance joining initial and final position of the body= displacement of the body

Given: radius of the semicircle = 7m

BC= diameter of the semicircle = 2x radius = 2x7m= 14m

Displacement (AC)= AB + BC

=10m + 14m

=24m

Hence, displacement from A to C is 24m

20. Correct Matching is: a-QR; b-PQST; c-P; d-Q

Distance: It is the actual path traversed by the object in the given time. It is always positive.

Distance travelled by the particle when it moves around a circular playground
= circumference of the circle
= $2\pi r$ (r= radius of the circle)

- Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. It can be positive, negative or zero

The displacement of an object in a time can be equal to or less than distance travelled but never greater than the distance travelled.

Displacement ≤ distance

- While travelling along the semicircular playground

Let A is the initial position and B is the final position of the body.
So, AB =shortest distance joining initial and final position of the body= displacement of the body
Let the radius of the playground=r
AB= diameter of the semicircular playground= 2r
Hence, displacement of the body = 2r
This displacement is positive.

SPEED AND VELOCITY
LEVEL-I

1. Correct option: (C)
Given: speed = 72km/h= 20m/s
Time = 2s
1km/h = $\frac{5}{18}$ m/s
72km/h= 72 x $\frac{5}{18}$ m/s
72km/h = 20m/s
As Speed = $\frac{\text{distance travelled}}{\text{time taken}}$ = 20m/s
20m/s= $\frac{\text{distance travelled}}{2}$
Distance travelled = 2s x 20m/s
= 40m

2. Correct option: (A)
Speed of an object in motion is defined as the ratio of the total path length/distance covered and the corresponding time taken by the object,
Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

3. Correct option: (B)
Speed = $\frac{\text{distance travelled}}{\text{time taken}}$
SI unit of distance is m
SI unit of time is s
So, SI unit of speed is m/s.

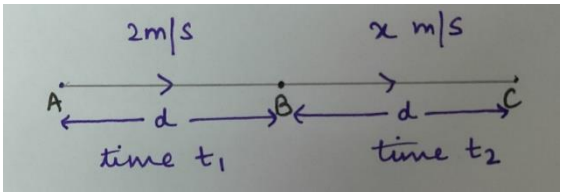
4. Correct option: (C)
1km/h = $\frac{5}{18}$ m/s
108km/h = 108 x $\frac{5}{18}$ m/s
= 30m/s

5.Correct option: (A)

Speed $= \frac{\text{distance travelled}}{\text{time taken}}$

As speed is inversely proportional to the time and directly proportional to distance.
So, more will be the distance travelled in less time, more will be the speed.
Hence, covering a large distance in short time is called high speed.

6. Correct option: (B)



Let the total distance of the journey = 2d
(a)For the first half of the journey (AB):
Given: speed = 2 ms^{-1}
Let distance = d
Time = t_1

Speed $= \frac{\text{distance travelled}}{\text{time taken}}$

Time $= \frac{\text{distance travelled}}{\text{speed}}$

$t_1 = \frac{d}{2}$ ----(1)

(b) For the remaining half of the journey (BC):
Let speed = x ms^{-1}
Distance = d
Time = t_2

Time $= \frac{\text{distance travelled}}{\text{speed}}$

$t_2 = \frac{d}{x}$ ----(2)

(a) Average speed $= \frac{\text{total distance}}{\text{total time}}$

Given: average speed = 3 ms^{-1}

Average speed $= \frac{d+d}{t_1+t_2}$

From equation 1 and 2

$$3\ ms^{-1} = \frac{2d}{\frac{d}{2}+\frac{d}{x}}$$

$$3\ ms^{-1} = \frac{2d}{d\left(\frac{1}{2}+\frac{1}{x}\right)}$$

$$3\ ms^{-1} = \frac{2}{\left(\frac{1}{2}+\frac{1}{x}\right)}$$

$$3\ ms^{-1} = \frac{2}{\frac{(x+2)}{2x}}$$

$$3\ ms^{-1} = \frac{2x\ 2x}{x+2}$$

$$3(x + 2) = 4x$$
$$3x + 6 = 4x$$
$$6 = 4x - 3x$$
$$6 = x$$

Or

$$x = 6\ ms^{-1}$$

Hence, speed with which the object travels remaining distance is 6 ms^{-1}.

7. Correct option: (C)

Given:

Displacement = 200m –west

Time= 2 s

Velocity $=\frac{\text{displacement}}{\text{time interval}}$

Velocity $=\frac{200\text{m}}{2\text{ s}}$

Velocity= 100 m/s

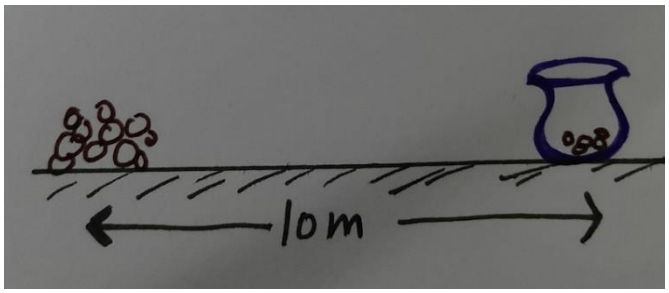
8. Correct option: (B)

1m/s= $\frac{18}{5}$ km/h

15m/s= 15 x$\frac{18}{5}$ km/h

15m/s= 54 km/h or kmph

9. Correct option: (A)



Distance travelled by the crow in one trip= distance from pebbles to +distance from pot to
pebbles

= 10m+10m

=20m

Distance travelled by the crow in 5 trips = 5 x Distance travelled by the crow in one trip

= 5 x 20m

= 100m

Total distance = 100m

Total time= 80s

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Average speed = $\frac{100}{80}$

= 1.25 m/s

10. Correct option: (B)

An odometer is an instrument used for measuring the distance travelled by a vehicle such as car.

11. Correct option: (C)

Speed $=\frac{\text{distance travelled}}{\text{time taken}}$

$=\frac{10\text{m}}{2s}$

$=5ms^{-1}$

12.Correct option: (C)

- **Calculation of distance for first 20 min**

Given: speed = 60kmh^{-1}

Time = 20 min

1 min=$\frac{1}{60}$h

20min= 20 x $\frac{1}{60}$h

20min= $\frac{1}{3}$h

As Speed =$\frac{\text{distance travelled}}{\text{time taken}}$= 60kmh^{-1}

60kmh^{-1}= $\frac{\text{distance travelled}}{\frac{1}{3}\text{h}}$

Distance travelled = $\frac{1}{3}$h x 60kmh^{-1}

= 20km

• **Calculation of distance for next 20 min**

Given: speed = 30kmh^{-1}

Time = 20 min

1 min=$\frac{1}{60}$h

20min= 20 x $\frac{1}{60}$h

20min= $\frac{1}{3}$h

As Speed =$\frac{\text{distance travelled}}{\text{time taken}}$= 30kmh^{-1}

30kmh^{-1}= $\frac{\text{distance travelled}}{\frac{1}{3}\text{h}}$

Distance travelled = $\frac{1}{3}$h x 30kmh^{-1}

= 10km

Total distance travelled = 20km +10km

Total distance travelled =30km

13.Correct option: (C)

Given: distance travelled by bike = 100km

Time taken =120 min

1 min=$\frac{1}{60}$h

120min= 120 x $\frac{1}{60}$h

120min= 2 h

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Average speed = $\frac{100\text{ km}}{2\text{h}}$

= 50kmph

14. Correct option: (B)

1 hour = 3600 seconds

1 km = 1000 m

45 kmh^{-1} = $\frac{45\text{x}1000}{3600}$ ms^{-1}

45 kmh^{-1} = 12.5 ms^{-1}

15. Correct option: (B)

Given: distance between the stations =20 km

Speed of the train= 60kmh^{-1}

Average speed = $\frac{\text{total distance}}{\text{total time}}$

60kmh^{-1} = $\frac{20\text{km}}{\text{time}}$

time = $\frac{20\text{km}}{60\text{kmh}^{-1}}$

time= $\frac{1}{3}$ h

1 h = 60 minutes

So, time= $\frac{1}{3} \times 60\ minutes$

= 20 minutes

So, time taken by the train to reach the next station is 20 minutes.

16.Correct option: (A)

Given: velocity of the jogger= 5m/s

Time = 20 seconds

Velocity = $\frac{\text{displacement}}{\text{time interval}}$

Displacement= velocity x time

= 5m/s x 20 s

= 100m

Hence, displacement of jogger is 100m.

17. Correct option: (A)

When an object is moving with a variable speed, then the average speed of the object is that constant speed with which the object covers the same distance in a given time as it does while moving with variable speed during the given time.

Average speed = $\frac{\text{total distance}}{\text{total time}}$

18. Correct option: (A)

An object is said to be moving with a uniform speed, if it covers equal distances in equal intervals of time. For example, if a car is moving on a straight road from place A to B, covers a distance of 80 km in 8 hours, and covers 10 km in every one hour, then the car is moving with a uniform speed =10km/h.

Uniform Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Given: uniform speed =10km/h

time = 2 h

Uniform Speed x time taken =distance travelled

10km/h x 2 h = distance

Distance = 20 km ------(1)

When an object is moving with a variable speed, then we have to calculate average speed. For example, if a car is moving on a straight road from place A to B and covers successive 20 km distances in timings 1 h, 1.5h, 2h and 0.75 h, the speed of the car at different locations is different, so we calculate the average speed.

Average speed = $\frac{\text{total distance}}{\text{total time}}$

given: time = 2h

from equation 1

total distance = 20km

Average speed = $\frac{20\ \text{km}}{2\text{h}}$

= 10km/h

Which is same as the uniform speed, when the speed is uniform, average speed is equal to the uniform speed.

19. Correct option: (A)

Velocity is the speed of an object in a particular direction.

Given: uniform speed of the body =u ms^{-1}

Direction of speed = east

Therefore, body is said to possess uniform velocity of u ms^{-1} towards east.

20. Correct option: (C)

Speed = 0.06 m/s

$1 m/s = \frac{18}{5}$ km/h

$0.06 m/s = 0.06 \times \frac{18}{5}$ km/h

$= \frac{6 \times 18}{100 \times 5}$ km/h

= 0.216 km/h

21. Correct option: (B)

Velocity is a vector quantity as it has both magnitude(speed) and it acts in a particular direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

22. Correct option: (C)

$\text{Velocity} = \frac{\text{displacement}}{\text{time taken}}$

Hence, rate of change of displacement is called velocity.

23.Correct option: (A)

$\text{Velocity} = \frac{\text{displacement}}{\text{time interval}}$

SI unit of displacement is m

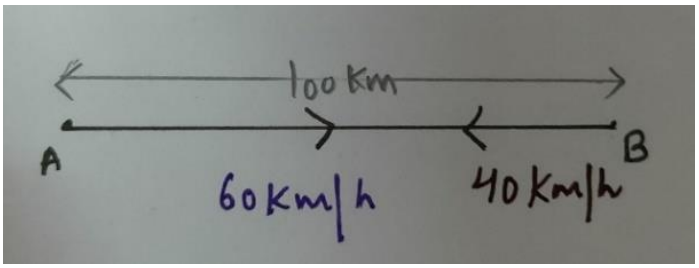
SI unit of time is s

So, SI unit of velocity is m/s.

24. Correct option: (48km/h**)**

Given: speed during going (AB) = 60km/h

Distance travelled = 100km



$\text{Speed} = \frac{\text{distance travelled}}{\text{time taken}}$

$\text{Time taken while going} = \frac{\text{distance}}{\text{speed}}$

$= \frac{100 \text{ km}}{60 \text{km/h}}$

$= \frac{5}{3}$ h

Speed while returning (BA) = 40km/h

distance travelled = 100km

$\text{Time taken to return} = \frac{\text{distance}}{\text{speed}}$

$= \frac{100 \text{ km}}{40 \text{km/h}}$

$= \frac{5}{2}$ h

Total distance = 100km+100km

= 200km

$\text{total time taken} = \frac{5}{3}\text{h} + \frac{5}{2}\text{h}$

$= \frac{25}{6}$ h **(L.C.M of 3 and 2 = 6)**

$\text{Average speed} = \frac{\text{total distance}}{\text{total time}}$

Average speed = $\frac{200}{\frac{25}{6}}$

$= \frac{200\times6}{25}$

Average speed for the whole journey = 48km/h

25.Correct option: (A)

speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Distance travelled by the particle when it goes from point A to C to B on a semicircle

= circumference of the semicircle

= πr (r= radius of the circle)

= 7π m (r = 7m)

=7x$\frac{22}{7}$m ($\pi = \frac{22}{7}$)

time taken= 11 s

speed = $\frac{7x22}{11x7}$ m/s

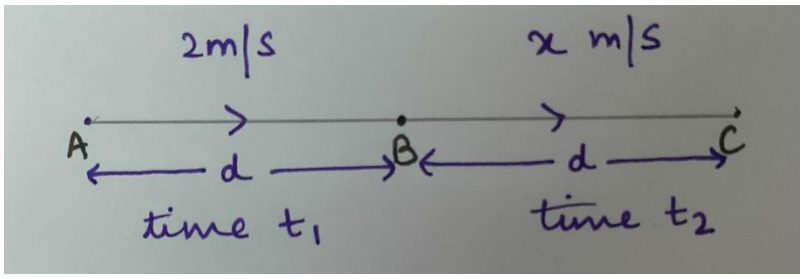
speed = 2m/s

26.Correct option: (A)

Instantaneous speed: when the speed of an object is variable/changing, then the object possesses different speeds at different instants. The speed of an object at a given instant of time is called its instantaneous speed.

Instantaneous speed of a vehicle is measured by speedometer.

27. Correct option: (B)



Let the total distance of the journey = 2d

(a)For the first half of the journey (AB):

Given: speed = 2 ms^{-1}

Let distance = d

Time = t_1

Speed =$\frac{\text{distance travelled}}{\text{time taken}}$

Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_1 = \frac{d}{2}$ ----(1)

(b) For the remaining half of the journey (BC):

Let speed = x ms^{-1}

Distance = d

Time = t_2

Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_2 = \frac{d}{x}$ ----(2)

(b) Average speed = $\frac{\text{total distance}}{\text{total time}}$

Given: average speed = 3 ms^{-1}

Average speed = $\frac{d+d}{t_1+t_2}$

From equation 1 and 2

$3\ ms^{-1} = \frac{2d}{\frac{d}{2}+\frac{d}{x}}$

$3\ ms^{-1} = \frac{2d}{d\left(\frac{1}{2}+\frac{1}{x}\right)}$

$3\ ms^{-1} = \frac{2}{\left(\frac{1}{2}+\frac{1}{x}\right)}$

$3\ ms^{-1} = \frac{2}{\frac{(x+2)}{2x}}$

$3\ ms^{-1} = \frac{2x\ 2x}{x+2}$

$3(x + 2) = 4x$

$3x + 6 = 4x$

$6 = 4x - 3x$

$6 = x$

Or

$x = 6\ ms^{-1}$

Hence, speed with which the object travels remaining distance is 6 ms^{-1}.

28.Correct option: (B)

Velocity $= \frac{\text{displacement}}{\text{time interval}}$

SI unit of displacement is m

SI unit of time is s

So, SI unit of velocity is m/s.

Velocity is the speed of an object in a particular direction.

Speed $= \frac{\text{distance travelled}}{\text{time taken}}$

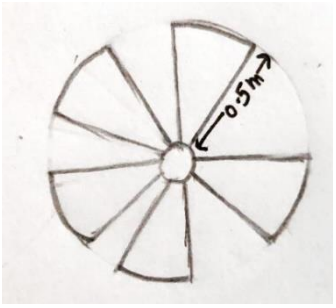
SI unit of distance is m

SI unit of time is s

So, SI unit of speed is m/s.

Hence, unit of velocity is same as speed.

29.Correct option: (A)



Given: radius of the circular path=0.5m

number of rotations=100

time taken = 50 seconds

speed = $\frac{\text{distance}}{\text{time taken}}$

Distance travelled by the particle when it goes along the circular path 1 time

= circumference of the circle

= $2\pi r$ (r= radius of the circle)

= 2π x 0.5 m (r = 0.5m)

Distance travelled by the particle when it goes along the circular path 100 times=100 x2π x 0.5 m

$= 100\pi$ m

speed = $\frac{100\pi}{50}$ m/s

$= 2\pi$ m/s

Hence, speed of the particle at the edge of the wing is 2π m/s

30.Correct option: (A)

speed = $\frac{\text{distance travelled}}{\text{time taken}}$

total distance = RQ + QP + PR

• To find the distance PR

Triangle RQP is a right-angled triangle

QP = base = 4m

RQ=perpendicular=3m

PR= hypotenuse

Using Pythagoras theorem in triangle ABC

$(hypotenuse)^2 = (base)^2 + (perpendicular)^2$

$(PR)^2 = (QP)^2 + (RQ)^2$

$(PR)^2 = (4)^2 + (3)^2$

$(PR)^2 = 16 + 9$

$(PR)^2 = 25$

PR = $\sqrt{25}$m

PR = 5 m

total distance = RQ + QP + PR

= 3m + 4m + 5m

Total distance =12m

Total time taken = 10 + 20 + 30 seconds

=60 seconds

speed = $\frac{\text{distance travelled}}{\text{time taken}}$

speed = $\frac{12}{60}$m/s

=0.2 m/s

Hence, speed of the person = 0.2 m/s

LEVEL-II

1. Correct option: (A)

speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Distance travelled by the particle when it goes from point A to O to B on a semicircle

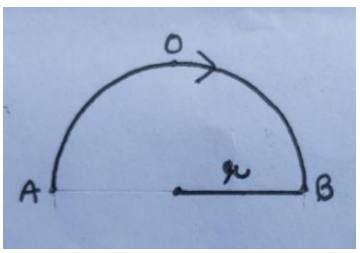
= circumference of the semicircle

= πr (r= radius of the circle)

= $2\pi\ m$ (r = 2m)

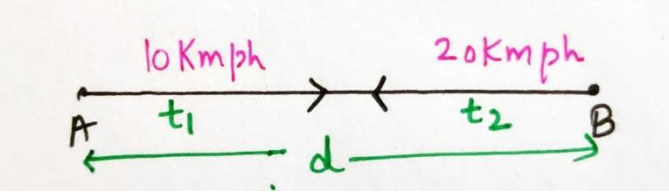
time taken= 2 s

speed = $\frac{2\pi}{2}$ m/s



= π m/s

2.Correct option: (A)



• Calculation of distance for A to B:

Given: constant speed = 10km/h

Distance= d km

Let time taken = t_1

Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_1 = \frac{d}{10}$ h - - - - - - **(1)**

• Calculation of distance for B to A:

Given: constant speed = 20km/h

Distance= d km

Let time taken = t_2

Time = $\frac{\text{distance travelled}}{\text{speed}}$

$t_2 = \frac{d}{20}$ h - - - - - - **(1)**

• Average speed for the whole trip (ABA)

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Total distance = d+d

=2d km

Total time = $t_1 + t_2$

Average speed = $\frac{2d}{t_1+t_2}$ km/h

Average speed = $\frac{2d}{\frac{d}{10}+\frac{d}{20}}$ km/h

Average speed = $\frac{2d}{d\left(\frac{1}{10}+\frac{1}{20}\right)}$ km/h

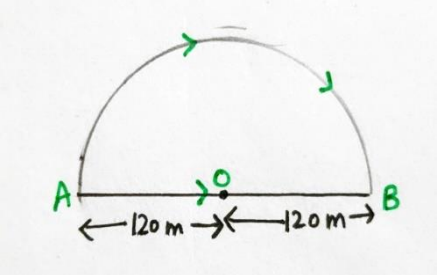
$= \frac{2}{\frac{1}{10}+\frac{1}{20}}$ km/h

$= \frac{2}{\frac{2+1}{20}}$ km/h (Lcm of 10 and 20=20)

$= \frac{2x\ 20}{3}$ km/h

Average speed = $\frac{40}{3}$ km/h or kmph

3.Correct option: (D)



Velocity = $\frac{\text{displacement}}{\text{time interval}}$

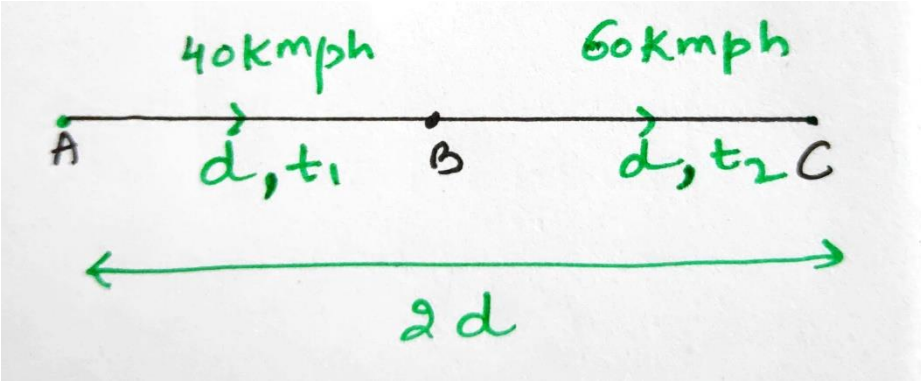
Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object.

Displacement of the cyclist is from A to O to B that is along the diameter of the semicircle
Given: radius of the semicircle = 120m
Diameter = 2r = 2x120m= 240m
Displacement = 240 m
Time = 2min
1 min = 60 sec
2 min = 2x60 sec
2 min = 120 sec
Velocity $=\frac{240}{120}$ m/s
Velocity =2m/s

4. Correct option: (D)

speed = $\frac{\text{distance travelled}}{\text{time taken}}$
Distance travelled by the particle when it completes one rotation along a circular path
= circumference of the circle
= $2\pi r$ m (r= radius of the circle)
time taken= t s
speed = $\frac{2\pi r}{t}$ m/s

5.Correct option: (A)



Let the total distance of the journey = 2d
(a)For the first half of the journey (AB):
Given: speed = 40 kmph
Let distance = d
Time = t_1
Speed $=\frac{\text{distance travelled}}{\text{time taken}}$
Time = $\frac{\text{distance travelled}}{\text{speed}}$
$t_1=\frac{d}{40}h$ ----(1)
(b) For the remaining half of the journey (BC):
speed = 60kmph
Distance = d
Time = t_2
Time = $\frac{\text{distance travelled}}{\text{speed}}$
$t_2=\frac{d}{60}h$ ----(2)
(c) Average speed = $\frac{\text{total distance}}{\text{total time}}$
Average speed = $\frac{d+d}{t_1+t_2}$

From equation 1 and 2

Average speed = $\frac{2d}{\frac{d}{40}+\frac{d}{60}}$ kmph

Average speed = $\frac{2d}{d\left(\frac{1}{40}+\frac{1}{60}\right)}$ kmph

Average speed = $\frac{2}{\left(\frac{1}{40}+\frac{1}{60}\right)}$ kmph (LCM OF 40 and 60 = 120)

Average speed= $\frac{2}{\frac{(3+2)}{120}}$ kmph

Average speed= $\frac{2x120}{5}$ kmph

Average speed= $\frac{2x120}{5}$ kmph

Average speed= 48 kmph

1km/h = $\frac{5}{18}$ m/s

48km/h= 48 x $\frac{5}{18}$ m/s

48km/h = 13.33m/s

Hence, average speed= 13.33m/s

6. Correct option: (A)

Average speed = $\frac{\text{total distance}}{\text{total time}}$

Distance travelled by the particle when it goes from point A to B on a semicircle

= circumference of the semicircle

= πr (r= radius of the circle)

= π m (r = 1m)

time taken= 1 s

Average speed = $\frac{\pi}{1}$ m/s

= π m/s

7.Correct option: (A), (B)

Given: uniform speed = $144kmh^{-1}=40ms^{-1}$

Time = 5s

1km/h = $\frac{5}{18}$ m/s

144km/h= 144 x $\frac{5}{18}$ m/s

144km/h = 40m/s

As Speed = $\frac{\text{distance travelled}}{\text{time taken}}$ = 40m/s

40m/s= $\frac{\text{distance travelled}}{5}$

Distance travelled = 5s x 40m/s

= 200m

1m =100 cm

200m = 20000cm

Hence, Distance travelled = 200m or 20000cm

8. Correct option: (A), (B), (C), (D)

Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

SI unit of distance is m

SI unit of time is s

So, SI unit of speed is m/s.

C.G.S unit of speed is cm/s

Other units of speed may be kmph or km/s

9. Correct option: (A), (B)

(A) Velocity is a vector quantity as it has both magnitude(speed) and it acts in a particular direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

(B) Velocity $=\frac{\text{displacement}}{\text{time interval}}$

Hence, rate of change of displacement is called velocity.

(C) Velocity $=\frac{\text{displacement}}{\text{time interval}}$

Hence, if displacement is zero then velocity is also zero.

(D) m/s is the unit of velocity.

10.Correct options: (A), (B), (C)

Velocity $=\frac{\text{displacement}}{\text{time interval}}$

SI unit of displacement is m

SI unit of time is s

So, SI unit of velocity is m/s.

C.G.S unit of velocity is cm/s

Higher unit of velocity is km/h

11. Correct option: (A), (B), (C)

(A) Velocity is a vector quantity as it has both magnitude(speed) and it acts in a particular direction.

For example, if velocity = 20 km/h towards north

Then 20 km/h is the magnitude(speed) and north is the direction.

(B) Velocity $=\frac{\text{displacement}}{\text{time interval}}$

SI unit of displacement is m

SI unit of time is s

So, SI unit of velocity is m/s.

(C) Velocity $=\frac{\text{displacement}}{\text{time interval}}$

Hence, if displacement is 0 then velocity is also 0.

(D) velocity is a vector quantity.

12. Correct option: (B)

Distance is the actual path traversed by the object in the given time. Distance travelled by an object can never be negative.

As Average speed $= \frac{\text{total distance}}{\text{total time}}$

So, average speed of an object can never be negative.

Whereas displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. It can be positive, negative or zero

The displacement of an object in a time can be equal to or less than distance travelled but never greater than the distance travelled.

As, Average velocity $=\frac{\text{total displacement}}{\text{time interval}}$

So, average velocity of an object in a time can be equal to or less than average speed but never greater than the average speed.

Displacement ≤ distance

Average Velocity ≤ average speed

Numerical ratio($\frac{\text{average velocity}}{\text{average speed}}$)

Can be equal to 1 or less than 1. Since, the denominator(speed) is either greater than or equal to the numerator(velocity).

13.Correct options: (A), (B)

1km/h = $\frac{5}{18}$ m/s

54km/h = 54 x $\frac{5}{18}$ m/s

54km/h = 15m/s

54km/h is also written as 54kmph

14. Correct option: (C)

Displacement is the shortest distance between the initial and final positions of the object in a given time. It is represented by a vector drawn from initial to final position of the object, during the given interval of time. It is a vector quantity as it possesses both magnitude and direction.

As the aero plane flies a distance of 324km towards south

So, displacement = 324km-south.

15. Correct option: (A)

Velocity = $\frac{\text{displacement}}{\text{time interval}}$

Given: displacement = 324 km-south

Time interval = 20 minutes

1 minute = $\frac{1}{60}$hour

20minutes=$\frac{20}{60}$ hour

= $\frac{1}{3}$hour

Velocity=$\frac{324}{\frac{1}{3}}$ km/h

= 324x 3 km/h

=972 km/h

16. Correct option: (C)

Given: displacement = 90 km-east

Time = 1 $\frac{1}{2}$hr = $\frac{3}{2}$hr

Velocity = $\frac{\text{displacement}}{\text{time interval}}$

Velocity = $\frac{90\times2}{3}$ km/hr

= $\frac{90}{\frac{3}{2}}$km/hr

= $\frac{90\times2}{3}$

=60 km/hr

Velocity of aero plane =60 km/hr

17.Correct option: (B)

Given: distance travelled = 540 m

Time taken = 1minute= 60 seconds

Speed = $\frac{\text{distance travelled}}{\text{time taken}}$

= $\frac{540\text{m}}{60\text{s}}$

Speed =9m/s

18.Correct option: (C)

distance covered by the race horse = 540m (given)

19.Correct option: (B)

Given: distance travelled = 540 m

Time taken = 1minute= 60 seconds

$$\text{Speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

$$= \frac{540\text{m}}{60\text{s}}$$

Speed =9m/s

$1\text{m/s} = \frac{18}{5}\ \text{km/h}$

$9\text{m/s} = 9 \times \frac{18}{5}\ \text{km/h}$

$= 32.4\ \text{km/h}$

Speed of the race horse = 32.4 km/h

20. Correct Matching is: a-Q; b-S; c-PR; d-PT

(a) 1km= 1000m

1hr= 3600s

$1\text{km/hr.} = \frac{1000}{3600}\ \text{m/s}$

$= \frac{5}{18}\ \text{m/s}$

(b) Instantaneous speed: when the speed of an object is variable/changing, then the object possesses different speeds at different instants. The speed of an object at a given instant of time is called its instantaneous speed.

Instantaneous speed of a vehicle is measured by speedometer.

(c) Average velocity: For a given motion of an object, the average velocity is that uniform velocity with which the object will cover the same displacement in a given interval of time as it does with actual variable velocity during the given interval of time

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time taken}}$$

SI unit of Average velocity is m/s.

(d)Uniform speed: An object is said to be moving with a uniform speed, if it covers equal distances in equal intervals of time. For example, if a car is moving on a straight road from place A to B, distance of 80 km in 4 hours, and covers 20 km in every one hour, then the car is moving with a uniform speed =20km/h. SI unit of speed is m/s.

ACCELERATION
LEVEL- I

1. Correct option: (A)

Acceleration of an object in motion is defined as the ratio of change in velocity and the corresponding time taken by the object.

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

Hence, rate of change of velocity =acceleration.

2. Correct option: (A)

Given: initial velocity(u) of car =10m/s

Final velocity(v) of car= 0m/s (because, car comes to rest)

Time taken = 5 s

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a = \frac{v-u}{t}$$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-10}{5}\ m/s^2$

$a = -2\ m/s^2$

acceleration $= -2 m/s^2$

3. Correct option: (B)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

The C.G.S unit of velocity = cm/s

The C.G.S unit of time = s

The C.G.S unit of acceleration = $\frac{cm}{s \times s}$

$= cm/s^2$

4. Correct option: (A)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

As it is given in the question that velocity of particle is uniform or constant means, velocity is not changing. so, change in velocity is zero.

Hence, acceleration of the particle is also zero.

5. Correct option: (A)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

If the velocity is increasing, then change in velocity will be positive and acceleration will be positive.

If the velocity is decreasing, then change in velocity will be negative and acceleration will be negative.

Negative acceleration is also called retardation.

Hence, decrease in velocity per unit time is called **retardation**.

6.Correct option: (B)

Vector quantities are those quantities which have both magnitude(size) and direction.

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Velocity is a vector quantity. Velocity is the speed of an object in a particular direction.

Acceleration is a vector quantity as it depends on the change in velocity.

7. Correct option: (C)

Given: initial velocity(u) of scooter =0m/s (because, scooter is starting from rest)

Final velocity(v) of scooter= 36km/h =10m/s

$1 km/h = \frac{5}{18}\ m/s$

$36 km/h = 36 \times \frac{5}{18}\ m/s$

$= 10 m/s$

Time taken = 10s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{10-0}{10}\ m/s^2$

$a = 1\ m/s^2$

acceleration $= 1\ m/s^2$

8.Correct option: (D)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

The C.G.S unit of velocity = cm/s

The C.G.S unit of time = s

The C.G.S unit of acceleration = $\frac{\text{cm}}{\text{s}\times\text{s}}$

$=\text{cm}/s^2$

The SI unit of velocity = m/s

The SI unit of time = s

The SI unit of acceleration = $\frac{\text{m}}{\text{s}\times\text{s}}$

$=\text{m}/s^2$

km/s^2 is also an unit of acceleration.

9. Correct option: (C)

An object is said to be in motion, if it changes its position with respect to surroundings with the passage of time and the rate of change of motion of an object is also called speed.

If this rate of change of motion is in specified direction, it is called velocity.

10. Correct option: (A)

An object is said to be moving with a uniform acceleration if its velocity changes by equal amounts in equal intervals of time.

11.Correct option: (C)

Negative acceleration is also called retardation.

12.Correct option: (A)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

The SI unit of velocity = m/s

The SI unit of time = s

The SI unit of acceleration = $\frac{\text{m}}{\text{s}\times\text{s}}$ $=\text{m}/\text{s}^2$

13. Correct option: (B)

Given: acceleration of car =3 m/s^2

Initial velocity(u) of car =0m/s (the car is starting from rest)

Final velocity(v) of car= 54m/s

To find: time taken to acquire velocity of 54m/s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$3 = \frac{54-0}{\text{time taken}}$

Time taken= $\frac{54}{3}$s

Time taken = 18 s

14.Correct option: (D)

Given: acceleration = $2\text{m}/\text{s}^2$

Time= 2s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Acceleration x time taken= change in velocity

$2\text{m}/\text{s}^2$ x 2s = change in velocity

4 m/s = change in velocity
change in velocity = 4m/s

15. Correct option: (C)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$
C.G.S is centimeter(cm) –gram(g)- second(s) system of units
The C.G.S unit of velocity = cm/s
The C.G.S unit of time = s
The C.G.S unit of acceleration = $\frac{\text{cm}}{\text{s}\times\text{s}}$
$=\text{cm}/s^2$
The SI unit of velocity = m/s
The SI unit of time = s
The SI unit of acceleration = $\frac{\text{m}}{\text{s}\times\text{s}}$
$=\text{m}/s^2$
km/s^2 is also an unit of acceleration.

16. Correct option: (C)

Speed = 0.06 m/s
1m/s= $\frac{18}{5}$ km/h
0.06m/s = 0.06 x $\frac{18}{5}$ km/h
= $\frac{6\times18}{100\times5}$ km/h
= 0.216 km/h

17.Correct option: (C)

initial velocity(u) of car =10m/s
Final velocity(v) of car=0m/s (car stops in 2seconds)
Time taken = 2s
Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$
$a = \frac{v-u}{t}$ *here*, a= acceleration
u= initial velocity
v = final velocity
$a = \frac{0-10}{2}\ \text{m/s}^2$
$a= -5\ \text{m/s}^2$
acceleration = $-\ 5\text{m/s}^2$
Negative acceleration is also called **retardation**.

18. Correct option: (B)

Acceleration of an object in motion is defined as the ratio of change in velocity and the corresponding time taken by the object.
Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$
If the velocity is increasing, then change in velocity will be positive and acceleration will be positive.

19.Correct option: (C)

Given: acceleration of particle $=5\text{m}/s^2$
Initial velocity(u) of particle =5m/s
Final velocity(v) of particle= 25m/s
To find: time taken
Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$5 = \frac{25-5}{\text{time taken}}$

Time taken= $\frac{25-5}{5}$s

Time taken= $\frac{20}{5}$s

Time taken = 4s

20.Correct option: (C)

initial velocity(u) of car =10m/s

Final velocity(v) of car=0m/s (car stops in 20seconds)

Time taken = 20s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-10}{20}\ m/s^2$

$a = -0.5\ m/s^2$

acceleration = $-\ 0.5m/s^2$

Negative acceleration is also called **retardation**.

Retardation= $+0.5m/s^2$

21. Correct option: (D)

initial velocity(u) of vehicle =10m/s

Final velocity(v) of vehicle=0m/s (vehicle stops in 5seconds)

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-10}{5}\ m/s^2$

$a = -2\ m/s^2$

acceleration = $-\ 2m/s^2$

Negative acceleration is also called **retardation/deceleration**.

deceleration= $+2m/s^2$

22.Correct option: (C)

initial velocity(u) of car =20m/s

Final velocity(v) of car=0m/s (car stops in 5seconds)

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-20}{5}\ m/s^2$

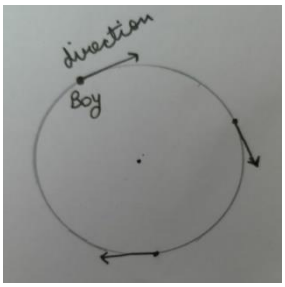
$a = -4m/s^2$

acceleration = $-\ 4m/s^2$

Negative acceleration is also called **retardation/deceleration**.
retardation= $+4m/s^2$

23. Correct option: (C)

As the boy is moving with a constant speed on merry-go-round, he is moving in a circular motion.
Although his speed is constant but direction is changing. So, his velocity is changing because velocity along a circular path is directed along the tangent to the circle. Thus, change in his velocity leads to acceleration. Hence, boy is in accelerated motion.



24.Correct option: (A)

initial velocity(u) of car =90m/s
Final velocity(v) of car=0m/s (car stops in 3seconds)
Time taken = 3s
Average acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-90}{3} m/s^2$

$a = -30m/s^2$

average acceleration = $- 30m/s^2$
Negative acceleration is also called retardation**/deceleration**.
Average deceleration= $+30m/s^2$

25. Correct option: (C)

initial velocity(u) of car =25m/s
Final velocity(v) of car=10m/s
Time taken = 5s
Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{10-25}{5} m/s^2$

$a = \frac{-15}{5} m/s^2$

$a = -3m/s^2$

acceleration = $- 3m/s^2$

26. Correct option: (B)

initial velocity(u) of body =10m/s
Final velocity(v) of body=40m/s
Time taken = 2s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{40-10}{2}\ m/s^2$

$a = \frac{30}{2}\ m/s^2$

$a = 15 m/s^2$

acceleration = $15 m/s^2$

27.Correct option: (A)

Given: initial velocity(u) of car =18km/h = 5m/s

Final velocity(v) of car= 72km/h= 20m/s

1km/h = $\frac{5}{18}$ m/s

18km/h = 18 x $\frac{5}{18}$ m/s

= 5m/s

72km/h = 72 x $\frac{5}{18}$ m/s

= 20m/s

Time taken = 30s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{20-5}{30}\ m/s^2$

$a = \frac{15}{30} m/s^2$

acceleration =$0.5\ m/s^2$

28.Correct option: (A)

Given: initial velocity(u) of car= 108km/h

Final velocity(v) of car= 36km/h

Time taken = 4 s

1second=$\frac{1}{3600}$h

4second=$\frac{4}{3600}$h

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{36-108}{4/3600}\ km/hr^2$

$a = \frac{-72x3600}{4}\ km/hr^2$

$a = -64800\ km/hr^2$

acceleration =$-64800\ km/hr^2$

deceleration = $64800\ km/hr^2$

Negative acceleration is also called **retardation/deceleration**.

29.Correct option: (D)

• Calculation of acceleration of first car

Given: initial velocity(u) of car =0m/s (because, car is starting from rest)

Final velocity(v) of car= 18km/h =5m/s

1km/h = $\frac{5}{18}$ m/s

18km/h = 18 x $\frac{5}{18}$ m/s

= 5m/s

Time taken = 0.5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{5-0}{0.5}\ m/s^2$

$a= \frac{5x10}{5} m/s^2$

$a= \frac{50}{5} m/s^2$

$a= 10m/s^2$

acceleration of first car =10 m/s^2

• Calculation of acceleration of second car

Given: initial velocity(u) of car =0m/s (because, car is starting from rest)

Final velocity(v) of car= 36km/h =10m/s

1km/h = $\frac{5}{18}$ m/s

36km/h = 36 x $\frac{5}{18}$ m/s

= 10m/s

Time taken = 0.5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{10-0}{0.5}\ m/s^2$

$a= \frac{10x10}{5} m/s^2$

$a= \frac{100}{5} m/s^2$

$a= 20m/s^2$

acceleration of second car =20 m/s^2

30.Correct option: (D)

Given: initial velocity(u) of cheetah =0m/s (because, cheetah is starting from rest)

Final velocity(v) of cheetah= 135km/h = $\frac{75}{2}$m/s

1km/h = $\frac{5}{18}$ m/s

135km/h = 135 x $\frac{5}{18}$ m/s

= $\frac{75}{2}$m/s

Time taken = 2s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{\frac{75}{2}-0}{2}\ m/s^2$

$a= \frac{75}{2x2} m/s^2$

$a= \frac{75}{4} m/s^2$

$a= \frac{75}{4} m/s^2$

acceleration of cheetah = $\frac{75}{4} m/s^2$

LEVEL-II

1. Correct option: (C)

Given: initial velocity(u) of car= 108km/h =30m/s

Final velocity(v) of car= 72km/h =20m/s

1km/h = $\frac{5}{18}$ m/s

108km/h = 108 x $\frac{5}{18}$ m/s

= 30m/s

72km/h= 72 x $\frac{5}{18}$ m/s

=20m/s

Time taken = 4 s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{20-30}{4}\ m/s^2$

$a= -2.5\ m/s^2$

acceleration =$-2.5\ m/s^2$

deceleration = $2.5\ m/s^2$

Negative acceleration is also called **retardation/deceleration**.

2.Correct option: (B)

initial velocity(u) of car =54km/h = 15m/s

Final velocity(v) of car= 0km/h=0m/s (car stops in 5seconds)

1km/h = $\frac{5}{18}$ m/s

54km/h = 54 x $\frac{5}{18}$ m/s

= 15m/s

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-15}{5}\ m/s^2$

a= -3 ms^{-2}

acceleration = - 3 ms^{-2}

Negative acceleration is also called **retardation**.

3.Correct option: (A)

Given: initial velocity(u) of scooter =0m/s (because, scooter is starting from rest)

Final velocity(v) of scooter= 54km/h =10m/s

1km/h = $\frac{5}{18}$ m/s

54km/h = 54x $\frac{5}{18}$ m/s

= 15m/s

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{15-0}{5}\ m/s^2$

$a= 3\ m/s^2$

acceleration = $3m/s^2$

4.Correct option: (C)

Given: initial velocity(u) of particle =10m/s

Final velocity(v) of particle= 30m/s

Time taken = 5 s

- **Calculation of acceleration**

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{30-10}{5}\ m/s^2$

$a = \frac{20}{5}\ m/s^2$

$a= 4m/s^2$

acceleration =$4m/s^2$

- **Calculation of time taken**

Given: initial velocity(u) of particle =30m/s

Final velocity(v) of particle= 40m/s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$4 = \frac{40-30}{t}$

$t= \frac{10}{4}$

t= 2.5s

hence, time taken by the particle = 2.5 seconds

5.Correct option: (C)

Given:

initial velocity(u) of body =5m/s

Final velocity(v) of body=35m/s

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{35-5}{5}\ m/s^2$

$a = \frac{30}{5}\ m/s^2$

$a = 6m/s^2$

acceleration = $6m/s^2$

6.Correct option: (A)

Given: initial velocity(u) of car =36km/h = 10m/s

Final velocity(v) of car= 72km/h= 20m/s

$1km/h = \frac{5}{18}\ m/s$

$36km/h = 36 \times \frac{5}{18}\ m/s$

$= 10m/s$

$72km/h = 72 \times \frac{5}{18}\ m/s$

$= 20m/s$

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{20-10}{5}\ m/s^2$

$a = \frac{10}{5} m/s^2$

$a = 2m/s^2$

acceleration =2 m/s^2

7. Correct option: (A)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

As it is given in the question that velocity of particle is uniform or constant means, velocity is not changing. so, change in velocity is zero.

Hence, acceleration of the particle is also $0m/s^2$.

8.Correct option: (A), (B)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

The C.G.S unit of velocity = cm/s

The C.G.S unit of time = s

The C.G.S unit of acceleration = $\frac{cm}{s \times s}$

$= cm/s^2$

The SI unit of velocity = m/s

The SI unit of time = s

The SI unit of acceleration = $\frac{m}{s \times s}$

$= m/s^2$

9. Correct option: (A), (B), (C), (D)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Velocity is a vector quantity as it has both magnitude(speed) and direction.

So, Acceleration is a vector quantity as it depends on the change in velocity.

Velocity can change in following ways:

- Its magnitude (speed) changes.
- Its direction changes.
- Its magnitude(speed) change but direction does not change.
- Its direction changes but magnitude(speed) does not change.

So, a body can do acceleration due to any of the above reasons.

10. Correct option: (A), (B), (C)

Vector quantities are those quantities which have both magnitude(size) and direction.

(a) Displacement is the shortest distance between two positions of the object and its direction is from initial to final position of the object. Hence it is distance measured in a particular direction.so it is a vector quantity.

(b) Velocity =$\frac{\text{displacement}}{\text{time interval}}$

Velocity is also a vector quantity as it depends on displacement. Velocity is the speed of an object in a particular direction.

(c) Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Acceleration is a vector quantity as it depends on the change in velocity.

(d) Distance is the actual path traversed by the object in the given time. It has just the magnitude and no direction so it is a scalar quantity.

11.Correct option: (A), (C)

Given: acceleration = $4ms^{-2}$

Time= 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a \times t = v - u$

$4ms^{-2} \times 5s = v-u$

$20ms^{-1} = v-u$

$v-u = 20ms^{-1}$

initial velocity (u) = 0ms^{-1} (car is starting from rest)

so, velocity at the end of 5 sec = $20ms^{-1}$ or 72 kmph

$1m/s = \frac{18}{5}$ km/h

$20m/s = 20 \times \frac{18}{5}$ km/h

$= 72$ km/h

12.Correct option: (B), (C), (D)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

C.G.S is centimeter(cm) –gram(g)- second(s) system of units

The C.G.S unit of velocity = cm/s

The C.G.S unit of time = s

The C.G.S unit of acceleration = $\frac{cm}{s \times s}$

$=cm/s^2$

The SI unit of velocity = m/s

The SI unit of time = s

The SI unit of acceleration = $\frac{m}{s \times s}$

$=m/s^2$

Km/h and m/s are the units of velocity.

13.Correct option: ($5\ m/s^2$ **)**

Given: initial velocity(u) of car = 5m/s

Final velocity(v) of car= 30m/s

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{30-5}{5}\ m/s^2$

$a = \frac{25}{5} m/s^2$

$a = 5m/s^2$

acceleration $=5\ m/s^2$

14.Correct option: (A)

initial velocity(u) of car =50m/s

Final velocity(v) of car=0m/s (car stops in 20 seconds)

Time taken =20s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-50}{20} m/s^2$

$a = -2.5\ m/s^2$

acceleration = $-2.5\ m/s^2$

Negative acceleration is also called **retardation**.

15.Correct option: (D)

Given: initial velocity(u) of car =9km/h = 2.5m/s

Final velocity(v) of car= 36km/h= 10m/s

Change in velocity = v-u = 10m/s -2.5 m/s = 7.5m/s

16.Correct option: (C)

Given: initial velocity(u) of car =9km/h = 2.5m/s

Final velocity(v) of car= 36km/h= 10m/s

1km/h = $\frac{5}{18}$ m/s

9km/h = 9 x $\frac{5}{18}$ m/s

= 2.5m/s

36km/h = 36 x $\frac{5}{18}$ m/s

= 10m/s

Time taken = 10s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{10-2.5}{10}\ m/s^2$

$a= \frac{7.5}{10} m/s^2$

$a= 0.75 m/s^2$

acceleration $=0.75 m/s^2$

17. Correct option: (A)

Given: initial velocity(u) of car at point A= 10m/s

Final velocity(v) of car at point B= 20m/s

Time taken = 5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{20-10}{5}\ m/s^2$

$a= \frac{10}{5} m/s^2$

$a= 2 m/s^2$

acceleration $=2\ m/s^2$

18.Correct option: (B)

Given: acceleration of car $=-3 m/s^2$ (retardation)

Initial velocity(u) of car =30m/s

Final velocity(v) of car= 0m/s (car is coming to rest)

To find: time taken

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$-3 = \frac{0-30}{\text{time taken}}$

Time taken= $\frac{0-30}{-3}$s

Time taken= 10s

19.Correct option: (C)

initial velocity(u) of train =100m/s

Final velocity(v) of train=0m/s (train stops in 5seconds)

Time taken =5s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{0-100}{5}\ m/s^2$

$a= -20\ m/s^2$

acceleration = $-20\ m/s^2$

Negative acceleration is also called **retardation**.

retardation=$20\ ms^{-2}$

20. Correct Matching is: a-Q; b-R; c-PQR; d-P

• Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

• If the velocity is increasing(v>u) then change in velocity will be positive and acceleration will be positive.

• If the velocity is decreasing(v<u) then change in velocity will be negative and acceleration will be negative.

Negative acceleration is also called **retardation**.

• An object is said to be moving with a uniform acceleration if its velocity changes by equal amounts in equal intervals of time.

• Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

Time is represented as [T]

$[M^0L^1T^{-1}]$ is the dimensional formula of velocity.

Acceleration = $\frac{[M^0L^1T^{-1}]}{[T]}$

Acceleration =$[M^0L^1T^{-1-1}]$

Acceleration =$[M^0L^1T^{-2}]$

$[M^0L^1T^{-2}]$ is the dimensional formula of acceleration.

EQUATIONS OF MOTION
LEVEL-I

1.Correct options: (A), (B), (C), (D)

$v=v_0 + at$ is a linear equation as it is of the form y**=mx + c.**

This equation of motion is true for uniform motion/uniformly accelerated motion or for any type of motion along straight line.

2.Correct option: (C)

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement of the object

Given: $u^2 = -2aS$

Putting value of $u^2 = -2aS$ in equation (1)

$v^2 - (-2aS) = 2aS$

$v^2 + 2aS = 2aS$

$v^2 = 2aS - 2As$

$v^2 = 0$

v = 0m/s

hence, final velocity = 0m/s

3.Correct option: (12.5 m**)**

Given: initial velocity(u) = 36kmph=10m/s

$1km/h = \frac{5}{18}\ m/s$

$36km/h = 36\ x\ \frac{5}{18}\ m/s$

$= 10 m/s$

final velocity(v)= $0 m/s$ (bus stops due to applied brakes)

$a = -4\ m/s^2$ (negative sign shows retardation)

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS$ ------(1) Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$

$(0)^2 - (10)^2 = 2x(-4)xS$

0-100 = -8 XS

-100=-8S

$\frac{100}{8} = S$

$12.5\ m = S$

Distance travelled = 12.5 m

4.Correct option: (C)

The second equation of motion gives Position-time relation given as follows

$S = ut + \frac{1}{2}at^2$------(1) here, u= initial velocity of the object

a = acceleration of the object

t= time taken

S= displacement of the object

Given: $S = 2t - \frac{1}{2}3t^2$

Comparing the given equation with equation (1)

Acceleration(a)= $-3\ m/s^2$

5.Correct option: (C)

The first equation of motion gives Velocity time relation given as follows

v=u + at --------(1)

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

given: v = 2 + 4t

comparing the given equation with equation (1)

$a = 4m/s^2$

hence, acceleration(a)= $4m/s^2$

6.Correct option: (A)

The second equation of motion gives Position-time relation given as follows

$S = ut + \frac{1}{2}at^2$------(1) u= initial velocity of the object

a = acceleration of the object

t= time taken

S= displacement of the object

Given: $S = 2t + \frac{1}{2}3t^2$

Comparing the given equation with equation (1)

Initial velocity(u)= $2m/s$

7.Correct option: (C)

Given: v= 5 + 8t

Put t = 5 in the given equation

V= 5 + 8x5

V= 5 + 40

V=45 m/s

8.Correct option: (A)

The first equation of motion is as follows

v =u + at

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

It gives the velocity of an object at any time t, when moving with initial velocity u and uniform acceleration a.

Hence, it gives the velocity-time relation

9.Correct option: (D)

Displacement in the nth second of motion is given by

$S_{nth} = u + \frac{1}{2}a(2n-1)$---------(1)

u= initial velocity of the object

a = acceleration of the object

S_{nth} = displacement of the object in nth second

Given: $S_{nth} = \frac{1}{2}a(2n-1)$

Comparing the given equation with the equation (1)

u= 0m/s

hence, initial velocity u=0m/s

10.Correct option: (A)

The second equation of motion gives Position-time relation given as follows

$S = ut + \frac{1}{2}at^2$

u= initial velocity of the object

a = acceleration of the object

t= time taken

S= displacement of the object

11.Correct option: (C)

The second equation of motion gives Position-time relation given as follows

$S = ut + \frac{1}{2}at^2$------(1)

u= initial velocity of the object

a = acceleration of the object

t= time taken

S= displacement of the object

Given: $S = \frac{1}{2}at^2$

Comparing the given equation with equation (1)

Initial velocity(u)= 0m/s

12. Correct option: (A)

From the first equation of motion

Velocity time relation is given as

v=u + at --------(1)

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

given: v = 2 + 4t
comparing the given equation with equation (1)
u= 2m/s
hence, initial velocity(u) = 2m/s

13.Correct option: (A)

Displacement = average velocity x time

Average velocity $=\frac{[\text{initial velocity (u)}+\text{final velocity (v)}]}{2}$

Displacement (S)= $\frac{[\text{initial velocity (u)}+\text{final velocity (v)}]}{2}$ x time (t)

$S=\frac{(u+v)}{2}xt$

14.Correct option: (B)

The second equation of motion gives displacement-time relation given as follows

$S = ut + \frac{1}{2}at^2$ u= initial velocity of the object
a = acceleration of the object
t= time taken
S= displacement of the object

15.Correct option: (A)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

As it is given in the question that velocity of particle is uniform or constant means, velocity is not changing. so, change in velocity is zero.
Hence, acceleration of the particle is also zero.

16.Correct option: (C)

Displacement in the nth second of motion is given by

$S_{nth} = u + \frac{1}{2}a(2n-1)$---------(1) u= initial velocity of the object
a = acceleration of the object
S_{nth} = displacement of the object in nth second

Given: u=0 m/s (body starts from rest)
n= 5
putting the values in equation (1)

$S_{5th} = 0 + \frac{1}{2}a(2x5-1)$

$S_{5th} = \frac{1}{2}a(10-1)$

$S_{5th} = \frac{1}{2}a(9)$

$S_{5th} = \frac{9a}{2}$

Hence, distance travelled in 5th second $=\frac{9a}{2}$

17.Correct option: (A)

Given: initial velocity(u) = 15m/s
final velocity(v)= 0m/s (car stops due to applied brakes)
distance travelled(S)= 3m

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$

$(0)^2 - (15)^2 = 2xax3$

0-225 = 6a

-225=6a

$-\frac{225}{6} = a$

$-37.5\ m/s^2 = a$

acceleration=$-37.5\ m/s^2$

Negative acceleration is also called retardation.

Hence, retardation =$37.5\ m/s^2$

18.Correct option: (A)

Average Velocity =$\frac{\text{total displacement}}{\text{total time}}$

Average Velocity x time =Total displacement

Or displacement = average velocity x time

19.Correct option: (C)

Displacement in the nth second of motion is given by

$S_{nth} = u + \frac{1}{2}a(2n-1)$---------(1) u= initial velocity of the object

a = acceleration of the object

S_{nth} = displacement of the object in nth second

Given: a=$6ms^{-2}$

u=0 m/s (body starts from rest)

n= 3

putting the values in equation (1)

$S_{3rd} = 0 + \frac{1}{2}6(2x3-1)$

$S_{3rd} = 3(6-1)$

$S_{3rd} = 3x5$

$S_{3rd} = 15m$

Hence, distance travelled in the 3rd second =15m

20.Correct option: ($5\ ms^{-2}$**)**

Given: initial velocity(u) = 0m/s (bike starts from rest)

final velocity(v)= $72kmh^{-1} = 20ms^{-1}$

$1km/h = \frac{5}{18}\ m/s$

$72km/h = 72\ x\ \frac{5}{18}\ m/s$

$= 20m/s$

distance travelled(S)= 40m

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$

$(20)^2 - (0)^2 = 2xax40$
400 = 80a
$\frac{400}{80} = a$
$5\ ms^{-2} = a$
acceleration=$5\ ms^{-2}$

21.Correct option: (C)

Given: initial velocity(u) =$22.4ms^{-1}$
Final velocity(v)=$0ms^{-1}$ (car stops after sometime)
Time = 2.55s

• Calculation of the acceleration

Using the First equation of motion
v=u + at --------(1)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

putting the given values in equation 1
0= 22.4+ a x 2.55
-22.4 = 2.55a
$-\frac{22.4}{2.55} = a$
a=-8.78m/s^2
negative sign shown the retardation of the car

• Calculation of the skidding distance of the car

using the third equation of motion
$v^2 - u^2 = 2aS$ ------(2)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object

Putting the values in equation (2)
$(0)^2 - (22.4)^2 = 2x(-8.78)x\ S$
-501.76 = -17.56x S
$\frac{501.76}{17.56} = S$
28.6m = S
Hence, skidding distance of the car= 28.6 m

22.Correct option: (A)

Given: initial velocity(u) =5m/s
Final velocity(v)=15m/s
Time = 5s

• Calculation of the acceleration

Using the First equation of motion
v=u + at --------(1)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

putting the given values in equation 1
15 = 5 + a x 5
15-5 = 5a
10 = 5a
$\frac{10}{5} = a$
a=2 m/s^2
• Calculation of the displacement of the body
using the third equation of motion
$v^2 - u^2 = 2aS$ ------(2)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object

Putting the values in equation (2)
$(15)^2 - (5)^2 = 2x2x\,S$
225-25 = 4S
200=4S
$\frac{200}{4} = S$
50m = S
Hence, displacement of the body= 50 m

23.Correct option: (C)
Given: initial velocity (u) = 1m/s
Time taken(t)=2s
Distance travelled(S)= 5m
To find: final velocity (v)
• Calculation of acceleration
Using The second equation of motion
$S = ut + \frac{1}{2}at^2$ ---------(1)

u= initial velocity of the object
a = acceleration of the object
t= time taken
S= displacement/distance travelled by the object

putting the values in equation (1)
$5 = 1x\,2 + \frac{1}{2}ax(2)^2$
$5 = 2 + \frac{1}{2}$ x a x4
5 = 2 + 2a
5-2 =2a
3 =2a
$\frac{3}{2} = a$
a = 1.5 m/s^2
• Calculation of final velocity(v)
Using the First equation of motion
v = u + at
putting the values of u, a and t
v = 1 + 1.5 x 2

v= 1 + 3
v = 4 m/s
hence, final velocity (v) = 4m/s

24.Correct option: (D)

Given: initial velocity(u) =12.5m/s =45kmph

$1\text{m/s} = \frac{18}{5}$ kmph

$12.5\text{m/s} = 12.5 \text{ x } \frac{18}{5}$ kmph

12.5m/s =45kmph

Final velocity(v)=135kmph

Time = 10min

$1\text{min} = \frac{1}{60}\text{h}$

$10\text{min} = \frac{10}{60}\text{h}$

$10 \text{ min} = \frac{1}{6}\text{h}$

• Calculation of the acceleration

Using the First equation of motion

v=u + at --------(1)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

putting the given values in equation 1

$135 = 45 + a \text{ x } \frac{1}{6}$

$135\text{-}45 = \frac{1}{6}a$

$90 = \frac{1}{6}a$

$540 = a$

$a=540\text{km/h}^2$

• Calculation of the distance travelled by the body

using the third equation of motion

$v^2 - u^2 = 2aS$ ------(2)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object

Putting the values in equation (2)

$(135)^2 - (45)^2 = 2\text{x}540\text{x S}$

18225-2025 = 1080 S

16200=1080S

$\frac{16200}{1080} = \text{S}$

15km = S

Hence, distance travelled by the body= 15km

25. Correct option: (A)

Given: initial velocity(u) = 4m/s

Acceleration(a) = 3m/s^2

Time taken(t)=6s

The second equation of motion gives displacement-time relation given as follows

$S = ut + \frac{1}{2}at^2$ ---------(1)

u= initial velocity of the object

a = acceleration of the object

t= time taken

S= displacement/distance travelled by the object

putting the values in equation (1)

$S = 4 \text{ x } 6 + \frac{1}{2}3x(6)^2$

$S = 24 + \frac{1}{2} \text{ x } 3 \text{ x} 36$

$S = 24 + 54$

S= 78m

Hence, distance travelled by the cyclist = 78m

26.Correct option: (B)

Given: initial velocity(u) = 0km/h

Final velocity(v)=180kmph

Time taken (t)=160min = $\frac{8}{3}$h

1 min= $\frac{1}{60}$h

160min= $\frac{160}{60}$h

= $\frac{8}{3}$h

• Calculation of acceleration

The first equation of motion gives Velocity time relation given as follows

v=u + at --------(1)

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

given: u= 0kmph (car starts from rest)

v=180 kmph

t= $\frac{8}{3}$h

putting values in equation (1)

$180=0 + ax \frac{8}{3}$

$180= \frac{8}{3}a$

$180= \frac{8}{3}a$

$\frac{180x3}{8} = a$

$67.5 = a$

Acceleration = $67.5kmh^{-2}$

• Calculation of length of the lap

The second equation of motion gives displacement-time relation given as follows

$S = ut + \frac{1}{2}at^2$ ---------(2)

u= initial velocity of the object

a = acceleration of the object

t= time taken

S= displacement/distance travelled by the object

putting the values in equation (2)

$S = 0x\,6 + \frac{1}{2}x67.5x\left(\frac{8}{3}\right)^2$

$S = \frac{1}{2}\,x\,67.5\,x\frac{64}{9}$

S =240km

Hence, distance travelled by the car = 240km

27. Correct option: (A)

Displacement in the nth second of motion is given by

$S_{nth} = u + \frac{1}{2}a(2n-1)$---------(1)

u= initial velocity of the object

a = acceleration of the object

S_{nth} = displacement of the object in nth second

Given: $a=2m/s^2$

u=0 m/s (body starts from rest)

n= 5

putting the values in equation (1)

$S_{5th} = 0 + \frac{1}{2}2(2x5-1)$

$S_{5th} = 1(10-1)$

$S_{5th} = 1x\,9$

$S_{5th} = 9m$

Hence, distance travelled in the 5^{th}second =9m

28. Correct option: (D)

Displacement in the nth second of motion is given by

$S_{nth} = u + \frac{1}{2}a(2n-1)$---------(1)

u= initial velocity of the object

a = acceleration of the object

S_{nth} = displacement of the object in nth second

Given: $a=6m/s^2$

u=0 m/s (body starts from rest)

n= 12

putting the values in equation (1)

$S_{12th} = 0 + \frac{1}{2}6(2x12-1)$

$S_{12th} = 3(24-1)$

$S_{12th} = 3x\,23$

$S_{12th} = 69m$

Hence, distance travelled in the 12^{th}second =69m

29. Correct option: (D)

Displacement in the nth second of motion is given by

$S_{nth} = u + \frac{1}{2}a(2n-1)$---------(1)

u= initial velocity of the object

a = acceleration of the object

S_{nth} = displacement of the object in nth second

Given: $a=4m/s^2$

u=10m/s

n= 8

putting the values in equation (1)

$S_{8th} = 10 + \frac{1}{2}4(2x8-1)$

$S_{8th} = 10 + 2(16-1)$

$S_{8th} = 10 + 2x\,15$
$S_{8th} = 10 + 30$
$S_{8th} = 40m$
Hence, distance travelled in the 8^{th}second =40m

30.Correct option: (C)

Displacement in the nth second of motion is given by

$S_{nth} = u + \frac{1}{2}a(2n-1)$---------(1)

u= initial velocity of the object
a = acceleration of the object
S_{nth} = displacement of the object in nth second

Given: $a=6m/s^2$
u=3m/s
n= 5
putting the values in equation (1)

$S_{5th} = 3 + \frac{1}{2}6(2x5-1)$
$S_{5th} = 3 + 3(10-1)$
$S_{5th} = 3 + 3x\,9$
$S_{5th} = 3 + 27$
$S_{5th} = 30m$
Hence, distance travelled in the 5^{th}second =30m

LEVEL-II

1.Correct option: (D)

Given: initial velocity(u) of bus =20km/h =$\frac{50}{9}$ m/s
Final velocity(v) of bus= 50km/h= $\frac{125}{9}$m/s

1km/h = $\frac{5}{18}$ m/s
20km/h = 20 x $\frac{5}{18}$ m/s
= $\frac{50}{9}$m/s
50km/h = 50 x $\frac{5}{18}$ m/s
= $\frac{125}{9}$m/s
Time taken = 10s
Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration
u= initial velocity
v = final velocity

$a = \frac{\frac{125}{9}-\frac{50}{9}}{10}\ ms^{-2}$
$a = \frac{125-50}{10x9}\ ms^{-2}$
$a = \frac{75}{90}\ ms^{-2}$
$a = 0.83ms^{-2}$
acceleration $=0.83ms^{-2}$

2.Correct option: (A)

Given: acceleration of car =-2.5m/s^2 (retardation)

Time taken=10 s

Final velocity(v) of car= 0m/s (car is coming to rest)

To find: initial velocity(u)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$-2.5 = \frac{0-u}{10}$

$-2.5 \times 10 ms^{-1} = -u$

$25 ms^{-1} = u$

Hence, initial velocity(u) of the car = $25 ms^{-1}$

3.Correct option: (C)

Given: initial velocity(u) of bus =36km/h = 10m/s

Final velocity(v) of bus= 72km/h= 20m/s

1km/h = $\frac{5}{18}$ m/s

36km/h = 36 x $\frac{5}{18}$ m/s

= 10m/s

72km/h = 72 x $\frac{5}{18}$ m/s

= 20m/s

Time taken = 10s

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{20-10}{10}\ ms^{-2}$

$a = 1\ ms^{-2}$

acceleration =1 ms^{-2}

4.Correct option: (D)

The first equation of motion gives Velocity time relation given as follows

v=u + at --------(1)

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

given: u= $20 ms^{-1}$

a= $4 ms^{-2}$

t=2seconds

putting values in equation (1)

v=20 + 4 x 2

v= 20 + 8

v = $28 ms^{-1}$

hence, final velocity =$28 ms^{-1}$

5.Correct option: (A)

The first equation of motion gives Velocity time relation given as follows

$v=u + at$ --------(1)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

given: $u= 20ms^{-1}$

$v=0\ ms^{-1}$ (motor cycle comes to rest)

t=5seconds

putting values in equation (1)

$0=20 + ax\ 5$

$0= 20 + 5a$

$-20=5a$

$-\frac{20}{5} = a$

$-4ms^{-2} = a$

Acceleration = $-4ms^{-2}$

Negative acceleration is also called retardation

hence, retardation=$4ms^{-2}$

6.Correct option: (C)

given: final velocity(v)=$0\ ms^{-1}$ (car comes to rest)

time taken(t)=25seconds

acceleration(a)= $-0.4\ ms^{-2}$ (negative sign for retardation)

- **Calculation of initial velocity**

The first equation of motion gives Velocity time relation given as follows

$v=u + at$ --------(1)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

putting values in equation (1)

$0=u + (-0.4)\ x\ 25$

$0= u – (0.4)\ x\ 25$

$-u=-10$

$u = 10\ ms^{-1}$

- **Calculation of distance travelled**

Given: initial velocity(u) = 10m/s (calculated above)
final velocity(v)= 0m/s (car stops after sometime)

acceleration(a) =$-0.4m/s^2$ (negative sign for deceleration)

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS \quad ------(2)$ Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object

putting the values in equation (2)

$v^2 - u^2 = 2aS$

$(0)^2 - (10)^2 = 2x(-0.4)xS$

0-100= -0.8S
-100=-0.8S
$\frac{100}{0.8} = S$
$125m = S$
Distance travelled by the car =125m

7.Correct option: (A)

Given: initial velocity(u) = 6m/s
final velocity(v)= 0m/s (bus stops due to applied brakes)
distance travelled(S)= 3m
Third equation of motion gives the relation between velocity and displacement given as follows
$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object
putting the values in equation (1)
$v^2 - u^2 = 2aS$
$(0)^2 - (6)^2 = 2xax3$
0-36 = 6a
-36=6a
$-\frac{36}{6} = a$
$-6\ m/s^2 = a$
acceleration=$-6\ m/s^2$

8.Correct options: (B), (C), (D)

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$
$a = \frac{v-u}{t}$ *here,* a= acceleration
u= initial velocity
v = final velocity
we can see from the above relation that the acceleration depends on the change in velocity of an object.

- If velocity of an object is 0 then acceleration must be 0.
- If the velocity of particle is uniform or constant means, velocity is not changing. so, change in velocity is zero.

Hence, acceleration of the particle is also zero.

- Acceleration may be 0 when velocity is non zero. This can happen if the velocity is uniform.

9.Correct option: (B)

Given: initial velocity(u) = 4m/s
Acceleration(a) = $1.2m/s^2$
time taken(t)=7s
The second equation of motion gives displacement-time relation given as follows
$S = ut + \frac{1}{2}at^2$ ---------(1) u= initial velocity of the object
a = acceleration of the object
t= time taken
S= displacement/distance travelled by the object
putting the values in equation (1)

$S = 4 \times 7 + \frac{1}{2} x 1.2 x (7)^2$

$S = 28 + \frac{1}{2} \times 1.2 \times 49$

$S = 28 + 29.4$

$S = 57.4m$

Hence, distance travelled = 57.4m

10.Correct option: (A)

Given: x =initial velocity

b = acceleration

SI unit of x = m/s

SI unit of b =m/s^2

SI unit of $\frac{x}{b} = \frac{m/s}{m/s^2}$

SI unit of $\frac{x}{b}$ = seconds

SI unit of time is seconds

Hence, $\frac{x}{b}$ represents time

11.Correct option: (A), (B)

given: initial velocity (u)= 36kmph = $10ms^{-1}$

$1km/h = \frac{5}{18} m/s$

$36km/h = 36 \times \frac{5}{18} m/s$

$= 10m/s$

Final velocity(v)=0 ms^{-1} (cycle comes to rest)

Time taken (t)=2seconds

- **Calculation of retardation**

The first equation of motion gives Velocity time relation given as follows

$v = u + at$ --------(1)

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

putting values in equation (1)

$0 = 10 + a \times 2$

$0 = 10 + 2a$

$-10 = 2a$

$-\frac{10}{2} = a$

$-5ms^{-2} = a$

Acceleration = $-5ms^{-2}$

Negative acceleration is also called retardation

hence, retardation=$5ms^{-2}$

- **Calculation of distance travelled**

Given: initial velocity(u) = $10m/s$

final velocity(v)= $0m/s$ (cyclist stops after sometime)

acceleration(a) =$-5m/s^2$ (negative sign for deceleration)

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS$ ------(2) Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object
S= displacement/distance travelled by the object

putting the values in equation (2)
$v^2 - u^2 = 2aS$
$(0)^2 - (10)^2 = 2x(-5)xS$
0-100= -10S
-100=-10S
$\frac{100}{10} = S$
10m = S
Distance travelled by the cyclist =10m

12.Correct option: (A), (C)

given: final velocity(v)= 72kmph= $20ms^{-1}$
$1km/h = \frac{5}{18} m/s$
$72km/h = 72 \times \frac{5}{18} m/s$
$= 20m/s$
Initial velocity(u)=0 ms^{-1} (car starts from rest)
Time taken(t)=10 sec
The first equation of motion gives Velocity time relation given as follows
v=u + at --------(1)
Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

putting values in equation (1)
20=0 + ax 10
20 =0+ 10a
20=10a
$\frac{20}{10} = a$
$2m/s^2 = a$
Acceleration = $2m/s^2$ or $200cm/s^2$
1m = 100 cm
$2m/s^2 = 200cm/s^2$

13.Correct option: (A), (C)

Given: initial velocity(u) = 4m/s
final velocity(v)= 0m/s (car stops after sometime)
acceleration(a) =$-8m/s^2$ (negative sign for retardation)

- **Calculation of the distance travelled**

Third equation of motion gives the relation between velocity and displacement given as follows
$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object

putting the values in equation (1)
$v^2 - u^2 = 2aS$

$(0)^2 - (4)^2 = 2x(-8)xS$
0-16= -16S
-16=-16S
$\frac{16}{16} = S$
1m = S
Distance travelled by the car=1m or 100cm

14.Correct option: (A)

given: initial velocity(u)= 225kmph= $62.5ms^{-1}$
$1km/h = \frac{5}{18} m/s$
$225km/h = 225 \times \frac{5}{18} m/s$
$= 62.5m/s$
Final velocity(v)=0 ms^{-1} (aeroplane comes to rest)
time taken(t)=2minutes = 2 x 60 seconds =120 seconds
The first equation of motion gives Velocity time relation given as follows
v=u + at --------(1)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

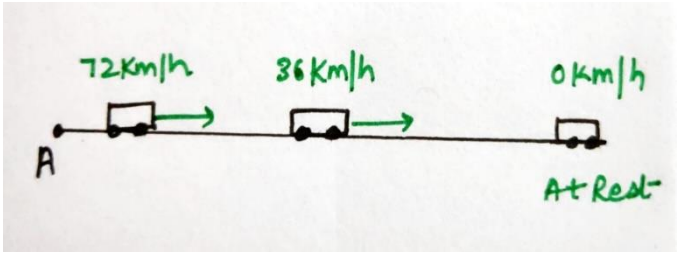
putting values in equation (1)
0=62.5 + ax 120
0= 62.5 + 120a
-62.5=120a
$-\frac{62.5}{120} = a$
$-0.5ms^{-2} = a$
Acceleration = $-0.5ms^{-2}$

15.Correct option: (B)

Given: initial velocity(u) = 62.5m/s
final velocity(v)= 0m/s (aero plane stops after sometime)
acceleration(a) =$-0.5m/s^2$ (negative sign for deceleration)
Third equation of motion gives the relation between velocity and displacement given as follows
$v^2 - u^2 = 2aS$ $------(1)$ Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object

putting the values in equation (1)
$v^2 - u^2 = 2aS$
$(0)^2 - (62.5)^2 = 2x(-0.5)xS$
0-3906.25= -1S
-3906.25=-1S
$\frac{3906.25}{1} = S$
3906.25m = S
Length of the runway is =3906.25m

16.Correct option: (C)



Given: initial velocity(u) of car= 72km/h =20m/s

Final velocity(v) of car= 36km/h =10m/s

Distance travelled = 25m

1km/h = $\frac{5}{18}$ m/s

72km/h = 72 x $\frac{5}{18}$ m/s

= 20m/s

36km/h= 36 x $\frac{5}{18}$ m/s

=10m/s

- **Calculation of the acceleration**

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS$ ------(1) Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$

$(10)^2 - (20)^2 = 2xax25$

100-400 = 50a

-300=50a

$-\frac{300}{50} = a$

$-6\ m/s^2 = a$

acceleration=$-6\ m/s^2$

- calculation of the time taken

The first equation of motion gives Velocity time relation given as follows

v=u + at --------(2)

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

putting the values in equation (2)

0=20 + (-6) x t (v= 0m/s because the car finally stops)

0= 20 - 6t

-20=-6t

$\frac{20}{6} = t$

3.33seconds = t

Hence, time taken for the car to stop= 3.33 seconds

17.Correct option: (A)

Given: initial velocity(u) = 0m/s (body starts from rest)

final velocity(v)= 20m/s

acceleration(a) =2m/s^2

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$

$(20)^2 - (0)^2 = 2x2xS$

400= 4S

400=4S

$\frac{400}{4} = S$

$100m = S$

Distance travelled by the body=100m

18.Correct option: (A)

Given: initial velocity(u) = 0m/s (body starts from rest)

final velocity(v)= 20m/s

acceleration(a) =4m/s^2

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$

$(20)^2 - (0)^2 = 2x4xS$

400= 8S

400=8S

$\frac{400}{8} = S$

$50m = S$

Distance travelled by the body=50m

19.Correct option: (A)

Given: initial velocity(u) = 20m/s

final velocity(v)= 0m/s (body stops after sometime)

acceleration(a) =$-4m/s^2$ (negative sign for deceleration)

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS \quad ------(1)$ Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$
$(0)^2 - (20)^2 = 2x(-4)xS$
0-400= -8S
-400=-8S
$\frac{400}{8} = S$
$50m = S$
Distance travelled by the body=50m

20. Correct Matching is: a-RT; b-Q; c-S; d-P

(a) The first equation of motion gives Velocity time relation given as follows
v=u + at --------(1)

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
t= time taken

rearranging equation 1
$a = \frac{v-u}{t}$

(b) The second equation of motion gives displacement-time relation given as follows
$S = ut + \frac{1}{2}at^2$

u= initial velocity of the object
a = acceleration of the object
t= time taken
S= displacement/distance travelled by the object

(c)Third equation of motion gives the relation between velocity and displacement given as follows
$v^2 - u^2 = 2aS$

Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object

(d)Displacement in the nth second of motion is given by
$S_{nth} = u + \frac{1}{2}a(2n-1)$

u= initial velocity of the object
a = acceleration of the object
S_{nth} = displacement of the object in nth second

3. Newton's Laws of Motion

Solutions

LEVEL-I

1.Correct option: (A)

Field forces are the forces which can create push or pull even at a distance and it can happen even if two objects are not in physical contact with each other.

For example: gravitational force, electrostatic force, magnetic force are examples of field forces.

2.Correct option: (A)

Balanced forces: if the resultant of all the forces acting on a body is zero, the forces are called balanced forces.

Balanced forces cannot produce motion in a stationary object or stop a moving object, they can however, change the shape of the body. Hence, when a toothpaste tube is squeezed, its shape changes due to the balanced forces applies by our hands on the tube.

3.Correct option: (B)

The earth attracts everybody/object towards its center with a certain force which depends on the mass(m) of the body and the acceleration due to gravity(g) at that place. The weight of the body is the force with which it is attracted towards the center of the earth.

Or, the gravitational force of attraction of the earth on a body is called its weight(W)

W= m x g

4.Correct option: (A)

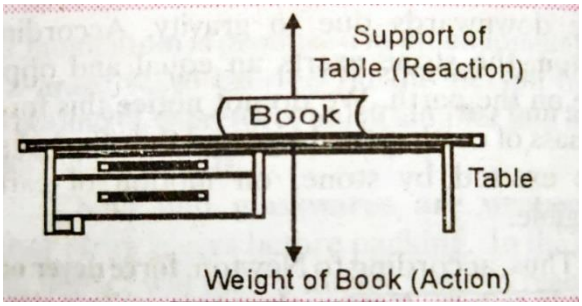
The earth attracts everybody/object towards its center with a certain force which depends on the mass(m) of the body and the acceleration due to gravity(g) at that place. The weight of the body is the force with which it is attracted towards the center of the earth.

Or, the gravitational force of attraction of the earth on a body is called its weight(W)

W= mg

5.Correct option: (A)

The force that can cause or change the motion of an object by touching it is called **<u>contact force</u>.** Contact force occurs due to the contact between two different objects.



Normal reaction is the force acting perpendicularly to two surfaces in contact with each other. hence, it is a contact force.

6.Correct option: (A)

The earth attracts everybody/object towards its center with a certain force which depends on the mass(m) of the body and the acceleration due to gravity(g) at that place. The weight of the body is the force with which it is attracted towards the center of the earth.

Or, the gravitational force of attraction of the earth on a body is called its weight(W)

W= m x g

As acceleration due to gravity(g) acts in the vertically downward direction so weight also acts in vertically downward direction.

7.Correct option: (B)

The force that can cause or change the motion of an object by touching it is called **contact force.** Contact force occurs due to the contact between two different objects.

8.Correct option: (C)

Unbalanced forces: if the resultant of all the forces acting on a body is not zero, the forces are called unbalanced forces.

when unbalanced forces act on a body, they produce a change in its state of rest or of uniform motion. That is, unbalanced forces can move a stationary object or they can stop a moving object.

9.Correct option: (C)

According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.

The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.

Hence, newton's first law defines **inertia.**

10.Correct option: (A)

Balanced forces: if the resultant of all the forces acting on a body is zero, the forces are called balanced forces.

Balanced forces cannot produce motion in a stationary object or stop a moving object, they can however, change the shape of the body. Hence, when a rubber balloon held between the hands is pressed, its shape changes due to the balanced forces applies by our hands on the balloon.

11.Correct option: (B)

The tendency of a body to remain in the state of uniform motion or inability of the body to change its state of uniform motion by itself, is called inertia of motion.

When a running car stops suddenly, the passengers are jerked forward due to the inertia of motion.

As the car is stopped suddenly, the lower body stop but the rest of the body continues to move forward due to inertia of motion. That is why passengers are thrown forward.

12.Correct option: (A)

The tendency of a body to remain at rest or inability of the body to change its state of rest by itself, is called inertia of rest.

When a stationary car starts suddenly, the passengers are jerked backwards due to the inertia of rest.

As the car starts suddenly, the lower body move with the car but the rest of the body continues to remain at rest due to inertia of rest. That is why passengers are jerked backwards.

13.Correct option: (C)

Inertia of direction is the inability of the body to change its direction of motion by itself. That is, a body continues to move along the same straight line unless compelled by some external force to change it.

The rotating wheels of the vehicle throw out the mud tangentially from the wheel due to the inertia of direction.

14.Correct option: (C)

Inertia is that property of a body due to which it resists a change in its state of rest or of uniform motion. Greater the inertia of the body, greater will be the force required to bring a change in its state of rest or of uniform motion. In fact, mass is a measure of inertia of the body. If a body has more mass, it has more inertia.

15.Correct option: (A)
Inertia of direction is the inability of the body to change its direction of motion by itself. That is, a body continues to move along the same straight line unless compelled by some external force to change it.
When a bus suddenly takes a turn, the passengers are thrown outwards. This is because the person tries to maintain his/her direction of motion due to inertia of direction while the bus turns.
16.Correct option: (B)
The tendency of a body to remain in the state of uniform motion or inability of the body to change its state of uniform motion by itself, is called inertia of motion.
A fan has the tendency to continue its circular motion even after it is switched off due to the inertia of motion. Thus, the fan continues to move for some time even after switching it off.
17. Correct option: (D)
According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.
The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.
Thus, a force is not required to keep the particle moving with uniform velocity.
18.Correct option: (B)
Inertia is that property of a body due to which it resists a change in its state of rest or of uniform motion. Greater the inertia of the body, greater will be the force required to bring a change in its state of rest or of uniform motion. In fact, mass is a measure of inertia of the body. If a body has more mass, it has more inertia.
19.Correct option: (B)
Inertia is that property of a body due to which it resists a change in its state of rest or of uniform motion. Greater the inertia of the body, greater will be the force required to bring a change in its state of rest or of uniform motion. In fact, mass is a measure of inertia of the body. If a body has more mass, it has more inertia.
Hence, body of mass 4 kg will have more inertia.
20.Correct option: (C)
According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.
The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.
Hence, newton's first law defines **inertia.**
21. Correct option: (C)
According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.
The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.
Hence, newton's first law defines **inertia.**
22.Correct option: (D)
force is the physical cause which changes or tends to change the state of rest or the state of motion of a body.

23. Correct option: (A)

According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.

The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.

Hence, newton's first law defines **inertia.**

24.Correct option: (C)

The tendency of a body to remain at rest or inability of the body to change its state of rest by itself, is called inertia of rest.

When a stationary horse starts moving suddenly, the rider falls backwards due to the inertia of rest.

As the horse starts moving suddenly, the lower part of the body of the rider (which is in contact with the horse) move with the horse but the upper body continues to remain at rest due to inertia of rest. That is why the rider falls backwards.

25.Correct option: (C)

The tendency of a body to remain in the state of uniform motion or inability of the body to change its state of uniform motion by itself, is called inertia of motion.

When a running train stops suddenly, the passengers are jerked forward due to the inertia of motion.

As the train stopped suddenly, the lower part of the body in contact with the seat, stop but the rest of the body continues to move forward due to inertia of motion. That is why passengers are thrown forward.

26. Correct option: (D)

Inertia is that property of a body due to which it resists a change in its state of rest or of uniform motion.

27. Correct option: (B)

A man getting down a running bus fall forward. This is because his feet come to rest on touching the ground and the remaining body continues to move due to inertia of motion.

The tendency of a body to remain in the state of uniform motion or inability of the body to change its state of uniform motion by itself, is called inertia of motion.

28. Correct option: (D)

When the person throws the ball vertically upwards, it follows a parabolic path.

Horizontal velocity of the ball and the person is same so both will cover equal horizontal distance in a given interval of time and after following the parabolic path, the ball falls exactly in the hand which threw it up.

29. Correct option: (A)

Inertia of direction is the inability of the body to change its direction of motion by itself. That is, a body continues to move along the same straight line unless compelled by some external force to change it.

When a bus suddenly takes a turn, the passengers are thrown outwards. This is because the person tries to maintain his/her direction of motion due to inertia of direction while the bus turns.

30. Correct option: (D)

An athlete runs a certain distance before taking a long jump. This is because velocity acquired by running is added to the velocity of the athlete at the time of jump. Hence, he can jump over a longer distance due to the large inertia of motion.

LEVEL-II

1. Correct option: (C)

Inertia of direction is the inability of the body to change its direction of motion by itself. That is, a body continues to move along the same straight line unless compelled by some external force to change it.

The rotating wheels of the vehicle throw out the mud tangentially from the wheel due to the inertia of direction.

2.Correct option: (C)

According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.

The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.

Hence, newton's first law defines **inertia.**

When a hanging carpet is beaten with a stick, on hitting with a stick, the carpet comes in motion; but because of inertia, the dust particles remain at rest. Due to this, dust particles get removed from the carpet.

3.Correct option: (D)

The force due to which earth attracts/pulls all bodies towards itself is called gravitational force.

The earth attracts every other object lying near or on its surface towards its center. The origin of this force between the earth and a body is due to interaction between their masses.

4.Correct option: (A)

The force which drives an electric fan is an electric force.

Electric force is the force of attraction or repulsion between any two charged objects.

5.Correct option: (A)

The force applied by using our muscles like legs/arms/hands etc. Is called muscular force. It is a type of contact force.

6.Correct option: (C)

The tendency of a body to remain in the state of uniform motion or inability of the body to change its state of uniform motion by itself, is called inertia of motion.

When a running train stops suddenly, the passengers are jerked forward due to the inertia of motion.

As the train stopped suddenly, the lower part of the body in contact with the seat, stop but the rest of the body continues to move forward due to inertia of motion. That is why passengers are thrown forward.

7.Correct option: (A)

The tendency of a body to remain in the state of uniform motion or inability of the body to change its state of uniform motion by itself, is called inertia of motion.

A passenger of the bus should jump out running in the direction of the motion of the bus because his feet come to rest on touching the ground and the remaining body continues to move due to inertia of motion.

8. Correct option: (A)

According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.

The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.

Hence, newton's first law defines **inertia.**

When we shake the branch of a tree, the ripe fruits get detached from the branch. This is because the branch comes in motion and the fruits tend to remain at rest due to inertia of rest. Hence, they get detached from the branch.

9.Correct option: (C)

Balanced forces: if the resultant of all the forces acting on a body is zero, the forces are called balanced forces.

Balanced forces cannot produce motion in a stationary object or stop a moving object.

For example, in a tug of war, that is in rope pulling between two teams, if the resultant of forces applied by the two teams is zero, then the rope does not move in either direction. The forces exerted by the two teams are balanced. From this we conclude that if a number of balanced forces act on a stationary body, the body continues to remain in its stationary position.

10. Correct option: (B)

The tendency of a body to remain at rest or inability of the body to change its state of rest by itself, is called inertia of rest.

When a stationary horse starts moving suddenly, the rider falls backwards due to the inertia of rest.

As the horse starts moving suddenly, the lower part of the body of the rider (which is in contact with the horse) move with the horse but the upper body continues to remain at rest due to inertia of rest of the rider. That is why the rider falls backwards.

11.Correct option: (B)

The tendency of a body to remain in the state of uniform motion or inability of the body to change its state of uniform motion by itself, is called inertia of motion.

When the brakes are applied the water tanker stops suddenly, the water in the tanker moves forward due to the inertia of motion.

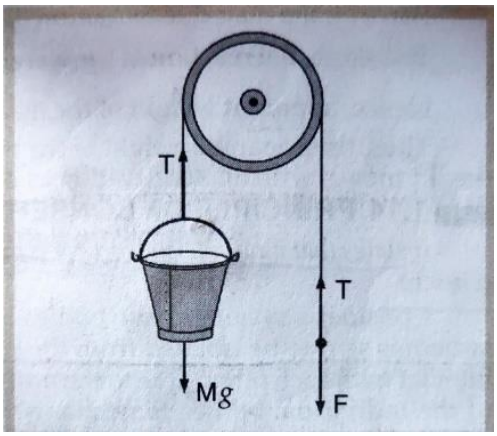
12.Correct option: (B), (C), (D)

Magnetic force can be defined as the attractive or repulsive force that is exerted between the poles of a magnet and electrically charged moving particles.

Magnetic force is a **field force** as it is able to exert an influence into the space around itself. This force comes into play even when a magnet and other objects (like iron) are kept at a **distance** and are **not in contact** with each other.

Hence, magnetic force is a non- contact force, force at a distance and a field force.

13.Correct option: (A), (B)



While pulling a bucket of water from a well the forces acting are:

(a) Weight (mg) of the bucket and water in the vertically downward direction.

(b) Tension (T) in the rope trying to pull the bucket in the vertically upward direction.

(A force that is transmitted through a string, rope, cable or wire when it is pulled tightly by the object on the opposite end is a tension force).

14. Correct option: (A), (B), (C), (D)

Force is a push or pull which either changes or tends to change the state of rest or uniform motion of a body.

Force as an external effort in the form of push or pull, which

(i) Produces or tries to produce motion in a body at rest.

(ii) Stops or tries to stop a moving body.

(iii) Increases or decreases the magnitude of velocity of a moving body.

(iv) Changes or tries to change the direction of motion of the body.

Force as an external effort has the capability to change the shape and size of an object.

- When a piece of metal is beaten by a hammer, its shape/dimension changes.
- When we pull a rubber band, its length increases due to elasticity and when this force is removed, then it may or may not regain their shape. Hence, the applied force has changed the size of the rubber band.

15.Correct options(A), (B), (C), (D)

- **Muscular force** is the force exerted by using muscles of the body parts such as the arms or legs. It is a force produced by muscle action and is a contact force because there is contact between the surfaces.
- **Mechanical force** is one that involves direct contact between two objects (one providing the force and the other in a state of rest or motion) that results in a change in the object's condition (state of rest or state of motion). The force generated by the means of a machine is a type of mechanical force.
- **Electrostatic force** is an attractive or repulsive force between particles which is caused due to their electric charges. The electric force between stationary charged bodies is conventionally known as the electrostatic force.
- **Magnetic force** can be defined as the attractive or repulsive force that is exerted between the poles of a magnet and electrically charged moving particles.

16. Correct options(C), (D)

According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.

The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.

Hence, newton's first law defines **inertia.**

Inertia is that property of a body due to which it resists a change in its state of rest or of uniform motion. Greater the inertia of the body, greater will be the force required to bring a change in its state of rest or of uniform motion. In fact, **mass is a measure of inertia of the body**. If a body has more mass, it has more inertia.

Newton's first law defines force qualitatively as an external effort in the form of push or pull, which

(i) Produces or tries to produce motion in a body at rest.

(ii) Stops or tries to stop a moving body.

(iii) Increases or decreases the magnitude of velocity of a moving body.

(iv) Changes or tries to change the direction of motion of the body.

If the net external force on a body is zero, its acceleration is zero. Acceleration can be non-zero only if there is net external force on the body.

17. Correct option: (A)

The tendency of a body to remain at rest or inability of the body to change its state of rest by itself, is called inertia of rest.

When a stationary car starts suddenly, the person falls backwards due to the inertia of rest.

As the car starts suddenly, the lower body of the person move with the car but the rest of the body continues to remain at rest due to inertia of rest. That is why the person falls backward backwards.

18.Correct option: (C)

Inertia of direction is the inability of the body to change its direction of motion by itself. That is, a body continues to move along the same straight line unless compelled by some external force to change it.

When A stone tied to one end of a string is whirled and the string is cut suddenly, the stone flies off along the tangent to the circle. this is because the pull in the string was forcing the stone to move in a circle. as soon as the string breaks the pull vanishes. The stone in a bid to move along the straight line flies off tangentially.

19. Correct Matching is: a-T; b-PS; c-R; d-Q

- **Muscular force** is the force exerted by using muscles of the body parts such as the arms or legs. It is a force produced by muscle action and is a contact force because there is contact between the surfaces. Pushing a lawn roller requires muscular force.
- **Mechanical force** is one that involves direct contact between two objects (one providing the force and the other in a state of rest or motion) that results in a change in the object's condition (state of rest or state of motion). The force generated by the means of a machine is a type of mechanical force.

Force used to run a motor car engine and force used to run a steam engine are mechanical forces.

- **Gravitational force** is the force of attraction on a body by earth. Example: The leaves and fruits fall from a tree downwards towards the ground. A stone fall downwards due to the gravitational force of earth.
- **Electrostatic force** is an attractive or repulsive force between particles which is caused due to their electric charges. The electric force between stationary charged bodies is conventionally known as the electrostatic force. Charged comb attracts small pieces of paper due to electrostatic force of attraction between them.

Newton's Second Law of Motion

LEVEL-I

1.Correct option: (A)

Newton's second law gives the relationship between 'force' and 'acceleration'. When a force acts on a body, it produces acceleration in the body, the acceleration produced may be positive or negative.

Force = mass x acceleration

F = m x a

2.Correct option: (A)

Absolute unit of force: in SI, the absolute unit of force is newton(N).

Gravitational unit of force: In SI, the gravitational unit of force is kilogram force (Kgf)

1Kgf = 9.8 N

3.Correct option: (B)

Momentum is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.

4.Correct option: (C)

Linear Momentum is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.

5.Correct option: (B)

Linear Momentum is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.

As linear momentum is equal to a scalar(mass) time a vector(velocity), it is therefore a vector quantity and is denoted by $\vec{p}$.

6.Correct option: (B)

Newton's second law of motion states that the time rate of change of momentum of a body is directly proportional to the external force applied on the body and the change in momentum takes place in the direction of the force.

Let $\vec{p}$ be the momentum of the body and $\vec{F}$ be the external force acting on it. Then according to the newton's second law of motion,

Force $\propto$ time rate of change of momentum

$$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$$

$$\vec{F} = \frac{\text{final momentum(mv)} - \text{initial momentum(mu)}}{t}$$

here, u= initial velocity

V= final velocity

Change in momentum= force x time taken

SI unit of force = newton(N)

SI unit of time = seconds(s)

SI unit of momentum = Ns

7.Correct option: (B)

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

The SI unit of velocity = m/s

The SI unit of time = s

The SI unit of acceleration = $\frac{m}{s \times s}$

$= m/s^2$

8.Correct option: (C)

Absolute unit of force: in SI, the absolute unit of force is newton(N).

In C.G.S, the absolute unit of force is dyne.

$1N = 10^5 dyne$

9.Correct option: (A)

Newton's second law of motion states that the time rate of change of momentum of a body is directly proportional to the external force applied on the body and the change in momentum takes place in the direction of the force.

Let $\vec{p}$ be the momentum of the body and $\vec{F}$ be the external force acting on it. Then according to the newton's second law of motion,

Force $\propto$ time rate of change of momentum

$$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$$

$$\vec{F} = \frac{\text{final momentum(mv)} - \text{initial momentum(mu)}}{t}$$

here, u= initial velocity

V= final velocity

Change in momentum= force x time taken

SI unit of force = newton(N)

SI unit of time = seconds(s)

SI unit of momentum = Ns

10.Correct option: (B)

Newton's second law gives the relationship between 'force' and 'acceleration'. When a force acts on a body, it produces acceleration in the body, the acceleration produced may be positive or negative.

Force = mass x acceleration

F = m x a

$1\ N = 1\ kg \times 1\ ms^{-2} = 1kg\ ms^{-2}$

Thus, 1 N is that much force, which produces an acceleration of $1ms^{-2}$ in a body of mass 1 kg.

11.Correct option: (A), (C)

Linear Momentum is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.

As linear momentum is equal to a scalar(mass) time a vector(velocity), it is therefore a vector quantity and is denoted by $\vec{p}$.

Momentum(p) = mass(m) x velocity(v)

It is clear from the above relation that momentum is directly proportional to the mass and velocity(speed) of a body, so a body will have a large momentum: (a) if its mass is large, or (b) if its velocity(speed) is large, or (c) if both its mass and velocity(speed) are large.

12.Correct option: (B)

Given: mass(m)= 5kg

Velocity(v)= 5m/s

Momentum(p) = mass(m) x velocity(v)

p = 5 x 5 kg m/s

$p = 25\ kg\ ms^{-1}$

hence, momentum of the body = $25\ kg\ ms^{-1}$

13.Correct option: (A)

Given: acceleration= $10ms^{-2}$

Force(F)=20N

To find: mass of the body

From the newton's second law of motion

Force = mass x acceleration

F = m x a

$m = \frac{F}{a}$

$m = \frac{20}{10}\ ms^{-2}$

m=2 kg

hence, mass of the body =2 kg

14.Correct option: (D)

Given: acceleration= $2ms^{-2}$

Force(F)=24N

To find: mass of the body

From the newton's second law of motion

Force = mass x acceleration

F = m x a

$m = \frac{F}{a}$

$m = \frac{24}{2}\ kg$

m=12 kg

hence, mass of the body =12 kg

15. Correct option: (B)

Given: mass(m) = 20 kg

Force(F)=2kgf

1kgf = 10 N

2kgf = 20N

To find: acceleration of the body

From the newton's second law of motion

Force = mass x acceleration

$F = m \times a$

$a = \frac{F}{m}$

$a = \frac{20}{20}\ ms^{-2}$

$a = 1\ ms^{-2}$

hence, acceleration = $1\ ms^{-2}$

16.Correct option: (B)

Linear Momentum is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.

As linear momentum is equal to a scalar(mass) time a vector(velocity), it is therefore a vector quantity and is denoted by $\vec{p}$.

Momentum(p) = mass(m) x velocity(v)

$\vec{p} = m\vec{v}$

Consider two bodies 1 and 2

Consider that body 1 is lighter

Let the mass of body 1 = m_1

Velocity of body 1 =v_1

Momentum of body 1 = $m_1 v_1$

Let the mass of body 2 =m_2

Velocity of body 2 =v_2

Momentum of body 2 = $m_2 v_2$

Let $m_1 < m_2$

It is given that two bodies have same linear momentum.

So

$m_1 v_1 = m_2 v_2$

Or

$\frac{m_1}{m_2} = \frac{v_2}{v_1}$

$as\ m_1 < m_2$

Then $v_1 > v_2$

Hence, lighter body has more velocity

Now,

Kinetic energy = $\frac{1}{2}\ mv^2$

Kinetic energy is directly proportional to square of velocity.

Hence, lighter body has greater kinetic energy.

17.Correct option: (B)

A spring balance is used to measure the weight of a body.

(Weight is defined as the force exerted on a body by gravity).

Its working is based on the **Hooke's law** in elasticity which states that the extension produced in a spring is directly proportional to the force applied, provided the spring is not



loaded beyond the elastic limit.

18.Correct option: (A)

Linear Momentum is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.

Momentum(p) = mass x velocity

19. Correct option: (A)

Newton's second law gives the relationship between 'force' and 'acceleration'. When a force acts on a body, it produces acceleration in the body, the acceleration produced may be positive or negative.

Force = mass x acceleration

F = m x a

20.Correct option: (B)

Newton's second law of motion states that the time rate of change of momentum of a body is directly proportional to the external force applied on the body and the change in momentum takes place in the direction of the force.

Let $\vec{p}$ be the momentum of the body and $\vec{F}$ be the external force acting on it. Then according to the newton's second law of motion,

$\text{Force} \propto \text{time rate of change of momentum}$

$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$

$\vec{F} = \frac{\text{final momentum(mv)} - \text{initial momentum(mu)}}{t}$ here, u= initial velocity

V= final velocity

21.Correct option: (C)

Given: mass(m) = 100 kg

Acceleration(a)=$1.5 ms^{-2}$

From the newton's second law of motion

Force = mass x acceleration

F = m x a

F = 100 x 1.5 N

F = 150 N

Hence, force exerted by the bullock = 150N

22.Correct option: (C)

Momentum(p) = mass x velocity

SI unit of mass = kg

SI unit of velocity = m/s
So, SI unit of momentum = kg m/s

23.Correct option: (D)

Newton's second law gives the relationship between 'force' and 'acceleration'. When a force acts on a body, it produces acceleration in the body, the acceleration produced may be positive or negative.
From the newton's second law of motion
Force = mass x acceleration
F = m x a
Or

$$a = \frac{F}{m}$$

as acceleration(a) is directly proportional to the force applied
hence, acceleration of a body can be doubled by doubling the force acting on it.

24.Correct option: (B)

Given: mass(m)= 15kg
Velocity(v)= 5m/s
Momentum(p) = mass(m) x velocity(v)
p = 15 x 5 kg m/s
p = 75 kg m/s
hence, momentum of the body = 75 kg m/s

25. Correct option: (A)

Given: mass(m)= 10kg
Velocity(v)= 10m/s
Momentum(p) = mass(m) x velocity(v)
p = 10 x 10 kg m/s
p = 100 kg m/s

LEVEL-II

1.Correct option: (D)

Force is external a push or pull which either changes or tends to change the state of rest or uniform motion of a body.
Force as an effort in the form of push or pull, which

(v) Produces or tries to produce motion in a body at rest.
(vi) Stops or tries to stop a moving body.
(vii) Increases or decreases the magnitude of velocity of a moving body.
(viii) Changes or tries to change the direction of motion of the body.

Force as an external effort has the capability to change the shape and size of an object.

- When a piece of metal is beaten by a hammer, its shape changes.
- When we pull a rubber band, its length increases due to elasticity and when this force is removed, then it may or may not regain their shape. Hence, the applied force has changed the size of the rubber band.

2.Correct option: (B)

Newton's second law gives the relationship between 'force' and 'acceleration'. When a force acts on a body, it produces acceleration in the body, the acceleration produced may be positive or negative.
From the newton's second law of motion
Force = mass x acceleration

F = m x a

Or

$a = \frac{F}{m}$

since, the acceleration produced is inversely proportional to the mass of the body, it means that it will be easier to move light bodies (having less mass) than heavier bodies (having large mass).

3. Correct option: (B), (C)

Newton's second law of motion states that the time rate of change of momentum of a body is directly proportional to the external force applied on the body and the change in momentum takes place in the direction of the force.

Let $\vec{p}$ be the momentum of the body and $\vec{F}$ be the external force acting on it. Then according to the newton's second law of motion,

$\text{Force} \propto \text{time rate of change of momentum}$

$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$

$\vec{F} = \frac{\text{final momentum(mv)} - \text{initial momentum(mu)}}{t}$ here, u= initial velocity

V= final velocity

4.Correct option: (B)

According to the newton's second law of motion,

$\text{Force} \propto \text{time rate of change of momentum}$

$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$

A fast-moving cricket ball has a larger momentum. In catching this cricket ball, its momentum has to be reduced to zero. Now, when a fielder moves his hands backwards after catching the fast ball, then the time taken to reduce the momentum of the ball to zero is increased. Due to more time taken to stop the ball, the rate of change of momentum of the ball is decreased and hence a small force is exerted on the hands of the fielder. So, the hands of the fielder do not get hurt.



5.Correct option: (B)

Weight is defined as the force exerted on a body by gravity.

Weight is often expressed as

W= mg----------(1)

Where, w= weight of the body

m = mass of the body

g= acceleration due to gravity = $9.8 m/s^2$

given: force = 5N

weight = 9.8N or $9.8 kg\, m/s^2$

from equation 1
$9.8\ kg\ m/s^2 = m \times 9.8\ m/s^2$
m = 1kg
mass = 1 kg
To find: acceleration of the body
From the newton's second law of motion
Force = mass x acceleration
F = m x a
$a = \frac{F}{m}$
$a = \frac{5}{1}\ m/s^2$
$a = 5 m/s^2$
hence, acceleration = $5\ m/s^2$

6.Correct option: (C)

Newton's second law gives the relationship between 'force' and 'acceleration'. When a force acts on a body, it produces acceleration in the body, the acceleration produced may be positive or negative.
From the newton's second law of motion
Force = mass x acceleration
F = m x a
Or
$a = \frac{F}{m}$
since, the acceleration produced is directly proportional to the force acting on the body. Hence, a body will continue to have acceleration until resultant force on it is zero.

7.Correct option: (A)

Given: force(F) = 12N
Time taken = 4s
To find: change in linear momentum
According to the newton's second law of motion
$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$
Or
Change in linear momentum = force x time taken
$= 12 \times 4\ \ kg\ ms^{-1}$
$= 48\ \ kg\ ms^{-1}$

8.Correct option: (C)

Given: mass(m) = 20 g
$1g = \frac{1}{1000} kg$
$20g = \frac{20}{1000} kg$
Acceleration(a)=$5cms^{-2}$
$1cm = \frac{1}{100} m$
$5cm = \frac{5}{100} m$
From the newton's second law of motion
Force = mass x acceleration
F = m x a

$F = \frac{20}{1000} \times \frac{5}{100} N$

$F = \frac{100}{100000} N$

$F = \frac{1}{1000} N$

$F = 10^{-3} N$

Hence, force = $10^{-3} N$

9.Correct option: (A)

- **To calculate the force** F_1

Given: mass (m_1) = 2 kg

Acceleration (a_1)=$2.5 ms^{-2}$

From the newton's second law of motion

Force = mass x acceleration

$F_1 = m_1 \times a_1$

$F_1 = 2 \times 2.5$ N

$F_1 = 5N$

- **To calculate the force** F_2

Given: mass (m_2) = 5 kg

Acceleration (a_2)=$2 ms^{-2}$

From the newton's second law of motion

Force = mass x acceleration

$F_2 = m_2 \times a_2$

$F_2 = 5 \times 2$ N

$F_2 = 10N$

Hence, forces F_1 *and* F_2 *are* 5N ,10N

10.Correct option: (D)

- **To calculate the force**

Given: mass(m) = 2 kg

Acceleration(a)=$5 ms^{-2}$

From the newton's second law of motion

Force = mass x acceleration

F = m x a

F = 2 x 5 N

F = 10 N

Hence, force= 10N

- **To calculate the acceleration**

Given: force is same as before = 10N (calculated above)

Mass(m)= 1.5 kg

To find: acceleration of the body

From the newton's second law of motion

Force = mass x acceleration

F = m x a

$a = \frac{F}{m}$

$a = \frac{10}{1.5} ms^{-2}$

$a = 6.6 ms^{-2}$

hence, acceleration = $6.6\ ms^{-2}$

11.Correct option: (A)

Given: mass(m) of the body = 2 kg
Initial velocity(u) = 4m/s
Final velocity(v)= 8m/s
Initial momentum = mu
Final momentum = mv
Change in momentum = final momentum- initial momentum
= mv – mu
=m(v-u)
= 2(8-4) kg m/s
= 2(4) kg m/s
= 8kg m/s

12.Correct option: (A), (B)

Linear Momentum is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.
As linear momentum is equal to a scalar(mass) time a vector(velocity), it is therefore a vector quantity and is denoted by $\vec{p}$.
Momentum(p) = mass(m) x velocity(v)
$\vec{p} = m\vec{v}$

- **Impulse:** impulse of a force is measured by the change in momentum.

Impulse = force x time - - - - - (1)
according to the newton's second law of motion,
Force $\propto$ time rate of change of momentum
$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}} - - -$ (2)
Comparing 1 and 2
We can say that impulse is measured by change in momentum.
Impulse acts in the direction of the change in momentum.

- $a = \frac{F}{m}$

From equation 2
$a = \frac{1}{m}\frac{\text{change in momentum}}{\text{time taken}}$
Hence, acceleration has same direction as the change in momentum.

- Linear momentum is not conserved in uniform circular motion because

The linear momentum doesn't remain constant as
momentum = mass × velocity
and the direction of velocity keeps on changing in uniform circular motion.

13.Correct option: (A), (C)

- **Calculation of acceleration**

initial velocity(u) of particle =5m/s
Final velocity(v) of particle=6m/s
Time taken = 2s
Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here,* a= acceleration

u= initial velocity

v = final velocity

$a = \frac{6-5}{2}\ m/s^2$

$a = 0.5\ m/s^2$

acceleration = $0.5 m/s^2$

- **To calculate the mass of the particle**

Given: force(F)= 0.6 N

Acceleration (a)= $0.5 m/s^2$

From the newton's second law of motion

Force = mass x acceleration

F = m x a

$m = \frac{F}{a}$

$m = \frac{0.6}{0.5}$ kg

m=1.2 kg

hence, mass of the particle =1.2 kg

1 kg = 1000 g

1.2 kg= 1.2 x1000g=1200g

So, mass of the particle = 1.2kg or 1200g

14.Correct option: (B), (C)

Newton's second law of motion states that the time rate of change of momentum of a body is directly proportional to the external force applied on the body and the change in momentum takes place in the direction of the force.

Let $\vec{p}$ be the momentum of the body and $\vec{F}$ be the external force acting on it. Then according to the newton's second law of motion,

Force $\propto$ time rate of change of momentum

$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$

$\vec{F} = \frac{\text{final momentum(mv)} - \text{initial momentum(mu)}}{t}$

here, u= initial velocity

V= final velocity

15. Correct option: (A)

Given: mass(m) = 1.5 kg

Acceleration(a)=$2.4 ms^{-2}$

From the newton's second law of motion

Force = mass x acceleration

F = m x a

F = 1.5 x 2.4 N

F = 3.6 N

Hence, force = 3.6 N

16. Correct option: (D)

Given: mass(m) = 2 kg

Force(F)=15N

From the newton's second law of motion

Force = mass x acceleration

F = m x a

$a = \frac{F}{m}$

$a = \frac{15}{2}$ ms^{-2}

a =$17.5 ms^{-2}$

hence, acceleration = $7.5 ms^{-2}$

17.Correct option: (C), (D)

Given: mass(m) = 600 g

1g = $\frac{1}{1000}$kg

600g = $\frac{600}{1000}$kg

Acceleration(a)=10ms^{-2}

From the newton's second law of motion

Force = mass x acceleration

F = m x a

F = $\frac{600}{1000}$ x10 N

F = $\frac{600}{100}$N

F = 6 N

F= 6N

Hence, force = 6 N or 6kgms^{-2}

18.Correct option: (C)

Given: mass(m) of the body = 3kg

Initial velocity(u) = 10m/s

Initial momentum = m x u

= 3 x 10 kg m/s

= 30kg m/s

Initial momentum = 30 kg m/s

19. Correct option: (D)

Given: mass(m) of the body = 3kg

final velocity(v) = 20m/s

final momentum = m x v

= 3 x 20 kg m/s

= 60kg m/s

final momentum = 60 kg m/s

20. Correct option: (B)

- **To calculate initial momentum**

Given: mass(m) of the body = 3kg

Initial velocity(u) = 10m/s

Initial momentum = m x u

= 3 x 10 kg m/s

= 30kg m/s

Initial momentum = 30 kg m/s

- **To calculate final momentum**

final velocity(v) = 20m/s

final momentum = m x v

= 3 x 20 kg m/s

= 60kg m/s

final momentum = 60 kg m/s

- **To calculate change in momentum**

Initial momentum = 30 kg m/s

Final momentum = 60 kg m/s

Change in momentum = final momentum- initial momentum

= 60 kg m/s – 30 kg m/s

= 30 kg m/s

21. Correct Matching is: a-Q; b-S; c-P; d-RT

- Force = mass x acceleration

  F = m x a

  1 N = 1 kg x 1 ms^{-2} = 1kg ms^{-2}

- According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.

The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called

inertia.

Hence, newton's first law defines **inertia.**

- **Newton's second law of motion** states that the time rate of change of momentum of a body is directly proportional to the external force applied on the body and the change in momentum takes place in the direction of the force. This law gives quantitative explanation of force.

Let $\vec{p}$ be the momentum of the body and $\vec{F}$ be the external force acting on it. Then according to

the newton's second law of motion,

Force $\propto$ time rate of change of momentum

$\vec{F} = \frac{\text{change in momentum}}{\text{time taken}}$

$\vec{F} = \frac{\text{final momentum(mv)} - \text{initial momentum(mu)}}{t}$ here, u= initial velocity

v= final velocity

- **Momentum** is defined as the total quantity of motion contained in a body and it is measured as the product of the mass of the body and its velocity.

As momentum is equal to a scalar(mass) time a vector(velocity), it is therefore a vector quantity

and is denoted by $\vec{p}$.

Momentum(p) = mass(m) x velocity(v)

SI unit of mass is kg and SI unit of velocity is m/s

Hence, SI unit of momentum is kg m/s

NEWTON'S THIRD LAW OF MOTION
LEVEL-I

1.Correct option: (B)

According to the law of conservation of momentum: when two or more bodies act upon one another, their total momentum remains constant (or conserved) provided no external forces are acting. The law of conservation of momentum means that whenever one body gains momentum, then some other body must lose and equal amount of momentum. This law can also be stated as: momentum is never created or destroyed. The law of conservation of momentum is also known as the principle of conservation of momentum. It is in accordance with the Newton's third law of motion which says that action and reaction(forces) are equal and opposite.

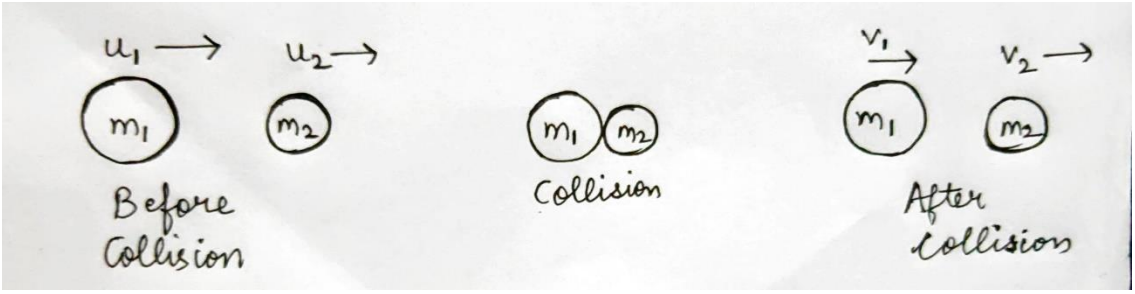
Consider two bodies are moving in the same direction but with different velocities.

Let

Mass of the body 1=m_1

Initial velocity of body 1 =u_1

Initial momentum of body 1 = m_1u_1
Mass of the body 2=m_2
Initial velocity of body 2 =u_2
Initial momentum of body 2 = m_2u_2
Thus, the total momentum of the two bodies before collision=$m_1u_1 + m_2u_2$



Suppose the two bodies collide, due to collision, the velocities of both the bodies will change
Let
Final velocity of body 1 =v_1
Final momentum of body 1= m_1v_1
Final velocity of body 2 =v_2
Final momentum of body 2 = m_2v_2
Thus, the total momentum of the two bodies after collision=$m_1v_1 + m_2v_2$
Suppose during the collision, the body 1 exerts a force F_1 on the body 2, the body 2 exerts a force F_2 on the body 1

- Calculation of the forces F_1 *and* F_2

When the force F_1 acts on the body 2 for a time t, then the velocity of the body 2 changes from u_2 *to* v_2
So,

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a_2 = \frac{v_2 - u_2}{t}$$

But force = mass x acceleration
So, the force F_1 exerted by the body 1 on 2 is given by

$$F_1 = m_2\frac{v_2 - u_2}{t} \text{----------(1)}$$

When the force F_2 acts on the body 1 for a time t, then the velocity of the body 1 changes from u_1 *to* v_1
So,

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a_1 = \frac{v_1 - u_1}{t}$$

But force = mass x acceleration
So, the force F_1 exerted by the body 1 on 2 is given by

$$F_2 = m_1\frac{v_1 - u_1}{t} \text{----------(2)}$$

Force F_1 exerted by body 1 is the **action** and the force F_2 exerted by the body 2 is the **reaction.**
According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.
So $F_1 = -F_2$
Putting the values from equation 1 and 2 we get,

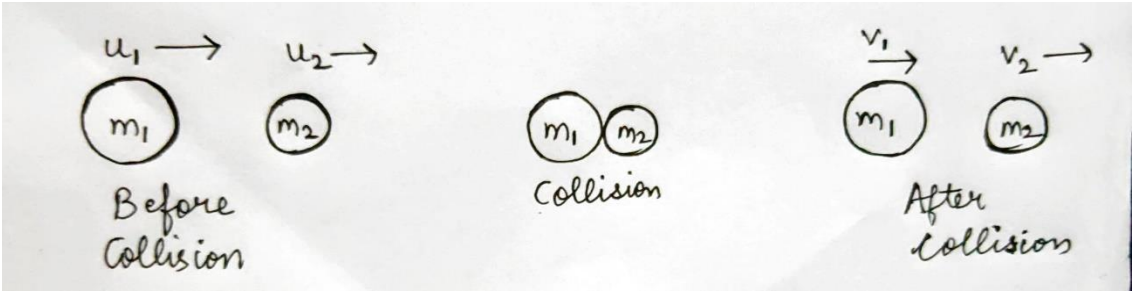
$$m_2\frac{v_2 - u_2}{t} = m_1\frac{v_1 - u_1}{t}$$

$m_2(v_2 - u_2) = -m_1(v_1 - u_1)$
$m_2v_2 - m_2u_2 = -m_1v_1 - m_1u_1$
$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
Total momentum before collision = total momentum after collision
Thus, means that the momentum of the two bodies remains conserved.
Hence, newton's third law is law of conservation of momentum.

2. Correct option: (C)

According to the law of conservation of momentum: when two or more bodies act upon one another, their total momentum remains constant (or conserved) provided no external forces are acting. The law of conservation of momentum means that whenever one body gains momentum, then some other body must lose and equal amount of momentum. This law can also be stated as: momentum is never created or destroyed. The law of conservation of momentum is also known as the principle of conservation of momentum. It is in accordance with the Newton's third law of motion which says that action and reaction(forces) are equal and opposite.

Consider two bodies are moving in the same direction but with different velocities.
Let
Mass of the body 1=m_1
Initial velocity of body 1 =u_1
Initial momentum of body 1 = m_1u_1
Mass of the body 2=m_2
Initial velocity of body 2 =u_2
Initial momentum of body 2 = m_2u_2
Thus, the total momentum of the two bodies before collision=$m_1u_1 + m_2u_2$



Suppose the two bodies collide, due to collision, the velocities of both the bodies will change
Let
Final velocity of body 1 =v_1
Final momentum of body 1= m_1v_1
Final velocity of body 2 =v_2
Final momentum of body 2 = m_2v_2
Thus, the total momentum of the two bodies after collision=$m_1v_1 + m_2v_2$
Suppose during the collision, the body 1 exerts a force F_1 on the body 2, the body 2 exerts a force F_2 on the body 1

- Calculation of the forces F_1 *and* F_2

When the force F_1 acts on the body 2 for a time t, then the velocity of the body 2 changes from u_2 *to* v_2
So,

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a_2 = \frac{v_2 - u_2}{t}$$

But force = mass x acceleration
So, the force F_1 exerted by the body 1 on 2 is given by

$F_1 = m_2 \frac{v_2 - u_2}{t}$----------(1)

When the force F_2 acts on the body 1 for a time t, then the velocity of the body 1 changes from u_1 *to* v_1

So,

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a_1 = \frac{v_1 - u_1}{t}$$

But force = mass x acceleration

So, the force F_1 exerted by the body 1 on 2 is given by

$F_2 = m_1 \frac{v_1 - u_1}{t}$----------(2)

Force F_1 exerted by body 1 is the **action** and the force F_2 exerted by the body 2 is the **reaction.**

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

So $F_1 = -F_2$

Putting the values from equation 1 and 2 we get,

$$m_2 \frac{v_2 - u_2}{t} = m_1 \frac{v_1 - u_1}{t}$$

$$m_2(v_2 - u_2) = -m_1(v_1 - u_1)$$

$$m_2v_2 - m_2u_2 = -m_1v_1 - m_1u_1$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Total momentum before collision = total momentum after collision

Thus, means that the momentum of the two bodies remains conserved.

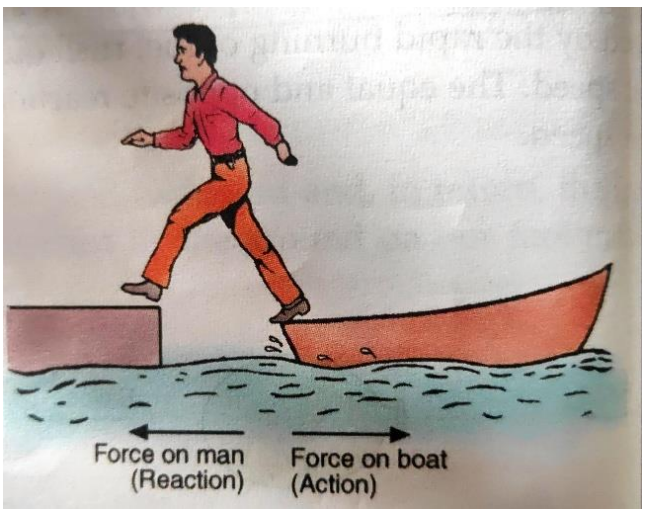
Hence, newton's third law is law of conservation of momentum.

3. Correct option (A), (D)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

When we jump out of a boat standing in water, the boat moves backwards, away from us. This is due to the fact that to step out of the boat, we press the boat with our foot in the backward direction. Our push on the boat is the action(force). The boat exerts an equal force on us in the forward direction which enables us to move forward. This force exerted by the boat on us is reaction(force). Since the boat is floating on water and not fixed, it moves backward due to the action force exerted by us.

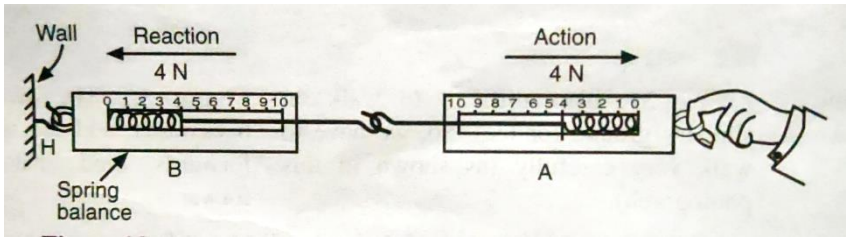


4.Correct option: (D)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

Action and reaction are the forces which always exist in pairs, are equal in magnitude and always act in opposite direction.



When we pull the spring balance A, it exerts a force of 4 N on the spring balance B. The spring balance B pulls the spring balance A with an equal force of 4 N, but in opposite direction. In other words, when balance A exerts a force of action on balance B, then balance B exerts an equal and opposite force of reaction on balance A. Both the spring balance show same reading of 4N, we conclude that the action and reaction forces are equal in magnitude and opposite in direction.

5.Correct option: (A)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

When we kick a stone, we exert a force(action) on it and the stone exerts an equal and opposite force(reaction) on our feet due to which we get hurt.

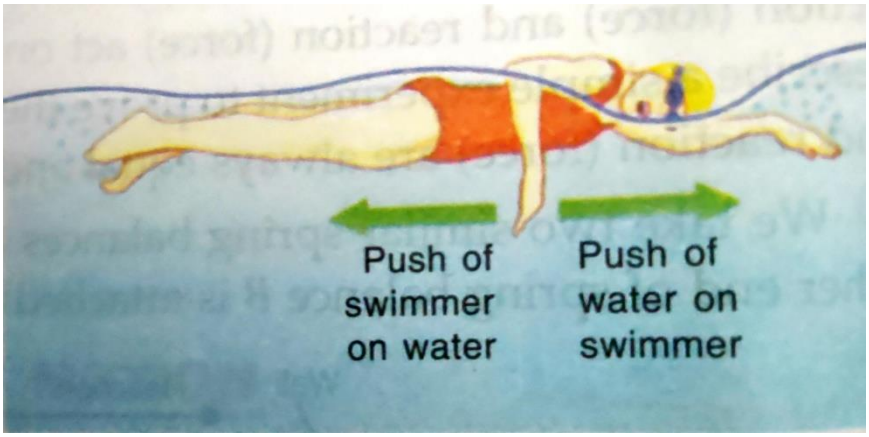
Hence, it is the reaction force of the stone due to which we get hurt.

6.Correct option: (C)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

A swimmer pushes the water backwards(action). The water pushes the swimmer forward(reaction) with the same force. Hence, the swimmer swims.



7. Correct option: (C)

According to the law of conservation of momentum: when two or more bodies act upon one another, their total momentum remains constant (or conserved) provided no external forces are acting. The law of conservation of momentum means that whenever one body gains momentum, then some other body must lose and equal amount of momentum. This law

can also be stated as: momentum is never created or destroyed. The law of conservation of momentum is also known as the principle of conservation of momentum.

8.Correct option: (D)

Both action and reaction are forces.

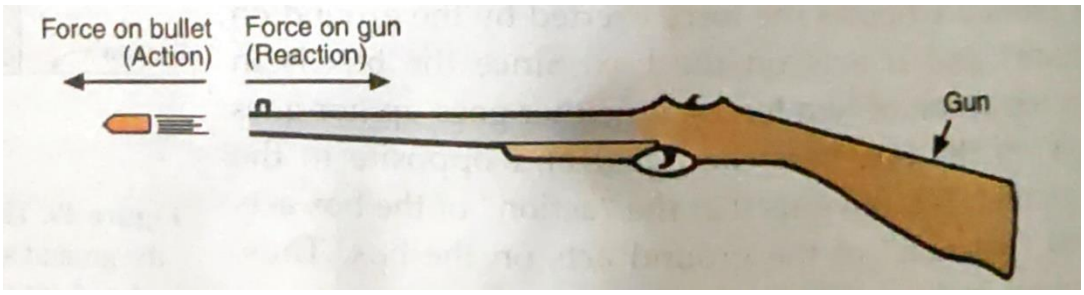
The term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

9.Correct option: (C)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

When a bullet is fired from a gun, the force sending the bullet forward(action) is equal to the force sending the gun backward(reaction). But due to high mass of the gun, it moves only a little distance backward and gives a backward jerk or kick to the shoulder of the gunman. The gun is said to have recoiled.



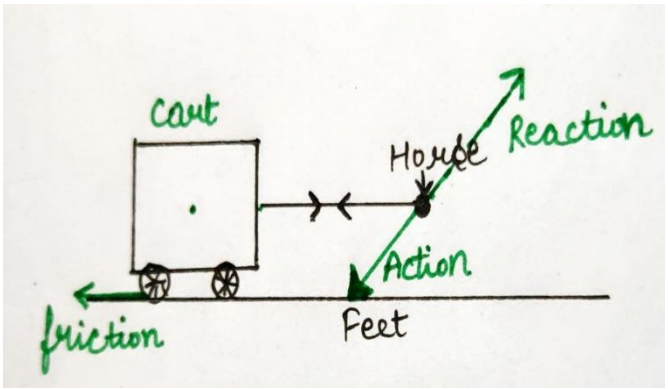
10.Correct option: (B)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

There is frictional force present between the wheels of the cart and the ground surface in contact. If the horse is able to overcome the frictional forces present, the cart will move.

To make the cart move, the horse bends forward and pushes the ground with its feet in the backward direction(action). The ground exerts an equal and opposite force(reaction) on the feet of the horse in the forward direction. When the forward reaction to the backward push of the horse is greater than the opposing frictional force of the wheels, the cart moves.

Hence, the forward reaction(force) of the ground on the horse, helps the cart and the horse to move forward.

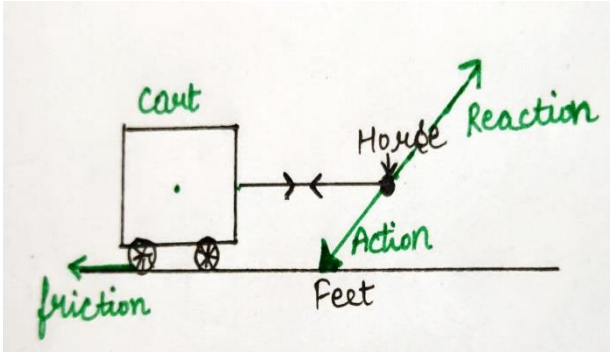


11. Correct option: (B)

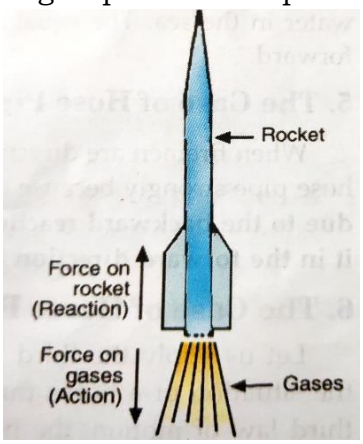
According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

There is frictional force present between the wheels of the cart and the ground surface in contact. If the horse is able to overcome the frictional forces present, the cart will move.

To make the cart move, the horse bends forward and pushes the ground with its feet in the backward direction(action). The ground exerts an equal and opposite force(reaction) on the feet of the horse in the forward direction. When the forward reaction to the backward push of the horse is greater than the opposing frictional force of the wheels, the cart moves.
Hence, the forward reaction(force) of the ground on the horse, helps the cart and the horse to move forward.



12. Correct option: (C)
According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.
Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.
In a rocket, the hot gases produced by the rapid burning of fuel rush out of a jet at the bottom of the rocket at a very high speed. The equal and opposite reaction force of the



downward going gases pushes the rocket upward with a high speed.
13. Correct option: (D)
According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.
When we kick a stone, we exert a force(action) on it and the stone exerts an equal and opposite force(reaction) on our feet due to which we get hurt.
Hence, it is the reaction force of the stone due to which we get hurt.
14. Correct option: (C)
When a body is travelling at constant velocity, it will have zero acceleration which means no force is acting on it or the net force acting on it is zero.
15.Correct option: (B)
The law of conservation of momentum means that whenever one body gains momentum, then some other body must lose and equal amount of momentum.
Applying the law of conservation of momentum on the gun bullet system
Momentum of the bullet = momentum of the gun
As, Momentum(p) = mass(m) x velocity(v)

Mass of bullet x velocity of bullet = mass of gun x recoil velocity of gun----------(1)

Given: Mass of the bullet = 100g = $\frac{100}{1000}$kg = 0.1kg

Velocity of the bullet = 100 ms^{-1}

Momentum of the bullet = Mass of bullet x velocity of bullet

= 0.1 x 100

= 10 kg ms^{-1}

Given: mass of the gun = 10 kg

To find: recoil velocity of the gun

Now, from equation 1

Mass of bullet x velocity of bullet = mass of gun x recoil velocity of gun----------(1)

10 kg ms^{-1}= 20 kg x recoil velocity of the gun

Recoil velocity of the gun = $\frac{10}{20}ms^{-1}$

=0.5ms^{-1}

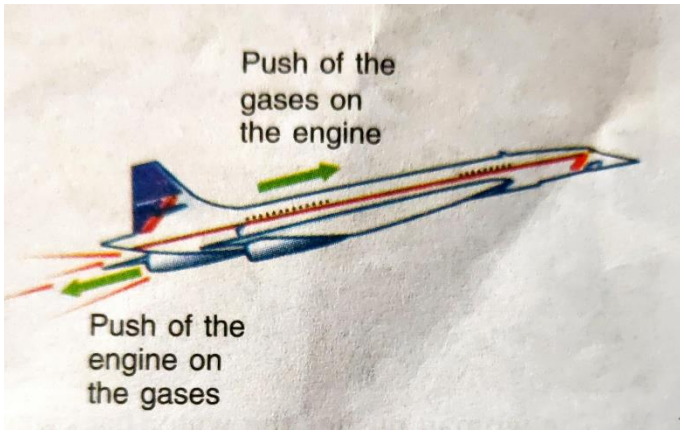
Recoil velocity of the gun= -0.5ms^{-1}

Negative sign is taken because Recoil velocity of gun is in opposite direction to that of the velocity of bullet.

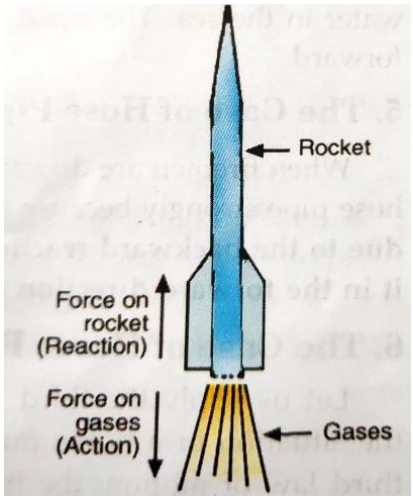
16.Correct option: (D)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

In the jet aero plane, the hot gases obtained by the rapid burning of fuel rush out of a jet (a nozzle) at the rear end (back end) of the aero plane at a greater speed. The equal and opposite reaction of the backward going gases pushes the aircraft forward at a greater speed



- In a rocket, the hot gases produced by the rapid burning of fuel rush out of a jet at the bottom of the rocket at a very high speed. The equal and opposite reaction force of the downward going gases pushes the rocket upward with a high speed.



- When a bullet is fired from a gun, the force sending the bullet forward(action) is equal to the force sending the gun backward(reaction). But due to high mass of the gun, it moves only a little distance backward and gives a backward jerk or kick to the shoulder of the gunman. The gun is said to have recoiled.

17.Correct option: (- 2 Ns**)**

Impulse= force x time

As force = mass x acceleration

So,

Impulse = mass x acceleration x time

I = (ma)t

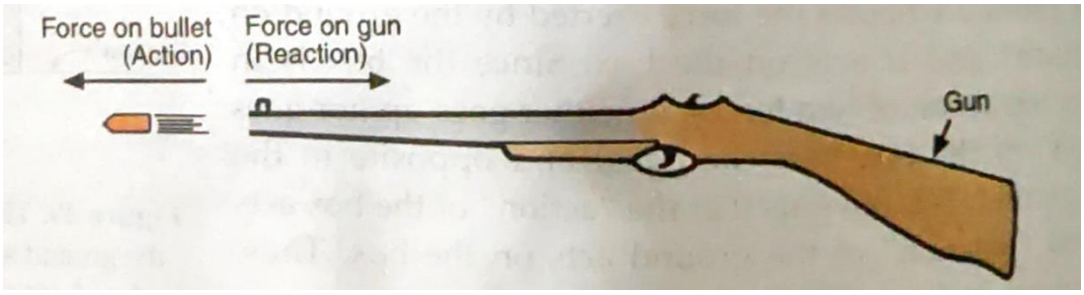
Given: mass of the hammer = 2kg

initial velocity of the nail = velocity of the hammer(u) = 1m/s

Final velocity of the nail(v)= 0 m/s (nail comes to rest)

Distance covered (S) = 10cm = 0.1 m

- To calculate the acceleration



Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2aS$ ------(1) Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation (1)

$v^2 - u^2 = 2aS$

$(0)^2 - (1)^2 = 2xax0.1$

0-1 = 0.2a

-1=0.2a

$-\frac{1}{0.2} = a$

$-5\ m/s^2 = a$

acceleration=$-5\ m/s^2$

- To calculate the time taken

Using The first equation of motion

v=u + at --------(2)

Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

t= time taken

putting the values in equation (2)

0=1+ (-5) x t

0= 1- 5t

-1=-5t

$\frac{1}{5} = t$

0.2seconds = t

Time = 0.2 seconds

- To calculate the impulse

Putting the value of m, a and t in following equation

I = (ma) t

I = 2 x (-5) x 0.2

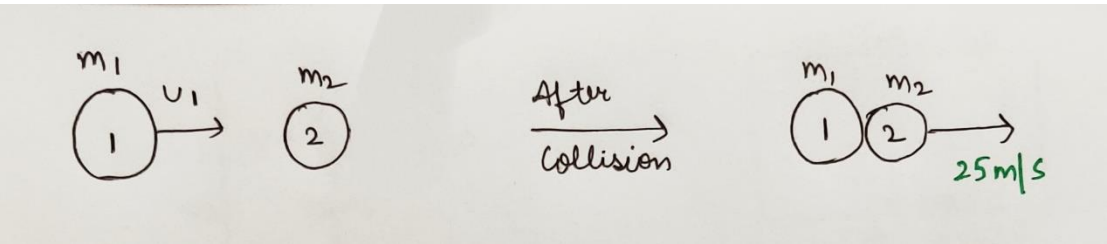
I= -2Ns

Impulse= - 2 Ns

18.Correct option: (A)

The term action means the force exerted by first body on the second body.

19.Correct option: (1.2 kg**)**



Given:

Mass of the body 1=m_1 =2kg

Initial velocity of body 1 =u_1= 40m/s

Initial velocity of body 2 =u_2= 0m/s (body 2 is at rest)

let Mass of body 2 =m_2

Combined velocity of body 1 and 2 after collision= 25m/s

Initial momentum of body 1 = m_1u_1

= 2 x 40

=80 kg m/s

Initial momentum of body 2 = m_2u_2

= 0 kg m/s

Thus, the total momentum of the two bodies before collision=$m_1u_1 + m_2u_2$

=80 kg m/s---------(1)

the total momentum of the two bodies after collision=combined mass of two bodies x combined Velocity

Combined mass of two bodies = 2kg + m_2

Combined velocity of body 1 and 2 after collision= 25m/s

the total momentum of the two bodies after collision= (2kg + m_2)25m/s--------(2)

according to the law of conservation of momentum

Total momentum before collision = total momentum after collision

From equation 1 and 2 we have

80 kg m/s = (2kg + m_2)25m/s

$\frac{80}{25}$kg = 2kg + m_2

3.2 kg – 2kg = m_2

1.2 kg = m_2

Hence, mass of the second body = 1.2 kg

20.Correct option: (C)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

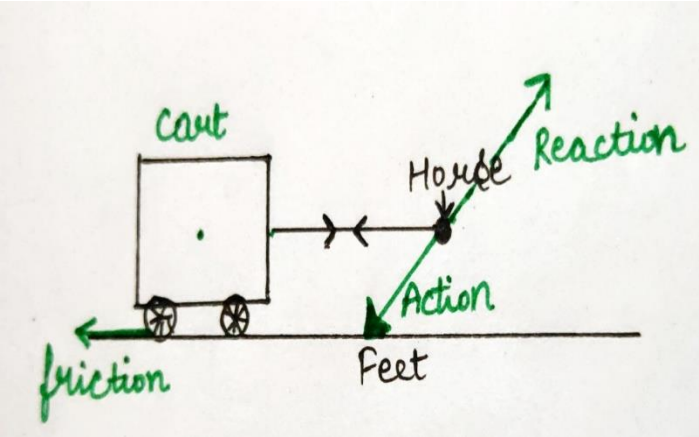
21. Correct option: (B)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

There is frictional force present between the wheels of the cart and the ground surface in contact. If the horse is able to overcome the frictional forces present, the cart will move.

To make the cart move, the horse bends forward and pushes the ground with its feet in the backward direction(action). The ground exerts an equal and opposite force(reaction) on the feet of the horse in the forward direction. When the forward reaction to the backward push of the horse is greater than the opposing frictional force of the wheels, the cart moves.

Hence, the forward reaction(force) of the ground on the horse, helps the cart and the horse to move forward.

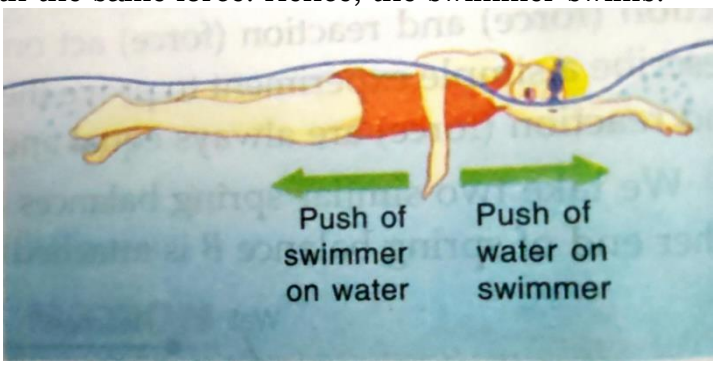


22. Correct option: (C)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

A swimmer pushes the water backwards(action). The water pushes the swimmer forward(reaction) with the same force. Hence, the swimmer swims.



23. Correct option: (D)

According to the law of conservation of momentum: when two or more bodies act upon one another, their total momentum remains constant (or conserved) provided no external forces are acting. The law of conservation of momentum means that whenever one body gains momentum, then some other body must lose and equal amount of momentum. This law

can also be stated as: momentum is never created or destroyed. The law of conservation of momentum is also known as the principle of conservation of momentum. It is in accordance with the Newton's third law of motion which says that action and reaction(forces) are equal and opposite.

Consider two bodies are moving in the same direction but with different velocities.

Let

Mass of the body 1=m_1

Initial velocity of body 1 =u_1

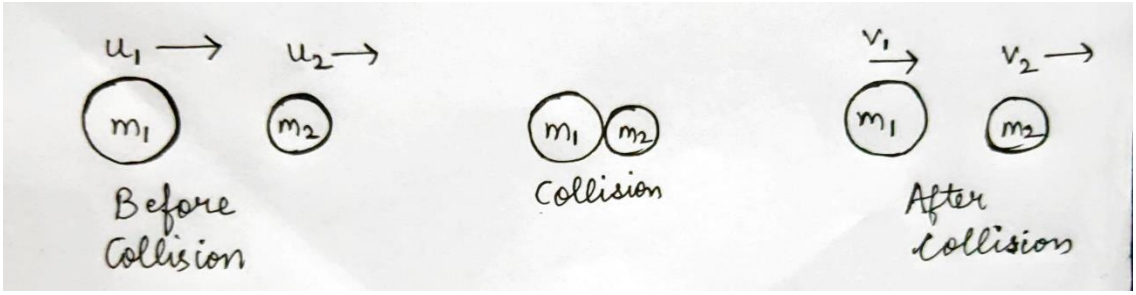
Initial momentum of body 1 = m_1u_1

Mass of the body 2=m_2

Initial velocity of body 2 =u_2

Initial momentum of body 2 = m_2u_2

Thus, the total momentum of the two bodies before collision=$m_1u_1 + m_2u_2$



Suppose the two bodies collide, due to collision, the velocities of both the bodies will change

Let

Final velocity of body 1 =v_1

Final momentum of body 1= m_1v_1

Final velocity of body 2 =v_2

Final momentum of body 2 = m_2v_2

Thus, the total momentum of the two bodies after collision=$m_1v_1 + m_2v_2$

Suppose during the collision, the body 1 exerts a force F_1 on the body 2, the body 2 exerts a force F_2 on the body 1

- Calculation of the forces F_1 *and* F_2

When the force F_1 acts on the body 2 for a time t, then the velocity of the body 2 changes from u_2 *to* v_2

So,

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a_2 = \frac{v_2 - u_2}{t}$$

But force = mass x acceleration

So, the force F_1 exerted by the body 1 on 2 is given by

$$F_1 = m_2 \frac{v_2 - u_2}{t} \text{----------(1)}$$

When the force F_2 acts on the body 1 for a time t, then the velocity of the body 1 changes from u_1 *to* v_1

So,

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

$$a_1 = \frac{v_1 - u_1}{t}$$

But force = mass x acceleration

So, the force F_1 exerted by the body 1 on 2 is given by

$$F_2 = m_1 \frac{v_1 - u_1}{t} \text{----------(2)}$$

Force F_1 exerted by body 1 is the **action** and the force F_2 exerted by the body 2 is the **reaction.**

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

So $F_1 = -F_2$

Putting the values from equation 1 and 2 we get,

$$m_2 \frac{v_2 - u_2}{t} = m_1 \frac{v_1 - u_1}{t}$$

$$m_2(v_2 - u_2) = -m_1(v_1 - u_1)$$

$$m_2v_2 - m_2u_2 = -m_1v_1 - m_1u_1$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Total momentum before collision = total momentum after collision

Thus, means that the momentum of the two bodies remains conserved.

Hence, newton's third law is law of conservation of momentum.

24.Correct option: (C)

Given: weight of the bird = 2 kg

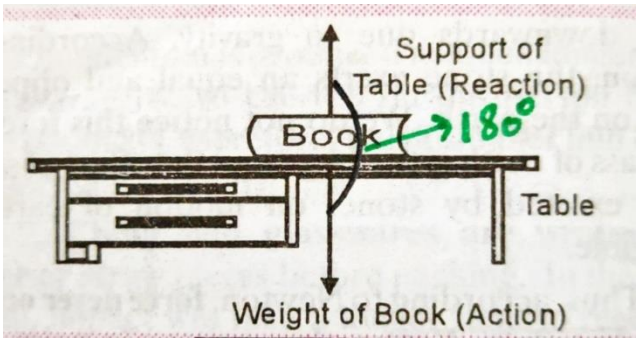
Weight of the cage= 1kg

When the bird flies the weight of the bird and the cage assembly = weight of bird + weight of cage

= 2kg + 1kg

=3kg

25.Correct option: (D)



According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

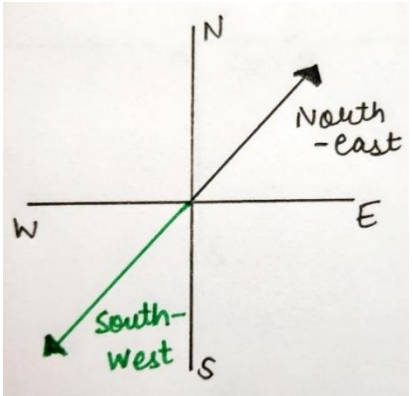
Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

The action and reaction are equal and opposite in direction.

Hence the angle between them is 180˚.

26.Correct option: (C)

To balance the force acting on the object in north-east direction, the second force must be applied in the south- west direction.



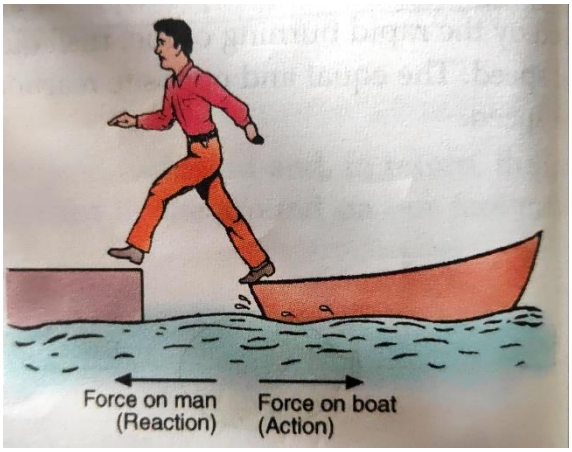
LEVEL-II

1.Correct option: (A)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

When we jump out of a boat standing in water, the boat moves backwards, away from us. This is due to the fact that to step out of the boat, we press the boat with our foot in the backward direction. Our push on the boat is the action(force). The boat exerts an equal force on us in the forward direction which enables us to move forward. This force exerted by the



boat on us is reaction(force). Since the boat is floating on water and not fixed, it moves backward due to the action force exerted by us.

2.Correct option: (C)

- Weight is defined as the force exerted on a body by gravity.

  Weight is often expressed as

  W= mg

  Where, w= weight of the body

  m = mass of the body

  g= acceleration due to gravity

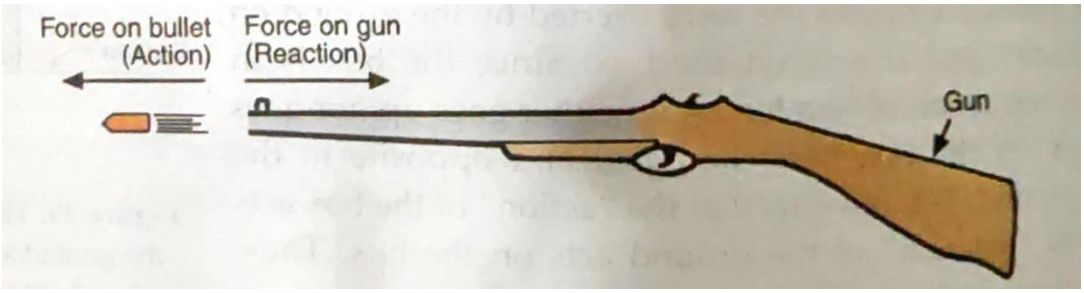
when the person takes stop on the left side, the weight of the man still acts in the downward direction so his weight will remain the same because there is no change in his mass or acceleration due to gravity.

3. Correct option: (C)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

When a bullet is fired from a gun, the force sending the bullet forward(action) is equal to the force sending the gun backward(reaction). But due to high mass of the gun, it moves



only a little distance backward and gives a backward jerk or kick to the shoulder of the gunman. The gun is said to have recoiled.

4.Correct option: (C)

Given: Mass of the car A=m_1= 1500kg

Initial velocity of car A =u_1 = 25m/s

Initial momentum of car A = m_1u_1

=1500 x 25 kg m/s

= 37500 kg m/s

Mass of the car B=m_2 =1000kg

Initial velocity of car B =u_2=15m/s

Initial momentum of car B= m_2u_2

=1000 x 15 kg m/s

=15000 kg m/s

Thus, the total momentum of the two bodies before collision=$m_1u_1 + m_2u_2$

=37500 + 15000 kg m/s

=52500 kg m/s

Suppose the two bodies collide, due to collision, the velocities of both the bodies will change

Let

Final velocity of car A =v_1=20m/s

Final momentum of car A= m_1v_1

= 1500 x 20 kg m/s

= 30000 kg m/s

Final velocity of car B =v_2

Final momentum of car B= m_2v_2= 1000 v_2

Thus, the total momentum of the two bodies after collision=$m_1v_1 + m_2v_2$ = 30000 kg m/s + 1000$v_2$kg m/s

According to the law of conservation of momentum

Total momentum before collision = total momentum after collision

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

52500 kg m/s = 30000 kg m/s + 1000$v_2$kg m/s

52500 kg m/s – 30000 kg m/s = 1000$v_2$kg m/s

22500 kg m/s = 1000$v_2$kg m/s

$\frac{22500}{1000} = v_2$

22.5 m/s =v_2

Hence, velocity of the car B after collision = 22.5 m/s

5. Correct option: (C)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

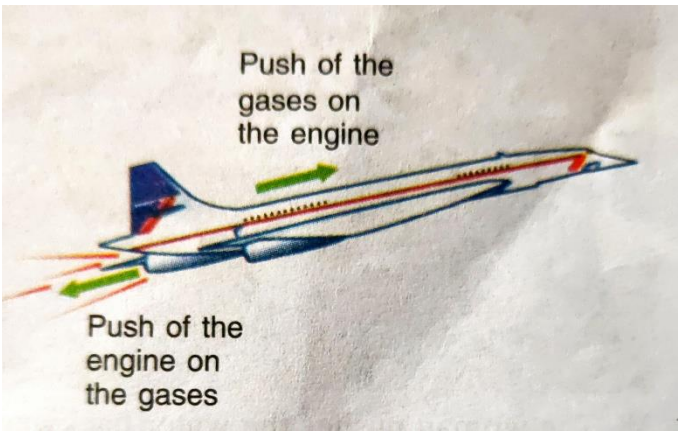
Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

When a shell is fired from a cannon, the force sending the shell forward(action) is equal to the force sending the cannon backward(reaction). But due to high mass of the cannon, it moves only a little distance backward. The cannon is said to have recoiled.

6.Correct option: (A)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

- In the jet aero plane, the hot gases obtained by the rapid burning of fuel rush out of a jet (a nozzle) at the rear end (back end) of the aero plane at a greater speed. The equal and opposite reaction of the backward going gases pushes the aircraft forward at a greater speed and also the thrust of the jet compensates the force of gravity.



7. Correct option: (D)

According to the newton's first law of motion: A body at rest will remain at rest, and a body in motion will continue in motion in a straight line with uniform speed, unless it is compelled by an external force to change its state of rest or of uniform motion.

The tendency of a body to remain at rest or, if moving, to continue moving in a straight line, is called inertia.

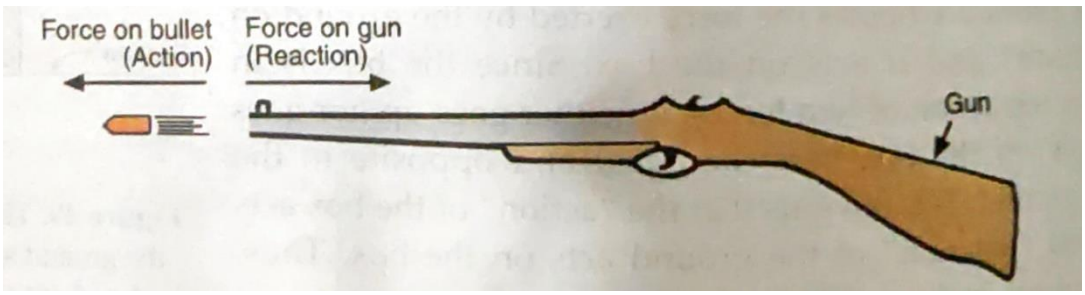
Thus, a force is not required to keep the particle moving with uniform velocity.

8.Correct options(A), (B), (D)

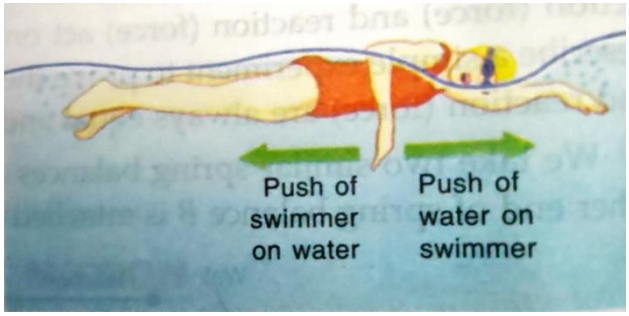
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Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

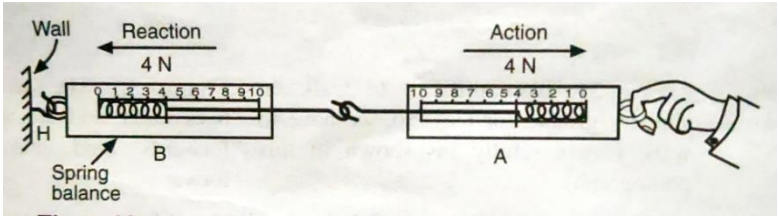
- When a bullet is fired from a gun, the force sending the bullet forward(action) is equal to the force sending the gun backward(reaction). But due to high mass of the gun, it moves only a little distance backward and gives a backward jerk or kick to the shoulder of the gunman. The gun is said to have recoiled.



- A swimmer pushes the water backwards(action). The water pushes the swimmer forward(reaction) with the same force. Hence, the swimmer swims.



- Action and reaction are the forces which always exist in pairs, are equal in magnitude and always act in opposite direction.



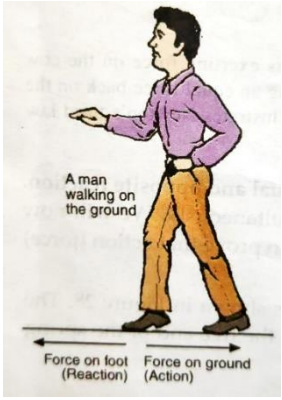
When we pull the spring balance A, it exerts a force of 4 N on the spring balance B. The spring balance B pulls the spring balance A with an equal force of 4 N, but in opposite direction. In other words, when balance A exerts a force of action on balance B, then balance B exerts an equal and opposite force of reaction on balance A.

9.Correct option: (A), (B), (C), (D)

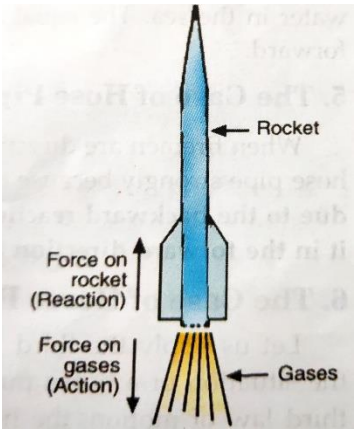
According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

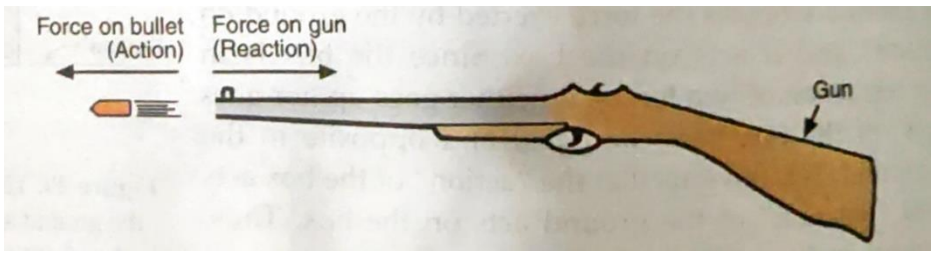
- During the rowing of a boat, the boatman pushes the water backwards with the oars. The water exerts an equal and opposite push on the boat which makes the boat move forward. In fact, harder the boatman pushes back the water with oars, greater is the reaction force exerted by the water and faster the boat moves forward.
- When a man walks on a road, then the foot of the man pushes the ground backwards(action) and in return, the ground pushes the foot of the man forward(reaction). The forward reaction exerted by the ground on the foot of the man makes the man walk forward.



- In a rocket, the hot gases produced by the rapid burning of fuel rush out of a jet at the bottom of the rocket at a very high speed. The equal and opposite reaction force of the downward going gases pushes the rocket upward with a high speed.



- When a bullet is fired from a gun, the force sending the bullet forward(action) is equal to the force sending the gun backward(reaction). But due to high mass of the gun, it moves only a little distance backward and gives a backward jerk or kick to the shoulder of the gunman. The gun is said to have recoiled.

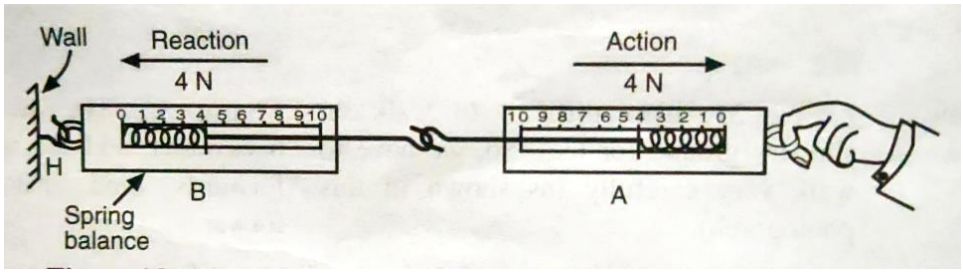


10.Correct options(B), (C)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

- Action and reaction are the forces which always exist in pairs, are equal in magnitude and always act in opposite direction.



When we pull the spring balance A, it exerts a force of 4 N on the spring balance B. The spring balance B pulls the spring balance A with an equal force of 4 N, but in opposite direction. In other words, when balance A exerts a force of action on balance B, then balance B exerts an equal and opposite force of reaction on balance A.

11.Correct option: (B)

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

12.Correct option: (A)

It is given that the force applied by the man on the car =5N

And we know force exerted by first body on the second is called action

So, value of action = 5 N

13.Correct option: (D)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

The term reaction means the force exerted by the second body on the first.

So, value of reaction is also 5N as action and reaction are equal and opposite.

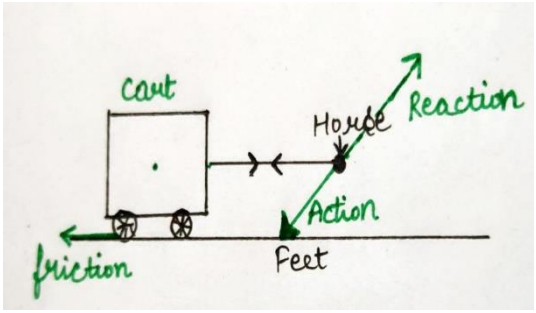
14. Correct option: (B)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

There is frictional force present between the wheels of the cart and the ground surface in contact. If the horse is able to overcome the frictional forces present, the cart will move.

To make the cart move, the horse bends forward and pushes the ground with its feet in the backward direction(action). The ground exerts an equal and opposite force(reaction) on the feet of the horse in the forward direction. When the forward reaction to the backward push of the horse is greater than the opposing frictional force of the wheels, the cart moves.

Hence, the forward reaction(force) of the ground on the horse, helps the cart and the horse to move forward.

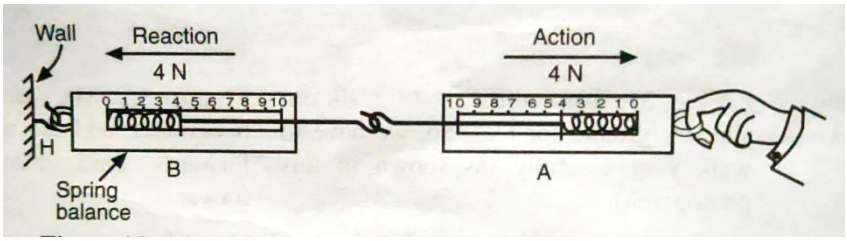


15. Correct option: (D)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

Action and reaction are the forces which always exist in pairs, are equal in magnitude and always act in opposite direction.



When we pull the spring balance A, it exerts a force of 4 N on the spring balance B. The spring balance B pulls the spring balance A with an equal force of 4 N, but in opposite direction. In other words, when balance A exerts a force of action on balance B, then balance B exerts an equal and opposite force of reaction on balance A. Both the spring balance show same reading of 4N, we conclude that the action and reaction forces are equal in magnitude and opposite in direction.

16.Correct option: (B)

According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

Here, the term action means the force exerted by one body on the other body, while the term reaction means the force exerted by the second body on the first.

Action and reaction are the forces which are equal in magnitude but opposite in direction

So, action = - reaction

17. Correct Matching is: a –P; b-T; c-Q; d-R

(a) Change in momentum = final momentum- initial momentum = mv – mu

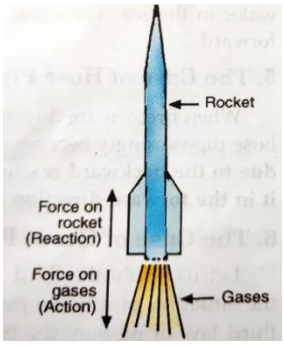
(b) Weight is defined as the force exerted on a body by gravity.

Weight is often expressed as W= mg----------(1)

Where, w= weight of the body

m = mass of the body

g= acceleration due to gravity = $9.8 m/s^2$



(c) Action and reaction are the forces which are equal in magnitude but opposite in direction So, action = - reaction

(d) According to the Newton's third law of motion, to every action, there is always and equal and opposite reaction.

In a rocket, the hot gases produced by the rapid burning of fuel rush out of a jet at the bottom of the rocket at a very high speed. The equal and opposite reaction force of the downward going gases pushes the rocket upward with a high speed.

4. Work Energy and Power

Solutions

WORK

LEVEL-I

1.Correct option: (C)

Whenever a force makes a body move, then work is said to be done. For example, a horse pulling a cart is doing work.

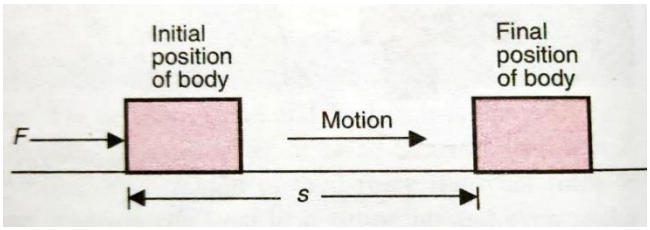
2. Correct option: (A)

For doing work, energy is required. Then the work is done by the human beings or animals (like horses, donkeys etc.), then energy for doing work is supplied by the food which they eat. And when the work is done by machines, then energy is supplied by the fuels such as petrol and diesel etc. or by electricity.

When work is done, an equal amount of energy is used up.

3. Correct option: (A)

Work done in moving a body is equal to the product of force exerted on the body and the



distance moved by the body in the direction of the force.

Work = force x distance moved in the direction of force

Or

Work = force x displacement in the direction of the force

W = F x S

Where, W= work done

F = force applied

S = distance / displacement

SI unit of force = newton (N)

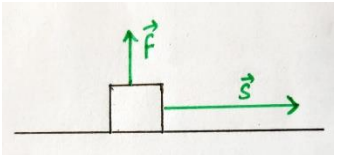
SI unit of distance = metre (m)

So, SI unit of work = Nm

1 joule = 1 newton x 1m

The SI unit of work is called **joule.**

4. Correct option: (C)



W= F cos θ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

Given: θ = 90˚

W = F cos90˚ x S (cos 90˚ = 0)

W= F x 0 x S
W = 0 J
Hence, work done is zero.

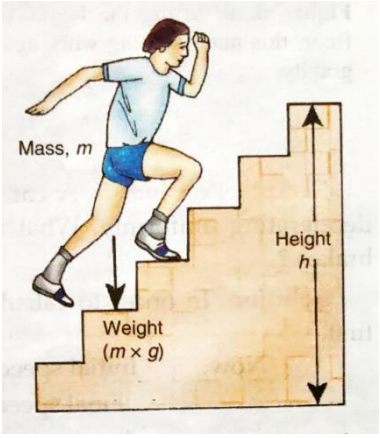
5. Correct option: (A)

Work done in moving a body is equal to the product of force exerted on the body and the distance moved by the body in the direction of the force.
Work = force x distance moved in the direction of force
Or
Work = force x displacement in the direction of the force
W = F x S
Where, W= work done
F = force applied
S = distance / displacement

6. Correct option: (B)

Work = force x distance moved in the direction of force
Or
Work = force x displacement in the direction of the force
W = F x S
Where , W= work done
F = force applied
S = distance / displacement
C.G.S unit of force = dyne
C.G.S unit of distance =cm
So, C.G.S unit of work = dyne cm
The C.G.S unit of work is called **erg.**
1 erg = 1 dyne x 1cm
Work done is said to be one erg, when a force of one dyne actually moves a body through a distance of one cm in the direction of the applied force.

7. Correct option: (C)



Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.
Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = $9.8 m/s^2$
h= height through which the body is lifted

given: mass of the man = 80 kg
mass of the load = 20 kg
total mass (m) = 80 kg+ 20 kg= 100kg
height of 1 stair= 25 cm = 0.25 m (1cm=0.01m)
total stairs = 40
total height(h) = 40 x 0.25 m
= 10m
Time = 20 sec
Putting the values in equation 1
W = 100 x 9.8 x 10 J
W = 9.8 x 1000 J
W = 9800 J
Work done = 9800 J

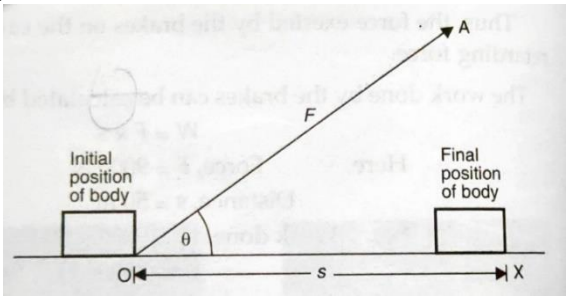
8. Correct option: (B)

Work done in moving a body is equal to the product of force exerted on the body and the distance moved by the body in the direction of the force.
Work = force x distance moved in the direction of force
Or
Work = force x displacement in the direction of the force
W = F x S
Where , W= work done
F = force applied
S = distance / displacement
SI unit of force = newton (N)
SI unit of distance = metre (m)
So, SI unit of work = newton meter (Nm)

9. Correct option: (C)

Given: force applied= 60N
Distance moved in the direction of force = 20m
Work = force x displacement in the direction of the force
W = F x S
W= 60 x 20 J
W= 1200 J
Hence, work done = 1200 J

10. Correct option: (B)



When a body moves at an angle to the direction of the force. Then all the force F is not utilized in moving the body, only the horizontal component of the force F is effective force which is moving the body.
The work done in moving the body will be equal to the product of horizontal component of the force and the distance moved / displacement of the body.
In the above figure
Horizontal component of force = F cosθ

Hence, W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
Given: θ = 90˚
W = F cos90˚ x S (cos 90˚ = 0)
W= F x 0 x S
W = 0 J
Hence, no work is done

11. Correct option: (A)

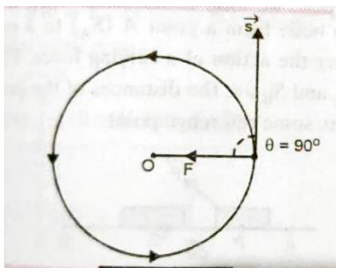
Given: force applied= 10N
Distance moved in the direction of force = 5m
Work = force x distance moved in the direction of force
W = F x S
W = 10 x 5 J
W = 50 J

12. Correct option: (B)



When a body moves at an angle to the direction of the force.
The work done in moving the body will be equal to the product of horizontal component of the force and the distance moved / displacement of the body.
Hence, W = F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
if the body moves in the direction opposite to the applied force then
θ = 180˚
W = F cos180˚ x S (cos 180˚ = -1)
W= F x (-1)x S
W = -FS
Hence, work done is negative.

13. Correct option: (C)



W= F cosθ x S

Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.

Angle θ between the centripetal Force acting along the radius of the circle and the direction of motion of body is 90˚.

So, θ = 90˚

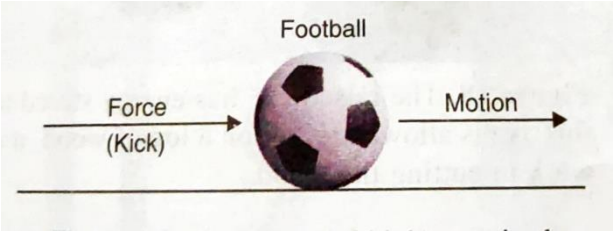
W = F cos90˚ x S (cos 90˚ = 0)

W= F x 0 x S

W = 0 J

Hence, work done is 0.

14. Correct option: (A)



W= F cosθ x S

Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.

If the body is displaced in the direction of the applied force. then,

θ = 0˚

W = F cos0˚ x S (cos 0˚ = 1)

W= F x 1 x S

W = FS

Hence, work done is positive.

15. Correct option: (C)

Whenever a force makes a body move, then work is said to be done. For example, a horse pulling a cart is doing work.

Work done in moving a body is equal to the product of force exerted on the body and the distance moved by the body in the direction of the force.

Work = force x displacement in the direction of the force

W= F X S

If the applied force fails to displace a body then

S = 0

Work done = Force x 0

Work done = 0

Hence work done will be zero.

16. Correct option: (B)

When a body moves at an angle to the direction of the force.

The work done in moving the body will be equal to the product of horizontal component of the force and the distance moved / displacement of the body.

Hence,

Hence, work done is negative.

17. Correct option: (C)

Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $9.8 m/s^2$or $10 m/s^2$

h= height through which the body is lifted

given: mass of water(m) = 200 kg

height (h) = 6m

g = $10 m/s^2$

putting the values in equation 1

W= 200 x 10 x 6

W = 12000J

Hence, work done = 12000 J

18. Correct option: (C)

Given: force applied= 40N

Displacement in the direction of force = 25cm = 0.25m (1cm = 0.01m)

Work = force x displacement in the direction of the force

W = F x S

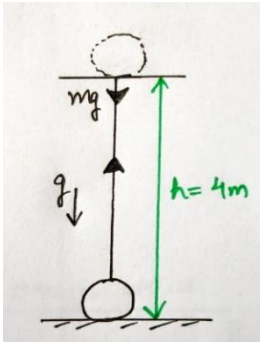
W= 40 x 0.25 J

W= $40 \times \frac{25}{100}$ J

W= 10 J

hence, work done = 10J

19. Correct option: (C)



Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10 m/s^2$

h= height through which the body is lifted

given: mass of ball(m) = 1 kg

height (h) = 4m

g = $10 m/s^2$

putting the values in equation 1

W= 1 x 10 x 4

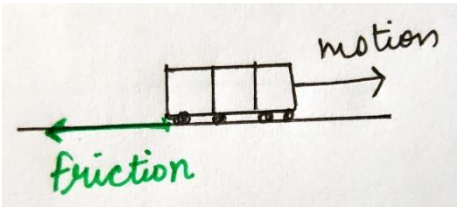
W = 40 J

Hence, work done = -40 J
Negative sign shows that work done by the force of gravity is negative.
Because the ball moves upwards and the force of gravitation is downwards.

20. Correct option: (B)

Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.
Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = $10m/s^2$
h= height through which the body is lifted
given: mass of 1 brick = 0.25 kg
mass of 12 bricks = 12 x 0.25 = 3 kg
total mass (m) = 3 kg
total height(h) = 10m
Putting the values in equation 1
W = 3 x 10 x 10 J
W = 300 J
Work done = 300 J

21. Correct option: (A)



Given: frictional force (f)= $5 \times 10^5 N$
Displacement in the direction of force(S) = 1km = 1000m
To find: work done by the train
$W = F \cos\theta \times S$
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
As the train moves opposite to the direction of frictional force
So, $\theta = 180^\circ$
$W = 5 \times 10^5 \cos 180^\circ \times 1000$ ($\cos 180^\circ = -1$)
$W = -5 \times 10^{5+3} J$
$W = -5 \times 10^8 J$
hence, work done = $-5 \times 10^8 J$

22. Correct option: (B)

Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = $10m/s^2$
h= height through which the body is lifted

given: mass of the man = 20 kg
mass of the load = 10 kg
total mass (m) = 20 kg+ 10 kg= 30kg
total height(h) = 100m
Putting the values in equation 1
W = 30 x 10 x 100 J
W = 30000 J (1K J =1000J)
W = 30 kJ
Work done = 30 KJ

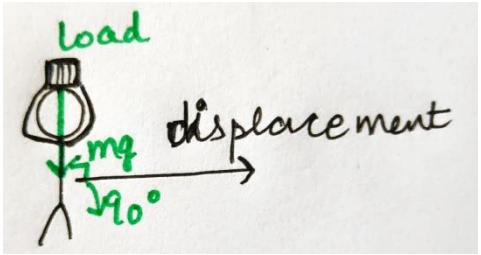
23. Correct option: (B)

Given: force applied= 5N
Displacement in the direction of force = 2m
Work = force x displacement in the direction of the force
W = F x S
W= 5 x 2 J
W= 10 J
Hence, work done = 10 J

24. Correct option: (C)

Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = 10 m/s^2
h= height through which the body is lifted
given: mass of luggage(m) = 15 kg
height (h) = 1.5m
g = 10 m/s^2
putting the values in equation 1
W= 15x 10 x 1.5
W = 225J
Hence, work done = 225 J

25. Correct option: (B)



When a coolie carries a load of 50kg on his head and moves horizontally on a road upto 100m, he does no work with respect to gravity because the force of gravity acts vertically downwards and the angle between the displacement of the luggage and the direction of the force is 90°
W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.

θ = 90˚
W = F cos90˚ x S (cos 90˚ = 0)
W= F x 0 x S
W = 0 J
Hence, work done is 0.

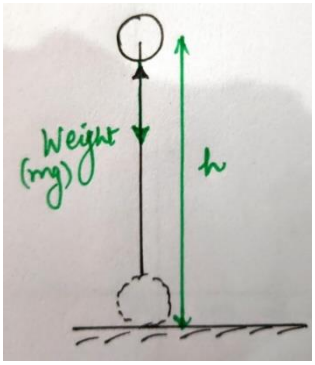
26. Correct option: (B)

Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = 10 m/s^2
h= height through which the body is lifted
given: mass of car(m) = 500 kg
height (h) = 4m
g = 10 m/s^2
putting the values in equation 1
W= 500x 10 x 4
W = 20000J (1K J =1000J)
W = 20kJ
Hence, work done = 20kJ

27. Correct option: (B)

Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = 10m/s^2
h= height through which the body is lifted
given:
mass (m) of the boy = 55kg
height of 1 stair= 0.15 m
total stairs = 40
total height(h) = 40 x 0.15 m
= 6m
Putting the values in equation 1
W = 55 x 10 x 6 J
W = 3300 J
Work done = 3300 J

28. Correct option: (A)



Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)

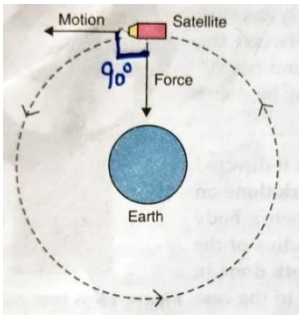
Where W = work done
m = mass of the body
g = acceleration due to gravity = $10m/s^2$
h= height through which the body is lifted

As W = F cosθ x S

Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.

when the body is lifted upwards then the work done by the gravity is taken as negative.
Acceleration due to gravity acts downwards hence, force of gravity is downwards and the object moves upwards.
θ = 180˚
W = F cos180˚ x S (cos 180˚ = -1)
W= F x (-1)x S
W = -FS
Hence, work done by the gravity is negative.

29. Correct option: (A)



When a body moves at an angle to the direction of the force.
The work done in moving the body will be equal to the product of horizontal component of the force and the distance moved / displacement of the body.
Hence, W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
Angle θ between the Force of gravity and the direction of motion of satellite is 90˚.
So, θ = 90˚
W = F cos90˚ x S (cos 90˚ = 0)
W= F x 0 x S
W = 0 J
Hence, work done is 0.

30. Correct option: (C)

Given: force applied= 10N
Distance moved in the direction of force = 4m
Work = force x displacement in the direction of the force
W = F x S
W= 10 x 4 J
W= 40J
Hence, work done = 40 J

LEVEL-II

1.Correct option: (C)

When a body moves at an angle to the direction of the force.

The work done in moving the body will be equal to the product of horizontal component of the force and the distance moved / displacement of the body.

Hence, $W = F \cos\theta \times S$

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

If $\theta = 90^\circ$

$W = F \cos 90^\circ \times S$ ($\cos 90^\circ = 0$)

$W = F \times 0 \times S$

$W = 0$ J

So, angle must be other than 90° so that work done is non zero.

Also, If S = 0 then work done will be zero.

So, body must have displaced in the direction of the force then only work is said to be done.

2. Correct option: (A)

Work done in lifting a body = weight of body x vertical distance

$W = m \times g \times h$----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10 m/s^2$

h= height through which the body is lifted

As

$W = F \cos\theta \times S$

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

when the body is lifted upwards then the work done by the gravity is taken as negative.

Acceleration due to gravity acts downwards hence, force of gravity is downwards and the the object moves upwards.

$\theta = 180^\circ$

$W = F \cos 180^\circ \times S$ ($\cos 180^\circ = -1$)

$W = F \times (-1) \times S$

$W = -FS$

Hence, work done by the gravity is negative.

3. Correct option: (A)

Whenever a force makes a body move, then work is said to be done. For example, a horse pulling a cart is doing work.

Work done in moving a body is equal to the product of force exerted on the body and the distance moved by the body in the direction of the force.

Work = force x displacement in the direction of the force

As the box on the head of the man is not moving

Hence, displacement (S) = 0

W=F X S

Work done = Force x 0
Work done = 0
Hence work done will be zero.

4. Correct option: (C)

Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = $10m/s^2$
h= height through which the body is lifted
given:
let mass of the body=m
height= h

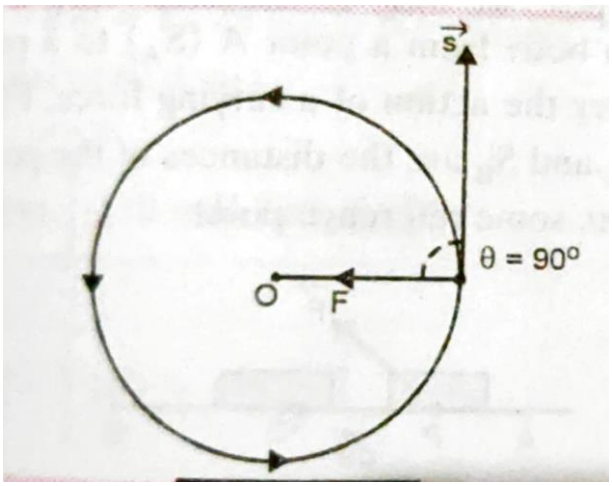
- Work done by person x

Putting the values in equation 1
W = m x g x h
W = 10 gh

- Work done by person y

Putting the values in equation 1
W = m x g x h
W = 10 gh
Hence, work done by person x and y is same.

5. Correct option: (C)

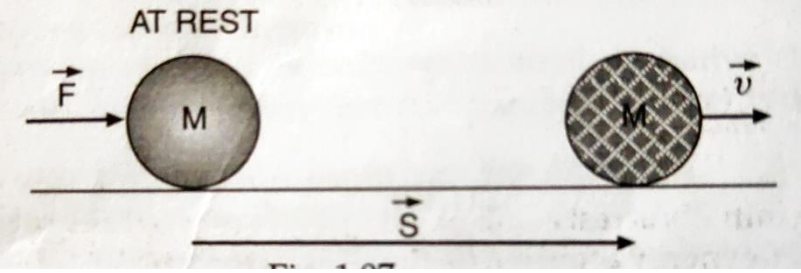


W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
Angle θ between the centripetal Force acting along the radius of the circle and the direction of motion of body is 90˚.
So,
θ = 90˚
W = F cos90˚ x S (cos 90˚ = 0)
W= F x 0 x S
W = 0 J
Hence, work done is 0.

6. Correct option: (B)

For doing work, energy is required. When work is done, an equal amount of energy is used up. For example, to cut a log of wood into small pieces, we have to raise the axe vertically above the log of wood and some work has to be done in raising the axe. If the axe is now allowed to fall on wood, it can do work in cutting the wood. Thus, the work done in raising the axe has been stored up in it, giving it the ability for doing work. It means energy of the axe has increased when work was done on it.

Consider a body of mass(m) initially at rest. Suppose a constant force F is applied on the body, till its velocity becomes v along the direction of the force. Let S be the displacement of the body during this time.



Initial kinetic energy of the body = $\frac{1}{2}mu^2 = 0$ (initial velocity(u) = 0m/s , body starts from rest)

Finial kinetic energy of the body = $\frac{1}{2}mv^2$

Change in kinetic energy = $\frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$

We know W= F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

θ = 0˚

W = F cos0˚ x S (cos 0˚ = 1)

W= F S----------(1)

If a is the acceleration produced in the body

Then according to newton's second law

F= ma----------(2)

Putting in equation 1

W =ma S ---------(3)

Using the Third equation of motion

$v^2 - u^2 = 2aS$ ------(4) Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation(4)

$v^2 - u^2 = 2aS$ (u = 0m/s body starts from rest)

$v^2 - 0 = 2aS$

$\frac{v^2}{2S} = a$

Putting value of a in equation 3

$W = m\frac{v^2}{2S} S$

Or

$W=\frac{mv^2}{2}$

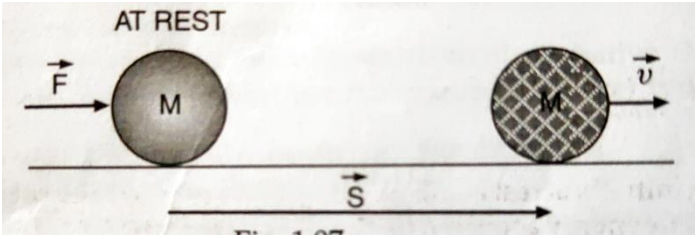
Or

W = change in kinetic energy

So the work done by external forces on a system equal the change in kinetic energy.

7. Correct option: (B)

Consider a body of mass(m) initially at rest. Suppose a constant force F is applied on the body, till its velocity becomes v along the direction of the force. Let S be the displacement of the body during this time.



Initial kinetic energy of the body = $\frac{1}{2}mu^2= 0$ (initial velocity(u) = 0m/s , body starts from rest)

Finial kinetic energy of the body =$\frac{1}{2}mv^2$

Change in kinetic energy = $\frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$

We know W= F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

$\theta = 0^\circ$

W = F cos0˚ x S (cos 0˚ = 1)

W= F S----------(1)

If a is the acceleration produced in the body

Then according to newton's second law

F= ma----------(2)

Putting in equation 1

W = ma S ---------(3)

Using the Third equation of motion

$v^2 - u^2 = 2aS$ ------(4) Here, v = final velocity of the object

u= initial velocity of the object

a = acceleration of the object

S= displacement/distance travelled by the object

putting the values in equation(4)

$v^2 - u^2 = 2aS$ (u = 0m/s body starts from rest)

$v^2 - 0 = 2aS$

$\frac{v^2}{2S} = a$

Putting value of a in equation 3

$W = m\frac{v^2}{2S} S$

Or

$W=\frac{mv^2}{2}$

Or

W = change in kinetic energy

Internal forces can only vibrate the molecules or atoms leading to the movement of the particles.

So the work done by all (external and internal forces) on a system equal the change in kinetic energy.

8. Correct option: (A)

Work done in carrying a body from one point to another does not depend on the path taken.

So, work done in moving the body from point A to C = work done in moving the body from point A to B +

work done in moving the body from point B to C

= 20 J + 15 J

= 35J

9. Correct option: (A), (B)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

the work gets stored up in the body as potential energy. Thus,

potential energy(P.E) = m x g x h

when the metal ball is resting on the plate no work is done by it as its vertical displacement(h) is 0.

When the ball is raised to some height then it is able to do some work because it gains some potential energy due to its height above the earth's surface.

10. Correct option: (D)

For doing work, energy is required. When work is done, an equal amount of energy is used up.

Hence, the amount of work done by an object on the other object depends on the energy it has.

11. Correct option: (A), (B)

For doing work, energy is required. When work is done, an equal amount of energy is used up.

Whenever a force makes a body move, then work is said to be done.

Work done in moving a body is equal to the product of force exerted on the body and the distance moved by the body in the direction of the force.

Work = force x displacement in the direction of the force

W= F X S

In the options(A), (B)

The displacement is non- zero hence some work is done by the man and the girl. Hence some energy is spent.

In the option(C) and (D) applied force fails to displace a body then

S = 0

Work done = Force x 0

Work done = 0

Hence work done will be zero.

12. Correct option: (A), (B)

W= F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

Velocity and acceleration are in the direction of the Displacement of the body.

If θ = 90˚

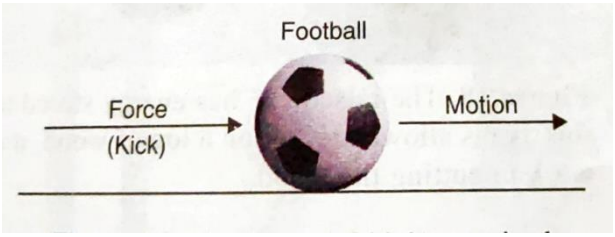
W = F cos90˚ x S (cos 90˚ = 0)

W= F x 0 x S

W = 0 J

Hence, work done is 0.

13. Correct option: (C)



W= F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

If the body is displaced in the direction of the applied force. then,

θ = 0˚

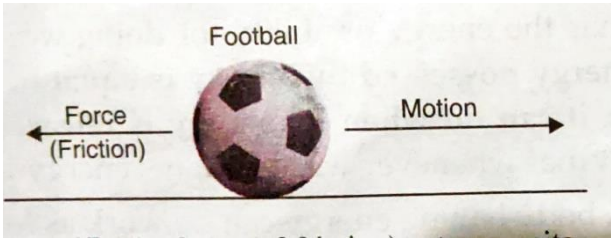
W = F cos0˚ x S (cos 0˚ = 1)

W= F x 1 x S

W = FS

Hence, work done is positive.

14. Correct option: (B)



W = F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

if the body moves in the direction opposite to the applied force then

θ = 180˚

W = F cos180˚ x S (cos 180˚ = -1)

W= F x (-1)x S
W = -FS
Hence, work done is negative.

15. Correct option: (D)

Whenever a force makes a body move, then work is said to be done. For example, a horse pulling a cart is doing work.
Work done in moving a body is equal to the product of force exerted on the body and the distance moved by the body in the direction of the force.
Work = force x displacement in the direction of the force
W= F X S
If the applied force does not displace a body from its position. then
S = 0
Work done = Force x 0
Work done = 0
Hence work done will be zero.

16. Correct option: (B)

Whenever a force makes a body move, then work is said to be done. For example, a horse pulling a cart is doing work.
Work done in moving a body is equal to the product of force exerted on the body and the distance moved by the body in the direction of the force.
Work = force x displacement in the direction of the force
As the man is not moving (stationary).
Hence, displacement (S) = 0
W=F X S
Work done = Force x 0
Work done = 0
Hence work done will be zero.

17. Correct option: (B)

When the man supports a man of 20kg on his hand and moves horizontally a distance of 30m, he does no work with respect to gravity because the force of gravity acts vertically downwards and the angle between the displacement of the man(weight) and the direction of the force is 90˚
W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
θ = 90˚
W = F cos90˚ x S (cos 90˚ = 0)
W= F x 0 x S
W = 0 J
Hence, work done is 0.

18. Correct option: (A)

W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

If the man moves a distance of 30 m vertically upwards in the direction of the force applied by him. then,

θ = 0˚

Given: mass of man = 50 kg

Mass of other man on his hand = 20kg

Total mass = 50kg +20kg =70 kg

Force = mass x acceleration due to gravity

F=mg

F= 70kg x 9.8m/s^2

F=686N

S=30m

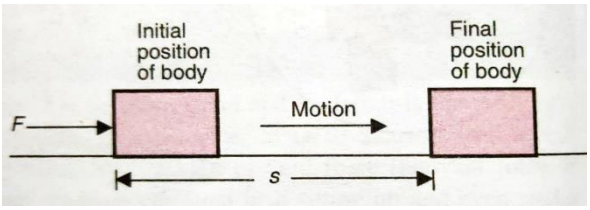
W = 686x cos0˚ x 30 J (cos 0˚ = 1)

W= 20,580 J

Hence, work done=20,580 J

19. Correct Matching is: **a-ST; b-QR; c-QR; d-P; e-P**

(a)



W= F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

If the body is displaced in the direction of the applied force. then,

θ = 0˚

W = F cos0˚ x S (cos 0˚ = 1)

W= F x 1 x S

W = FS

Hence, work done is positive.

(b) W = F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

when the body is lifted upwards then the work done by the gravity is taken as negative.

Acceleration due to gravity acts downwards hence, force of gravity is downwards and the the object moves upwards.

θ = 180˚

W = F cos180˚ x S (cos 180˚ = -1)

W= F x (-1)x S

W = -FS
Hence, work done by the gravity is negative.
(c)



W = F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
if the body moves in the direction opposite to the applied force then
θ = 180˚
W = F cos180˚ x S (cos 180˚ = -1)
W= F x (-1)x S
W = -FS
Hence, work done is negative.
(d) W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
If θ = 90˚
W = F cos90˚ x S (cos 90˚ = 0)
W= F x 0 x S
W = 0 J
Hence, work done is 0.
(e) Work = force x displacement in the direction of the force
W= F X S
If the applied force does not displace a body from its position. then
S = 0
Work done = Force x 0
Work done = 0
Hence work done will be zero.
20. Correct Matching is: a-T; b-S; c-R; d-P
(a) Work = force x displacement in the direction of the force
W = F x S
Where , W= work done
F = force applied
S = distance / displacement
C.G.S unit of force = dyne
C.G.S unit of distance =cm
So, C.G.S unit of work = dyne cm

The C.G.S unit of work is called **erg.**

1 erg = 1 dyne x 1cm

Work done is said to be one erg, when a force of one dyne actually moves a body through a distance of one cm in the direction of the applied force.

(b) Electron volt (eV) is the unit of energy.

One electron volt is the energy gained by an electron when accelerated through a potential difference of 1 volt.

1 e.V = 1.6×10^{-19} J

Or

1 J= $\frac{1}{1.6 \times 10^{-19}}$ ev

1 J= 6.25×10^{18} ev

(c) **W = F x S**

Where , W= work done

F = force applied

S = distance / displacement

SI unit of force = newton (N)

SI unit of distance = metre (m)

So, SI unit of work = Nm

1 joule = 1 newton x 1m----------(1)

(1 joule = 10^7 ergs

1 kgwt = 9.8 N)

Multiplying 9.8 on both sides of equation 1

9.8 joule = 9.8 newton x 1m

Or

9.8×10^7 ergs= 1 kgwt m

(d) 1kW = 10^3 W

ENERGY
LEVEL-I

1. Correct option: (C)

Energy is the ability to do work.

2. Correct option: (A)

Work = force x displacement in the direction of the force

W = F x S

Where , W= work done

F = force applied

S = distance / displacement

SI unit of force = newton (N)

SI unit of distance = metre (m)

So, SI unit of work = Nm

The SI unit of work is called **joule.**

1 joule = 1 newton x 1m

Work done is said to be one joule, when a force of 1N actually moves a body through a distance of 1m in the direction of the applied force.

3. Correct option: (B)

Scalar quantities are those which have magnitude(size) only and no direction.

Energy is a scalar quantity as it has magnitude only.3

4. Correct option: (B)
The amount of energy possessed by a body is equal to the amount of work it can do when its energy is released.
Hence, energy is equivalent to work
The SI unit of work is called **joule.**
Hence, SI unit of energy is Joule.
Different Forms of Energy
5. Correct option: (D)
The energy of a body due to its change in shape or size is known as elastic potential energy
Elastic potential energy is associated with the state of compression and extension of an object.
Hence, a compressed spring and a stretched elastic band both possess elastic potential energy.
6. Correct option: (D)
When a fuels burn, chemical energy stored in the fuel is converted into heat energy.
7. Correct option: (D)
The potential energy of a body is due to its higher position above the earth and it is equal to the work done on the body, against gravity, in moving the body to that position.
Water behind a dam on a hill top possesses potential energy due to its position(height) above the ground.
8. Correct option: (A)
The potential energy of a body is due to its higher position above the earth and it is equal to the work done on the body, against gravity, in moving the body to that position.
Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.
Work done in lifting a body = weight of body x vertical distance
W = m x g x h
Where W = work done
m = mass of the body
g = acceleration due to gravity = $9.8 m/s^2$ or $10 m/s^2$
h= height through which the body is lifted
the work gets stored up in the body as potential energy. Thus,
potential energy(P.E) = m x g x h----------(1)
Point P is at maximum height(h) as compared to rest all points
Hence,
Potential energy is of ball is high at point P.
9. Correct option: (B)
The energy of a body due to its change in shape or size is known as elastic potential energy
Elastic potential energy is associated with the state of compression and extension of an object.
A stretched elastic band of the catapult possess elastic potential energy due to the change in its shape.
The marble in the catapult has also potential energy stored in it due to its position.
When the marble is released the potential energy gets converted into the kinetic energy of the marble and it hits the ball.
10. Correct option: (B)
When a substance is heated, the temperature of the substance increases and the atoms/ molecules in it start moving due to thermal energy gained by them and hence they attain some kinetic energy.

11. Correct option: (A)

The energy of a body due to its change in shape or size is known as elastic potential energy Elastic potential energy is associated with the state of compression and extension of an object.

A stretched elastic spring of the bow possess elastic potential energy due to the change in its shape.

When the arrow is released the potential energy gets converted into the kinetic energy of the arrow and it hits the target.

12. Correct option: (A)

The potential energy of a body is due to its higher position above the earth and it is equal to the work done on the body, against gravity, in moving the body to that position.

Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.

Work done in lifting a body = weight of body x vertical distance

W = m x g x h

Where W = work done

m = mass of the body

g = acceleration due to gravity = $9.8 m/s^2$ or $10 m/s^2$

h= height through which the body is lifted

the work gets stored up in the body as potential energy. Thus,

potential energy(P.E) = m x g x h----------(1)

Point A is at least height(h) as compared to rest all points

Hence,

Potential energy is of ball is least at point A.

13. Correct option: (B)

Energy can neither be created nor be destroyed. Whenever energy changes from one form to another, the total amount of energy remains constant.

In the given figure

Energy possessed by the ball at height(h) point 1=maximum potential energy = mgh

When the ball falls downwards the potential energy starts decreasing. The velocity of the ball constantly increases and therefore its kinetic energy ($\frac{1}{2}mv^2$) also increases.

At point 2 the ball possesses both the potential energy (mgh) and the kinetic energy($\frac{1}{2}mv^2$).

When the ball just reaches the ground at point 3, its potential energy becomes 0 and kinetic energy is maximum.

14. Correct option: (C)

The marble came to a stop because the frictional force acts in the direction opposite to the direction of the motion of the marble.

So, after some time when external forces are zero the marbles stopped due to the frictional force between the marble and the surface of the ground.

15. Correct option: (B)

Photosynthesis is a process utilized by plants to produce glucose using carbon dioxide, water and light energy. Thus light energy is converted into chemical energy and stored in the form of glucose.

16. Correct option: (A)

For doing work, energy is required. When work is done, an equal amount of energy is used up. For example, to cut a log of wood into small pieces, we have to raise the axe vertically above the log of wood and some work has to be done in raising the axe. If the axe is now allowed to fall on wood, it can do work in cutting the wood. Thus, the work done in raising

the axe has been stored up in it, giving it the ability for doing work. It means energy of the axe has increased when work was done on it.

17. Correct option: (A)

When a weight is attached to a spring and it is pulled down and released, then it performs an oscillatory motion.

Oscillatory motion: It is that motion in which a body moves/fluctuates to and fro or back and forth repeatedly about a fixed point (called mean position) in definite interval of time. In such a motion the body is confined within well defined limits (called extreme positions) on either side of mean position.

When the spring is pulled down, potential energy is stored in it due to change in its shape.

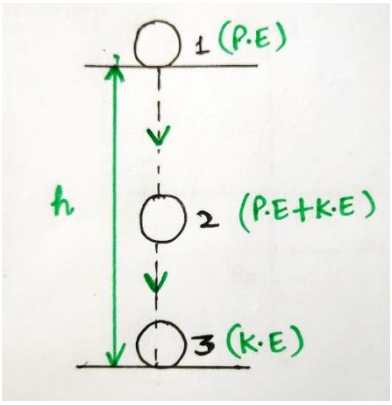
When the spring is released, the potential energy gets converted into the kinetic energy.

Kinetic energy is maximum in the middle of the movement (mean position)

Potential energy is maximum at the end points (extreme positions).

18. Correct option: (C)

Energy can neither be created nor be destroyed. Whenever energy changes from one form to another, the total amount of energy remains constant.



In the above figure

Energy possessed by the ball at height(h) point 1=maximum potential energy = mgh and kinetic energy is 0.

When the ball falls downwards the potential energy starts decreasing. The velocity of the ball constantly increases and therefore its kinetic energy ($\frac{1}{2}mv^2$) also increases.

At point 2 the ball possesses both the potential energy (mgh) and the kinetic energy($\frac{1}{2}mv^2$).

When the ball just reaches the ground at point 3, its potential energy becomes 0 and kinetic energy is maximum.

Hence, the sum of both the energies remains the same at any point of the journey.

19. Correct option: (B)

All the sources of heat may not be the light sources, they may or may not emit light.

Hence,

Observation Q is wrong.

20. Correct option: (B)

While drilling a hole in the wall. Heat energy and sound energy are of no use.

21. Correct option: (D)

(A) Energy can be stored in the form of potential energy.

(B) For doing work, energy is required. When work is done, an equal amount of energy is used up. For example, to cut a log of wood into small pieces, we have to raise the axe

vertically above the log of wood and some work has to be done in raising the axe. If the axe is now allowed to fall on wood, it can do work in cutting the wood. Thus, the work done in raising the axe has been stored up in it, giving it the ability for doing work. It means energy of the axe has increased when work was done on it.

(C) Whenever a force makes a body move, then work is said to be done and energy is used up in applying that force. For example, a horse pulling a cart is doing work.

22. Correct option: (C)

The energy possessed by a body by the virtue of its motion is called kinetic energy.

$\text{kinetic energy}(\text{K.E}) = \frac{1}{2}mv^2$----------(1)

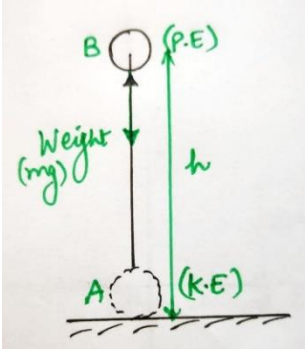
where m = mass of the body

v = velocity/speed of the body

from the equation 1 we can see that

kinetic energy depends on both the mass and the speed of the body.

23. Correct option: (D)



When a ball is thrown upwards its kinetic energy gets converted into potential energy.

When the ball is at point A

It has maximum kinetic energy

When the ball reaches the point B

It has maximum potential energy which came from the kinetic energy at point A.

So, potential energy at point B = kinetic energy at point A

24. Correct option: (D)

The energy possessed by a body by the virtue of its motion is called kinetic energy.

$\text{kinetic energy}(\text{K.E}) = \frac{1}{2}mv^2$----------(1)

where m = mass of the body

v = velocity/speed of the body

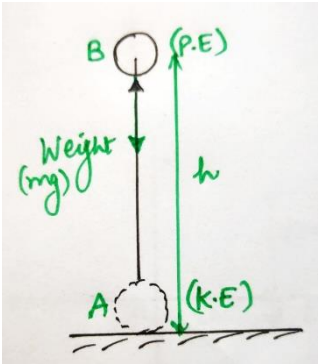
when the velocity becomes two time then

V = 2v

Let new kinetic energy= $\frac{1}{2}mV^2 = \frac{1}{2}m(2v)^2 = \frac{1}{2}m4v^2 = 4\frac{1}{2}mv^2$ = 4 K.E

Hence, kinetic energy becomes 4 times.

25. Correct option: (A)



When a ball is thrown upwards its kinetic energy gets converted into potential energy.
When the ball is at point A
It has maximum kinetic energy
When it is thrown upwards, its kinetic energy starts decreasing and potential energy starts increasing.
When the ball reaches the point B
It has maximum potential energy which came from the kinetic energy at point A.

26. Correct option: (B)
Mechanical energy is converted into electrical energy in a dynamo.

27. Correct option: (A)
Chemical energy stored in a cell is converted into electrical energy in case of an electric torch.

28. Correct option: (A)
Electric energy is converted into mechanical energy in an electric fan.

29. Correct option: (C)
The energy possessed by a body by the virtue of its motion is called kinetic energy.
kinetic energy(K. E) $= \frac{1}{2}mv^2$----------(1)
where m = mass of the body
v = velocity/speed of the body

30. Correct option: (B)
Work done in lifting a body = weight of body x vertical distance
W = m x g x h
Where W = work done
m = mass of the body
g = acceleration due to gravity = $9.8m/s^2$ or $10m/s^2$
h= height through which the body is lifted
the work gets stored up in the body as potential energy. Thus,
potential energy(P.E) = mgh

LEVEL- II

1.Correct option: (B)
When the car stops, the kinetic energy is converted into sound energy and the thermal energy.

2. Correct option: (A)
Energy can neither be created nor be destroyed. Energy just changes from one form to another. The total amount of energy remains constant.

3. Correct option: (C)
The potential energy of a body is due to its higher position above the earth and it is equal to the work done on the body, against gravity, in moving the body to that position.
Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.
Work done in lifting a body = weight of body x vertical distance
W = m x g x h
Where W = work done
m = mass of the body
g = acceleration due to gravity = $9.8m/s^2$ or $10m/s^2$
h= height through which the body is lifted
the work gets stored up in the body as gravitational potential energy. Thus,
Gravitational potential energy(P.E) = m x g x h

4. Correct option: (B)

kinetic energy(E_K) = $\frac{1}{2}mv^2$----------(1)

where m = mass of the body

v = velocity/speed of the body

given: mass (m) = 2kg

speed(v)= 3 m/s

putting the values in equation 1

kinetic energy(E_K) = $\frac{1}{2}$ x 2 x $(3)^2$

$=\frac{1}{2}$ x 2 x 9 J

= 9 J

Hence, kinetic energy of shopping cart = 9 J

5. Correct option: (A)

kinetic energy(E_K) = $\frac{1}{2}mv^2$----------(1)

where m = mass of the body

v = velocity/speed of the body

given: mass (m) = 20kg

speed(v)= 5 m/s

putting the values in equation 1

kinetic energy(E_K) = $\frac{1}{2}$ x 20 x $(5)^2$

$=\frac{1}{2}$ x 20 x 25 J

= 250 J

Hence, kinetic energy of skate boarder =250 J

6. Correct option: (C)

The amount of energy possessed by a body is equal to the amount of work it can do when its energy is released.

Hence, energy is equivalent to work

The SI unit of work is called **joule.**

Hence, SI unit of energy is Joule.

7. Correct option: (A)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h

Where W = work done

m = mass of the body

g = acceleration due to gravity =10m/s^2

h= height through which the body is lifted

the work gets stored up in the body as potential energy. Thus,

potential energy(P.E) = m x g x h----------(1)

given: mass of bucket(m) = 3 kg

g = 10 m/s^2

height(h) = 25m

to find: potential energy(P.E)

putting the values in equation 1

P.E= 3 x 10 x 25 J

P.E = 750 J

Hence, potential energy of bucket = 750 J

8. Correct option: (D)

The energy of a body due to its change in shape or size is known as elastic potential energy Elastic potential energy is associated with the state of compression and extension of an object.

A stretched spring or a rubber band possess elastic potential energy due to the change in its shape.

When a stretched spring or a rubber band is released the potential energy gets converted into the kinetic energy.

9. Correct option: (A), (B), (C), (D)

The energy possessed by a body by the virtue of its motion is called kinetic energy.

$\text{kinetic energy}(K.E) = \frac{1}{2}mv^2$----------(1)

where m = mass of the body

v = velocity/speed of the body

work done = change in kinetic energy

also work done = force x displacement in the direction of the force

W = F x S

Or

change in kinetic energy= force x displacement in the direction of the force

change in kinetic energy= F x S

- kinetic energy increases if the force acting on the particle moves the particle along/parallel its direction.

Also

W= F cosθ x S

Where, W= work done

F = force applied

S = distance / displacement

θ =angle between the direction of the applied force and the direction of motion of the body.

Change in kinetic energy = F cosθ x S

If θ< 90˚ then cos θ will be positive

And kinetic energy will increase.

- The potential energy of a body is due to its higher position above the earth and it is equal to the work done on the body, against gravity, in moving the body to that position.

Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.

Work done in lifting a body = weight of body x vertical distance

W = m x g x h

Where W = work done

m = mass of the body

g = acceleration due to gravity = $9.8m/s^2$ or $10m/s^2$

h= height through which the body is lifted

the work gets stored up in the body as gravitational potential energy. Thus,

Gravitational potential energy(P.E) = m x g x h

When the height(h) decreases then the potential energy of an object decreases and kinetic energy starts increasing.

- $(K.E) = \frac{1}{2}mv^2$--------(1)

Linear momentum(p) = m v

$p^2 = m^2v^2$

Using in equation 1

$K.E = \frac{p^2}{2m}$

Kinetic energy increases if the linear momentum is increasing.

10. Correct option: (B), (C), (D)

When force acts on a body.

- Its velocity may increase, K.E increases. For example when an object is pushed from its position of rest, it gains kinetic energy.
- Its velocity may decrease, K.E decreases. for example when brakes are applied to a car.its velocity starts decreasing and kinetic energy decreases.
- Its velocity may decrease and become 0, K.E becomes 0. for example when brakes are applied to a car. It stops after sometime and kinetic energy becomes zero.

11. Correct option: (A), (C)

The energy possessed by a body by the virtue of its motion is called kinetic energy.

$\text{kinetic energy}(K.E) = \frac{1}{2}mv^2$----------(1)

where m = mass of the body

v = velocity/speed of the body

work done = change in kinetic energy

work done = force x displacement in the direction of the force

W = F x S

Or

change in kinetic energy= force x displacement in the direction of the force

change in kinetic energy= F x S

It is given that both the bodies have the same kinetic energy. So the work done in bringing them to rest will also be same.

work done by the retarding force on m_1 and m_2 will be same.

- For body 1

kinetic energy= F x S1 (S1= displacement of body 1)

- For body 2

kinetic energy= F x S2 (S2= displacement of body 2)

as the kinetic energy is same for both the bodies. So

FS1 = FS2

S1 = S2

Hence, both the bodies will come to rest at the same distance.

12. Correct option: (A), (C)

Using the third equation of motion

$v^2 - u^2 = 2gh$ ------(1) Here, v = final velocity of the object

u= initial velocity of the object

g = acceleration due to gravity = $10 m/s^2$

h= displacement/distance travelled by the object vertically

$v^2 - (0)^2 = 2gh$ (u = 0m/s because body starts from rest)

$v^2 = 2gh$ --------(2)

- To calculate the kinetic energy

$\text{kinetic energy}(K.E) = \frac{1}{2}mv^2$----------(3)

where m = mass of the body

v = velocity/speed of the body

putting the value from equation 2

$\text{kinetic energy}(K.E) = \frac{1}{2} mx\ 2gh$

$\text{kinetic energy}(K.E) = mgh$

hence, kinetic energy is directly proportional to the height of its fall.

Using the first equation of motion

$v = u + gt$ ----- (4) Here, v = final velocity of the object

u= initial velocity of the object

g = acceleration due to gravity = $10 m/s^2$

t= time taken

$v = 0 + gt$ (u = 0m/s because body starts from rest)

$v = gt$----------(5)

- To calculate the kinetic energy

$\text{kinetic energy}(K.E) = \frac{1}{2} mv^2$----------(6)

where m = mass of the body

v = velocity/speed of the body

putting the value from equation 5

$\text{kinetic energy}(K.E) = \frac{1}{2} mx\ (gt)^2$

$\text{kinetic energy}(K.E) = \frac{1}{2} xm\ x\ g^2t^2$

hence, kinetic energy is directly proportional to the square of time of its fall.

13. Correct option: (A)

The potential energy of a body is due to its higher position above the earth and it is equal to the work done on the body, against gravity, in moving the body to that position.

Whenever the work is done against gravity, the amount of work done is equal to the product of weight of the body and the vertical distance through which the body is lifted.

Work done in lifting a body = weight of body x vertical distance

W = m x g x h

Where W = work done

m = mass of the body

g = acceleration due to gravity = $9.8 m/s^2$ or $10 m/s^2$

h= height through which the body is lifted

the work gets stored up in the body as potential energy. Thus,

potential energy(P.E) = m x g x h----------(1)

where m = mass of the body

g = acceleration due to gravity = $9.8 m/s^2$ or $10 m/s^2$

h= height of the body above a reference point, say the surface of the earth.

given: mass of box(m) = 5.8 kg

g = 10 m/s^2

potential energy(P.E)= 145 J

to find: height (h)

putting the values in equation 1

145= 5.8x 9.8x h

$\frac{145}{5.8\ x\ 9.8}$=h

2.55 m = h

Height of shelf= 2.55m

14. Correct option: (B)

potential energy(P.E) = m x g x h----------(1)

where m = mass of the body

g = acceleration due to gravity = $10m/s^2$
h= height of the body above a reference point, say the surface of the earth.

given: mass of child(m) = 30 kg
g = 10 m/s^2
height(h) = 15m
to find: potential energy(P.E)
putting the values in equation 1
P.E= 30 x 10 x 15 J
P.E = 4500 J
Hence, potential energy of child = 4500 J

15. Correct option: (B)

potential energy (P.E) = m x g x h----------(1)
where m = mass of the body
g = acceleration due to gravity = $10m/s^2$
h= height of the body above a reference point, say the surface of the earth.

given: mass of pole (m) = 50 kg
g = 10 m/s^2
height(h) = 12m
to find: potential energy(P.E)
putting the values in equation 1
P.E= 50 x 10 x 12 J
P.E = 6000 J
Hence, potential energy of pole= 6000 J

16. Correct option: (B)

given: mass (m) of the body = 3kg
height(h) = 1m

- To calculate the final velocity (v) on the reaching the ground

Third equation of motion gives the relation between velocity and displacement given as follows

$v^2 - u^2 = 2gh$ ------(1)

Here, v = final velocity of the object
u= initial velocity of the object
g = acceleration due to gravity = $10m/s^2$
h= displacement/distance travelled by the object vertically

putting the values in equation(1)
$v^2 - u^2 = 2gh$
$v^2 - (0)^2 = 2x10x1$ (u = 0m/s because body starts from rest)
$v^2 = 20$ --------(2)

- To calculate the kinetic energy

kinetic energy$(K.E) = \frac{1}{2}mv^2$----------(3)
where m = mass of the body
v = velocity/speed of the body
putting the values in equation 3

kinetic energy$(K.E) = \frac{1}{2}$ x 3 x 20
= 30 J

Hence, kinetic energy of body = 30 J

17. Correct option: (A)

The energy possessed by a body by the virtue of its motion is called kinetic energy.

$\text{kinetic energy}(K.E) = \frac{1}{2}mv^2$----------(1)

where m = mass of the body

v = velocity/speed of the body

given: mass (m) = 15kg

speed(v)= 50 m/s

putting the values in equation 1

$\text{kinetic energy}(K.E) = \frac{1}{2} \text{x } 15 \text{ x } (50)^2$

$= \frac{1}{2} \text{x } 15 \text{ x } 2500 \text{ J}$

$= 18750 \text{ J}$

Hence, kinetic energy of bicycle = 18750 J

18. Correct option: (B)

The energy possessed by a body by the virtue of its motion is called kinetic energy.

$\text{kinetic energy}(K.E) = \frac{1}{2}mv^2$----------(1)

where m = mass of the body

v = velocity/speed of the body

given: mass (m) = 10kg

velocity(v)= 2 m/s

putting the values in equation 1

$\text{kinetic energy}(K.E) = \frac{1}{2} \text{x } 10 \text{ x } (2)^2$

$= \frac{1}{2} \text{x } 10 \text{ x } 4 \text{ J}$

$= 20 \text{ J}$

Hence, kinetic energy of body = 20 J

19. Correct Matching is: a-R; b-P; c-R; d-Q

In the given curve

At point A and C the curve meets the x axis which means force has zero value at these points.

At point B, the curve is above the x axis in the first quadrant which means force has positive value.

At point E, the curve is below the x axis in the fourth quadrant which means force has a negative value.

POWER
LEVEL-I

1. Correct option: (D)

Power is defined as the rate of doing work.

$\text{power} = \frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

We know that when work is done, an equal amount of energy is consumed. Thus power is also defined as the rate at which energy is consumed.

$\text{power} = \frac{\text{energy consumed}}{\text{time taken}}$

$P = \frac{E}{t}$

SI unit of power is watt (W).

Given: power(P) = 2000W
Energy consumed(E)= 240000J
To find :time taken

$$\text{power} = \frac{\text{energy consumed}}{\text{time taken}}$$

$$P = \frac{E}{t}$$

$$t = \frac{E}{P}$$

putting the given values

$$t = \frac{240000}{2000}\text{seconds}$$

t= 120 seconds
1 minute = 60 seconds
2 minutes = 120 seconds
Hence time taken (t) = 2 minutes

2. Correct option: (D)

Given: work done(W)= 30 J
Time taken(t) = 5 sec
To find: power
Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

$$P = \frac{30}{5}W$$

P = 6 W
Hence, power = 6W

3. Correct option: (C)

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

SI unit of work is joule
SI unit of time is seconds
SI unit of power is Joules/second
SI unit power is called Watt (W).

4. Correct option: (B)

Given: power(P) = 100W
Energy consumed(E)= 1000J
To find: time taken

$$\text{power} = \frac{\text{energy consumed}}{\text{time taken}}$$

$$P = \frac{E}{t}$$

$$t = \frac{E}{P}$$

putting the given values

$$t = \frac{1000}{100}\text{seconds}$$

t= 10 seconds
Hence, time taken (t) = 10 seconds

5. Correct option: (A), (C)

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

SI unit of work is joule

SI unit of time is seconds

SI unit of power is Joules/second

SI unit power is called Watt (W).

6. Correct option: (A)

Given: force applied= 600N

Distance moved in the direction of force = 20m

Time taken(t) = 8minutes = 8 x 60 seconds = 480 seconds

Work = force x displacement in the direction of the force

W = F x S

W= 600 x 20 J

W= 12000 J

Hence, work done(W) = 12000 J

To find: power

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

$$P = \frac{12000}{480}W$$

P = 25W

Hence, power = 25W

7. Correct option: (C)

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

Work in relevance to time is related to power.

8. Correct option: (B)

Given: Energy consumed(E)= 90000J

Time taken(t) = 30 min = 30 x 60 sec = 1800 sec

To find: power

$$\text{power} = \frac{\text{energy consumed}}{\text{time taken}}$$

$$P = \frac{E}{t}$$

$$P = \frac{90000}{1800}W$$

P = 50W

Hence, power = 50 watt

9. Correct option: (A)

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$P = \frac{W}{t}$

SI unit of work is joule (J)

SI unit of time is seconds (s)

SI unit of power is Joules/second (J/s)

SI unit power is called Watt (W).

Hence 1 watt = 1 J/s

10. Correct option: (D)

Commercial unit(trade unit) of energy is kilowatt-hour (kWh)

1 kWh is the amount of energy consumed at the rate of 1kilowatt for 1hour.

1kWh = 1000 W x 3600 s (1Kw= 1000 W , 1h = 3600s)

From equation (1)

$1W = \frac{1J}{1s}$

$1KWh = 1000 \times \frac{1J}{1s} \times 3600\ s$

$1KWh = 36 \times 10^5 J$

11. Correct option: (C)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $9.8 m/s^2$or 10 m/s^2

h= height through which the body is lifted

given: mass of man(m) = 60 kg

height (h) = 10m

g = 10 m/s^2

putting the values in equation 1

W= 60 x 10 x 10

W = 6000J

Hence, work done = 6000 J

Time taken(t) = 30sec

$\text{power} = \frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

$P = \frac{6000}{30} W$

P = 200W

Hence, power = 200W

12. Correct option: (375W**)**

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10 m/s^2$

h= height through which the body is lifted

given:

mass (m) of the boy = 50kg

height of 1 step= 15cm = 0.15 m (1cm = 0.01 m)

total stairs = 45

total height(h) = 45 x 0.15 m
= 6.75m
Putting the values in equation 1
W = 50x 10 x 6.75 J
W = 3375 J
Work done = 3375 J
Time taken(t) = 9sec

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

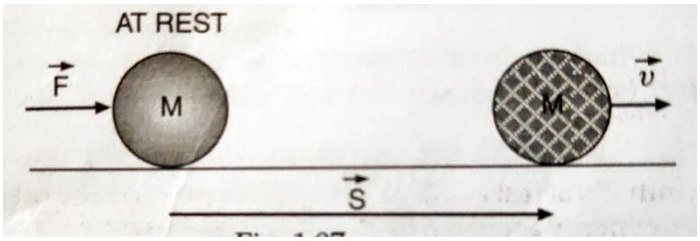
$$P = \frac{W}{t}$$

$$P = \frac{3375}{9} W$$

P = 375W
Hence, power = 375W

13. Correct option: (A)

Consider a body of mass(m) initially at rest. Suppose a constant force F is applied on the body, till its velocity becomes v along the direction of the force. Let S be the displacement of the body during this time.



Initial kinetic energy of the body = $\frac{1}{2}mu^2 = 0$ (initial velocity(u) = 0m/s , body starts from rest)
Finial kinetic energy of the body = $\frac{1}{2}mv^2$
Change in kinetic energy = $\frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$
We know W= F cosθ x S
Where, W= work done
F = force applied
S = distance / displacement
θ =angle between the direction of the applied force and the direction of motion of the body.
θ = 0˚
W = F cos0˚ x S (cos 0˚ = 1)
W= F S----------(1)
If a is the acceleration produced in the body
Then according to newton's second law
F= ma----------(2)
Putting in equation 1
W = ma S ---------(3)
Using the Third equation of motion
$v^2 - u^2 = 2aS \quad ------(4)$ Here, v = final velocity of the object
u= initial velocity of the object
a = acceleration of the object
S= displacement/distance travelled by the object
putting the values in equation(4)

$v^2 - u^2 = 2aS$ (u = 0m/s body starts from rest)

$v^2 - 0 = 2aS$

$\frac{v^2}{2S} = a$

Putting value of a in equation 3

$W = m\frac{v^2}{2S} S$

Or

$W=\frac{mv^2}{2}$

Or

W = change in kinetic energy

Work done = change in kinetic energy = $\frac{1}{2}mv^2$

Power(P) = $\frac{\text{work done}}{\text{time taken}}$

Power(P) = $\frac{\frac{1}{2}mv^2}{t}$

14. Correct option: (A)

Given: power of the pump(P) = 0.39Kw = 0.39 x 1000 W= 390 W

Height(h) = 10m

Time taken = 60s

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

$\text{power} = \frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get

$P = \frac{m \times g \times h}{t}$

$390 = \frac{m \times 10 \times 10}{60}$

$\frac{390 \times 60}{100}$ kg = m

234 kg = m

Hence, amount of water pumped = 234 kg

15. Correct option: (D)

Given: mass of water(m) = 2 kg

depth(h) = 10m

Time taken =1s

To find: power(P)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

Putting the given values and using equation 1

We get

$$P = \frac{m \times g \times h}{t}$$

$$P = \frac{2 \times 10 \times 10}{1} W$$

P = 200W

Hence, power delivered by the pump= 200 W

16. Correct option: (C)

Given: Energy consumed(E)= 4KJ = 4000J (1KJ = 1000J)

Time taken(t) = 2.5sec

To find: power

$$\text{power} = \frac{\text{energy consumed}}{\text{time taken}}$$

$$P = \frac{E}{t}$$

$$P = \frac{4000}{2.5} W$$

P=1600W

Hence, power delivered = 1600W

17. Correct option: (C)

Given: Mass(m) = 40kg

Time taken(t) =1 second

Velocity(v)= $3ms^{-1}$

Work done = change in kinetic energy = $\frac{1}{2}mv^2$

$$\text{Power(P)} = \frac{\text{work done}}{\text{time taken}}$$

$$\text{Power(P)} = \frac{\frac{1}{2}mv^2}{t}$$

$$\text{Power(P)} = \frac{\frac{1}{2} \times 40 \times (3)^2}{1}$$

Power(P)= $\frac{1}{2}$ x 40 x 9 W

Power(P)= 180 W

18. Correct option: (A)

Given: power of the pump(P) = 12Kw = 12x 1000 W= 12000 W

Height(h) = 10m

Time taken (t)= 1 min = 60s

To find: mass of water (m)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get

$P = \frac{m \times g \times h}{t}$

$12000 = \frac{m \times 10 \times 10}{60}$

$\frac{12000 \times 60}{100}$ kg = m

7200 kg = m

Hence, amount of water pumped = 7200 kg

19. Correct option: (A)

Given: power of the pump(P) = 2Kw = 2x 1000 W= 2000 W

Height(h) = 10m

Time taken (t)= 1 min = 60s

To find: mass of water (m)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

$power = \frac{work\ done}{time\ taken}$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get

$P = \frac{m \times g \times h}{t}$

$2000 = \frac{m \times 10 \times 10}{60}$

$\frac{2000 \times 60}{100}$ kg = m

1200 kg = m

Hence, amount of water pumped = 1200 kg

20. Correct option: (D)

Given: power(P) = 100W

Time taken(t) = 1min = 60 sec

To find: Energy consumed(E)

$power = \frac{energy\ consumed}{time\ taken}$

$P = \frac{E}{t}$

P x t =E

100 x 60 J = E

6000 J = E

Hence, energy transferred = 6000 J

21. Correct option: (D)

the gravitational force of attraction of the earth on a body is called its weight(W)

Weight= mx g
given: weight(mg) of the man =500N
weight(mg)of the load = 100N
total weight(mg)= 500N + 100N = 600N
total height(h) = 4m
time taken = 5 seconds
Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = $10m/s^2$
h= height through which the body is lifted
Putting the values in equation 1
W = 600 x 4 J
W = 2400 J
Work done = 2400 J
To find: power (P)

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

$$P = \frac{2400}{5}W$$

P=480 W
Hence, power = 480W

22. Correct option: (B)

Given: power(P) = 3 KW= 3 x 1000 W= 3000 W
Time taken(t) =20 sec
To find: work done(W)

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

P x t =W
3000 x 20 J = W
60000 J = W
60KJ = W (1KJ =1000J)
Hence, Work done = 60 KJ

23. Correct option: (A)

Given: power(P) of 1 fan = 120W= 0.120 KW (1W=0.001 KW)
Time taken(t) = 4 hours
To find: Energy consumed(E)

$$\text{power} = \frac{\text{energy consumed}}{\text{time taken}}$$

$$P = \frac{E}{t}$$

P x t =E
0.120 x 4 KWh= E
0.48 KWh= E
Hence, energy consumed by 1 fan = 0.48 KWh
Energy consumed by 5 fans = 0.48 x 5 = 2.4 KWh or kilowatt hours

24. Correct option: (A)

P= F v----------(1)

Where P= Power

F = force

v = velocity/ speed

if the velocity is constant, all the power of the agent is being used to overcome the resistive forces

given: speed of the motor boat(v) = 20m/s

resistive force(F) = 600N

putting the values in equation 1

P = 600 x 20 W

P = 12000 W

P=12KW (1KW=1000W)

Hence, power = 12KW

25. Correct option: (B)

Given: depth(h) = 200m

Mass (m) = 90 metric tonnes= 90000kg

1 metric tonnes= 1000 kg

90 metric tonnes = 90000 kg

Time taken(t) = 1hour = 3600 sec

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

Putting the values in equation 1

W = 90000 x 10 x 200 J

W = 180000000 J

Work done = 180000000 J

To find: power (P)

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

$$P = \frac{180000000}{3600} W$$

P=50,000 W

P= 50KW (1KW =1000W)

Hence, power = 50KW

26. Correct option: (A)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get $P = \frac{m \times g \times h}{t}$

- To calculate power supplied by crane P to lift a car

Given: time taken: 1min

$P_P = \frac{m \times g \times h}{1}$

$P_P = m \times g \times h$----------(2)

- To calculate power supplied by crane Q to lift a car

Given: time taken: 2min

$P_Q = \frac{m \times g \times h}{2}$---------(3)

Putting value of m x g x h from equation 2 in 3

$P_Q = \frac{P_P}{2}$

Or $2 P_Q = P_P$

Hence, power supplied by crane P is two times more than crane Q.

27. Correct option: (A)

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10 m/s^2$

h= height through which the body is lifted

$power = \frac{work\ done}{time\ taken}$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get $P = \frac{m \times g \times h}{t}$

- To calculate power of person A

Given: height(h) = 8m

time taken: 15 sec

$P_A = \frac{m \times 10 \times 8}{15}$

$P_A = 5.3m$ ----------(2)

- To calculate power of person B

Given: height(h) = 8m

time taken: 20sec

$P_B = \frac{m \times 10 \times 8}{20}$

$P_B = 4m$---------(3)

Hence, $P_A > P_B$

28. Correct option: (D)

P= F v----------(1)

Where P= Power

F = force

v = velocity/ speed

if the velocity is constant, all the power of the agent is being used to overcome the resistive forces
given: velocity(v) = 20m/s
resistive force(F) = 200N
putting the values in equation 1
P = 200 x 20 W
P =4000 W
Hence, power = 4000W

29. Correct option: (D)

P= F v----------(1)
Where P= Power
F = force
v = velocity/ speed
if the velocity is constant, all the power of the agent is being used to overcome the resistive forces
given: velocity(v) = 20m/s
power(P) =7500W
to find: retarding force(F)
putting the values in equation 1
7500 = F x 20
$\frac{7500}{20}$N = F
375N =F
Hence, retarding force = 375 N

30. Correct option: (D)

Given: mass of the load (M)= 50kg
Height(h)=40m
Time taken(t)= 25s
Power of the man(P)= 1568W
To find: mass of the man(m)
Work done in lifting a body = weight of body x vertical distance
W = m x g x h----------(1)
Where W = work done
m = mass of the body
g = acceleration due to gravity = $10m/s^2$
h= height through which the body is lifted

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

Putting the given values and using equation 1
We get $P = \frac{(M+m)\ x\ g\ x\ h}{t}$

$$1568 = \frac{(50 + m)\ x\ 10\ x\ 40}{25}$$

$\frac{1568\ x25}{10\ x\ 40}$= 50 +m
98 = 50 +m
98 – 50 = m
48 kg = m
Hence, mass of the man = 48kg

LEVEL-II

1. Correct option: (B)

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

Scalar quantities are those which have magnitude(size) only and no direction.

work is a scalar quantity as it has magnitude only.

Time is also a scalar quantity.

As, power is the ratio of two scalars so power is also a scalar quantity.

2. Correct option: (B)

In engineering, horse power (h.p.) is practical unit of power.

1h.p. = 746 W

3. Correct option: (B)

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

4. Correct option: (B)

Power is defined as the rate of doing work.

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

When 1 J work is done at the rate of 1 sec

Then

Power is said to be 1watt or 1W.

5. Correct option: (B)

Initial kinetic energy of the bullet= $\frac{1}{2}mu^2 = 0$ (initial velocity(u) = 0m/s , bullet starts from rest)

Finial kinetic energy of the bullet =$\frac{1}{2}mv^2$

Change in kinetic energy = $\frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$

Work done = change in kinetic energy =$\frac{1}{2}mv^2$

Work done by gun for firing n bullets = n $\frac{1}{2}mv^2$

Time taken = 1 sec

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$\text{power} = \frac{n\frac{1}{2}mv^2}{1} = n\ \frac{1}{2}mv^2$$

6. Correct option: (D)

Given: power(P) of 1 device = 500W= 0.500 KW (1W=0.001 KW)

Time taken(t) = 10 hours

To find: Energy consumed(E)

$$\text{power} = \frac{\text{energy consumed}}{\text{time taken}}$$

$$P = \frac{E}{t}$$

P x t =E
0.500 x 10 KWh= E
5 KWh= E
Hence, energy consumed by 1 device = 5KWh
Energy consumed by 4 devices = 5 x 4 KWh = 20 KWh or kilowatt hours

7. Correct option: (D)

Given: velocity(v) $=20ms^{-1}$
Area of cross section(A)= $10cm^2=10^{-3}m^2$ $(1cm^2 = 10^{-4}m^2)$
Work done = change in kinetic energy = $\frac{1}{2}mv^2$
Where m= mass of the water
Power(P) = $\frac{\text{work done}}{\text{time taken}}$
Power(P) = $\frac{\frac{1}{2}mv^2}{t}$----------(1)
We know that
Density = $\frac{\text{mass}}{\text{volume}}$
Or mass(m) = volume(V) x density
Density of water at $4°C(277K)$ **is** $10^3kg/m^3$
Hence,
Mass of water (m)= volume(V) x $10^3kg/m^3$
Also
Volume = area of cross section x distance
V= A x x
Hence, mass(m) = A x x x 10^3 ––––––––––––––––(2)
Using equation 2 in equation 1
We get
Power(P) = $\frac{\frac{1}{2}A \text{ x } x \text{ x } 10^3 \text{ v}^2}{t}$
And
$\frac{distance(x)}{time(t)} = speed(v)$
Power(P) = $\frac{1}{2}$x A x 10^3 v^3
Putting the given values
We get
Power(P) = $\frac{1}{2}$x 10^{-3} x 10^3 $(20)^3$
Power(P) = $\frac{1}{2}$ x 8000 W
Power(P) = 4000 W
Power(P) = 4 KW (1 KW = 1000W)

8. Correct option: (A)

Given: Power(P) = 200W
Mass(m) = 2kg
Depth(h) = 10m
Time taken(t) =1 second
g = acceleration due to gravity = $9.8m/s^2$
Total Work done = total energy = potential energy + kinetic energy= $mgh + \frac{1}{2}mv^2$
Power(P) = $\frac{\text{work done}}{\text{time taken}}$

Power(P) $=\frac{mgh+\frac{1}{2}mv^2}{t}$----------(1)

Putting the values in equation 1

$200 =\frac{2x\ 9.8\ x\ 10+\frac{1}{2}x2x(v)^2}{1}$

$200= 2x\ 9.8\ x\ 10 + \frac{1}{2}x2x(v)^2$

$200 = 196 + (v)^2$

$200 - 196 =(v)^2$

$4 =(v)^2$

$(2)^2=(v)^2$

$v = 2m/s$

hence, velocity = 2 m/s

9. Correct option: (A), (B)

Given: mass of water = 100 kg

Height(h) = 20m

Time taken(t) =10s

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $9.8m/s^2$

h= height through which the body is lifted

$\text{power} = \frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get

$P = \frac{m\ x\ g\ x\ h}{t}$

$P = \frac{100\ x\ 10\ x\ 20}{10}W$

P= 2000W

P= 2KW (1KW = 1000W)

$P= 2\ x\ 10^3W$

Hence, power of the pump= 2KW or $2\ x\ 10^3W$

10. Correct option: (A), (C)

- $\text{power} = \frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

Power is directly proportional to the work done.

- Scalar quantities are those which have magnitude(size) only and no direction.

work is a scalar quantity as it has magnitude only.

Time is also a scalar quantity.

As, power is the ratio of two scalars so power is also a scalar quantity.

11. Correct option: (A), (B), (C), (D)

- $\text{power} = \frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

Hence, power is inversely proportional to the time taken

- Power = $\frac{\text{work}}{\text{time}}$

Now,

work =force x displacement

Power = $\frac{\text{force x displacement}}{\text{time}}$

Length/displacement is represented as [L]

Dimensional formula of force = $[MLT^{-2}]$

Time is represented as [T]

So,

Power =$\frac{[MLT^{-2}]x[L]}{[T]}$

Power =$[ML^{1+1}T^{-2-1}]$

Power =$[ML^2T^{-3}]$

Dimensional formula of Power =$[ML^2T^{-3}]$

- P= $\overrightarrow{F}.\vec{v}$

Where P= Power

F = force

v = velocity/ speed

if the velocity is constant, all the power of the agent is being used to overcome the resistive forces

- power = $\frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

When time = 1 unit = 1 second

P= W

Hence, work done in a unit time is called power.

12. Correct option: (A), (B), (C), (D)

- power = $\frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

- P= F v

Where P= Power

F = force

v = velocity/ speed

if the velocity is constant, all the power of the agent is being used to overcome the resistive forces

- Power of n objects per second moving with a velocity v = $n\frac{1}{2}mv^2$

Let

Initial kinetic energy of the body = $\frac{1}{2}mu^2$= 0 (initial velocity(u) = 0m/s , body starts from rest)

Finial kinetic energy of the body =$\frac{1}{2}mv^2$

Change in kinetic energy = $\frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$

Work done = change in kinetic energy =$\frac{1}{2}mv^2$

Work done by n objects = $n\frac{1}{2}mv^2$

Time taken = 1 sec

Power(P) = $\frac{\text{work done}}{\text{time taken}}$

Power(P) = $\frac{n\frac{1}{2}mv^2}{1}$= n $\frac{1}{2}mv^2$

- In engineering, horse power (h.p.) is practical unit of power.

1h.p. = 746 W

13. Correct option: (A)

Amount of fuel used by a crane = work done by the car

- To calculate Work done by crane 1

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = 9.8m/s^2

h= height through which the body is lifted

Given: mass of car = 2000 kg

Height(h) = 30m

Putting values in equation 1

W = 2000 x 9.8 x 30 J

W = 588,000 J

W = 5.88 x 10^5 J

Hence, work done by car 1 = 5.88 x 10^5 J

Amount of fuel used =5.88 x 10^5 J

- To calculate Work done by crane 2

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = 9.8m/s^2

h= height through which the body is lifted

Given: mass of car = 2000 kg

Height(h) = 30m

Putting values in equation 1

W = 2000 x 9.8 x 30 J

W = 588,000 J

W = 5.88 x 10^5 J

Hence, work done by car 2 = 5.88 x 10^5 J

Amount of fuel used =5.88 x 10^5 J

14. Correct option: (B)

Given: mass of car = 2000 kg

Height(h) = 30m

Time taken(t) =1 minute = 60s

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = 9.8m/s^2

h= height through which the body is lifted

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get

$P = \frac{m \times g \times h}{t}$

$P = \frac{2000 \times 9.8 \times 30}{60}W$

P= 9800W

Hence, power supplied by crane 1 = 9800W

15. Correct option: (A)

Given: mass of car = 2000 kg

Height(h) = 30m

Time taken(t) =2 minute = 2 X 60s = 120 s

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $9.8m/s^2$

h= height through which the body is lifted

$power = \frac{work\ done}{time\ taken}$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get $P = \frac{m \times g \times h}{t}$

$P = \frac{2000 \times 9.8 \times 30}{120}W$

P= 4900W

Hence, power supplied by crane 2 = 4900W

16. Correct option: (A)

Given: mass of man M_1 = 90 kg

Height(h) = 10m

Time taken = 15 s

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

$power = \frac{work\ done}{time\ taken}$

$P = \frac{W}{t}$

Putting the given values and using equation 1

We get $P = \frac{m \times g \times h}{t}$

$P = \frac{90 \times 10 \times 10}{15}W$

P= 600W

Hence, power of man M_1 = 600W

17. Correct option: (D)

Given: mass of man $M_2 = 90$ kg

Height(h) = 10m

Time taken = 20s

Work done in lifting a body = weight of body x vertical distance

W = m x g x h----------(1)

Where W = work done

m = mass of the body

g = acceleration due to gravity = $10m/s^2$

h= height through which the body is lifted

$$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

Putting the given values and using equation 1

We get

$$P = \frac{m \times g \times h}{t}$$

$$P = \frac{90 \times 10 \times 10}{20} W$$

P= 450W

Hence, power of man $M_2 = 450W$

18. Correct option: (B)

power of man $M_1 = 600W$

power of man $M_2 = 450W$

$$\text{Ratio} = \frac{\text{power of man } M_1}{\text{power of man } M_2}$$

$$\text{Ratio} = \frac{600\ W}{450\ W}$$

$$\text{Ratio} = \frac{4}{3}$$

19. Correct Matching is: a-S; b-R; c-Q; d-P

(a) In engineering, horse power (h.p.) is practical unit of power.

1h.p. = 746 W

(b) $1W = \frac{1J}{1s}$--------(1)

1kWh = 1000 W x 3600 s (1Kw= 1000 W , 1h = 3600s)

From equation 1 $1W = \frac{1J}{1s}$

$1KWh = 1000 \times \frac{1J}{1s} \times 3600\ s$

$1KWh = 36 \times 10^5 J$

(c) Electron volt (eV) is the unit of energy.

One electron volt is the energy gained by an electron when accelerated through a potential difference of 1 volt.

$1\ e.V = 1.6 \times 10^{-19}$ J

(d) $\text{power} = \frac{\text{work done}}{\text{time taken}}$

$$P = \frac{W}{t}$$

SI unit of work is joule (J)

SI unit of time is seconds (s)

SI unit of power is Joules/second (J/s)

SI unit power is called Watt (W).

Hence 1 W = 1 J/s
1 joule = 10^7ergs
So,
1W = 10^7ergs/s

20. Correct Matching is: a-S; b-R; c-Q; d-P

(a)power = $\frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

Scalar quantities are those which have magnitude(size) only and no direction.
work is a scalar quantity as it has magnitude only.
Time is also a scalar quantity.
As, power is the ratio of two scalars so power is also a scalar quantity.

(b)Power = $\frac{\text{work}}{\text{time}}$

Now,
work =force x displacement

Power = $\frac{\text{force x displacement}}{\text{time}}$

Length/displacement is represented as [L]
Dimensional formula of force = $[MLT^{-2}]$
Time is represented as [T]
So,

Power =$\frac{[MLT^{-2}]x[L]}{[T]}$

Power =$[ML^{1+1}T^{-2-1}]$
Power =$[ML^2T^{-3}]$
Dimensional formula of Power =$[ML^2T^{-3}]$
(c) In engineering, horse power (h.p.) is practical unit of power.
1h.p. = 746 W
(d) Power is defined as the rate of doing work.

power = $\frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

C.G.S unit of work is ergs
C.G.S unit of time is seconds
C.G.S unit of power is erg/second

5. Pressure

Solutions

LEVEL-I

Thrust and Pressure & Atmospheric Pressure:

1. Correct option: (B) Thrust.

Thrust is the term for the force applied to an item perpendicular to the surface.
Newton is the SI unit for thrust.

2. Correct option: (A) Newton.

The International System of Units (SI) uses the newton (symbol: N) as its unit of force.

3. Correct option: (B) Dyne.

The C.G.S. unit is dyne, while the S.I. unit of thrust is newton (N).

4. Correct option: (B) Vector.

The force perpendicular to a body's surface is known as a thrust. Since thrust is described in terms of both magnitude and direction (since it only applies to bodies with surfaces perpendicular to them), it is regarded as a vector quantity.

5. Correct option: (A) Pressure.

Pressure is defined as force per unit of area. The force applied perpendicularly to an object's surface per unit area across which that force is dispersed is referred to as pressure.
When compared to atmospheric pressure, gauge pressure is the pressure.

6. Correct option: (A) Pascal.

One newton per square meter is equivalent to one pascal (abbreviated as Pa), the SI unit of pressure. $\left(\frac{N}{m^2}\right)$.

7. Correct option: (C) Hydrostatic pressure.

The pressure that a fluid is exerting at any one time due to gravity is known as hydrostatic pressure. As the weight of the fluid increases when a downward force is applied, hydrostatic pressure is proportional to the depth measured from the surface. Fluids apply pressure uniformly in all directions.

8. Correct option: (D) Mercury.

The substance used in a barometer is mercury. Below is a list of some of the qualities that make it a good option:

- a lot of density Mercury with a shorter length (76 cm) can balance the atmosphere's pressure.
- It reacts to pressure changes more quickly.
- Its freezing point is low.
- It is difficult to evaporate.
- Because it is bright, it is simple to read the mercury.

9. Correct option: (B) 1:10.

The SI measure of pressure is $\frac{N}{m^2}$, while the CGS unit is $\frac{dyne}{cm^2}$.

$1N=10^5 dyne$

$1m=10^0 cm$

So, the ratio of CGS unit to SI unit is given by as,

$$\frac{1\frac{\text{dyne}}{\text{cm}^2}}{\frac{\text{N}}{\text{m}^2}} = \frac{1\frac{\text{dyne}}{\text{cm}^2}}{\frac{10^5\ \text{dyne}}{(100)^2\text{cm}^2}} = \frac{1}{10}$$

So, the ratio of CGS unit to SI unit is given is 1:10.

10. Correct option: (A) $1.01\times10^5\text{Nm}^{-2}$ **.**

The Standard atmospheric pressure is$1.01\times10^5\text{Nm}^{-2}$.

11. Correct option: (C) 10m^2 **.**

Given:

F =40N

$$P = 4\frac{\text{N}}{\text{m}^2}$$

The pressure is given by as,

$$P = \frac{F}{A}$$

$$A = \frac{F}{P}$$

$$A = \frac{40}{4}$$

$$A = 10\text{m}^2$$

So, the area on which the force of water is acting is 10m^2 .

12. Correct option: (B) 2 atm.

Given:

atmospheric pressure = 1 atm

Pressure at the bottom = pressure due to water + pressure due to the atmosphere.

3atm = pressure due to water column +1atm

pressure due to water column =(3-1) atm

= 2atm.

So, calculate the pressure due to the liquid column in the tank is 2atm.

13. Correct option: A) 3200 Pa.

Given:

$F = 16\text{N}$

$$A = 50\text{cm}^2 = \frac{50}{10000}\text{m}^2$$

Now pressure is given by as,

$$a \quad P = \frac{F}{A}$$

$$P = \frac{16\times10000}{50} = 3200\text{Pa}$$

So, the pressure in Pascal is 3200 Pa.

Pressure Due to Liquid Column:

14. Correct option: (B) hdg **.**

Only the liquid's density and vertical depth have any bearing on the pressure that it exerts at a given spot. The container's shape has no bearing on it. The same liquid is contained in three jars of varying forms, and the pressure at the bottom of each vessel operates equally in all directions.

Thus, $P = hdg$, where h is the height, d is the density, and g is the acceleration due to gravity, is the formula for the pressure that a liquid of height h exerts.

15. Correct option: (C) 6400 Pa.

The density of alcohol column, $d = 0.8\text{gcm}^{-3} = 0.8 \times 1000 = 800\text{kgm}^{-3}$.

Size of the alcohol column's vertical column, $h = 0.8\text{m}$

Now pressure is given by,

$P = hdg$

$P = 0.8 \times 800 \times 10$

$P = 6400\text{Pa}$

So, the pressure is 6400 Pa.

16. Correct option: (A) 34 m.

The pressure exerted by water, $P = 333200\,\text{Pa}$

Density of water, $\rho = 1000\,\frac{\text{kg}}{\text{m}^3}$

Acceleration due to gravity, $g = 9.8\,\frac{\text{m}}{\text{s}^2}$.

Now, pressure is given by,

$P = h\rho g$

$333200 = h \times 1000 \times 9.8$

$h = \frac{333200}{1000 \times 9.8}$

$h = 34\text{m}$

Therefore, 34 metres of water column exert 333200 Pa of pressure.

17. Correct option: (D) 3920 Pa.

Height of the water column , $h = 40\text{cm} = \frac{40}{100}\,\text{m}$

Density of water, $\rho = 1000\,\frac{\text{kg}}{\text{m}^3}$

Acceleration due to gravity, $g = 9.8\,\frac{\text{m}}{\text{s}^2}$

Now, pressure is given by,

$P = h\rho g$

$P = \frac{40}{100} \times 1000 \times 9.8$

$P = 3920\,\text{Pa}$

So, water exerting hydrostatic pressure in a beaker's bottom is 3920 Pa.

18. Correct option: (C) 10.3 m.

There is just a change in the density and height of the fluid, therefore the pressure will be the same whether mercury or water is employed.

Density of mercury is $13.6\,\frac{\text{g}}{\text{cm}^2}$.

For, mercury, the height is 76 cm.

Pressure is given by,

$P = h\rho g$

$P = 76 \times 13.6 \times g$

Let's consider the height of the water column is h m .

Now pressure is given by,

$P = h\rho g$

$h \times 1 = 76 \times 13.6$

$h = 1033.6$

$h = 10.3\text{m}$

So, the height of the water column is 10.3 m.

19. Correct option: (A) $32.6 \times 10^3 \text{Pa}$ **.**

Given:

Area under the foot is, $A = 180\text{cm}^2 = \dfrac{180}{10000}\text{m}^2$

Force is given by,

$F = 60\text{kgwt} = 60 \times 9.8\text{N}$

$F = 60 \times \dfrac{98}{10} = 6 \times 98\text{N}$

The pressure is given by,

$P = \dfrac{F}{A}$

$P = \dfrac{6 \times 98 \times 10000}{180} = \dfrac{98}{3} \times 10^3 \text{Pa}$

$P = 32.6 \times 10^3 \text{Pa}$

So, the pressure exerted on the ground by the person is $32.6 \times 10^3 \text{Pa}$.

20. Correct option: (B) $2 \times 10^5 \text{Pa}$ **.**

$1\text{atm} = 10^5 \text{Pa}$

The required water column height to equal one atmosphere is: $\rho gh = 10^5$

$1000 \times 10 \times h = 10^5$

$h = 10\text{m}$

1 atm of pressure is equal to 10 m of water column.

2 atm of pressure is produced by 20 m of water.

As we know that, $1\text{atm} = 10^5 \text{Pa}$.

Therefore 2 atm in Pascal is,

$2\text{atm} = 2 \times 10^5 \text{Pa}$

So, the pressure on his body is nearly $2 \times 10^5 \text{Pa}$.

21. Correct option: (A) Barometer.

The barometer is the name of the device used to gauge atmospheric pressure. A manometer is used to gauge gas pressure. Temperature is measured with a thermometer.

22. Correct option: (A) $1.0129 \times 10^5 \text{Pa}$ **.**

Density of mercury is $13.6 \dfrac{\text{g}}{\text{cm}^2}$.

Acceleration due to gravity, $\text{g} = 980 \dfrac{\text{cm}}{\text{s}^2}$

For, mercury, the height is 76 cm.
Pressure is given by,

$P = h\rho g$

$P = 76 \times 13.6 \times 980$

$P = 1.0129 \times 10^5 \text{Pa}$

So, the pressure is $1.0129 \times 10^5 \text{Pa}$.

23. Correct option: (D) $1.225 \times 10^5 \frac{\text{dyne}}{\text{cm}^2}$

The water column's bottom has a pressure of,

$P = P_0 + \rho g h$

$P = 10^6 + 1 \times 980 \times 12.5$

$P = 1.225 \times 10^5 \frac{\text{dyne}}{\text{cm}^2}$

So, the pressure is $1.225 \times 10^5 \frac{\text{dyne}}{\text{cm}^2}$.

Pascal's law:

24. Correct option: (A) Pressure.

According to Pascal's law, pressure applied anywhere within a confined liquid propagates equally and undiminished in all directions.

25. Correct option: (D) Pascal's law.

The Pascal's law theory underlies the operation of every hydraulic device. Every hydraulic device works on the basis that a tiny force given to the smaller piston will cause a larger force to be applied to the larger piston.

26. Correct option: (A) Hydraulic press.

Hydraulic lifts are those that employ Pascal's Law to raise items with the aid of fluid. It utilises the law of Pascal.

27. Correct option: (A) $\frac{F_1}{A_1} = \frac{F_2}{A_2}$**.**

The pressure is given by, $P = \frac{F}{A}$

So, the correct mathematical expression for Pascal's law is given by as,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

28. Correct option: (C)20 N.

Given:

$P_1 = 10\text{N}$

$A_1 = 5\text{m}^2$

$A_2 = 10\text{m}^2$

Now, we know that,

$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

$\frac{10\text{N}}{5\text{m}^2} = \frac{F_2}{10\text{m}^2}$

$F_2 = 20\text{N}$

So, the upward thrust exerted by the piston is 20 N.

29. Correct option: (D) 10.

Given:

$F_1 = 1250\text{N}$

$A_2 = 625\text{cm}^2$

$A_2 = 5\text{cm}^2$

Now, we know that pressure is constant,

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\frac{1250}{625} = \frac{F_2}{5}$$

$F_2 = 10\,\text{N}$

So, the force 10 N on the smaller piston will support a load of 1250 N on the larger piston.

30. Correct option: (A) 7200 kg wt.

Pressure in a hydraulic press is transformed equally: $P_1 = P_2$

Force per unit area is used to measure pressure.

Therefore,

$$\frac{50 \times g}{A_1} = \frac{m \times g}{A_2}$$

$$\frac{50 \times 10}{(1.25)^2 x \times \pi} = \frac{m \times g}{(15)^2 x \times \pi}$$

$m = 7200\text{kg}$

The larger piston will exert a force of 7200 kg wt.

LEVEL-II

1. Correct option: (C) $10^7\,\text{Pa}$**.**

The pressure is given by,

$$P = \frac{F}{A}$$

$$P = \frac{500}{0.5 \times 10^{-4}}$$

$P = 10^7\,\text{Pa}$

So, the pressure acting on the area is $10^7\,\text{Pa}$.

2. Correct option: (D) Both B & C

Fluids are substances that have the ability to flow and don't have a set shape. Unless contained in a container, gases and liquids do not have a fixed shape and continue to flow. This implies that fluids can refer to both gases and liquids. Solids aren't fluids due of their inherent rigidity.

3. Correct option: (C) 2.5 m.

Given: Density, $\rho = 4\text{kgm}^{-3}$

Pressure, $P = 100\text{Pa}$

Now pressure is given by as,

$P = h\rho g$

$$h = \frac{P}{\rho g}$$

$$h = \frac{100}{4 \times 10} = 2.5\text{m}$$

So, the height of liquid column above that point is 2.5 m.

4. Correct option: (A) hdg **.**

Only the liquid's density and vertical depth have any bearing on the pressure that it exerts at a given spot. The container's shape has no bearing on it. The same liquid is contained in three jars of varying forms, and the pressure at the bottom of each vessel operates equally in all directions.

Thus, $P = hdg$, where h is the height, d is the density, and g is the acceleration due to gravity, is the formula for the pressure that a liquid of height h exerts.

5. Correct option: (B) 5 m.

Given: Density, $\rho = 12\text{kgm}^{-3}$

Pressure, $P = 600\text{Pa}$

Now pressure is given by as,

$P = h\rho g$

$h = \dfrac{P}{\rho g}$

$h = \dfrac{600}{12 \times 10} = 5\text{m}$

So, the height of liquid column above that point is 5 m.

6. Correct options: (A) and (B).

An aircraft moves through the air with the use of thrust. The force per area is known as pressure. The force of gravitational attraction is weight.

Thus, weight and thrust both symbolise force. Therefore, they will have same unit. .

7. Correct options: (A),(B), (C) and (D).

(A) In petrol stations, lifting cars is done using the hydraulic press technique. Based on Pascal's concept, a car hydraulic lift is an illustration of a force doubled by hydraulic press. The little cylinder's fluid must be pushed far farther than the height of the car.

(B) Oil is extracted from oil seeds using a hydraulic press, among other uses.

(C) Straw and cotton bales are compressed using hydraulic presses. Metals can also be punched with holes with it.

(D) A hydraulic press is a mechanical tool that shapes, deforms, and arranges various kinds of metals, plastics, rubber, and wood using the static pressure of a liquid, as specified by Pascal's principle. The mainframe, power system, and controls make up a hydraulic press's mechanism.

8. Correct options: (A) and (B).

The pressure is given by,

$P = \dfrac{F}{A}$

$P = \dfrac{5\text{N}}{5\text{m}^2}$

$P = 1\dfrac{\text{N}}{\text{m}^2}$

$\because 1\text{Pa} = 1\dfrac{\text{N}}{\text{m}^2}$

$P = 1\text{Pa}$

So, Correct options: (A) and (B).

9. Correct options: (B) and (C).

The pascal (Pa), which is equal to one Newton per square metre of force, is the unit of pressure in the SI system.

$1\text{Pa} = 1\dfrac{\text{N}}{\text{m}^2}$

10. Correct options: (B) and (C).

The pressure is given by,

$$P = \frac{F}{A}$$

$$P = \frac{400\text{N}}{0.04\text{m}^2}$$

$$P = 10^4 \frac{\text{N}}{\text{m}^2}$$

$$\because 1\text{Pa} = 1\frac{\text{N}}{\text{m}^2}$$

$$P = 10^4\,\text{Pa}$$

So, Correct options: (B) and (C).

11. Correct option: (C) 550N.

Given:

$$P = 27500\text{Pa}$$

$$A = 200\text{cm}^2 = 2\times10^{-2}\,\text{m}^2$$

The pressure is given by,

$$P = \frac{F}{A}$$

$$F = P \times A$$

$$F = 27500 \times 2 \times 10^{-2}$$

$$F = 550\text{N}$$

So, the magnitude of force is 550N.

12. Correct option: (D) 2000cm^2 **.**

Given:

$$P = 1500\text{Pa}$$

$$F = 330\text{N}$$

The pressure is given by,

$$P = \frac{F}{A}$$

$$A = \frac{F}{P}$$

$$A = \frac{300}{1500}$$

$$A = 0.2\text{m}^2 = 2000\text{cm}^2$$

So, the magnitude of A is 2000cm^2 .

13. Correct option: (A) 10,000 N.

$$P = 50{,}000\text{Pa}$$

$$A = 0.20\text{m}^2$$

The pressure is given by,

$$P = \frac{F}{A}$$

$$F = P \times A$$

$$F = 50{,}000 \times 0.2$$

$$F = 10000\text{N}$$

So, So, the magnitude of force is 10000 N.

14. Correct option: (B) 9 m.

Difference in pressure of water at ground floor and third floor= =120000-30000=90000Pa

We know that density of water is, d=1000kgm^{-3}

The pressure is given by,

$P = hdg$

$h = \frac{P}{dg}$

$h = \frac{90000}{1000 \times 10}$

$h = 9\text{m}$

So, the height of third floor is 9 m.

15. Correct option: (C)12 m.

We know that density of water is, ρ=1000kgm^{-3}

We know that, pressure is, $P = P_0 + \rho gh$

Now,

$P_0 + \rho gh_1 = 150000 \cdots\cdots (1)$

$P_0 + \rho gh_2 = 30,000 \cdots\cdots (2)$

Now from the equation (1) and (2), we get,

$\rho g(h_1 - h_2) = 120000$

$h_1 - h_2 = \frac{120000}{10 \times 1000}$

$h_1 - h_2 = 12\text{m}$

So, height of fourth floor is 12 m.

16. Correct option: (B) 200 Pa.

We know that density of liquid is, d=20 kgm^{-3}

The pressure is given by,

$P = hdg$

$P = 1 \times 20 \times 10$

$P = 200\text{Pa}$

So, the pressure exerted by the liquid on the bottom surface of vessel is 200 Pa.

17. Correct option: B) Both statements are correct.

Statement I: 1 Pascal is equal to 1Nm^{-2} :

This statement is true. The SI unit of pressure is the Pascal (Pa). Its unit of measurement is Newtons per square metre (Nm^{-2}), or one Newton of force per square metre. The fundamental idea of pressure as force dispersed over an area is reflected in this connection.

Statement II: Pressure is directly proportional to force applied:

This statement is also true. Physics dictates that pressure is, in fact, inversely proportional to applied force. If the area stays constant, the pressure increases as the force exerted on a surface increase. The equation $\text{Pressure} = \frac{\text{Force}}{\text{Area}}$ captures this relationship. The pressure exerted on the same area rises proportionately as the force does.

18. Correct option: (B) Both statements are correct.

Statement-I: Liquid pressure at a point in a liquid does not depends on shape of the vessel in which liquid is kept.

This statement is true. Pascal's principle states that a confined fluid's pressure at a specific depth is distributed equally in all directions. This means that the depth of the point and the liquid's density are the only factors affecting the pressure at a given location within a liquid, independent of the shape of the container.
The formula for expressing the pressure at a particular place in a fluid is,
$P = hdg$
Where, h is depth of the point from the surface, d is the density of the fluid and g is acceleration due to gravity.
As a result, the shape of the vessel in which the liquid is stored has no bearing on the pressure in the liquid at a given position.
Statement-II: Liquid pressure at point in a liquid on the density of liquid
The density of the liquid directly affects the pressure in a liquid at a given depth. This is because of the fundamental hydrostatic principle, according to which the weight of the fluid column above a given depth determines the pressure there. This assertion is true since pressure is a function of the fluid's weight, which is influenced by its density.
19. Correct Matching is (a)-(P),(b)-(R),(c)-(Q),(d)-(S)
(a) The air pressure acting on the liquid's free surface must be included when determining the total pressure in a liquid at depth. Total pressure within a liquid at depth $'h'$ is equal to the sum of atmospheric pressure and liquid column pressure.
Total pressure inside a fluid at a depth $'h'$ is $P_0 + \rho gh$.
(b) The formula for expressing the pressure at a particular place in a fluid is,
$P = hdg$
Where, h is depth of the point from the surface, d is the density of the fluid and g is acceleration due to gravity.
Thus, Thus, the pressure exerted by a liquid depends on density of the liquid, depth below the surface and acceleration due to gravity
(c) The formula for expressing the pressure at a particular place in a fluid is,
$P = hdg$
Where, h is depth of the point from the surface, d is the density of the fluid and g is acceleration due to gravity.
Thus, the pressure exerted by a liquid depends on density of the liquid, depth below the surface and acceleration due to gravity.
The static fluid pressure is independent of the liquid's volume, mass, total surface area, and container form.
(d) Pressure exerted sideways is known as lateral pressure.
The parallel force per unit area produced when fluid molecules collide with a surface is referred to as lateral pressure, also known as shear pressure. In addition to acting vertically, it also exerts a sideward force that affects buoyancy, fluid flow, and the stability of objects submerged in fluids, which helps to explain complex fluid dynamics phenomena.
20. Correct Matching is (a)-(R)(Q),(b)-(S),(c)-(P),(d)-(T).
(a) One newton per square metre is equivalent to one pascal (abbreviated as Pa), the SI unit of pressure.
(b) The barye (Ba), which is equivalent to $1\frac{\text{dyne}}{\text{cm}^2}$ or 0.1 Pa, is the CGS unit of pressure.
(c) $1\text{bar}=10^5\text{Pa}$
(d) $1\text{millibar}=10^2\text{Pa}$
The correct matching is (a)-(R)(Q),(b)-(S),(c)-(P),(d)-(T).

6. Light

Solutions

LEVEL – I

1.Correct option: (D)

• Light is a form of energy which causes in us the sensation of sight. Light travels in straight lines. The fact that a small source of light casts a sharp shadow of an opaque object tells us that light travels in straight line path. This is because if light could bend easily and go behind the opaque object, then no shadow could be formed.

• **According to the wave theory of light:** light consists of electromagnetic wave which do not require a material medium(like solid, liquid or gas) for their propagation.

• The speed of light waves in vacuum is very high being about $3x10^8$m/s, whereas speed of sound waves is 343 m/s which is less than speed of light.

2.Correct option: (C)

Water reflects only a portion of the incident light that falls on it. The rest of the light passes through the water and gets absorbed or scattered. As a result, only a part of the image of the clouds and the trees is reflected in water. This reflected image is always less bright than the actual object.

3. Correct option: (B)

• A plane mirror always form a virtual and erect image of the same size as the object.

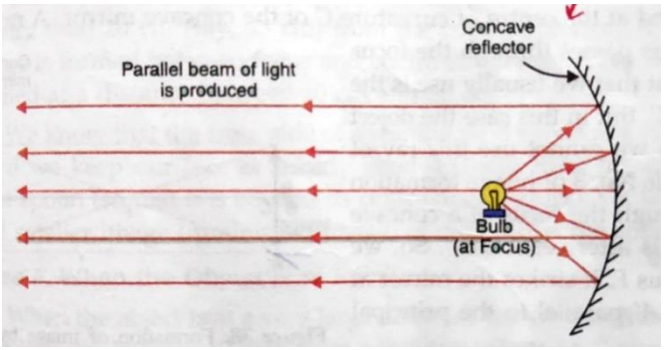
• A concave mirror can form a virtual and erect image larger than the object.

• A convex mirror can form only virtual and erect image smaller than the object.

as the face appears to be larger in the windowpane, it means the mirror could be concave mirror.

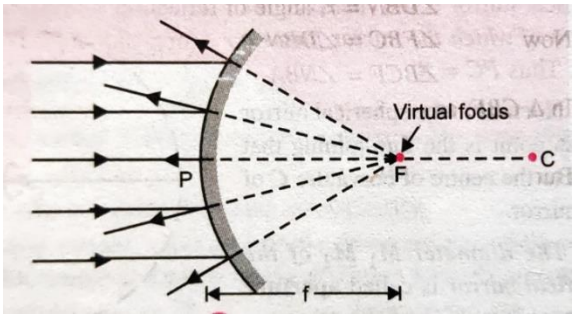
4. Correct option: (A)

A concave mirror can convert the diverging rays of light coming from an object into a parallel beam, when the object is placed at the focus of the concave mirror.



When a lighted bulb is placed at the focus of a concave mirror reflector, the diverging light rays of the bulb are collected by the concave reflector and then reflected to produce a strong, parallel-sided beam of light. This explains the use of concave reflectors in torches, car head-lights and search lights to produce a strong beam of light.

5. Correct option: (A)



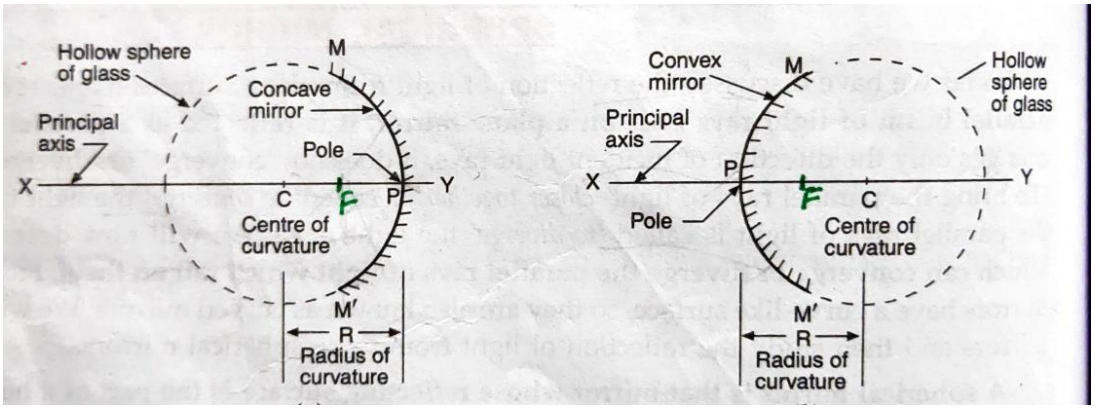
A convex mirror has its focus and centre of curvature behind it. Since no real rays of light can go behind the convex mirror, all the rays shown behind the convex mirror are virtual (or unreal) and hence they have been represented by dotted lines. In fact, no actual rays can pass through the focus and centre of curvature of a convex mirror.

whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished) When the distance of the object is changed from convex mirror, then only the position and size of the image changes.

6. Correct option: (D)

Focal length is a property of the spherical mirror. It does not depend on the light used. So, it is same for all the lights.

7. Correct option: (C)



for the spherical mirror having small aperture, the principal focus(F) lies exactly mid-way between the pole(P) and the centre of curvature (C). so, the focal length of a spherical mirror(a concave or a convex mirror) is equal to half of its radius of curvature.

If f is the focal length of a spherical mirror and R is the radius of curvature, then

$f = \frac{R}{2}$

8. Correct option: (B)

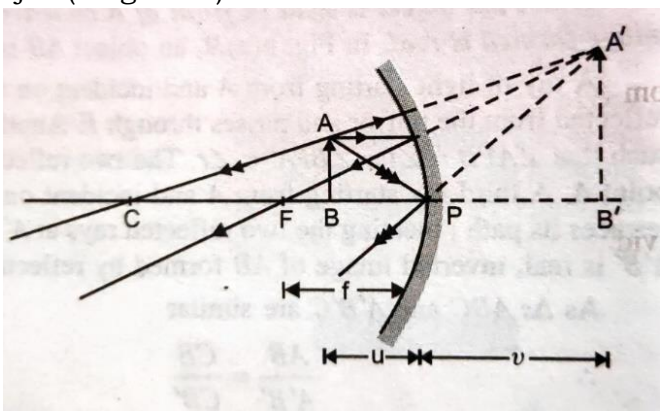
When the object is placed between the pole(P) and the focus(F) of a concave mirror,

Then the image formed is:

(i) behind the mirror

(ii) virtual and erect

(iii) larger than the object(magnified)

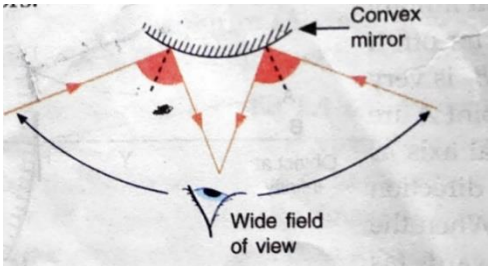


9. Correct option: (B)

Convex mirrors are used as rear-view mirrors in vehicles (like cars, trucks and buses) to see the traffic at the rear side (or back side). A driver prefers to use a convex mirror as rear-view mirror because of two reasons:

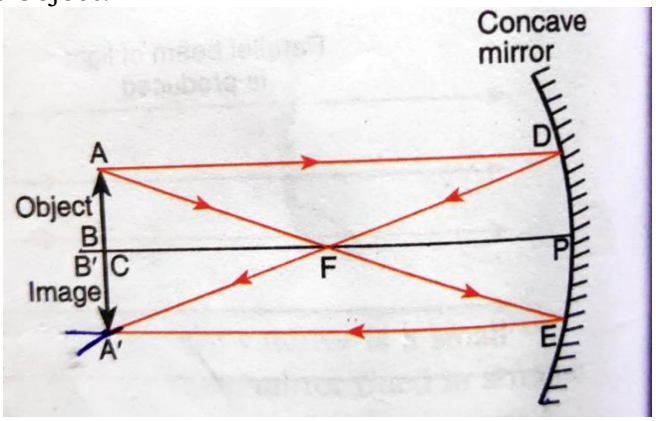
(a) A convex mirror always produces an erect (right side up) image of the objects.

(b) The image formed in a convex mirror is highly diminished or much smaller than the object, due to which a convex mirror gives a wide field of view (of the traffic behind.
A convex mirror enables a driver to view much larger area of the traffic behind him.



(a) A convex mirror gives a wide field of view (or large field of view).

10. Correct option: (D)
When a real object is placed at the centre of curvature (C) of a concave mirror, the image formed is:
(i) at the centre of curvature(C).
(ii) real and inverted
(iii) same size as the object.



As the object and image are at same distance from the mirror
So, distance between them is zero.

11.Correct option: (C)
The power of a mirror is the reciprocal of its focal length in metres.

$$power\ of\ a\ mirror = \frac{1}{focal\ length\ of\ the\ mirror\ in\ metres}$$

$$P = \frac{1}{f}$$

Given: focal length of the convex mirror(f)= 20cm = 0.20 m (focal length of convex mirror is positive)

$$power\ of\ a\ mirror = \frac{1}{0.20}$$

$$power\ of\ a\ mirror = \frac{100}{20} = 5$$

Unit of power is dioptre(D).
Hence, power od the convex mirror = +5D

12.Correct option: (D)
• Focal length is a property of the spherical mirror. It does not depend on the light used. So, it is same for all the lights.
• Focal length of a mirror is independent of the environment as it is a property of the mirror. So, even if the mirror is immersed in liquid, focal length will remain the same.

13.Correct option: (B)

Given: size of image is n times the size of the object

So, magnification = - n (negative sign for real and inverted image)

Magnification(m) $=\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

From equation 1

$-n = \frac{-f}{-f-u}$ (focal length of concave mirror is negative)

$-n(-f-u) = -f$

$n\,f + nu = -f$

$nu = -f + n\,f$

$u = \frac{-f+nf}{n}$

$u = \frac{(n-1)}{n}f$

Hence, distance of the object from the mirror = $\frac{(n-1)}{n}f$

14.Correct option: (B)

Given: object distance from focus = x_1

image distance from focus = x_2

so, $u = -(x_1 + f)$

$v = -(x_2 + f)$

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror

u=object distance from the mirror

v= image distance from the mirror

$-\frac{1}{f} = -\frac{1}{(x_1+f)} - \frac{1}{(x_2+f)}$ (focal length of concave mirror is negative)

$\frac{1}{f} = \frac{x_1+x_2+2f}{(x_1+f)(x_2+f)}$

$(x_1 + f)(x_2 + f) = f(x_1 + x_2 + 2f)$

$(x_1 + f)(x_2 + f) = f\,x_1 + f\,x_2 + 2f^2$

$x_1x_2 + f\,x_1 + f\,x_2 + f^2 = f\,x_1 + f\,x_2 + 2f^2$

$x_1x_2 = f^2$

$\sqrt{x_1x_2} = f$

15.Correct option: (A)

Given: Radius of curvature (R) = -6 m (radius of curvature is negative for concave mirror)

focal length (f) $= \frac{R}{2}$

focal length (f) $= \frac{-6}{2}$

Focal length(f) = -3m

Object distance (u) = - 92900000 miles

Diameter of the sun/height of the object h_o = 864100 miles

To find: height of the image h_i

As the sun is at a far off distance

We consider that the image is formed at the focus

Hence, image distance v = - 3m

$$\text{Magnification(m)} = \frac{\text{height of image}(h_i)}{\text{height of object}(h_o)} \text{----------(1)}$$

Also

$$\text{Magnification(m)} = -\frac{\text{image distance(v)}}{\text{object distance(u)}} \text{----------(2)}$$

Comparing equation 1 and 2

$$\frac{\text{height of image}(h_i)}{\text{height of object}(h_o)} = -\frac{\text{image distance(v)}}{\text{object distance(u)}}$$

Putting the values

We get

$$\frac{h_i}{864100} = -\frac{-3}{-92900000}$$

$$h_i = -\frac{3x\,864100}{92900000}$$

$$h_i = -0.028\text{m}$$

$$h_i = -28\text{mm}$$

Hence, height/size of image of the sun = 28mm

16.Correct option: (C)

Dentist mirror is a concave mirror

Given: object distance (u) = - 0.60cm

Magnification(m) = +4 (positive sign for virtual image)

We know that

$$\text{Magnification(m)} = -\frac{\text{image distance(v)}}{\text{object distance(u)}}$$

$$4 = -\frac{v}{-0.60}$$

v= 4 x 0.60

v= 2.4 cm

putting the values in the mirror formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Where f= focal length of the concave mirror

U=object distance from the mirror

V= image distance from the mirror

$$\frac{1}{f} = -\frac{1}{0.60} + \frac{1}{2.4}$$

$$\frac{1}{f} = -\frac{10}{6} + \frac{10}{24}$$ (lcm of 6 and 24 = 24)

$$\frac{1}{f} = \frac{-40 + 10}{24}$$

$$\frac{1}{f} = \frac{-30}{24}$$

$$f = -\frac{24}{30}$$

$$f = -0.8\text{ cm}$$ (negative sign because focal length of concave mirror is negative)

$$f = \frac{R}{2}$$

focal length(f) = -0.8 cm

$$-0.8 = \frac{R}{2}$$

R= -0.8 x 2 cm

R= -1.60cm

Hence, radius of curvature = -1.60cm

17.Correct option: (D)

Given: Magnification(m) = +4 (positive sign for virtual and erect image)

We know that

Magnification(m) = $-\frac{\text{image distance(v)}}{\text{object distance(u)}}$

$4 = -\frac{v}{u}$

Or

v= -4u----------(1)

According to the question

Distance between the object and the image = 90 cm

So,

-u+ v = 90 cm ---------(2) (object distance is negative as it is on the left of the mirror)

From equation 1

-u -4u = 90 cm

-5u = 90cm

$u = -\frac{90}{5}$cm

u= -18 cm

so,

from equation 1

v= - 4 x (-18)

v= 72 cm

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror

u=object distance from the mirror

v= image distance from the mirror

$\frac{1}{f} = -\frac{1}{18} + \frac{1}{72}$

$\frac{1}{f} = \frac{-4+1}{72}$ (lcm of 18 and 72 = 72)

$\frac{1}{f} = \frac{-3}{72}$

$f = -\frac{72}{3}$

$f = -24$ cm (negative sign because focal length of concave mirror is negative)

hence, it is a concave mirror of focal length = 24 cm

18.Correct option: (A)

Given: object distance (u) = 10cm (virtual object)

Focal length (f) = 15 cm

We know that

Magnification(m) =$\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

putting the given values in equation 1 we get

Magnification(m) =$\frac{15}{15-10}$

Magnification(m) = $\frac{15}{5} = 3$

as the magnification is positive so the image is virtual, erect and enlarged three times.

19.Correct option: (D)

Distance between the object and the wall = 1.5 m

As the image will be formed on the wall so

Distance between the object and the image = 1.5 m

Or

v = u + 1.5m----------(1)

Where v = image distance

u= object distance

Focal length(f) = -1m

To find: distance between mirror and the wall that is (u)

putting the values in the mirror formula

$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

Where f= focal length of the concave mirror

U=object distance from the mirror

V= image distance from the mirror

$-\frac{1}{1}=-\frac{1}{u}-\frac{1}{(u+1.5)}$

$-1=\frac{-u-1.5-u}{u(u+1.5)}$

$-u(u+1.5)=-2u-1.5$

$-u^2-1.5u+2u+1.5=0$

$u^2-0.5u-1.5=0$

Splitting the middle term

$u^2-1.5u+1u-1.5=0$

u(u-1.5) + 1(u-1.5) = 0

u-1.5 = 0 or u+1 = 0

u= 1.5m u=1 it is neglected because we already took negative sign in the formula so u can't be negative

Hence, hence distance of spherical mirror from the candle flame = 1.5 m

20. Correct option: (D)

When an object is placed between the pole(P) and the focus (F) of a concave mirror, the image formed is:

(i) behind the mirror,

(ii) virtual and erect, and

(iii) Magnified (larger than the object).

When the object is moved away from the object within the focus the image formed is larger than the object and it could be twice the size of the object.

21.Correct option: (B)

Given: object distance (u) = 20cm (virtual object)

Focal length (f) = 30 cm

We know that

Magnification(m) $=\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

putting the given values in equation 1 we get

Magnification(m) $=\frac{30}{30-20}$

Magnification(m) $=\frac{30}{10}=3$

as the magnification is positive so the image is virtual, erect and enlarged three times.

22. Correct option: (B)

Given: object distance (u) = -20cm

Magnification(m) = - 3 (negative sign for real image)

We know that

Magnification(m) = $-\frac{\text{image distance(v)}}{\text{object distance(u)}}$

$-3 = -\frac{v}{-20}$

v= -3 x 20

v= -60 cm

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror

U=object distance from the mirror

V= image distance from the mirror

$\frac{1}{f} = \frac{1}{-20} + \frac{1}{-60}$

$\frac{1}{f} = \frac{-1}{20} - \frac{1}{60}$ (lcm of 20 and 60 = 60)

$\frac{1}{f} = \frac{-3-1}{60}$

$\frac{1}{f} = \frac{-4}{60}$

$f = -\frac{60}{4}$

$f = -15$ cm (negative sign because focal length of concave mirror is negative)

hence, focal length of the concave mirror should be 15 cm.

23.Correct option: (C)

Given: object distance (u) = -20cm

Focal length(f) = 10cm (focal length is positive for convex mirror)

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror

u=object distance from the mirror

v= image distance from the mirror

$\frac{1}{10} = \frac{1}{-20} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{10} + \frac{1}{20}$ (lcm of 10 and 20 = 20)

$\frac{1}{v} = \frac{2+1}{20}$

$\frac{1}{v} = \frac{3}{20}$

$v = \frac{20}{3}$cm

As the convex mirror always forms a virtual image.

hence, virtual image is formed at a distance of $\frac{20}{3}$cm behind the mirror.

24.Correct option: (A)

Given: object distance (u) = -10cm

Radius of curvature (R) = -15 cm (radius of curvature is negative for concave mirror)

$\text{focal length (f)} = \frac{R}{2}$

$\text{focal length (f)} = \frac{-15}{2}$

Focal length(f) = -7.5cm

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror
u=object distance from the mirror
v= image distance from the mirror

$-\frac{1}{7.5} = \frac{1}{-10} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{10} - \frac{1}{7.5}$

$\frac{1}{v} = \frac{1}{10} - \frac{10}{75}$ (lcm of 10 and 75 = 150)

$\frac{1}{v} = \frac{15 - 20}{150}$

$\frac{1}{v} = \frac{-5}{150}$

$v = -\frac{150}{5}\text{cm}$

v= -30 cm

image distance(v) = -30 cm (negative sign shows image is formed in front of the mirror)

$\text{Magnification(m)} = -\frac{\text{image distance(v)}}{\text{object distance(u)}}$

$m = -\frac{-30}{-10}$

m= -3

The image is real and inverted because magnification is negative.

Hence, v= -30 cm, m= -3 (real, inverted)

25. Correct option: (D)

Given: object distance (u) = -20cm

Magnification(m) = - 4.5 (negative sign for real image)

We know that

$\text{Magnification(m)} = -\frac{\text{image distance(v)}}{\text{object distance(u)}}$

$-4.5 = -\frac{v}{-20}$

v= -4.5 x 20

v= -90cm

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror
u=object distance from the mirror
v= image distance from the mirror

$\frac{1}{f} = \frac{1}{-20} + \frac{1}{-90}$

$\frac{1}{f} = \frac{-1}{20} - \frac{1}{90}$ (lcm of 20 and 90 = 180)

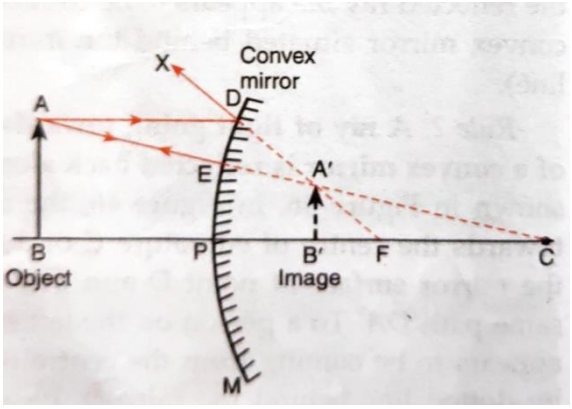
$\frac{1}{f} = \frac{-9 - 2}{180}$

$\frac{1}{f} = \frac{-11}{180}$

$f = -\frac{180}{11}$ (negative sign because focal length of concave mirror is negative)

hence, focal length = $\frac{180}{11}$cm

26. Correct option: (C)



A convex mirror has its focus and centre of curvature behind it. Since no real rays of light can go behind the convex mirror, all the rays shown behind the convex mirror are virtual (or unreal) and hence they have been represented by dotted lines. In fact, no actual rays can pass through the focus and centre of curvature of a convex mirror.

whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished) When the distance of the object is changed from convex mirror, then only the position and size of the image changes.

27. Correct option: (C)

Given: height of object(h_o) = 5cm

Object distance(u) = -1m = -100cm

Radius of curvature (R) = -20 cm (radius of curvature is negative for concave mirror)

focal length (f) $= \frac{R}{2}$

focal length (f) $= \frac{-20}{2}$

Focal length(f) = -10cm

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror

u=object distance from the mirror

v= image distance from the mirror

$-\frac{1}{10} = \frac{1}{-100} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{100} - \frac{1}{10}$

(lcm of 10 and 100 = 100)

$\frac{1}{v} = \frac{1-10}{100}$

$\frac{1}{v} = \frac{-9}{100}$

$v = -\frac{100}{9}$cm (negative sign shows image is formed in front of the mirror)

We know that Magnification(m) = $\frac{\text{height of image}(h_i)}{\text{height of object}(h_o)}$----------(1)

Also

Magnification(m) = $-\frac{\text{image distance(v)}}{\text{object distance(u)}}$----------(2)

Comparing equation 1 and 2

$\frac{\text{height of image}(h_i)}{\text{height of object}(h_o)} = -\frac{\text{image distance(v)}}{\text{object distance(u)}}$

$\frac{h_i}{5} = -\frac{-\frac{100}{9}}{100}$

$\frac{h_i}{5} = \frac{1}{9}$

$h_i = \frac{5}{9}$

$h_i = 0.55$ cm

Hence, height/size of the image = 0.55 cm

28.Correct option: (A)

according to the given question image should be doubled and inverted

hence, magnification should be equal to -2

given: focal length of the concave mirror(f) = -50cm

to find: object distance(u)

We know that Magnification(m) =$\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

putting he values in equation 1

-2 = $\frac{-50}{-50-u}$

-2(-50 – u) = -50

100 + 2u = -50

2u = -50 -100

2u = -150

u=$\frac{-150}{2} = -75$ cm

hence, object should be placed at a distance of 75 cm in front of the mirror.

29.Correct option: (A)

given: focal length of the concave mirror = - f

object distance(u)= -4f

height of object (h_o)= 6cm

We know that

Magnification(m) =$\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

putting he values in equation 1

m= $\frac{-f}{-f-(-4f)} = \frac{-f}{-f+4f} = \frac{-f}{3f} = -\frac{1}{3}$

We know that

Magnification(m) = $\frac{\text{height of image}(h_i)}{\text{height of object}(h_o)}$

$-\frac{1}{3} = \frac{\text{height of image}(h_i)}{6}$

$h_i = -\frac{6}{3} = -2$cm

Hence, height/length of the image = 2 cm (negative sign for inverted image)

30.Correct option: (D)

Mirror is unaffected by the effects of dipping. Convergence of a concave mirror is unaffected by any medium.

31.Correct option: (D)

Given: distance of object from the mirror= -30 cm

Focal length (f) = 30 cm (focal length is positive for convex mirror)

putting the values in the mirror formula

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$$

Where f= focal length of the concave mirror

u=object distance from the mirror

v= image distance from the mirror

$$\frac{1}{30}=\frac{1}{-30}+\frac{1}{v}$$

$$\frac{1}{v}=\frac{1}{30}+\frac{1}{30}$$

$$\frac{1}{v}=\frac{1+1}{30}$$

$$\frac{1}{v}=\frac{2}{30}$$

$$v=\frac{30}{2}=15\ \text{cm}$$

hence, the image is formed at a distance of 15 cm behind the mirror.

32.Correct option: (D)

A convex mirror has its focus and centre of curvature behind it. Since no real rays of light can go behind the convex mirror, all the rays shown behind the convex mirror are virtual (or unreal) and hence they have been represented by dotted lines. In fact, no actual rays can pass through the focus and centre of curvature of a convex mirror.

whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished) When the distance of the object is changed from convex mirror, then only the position and size of the image changes.

LEVEL-II

1. Correct option: (A)

Given: distance of object from the mirror=40 cm

Focal length (f) = 20 cm

We know that radius of curvature(R) = 2f

Distance of centre of curvature from the mirror = 2f

= 2x20 cm

= 40 cm

It means the object is placed at the centre of curvature of the concave mirror.

When an object is placed at the centre of curvature(C) of a concave mirror, the image formed is:

(i) At the centre of curvature,

(ii) Real and inverted, and

(iii) Same size as the object.

2.Correct option: (D)

A convex mirror has its focus and centre of curvature behind it. Since no real rays of light can go behind the convex mirror, all the rays shown behind the convex mirror are virtual

(or unreal) and hence they have been represented by dotted lines. In fact, no actual rays can pass through the focus and centre of curvature of a convex mirror.

whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished) When the distance of the object is changed from convex mirror, then only the position and size of the image changes.

3.Correct option: (A)

Given: size of image is half of the size of the object

So, magnification = $\frac{1}{2}$ (positive sign for virtual and erect image formed by convex mirror)

Focal length (f)= 10cm

Magnification(m) =$\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the mirror

From equation 1

$\frac{1}{2} = \frac{10}{10-u}$

$10 - u = 20$

u= 10-20

u= -10 cm

hence, the distance of the object from the mirror is 10 cm in front of the mirror.

4. Correct option: (B)

The focal length of the concave mirror will remain the same because the focal length does not depend on the external medium in which the mirror is held.

So, the focal length of concave mirror in water = 30cm

5.Correct option: (A)

Given: magnification(m) = -16 (image is real)

Distance between the object and the image = 120 cm

Or

v = u + 120----------(1)

Magnification(m) = $-\frac{\text{image distance(v)}}{\text{object distance(u)}}$

-16 = $-\frac{v}{u}$

v= 16u----------(2)

putting value in equation 1

16u= u + 120

16u – u = 120

15u = 120

u=$\frac{120}{15}$ = 8 cm

hence, the distance of flower from the mirror should be 8 cm.

6.Correct option: (C)

Given: size of image is 2 times the size of the object

Focal length(f)= - 20cm

• if the image is real and inverted

So, magnification = - 2 (negative sign for real and inverted image)

Magnification(m) =$\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

From equation 1

$-2 = \frac{-20}{-20-u}$

$-2(-20 - u) = -20$

40 + 2u = -20

2u = -20 -40

2u= -60

$u = \frac{-60}{2} = -30$ cm

Hence, distance of the object from the mirror = -30 cm

Also

Magnification(m) $= \frac{f-v}{f}$----------(2)

Where f= focal length

v= distance of image from the mirror

From equation 2

$-2 = \frac{-20-v}{-20}$

$-2\,x(-20) = -20 - v$

40 = -20-v

v = -20 -40

v= -60 cm

Hence, distance of the image from the mirror = -60 cm

distance between the object and the image = 60cm – 30cm = 30 cm

7. Correct option: (B)

• A plane mirror always form a virtual and erect image of the same size as the object.

• A concave mirror can form a virtual and erect image larger than the object.

• We know that whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished).

As the man observes his image as erect but smaller than his actual height

Hence, the mirror used is convex mirror.

8.Correct option: (B)

If a source of light is placed at the focus of a parabolic mirror, all the reflected rays advance parallel to the principal axis. So, the intensity remains high even at a large distance. Due to this reason, parabolic mirrors/reflectors are used in the headlights of automobiles.

9.Correct options: (B), (C), (D)

• **whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished)** When the distance of the object is changed from convex mirror, then only the position and size of the image changes.

• A concave mirror can form

(i) virtual, erect and magnified image

(ii) real and inverted image of same size, smaller in size, larger in size than the object.

10. Correct options: (A), (B)

• A plane mirror always form a virtual and erect image of the same size as the object for all positions of the object.

• **whatever be the position of object in front of a convex mirror, the image formed by a convex mirror is always behind the mirror, it is virtual, erect and smaller than the object (or diminished)** When the distance of the object is changed from convex mirror, then only the position and size of the image changes.

11.Correct options: (A), (B), (D)

• Focal length is a property of the spherical mirror. It does not depend on the light used. So, it is same for all the lights.

• focal length does not depend on the external medium in which the mirror is held.

• Spherical aberration is a phenomenon in which the outer parts of the mirror do not bring light rays into the same focus as the central part. Hence, image is not free from spherical aberrations

• Chromatic aberration is the failure of the lens to focus all the colors to the same point.

A curved mirror is free from chromatic aberrations because refraction doesn't take place in mirrors.

12. Correct options: (A), (B)

• When the object is placed between the pole(P) and the focus(F) of a concave mirror,

Then the image formed is:

(i) behind the mirror

(ii) virtual and erect

(iii) larger than the object(magnified)

for all the other positions of the object, the concave mirror forms a real and inverted image of the object.

13.Correct options: (A), (B)

Given: size of image is 2 times the size of the object

• if the image is real and inverted

So, magnification = - 2 (negative sign for real and inverted image)

Magnification(m) $=\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

From equation 1

$-2 = \frac{-f}{-f-u}$ (focal length is negative for concave mirror)

$-2(-f-u) = -f$

2 f + 2u = -f

2u = -f- 2 f

$u = \frac{-f-2f}{2}$

$u = \frac{-3f}{2}$

Hence, distance of the object from the mirror = $\frac{3f}{2}$

• if the image is virtual and erect

So, magnification = 2

Magnification(m) $=\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

From equation 1

$2 = \frac{-f}{-f-u}$ (focal length is negative for concave mirror)

$2(-f-u) = -f$

-2 f - 2u = -f

2u =-2 f + f

2u =-- f

$u = \frac{-f}{2}$

Hence, distance of the object from the mirror = $\frac{f}{2}$

14.Correct options: (A), (C), (D)

The concept of focus is defined only for the rays which have an angle of incidence close to zero degree. These rays are called paraxial rays. If the rays are not paraxial, the image is not precisely sharp. That is the focus is not precisely defined.

The pole, radius of curvature, and the principal axis of a spherical mirror do not depend on paraxial rays.

15.Correct option: (A)

Given: object distance (u) = -30cm

Focal length(f) = -20cm (focal length is negative for concave mirror)

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror

u=object distance from the mirror

v= image distance from the mirror

$-\frac{1}{20} = \frac{1}{-30} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{30} - \frac{1}{20}$

(lcm of 30 and 20 = 60)

$\frac{1}{v} = \frac{2-3}{60}$

$\frac{1}{v} = \frac{-1}{60}$

$v = -60cm$

v= -60 cm

image distance(v) = -60 cm (negative sign shows image is formed in front of the mirror)

16.Correct option: (B)

Magnification(m) = $-\frac{\text{image distance(v)}}{\text{object distance(u)}}$

$m = -\frac{-60}{-30}$

m= -2

hence magnification is 2

The image is real and inverted because magnification is negative.

17.Correct option: (A)

When the object is moved by a distance of 10 cm towards the mirror.

Given: object distance (u) = -20cm

Focal length(f) = -20cm (focal length is negative for concave mirror)

It means the object is placed at the focus of the concave mirror.

So, the image moves to the infinity.

18.Correct option: (B)

Given: Object distance(u) =-20 cm

Focal length(f) = 10cm (focal length is positive for convex mirror)

putting the values in the mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Where f= focal length of the concave mirror

u=object distance from the mirror

v= image distance from the mirror

$$\frac{1}{10} = \frac{1}{-20} + \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{10}$$

(lcm of 10 and 20 = 20)

$$\frac{1}{v} = \frac{2+1}{20}$$

$$\frac{1}{v} = \frac{3}{20}$$

position of image (v) = $\frac{20}{3}$cm (positive sign shows image is formed behind the mirror)

19. Correct option: (A)

$$\text{Magnification(m)} = -\frac{\text{image distance(v)}}{\text{object distance(u)}}$$

$$m = -\frac{\frac{20}{3}}{-20}$$

m= 1/3

hence, magnification is 1/3

20.Correct option: (C)

height of object(h_o) = 0.5cm

We know that

$$\text{Magnification(m)} = \frac{\text{height of image}(h_i)}{\text{height of object}(h_o)}$$

$$\frac{1}{3} = \frac{h_i}{0.5}$$

$$h_i = \frac{0.5}{3}$$

$$h_i = \frac{5}{30}$$

h_i = 0.167 cm

Hence, height of the image = 0.167 cm

21. Correct Matching is: a-Q; b-S; c-P; d-T

(a) given: object distance(u) = - f (object is placed before the convex mirror)

We know that

$$\text{Magnification(m)} = \frac{f}{f-u} \text{----------(1)}$$

Where f= focal length

u = distance of object from the pole of the mirror

putting he values in equation 1

$$m = \frac{f}{f-(-f)}$$

$$m = \frac{f}{f+f} = \frac{1}{2} = 0.5$$

magnification = 0.5

(b) given: object distance(u) = - 2f (object is placed before the concave mirror)

We know that

$$\text{Magnification(m)} = \frac{f}{f-u} \text{----------(1)}$$

Where f= focal length

u = distance of object from the pole of the mirror

putting he values in equation 1

$$m = \frac{-f}{-f-(-2f)}$$ (focal length of concave mirror is negative)

$$m = \frac{-f}{-f+2f} = -1$$

magnification = -1

(c) given: object distance(u) = - f (object is placed before the concave mirror)

We know that

Magnification(m) $=\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

putting he values in equation 1

m = $\frac{-f}{-f-(-f)}$ (focal length of concave mirror is negative)

m = $\frac{-f}{-f+f} = -\frac{1}{0} = -\infty$

magnification = -∞

(d) given: object distance(u) = - 2f (object is placed before the convex mirror)

We know that

Magnification(m) $=\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

putting he values in equation 1

m = $\frac{f}{f-(-2f)}$

m = $\frac{f}{f+2f} = \frac{1}{3} = 0.33$

magnification = 0.33

22. Correct Matching is: a- QT; b-PS; c-PT; d-RT

(a) magnification is positive when the image formed by the mirror is virtual

magnification is less than 1 when the image is smaller in size than the object.

A convex mirror can form a virtual image smaller in size than the object.

(b) magnification is negative when the image formed by the mirror is real.

magnification is equal to 1 when the image is equal in size to the object.

A concave mirror can form a real image equal in size to the object.

(c) magnification is positive when the image formed by the mirror is virtual.

magnification is more than 1 when the image is larger in size than the object.

A concave mirror can form a virtual image larger in size than the object.

(d) magnification is positive when the image formed by the mirror is virtual.

magnification is equal to 1 when the image is equal in size to the object.

A plane mirror can form a virtual image equal in size to the object.

23. Correct Answer is: 2

given: focal length of the concave mirror(f) = -10cm

object distance(u) = - 5cm

We know that

Magnification(m) $=\frac{f}{f-u}$----------(1)

Where f= focal length

u = distance of object from the pole of the mirror

putting he values in equation 1

m = $\frac{-10}{-10-(-5)}$

m = $\frac{-10}{-10+5} = \frac{-10}{-5}$

m = 2

hence magnification = 2

7. Electricity

Solutions

LEVEL – I

1. Correct option: (A)

When we rub two insulating substances against each other, we provide energy to overcome friction between them. This energy is used in removing electrons from one substance and transferring them to the other. The transfer takes place from the material in which electrons are held less tightly (with lower work function) to the material in which electrons are held more tightly (with higher work function), i.e., electrons are transferred from the material whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

For example, when we rub a glass rod with silk, electrons are transferred from glass rod to silk. The glass rod becomes positively charged and silk acquires an equal negative charge.

2.Correct option: (A)

electrons are transferred from the material whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

3.Correct option: (C)

In charging by induction, a charged body imparts some charge of opposite sign to another body without any actual contact between them.

4.Correct option: (C)

A gold leaf electroscope is an instrument which is used for detecting the presence of electric charge and its polarity (that is whether charge is negative or positive).

5.Correct option: (A)

The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell or a battery. Due to the chemical reactions going on inside the cell or battery, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell or battery is connected.

6.Correct option: (B)

A **primary cell** or battery is one that cannot easily be *recharged* after one use, and are discarded following discharge.

A **secondary cell** or battery is one that can be electrically recharged after use to their original pre-discharge condition, by passing current through the circuit in the opposite direction to the current during discharge.

7.Correct option: (C)

Electromotive force is defined as the maximum potential difference between the two electrodes of the cell when no current is drawn from the cell.

SI unit of electromotive force is volts(V).

8.Correct option: (D)

The full form of MCB is **Miniature Circuit Breaker**. It is a kind of electronic switch that functions automatically. It can detect a problem occurring inside a circuit by identifying the extreme amount of current flow in the circuit. If there is a large amount of power available in the loop, it can break the loop at any moment.

9.Correct option: (C)

The heating effect of electric current is utilised in electric fuse for protecting household wiring and electrical appliances. A fuse is a short length of a thin tin- plated copper wire

having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing). This prevents the fire in house (due to over-heating of wiring) and also prevents damage to various electrical appliances in the house due to excessive current flowing through them.

10.Correct option: (B)

- Similar charges (or Like charges) repel each other. For example, a positive charge repels a positive charge; and a negative charge repels a negative charge.
- Opposite charges (or Unlike charges) attract each other. For example, a positive charge attracts a negative charge.

11.Correct option: (B)

Electric cell is an instrument for the electricity generation where the electricity will generate after some chemical reaction takes place inside it. The generation is due to chemical reactions therefore the answer is chemical to electrical.

12. Correct option: (D)

Electrolysis is used extensively in metallurgical processes such as in **extraction** (electrowinning) or **purification** (electro refining) of metals from ores or compounds and in deposition of metals from solution (**electroplating**).

13. Correct option: (B)

During electrolysis of water, electricity is passed through acidified or salt water. Water undergoes decomposition and the process liberates hydrogen gas at the cathode and oxygen gas at the anode.

14.Correct option: (D)

Properties of electrolyte are:

- It conducts electric current as a result of its dissociation into ions.
- It generally ionizes into positive and negative ions when it is dissolved in a solvent.

15.Correct option: (D)

The electrodes decide which element or compound is deposited as a result of the chemical reaction triggered by the electric current passing through them.

An electrode is a nonmetallic component of a circuit that makes contact with an electrical conductor.

chemical reaction depends on Electrodes, Nature of electrolyte, magnitude of current.

16.Correct option: (A)

Drinking water is a good conductor of electricity because it contains small amounts of various salts dissolved in it.

17.Correct option: (B)

Electroplating is used to give objects made of a cheap metal coating of a more expensive metals to make them look more attractive. Less expensive metal are electroplated with more expensive metals like silver and gold to make Jewellery or ornaments. These ornaments have the appearance of silver or gold but they are much less expensive.

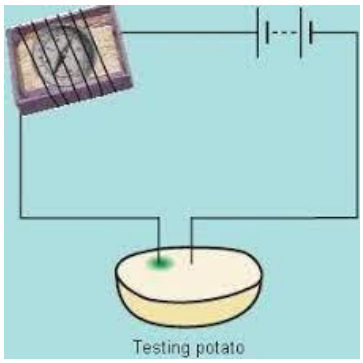
18.Correct option: (C)

Electrical conductivity of distilled water can be increased by adding some type of an acid or a salt to it. Vegetable oil, kerosene and sugar are bad conductors of electricity.

19.Correct option: (C)

Suppose we want to pass current through a potato. So, we connect ends of a battery by copper wire with potato by inserting in it. The electrons from negative potential of battery through copper wire reach in potato where they undergo in electrolysis (chemical reaction) inside the potato in the presence of potato juice (starch).

This process releases electrons in potato and increases the negative ions. When these negative ions reach the other end of copper wire (positive terminal) it reacts with part of the copper wire inside the potato and makes a compound of copper which is bluish-green in color.



This is the reason for greenish spots in potato on passing a current through it. It always appears around the copper wire connected to the positive terminal of the battery. This is an example that electricity causes chemical reactions. This chemical reaction causes color change.

20.Correct option: s(B)

The passage of an electric current through a solution causes chemical effects. The solution decomposes into positive and negative ions when electricity is passed through it. The process of decomposition of the solution is a chemical effect.

21.Correct option: s(C)

The process in which a layer of one metal is coated with another metal by electrolysis is called electroplating.

22.Correct option: (C)

The metallic rods or plates used in the electrolysis are called conductors or electroddes.

23.Correct option: (B)

When we rub two insulating substances against each other, we provide energy to overcome friction between them. This energy is used in removing electrons from one substance and transferring them to the other. The transfer takes place from the material in which electrons are held less tightly (with lower work function) to the material in which electrons are held more tightly (with higher work function), i.e., electrons are transferred from the material whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

For example, when we rub a glass rod with silk, electrons are transferred from glass rod to silk. The glass rod becomes positively charged and silk acquires an equal negative charge.

LEVEL-II

1. Correct option: (A)

The heating effect of electric current is utilised in electric fuse for protecting household wiring and electrical appliances. A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing). This prevents the fire in house (due to over-heating of wiring) and also prevents damage to various electrical appliances in the house due to excessive current flowing through them.

2.Correct option: (B)

The same charge is transferred to the gold leaves through the metal rod. The leaves repel each other and diverge as shown in. By measuring the divergence of leaves, the amount of charge on the body can be estimated.

Now, since the leaves are getting closer to each other it means the negative charge on the leaves is reducing which can happen if the rod is positively charged.

3.Correct option: (C)

When two charged bodies at different electric potentials are connected by a metal wire, then electric charges will flow from the body at higher potential to the one at lower potential (till they both acquire the same potential).

This flow of charges in the metal wire constitutes an electric current. It is the potential difference between the ends of the wire which makes the electric charges (or current) to flow in the wire.

We now know that the electric charges whose flow in a metal wire constitutes electric current are the **negative charges called electrons.**

4. Correct options: (A), (B)

The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell or a battery. Due to the chemical reactions going on inside the cell or battery, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell or battery is connected.

5.Correct options: (A),(C)

- Large Current is measured by an instrument called ammeter. The ammeter is always connected in series with the circuit in which the current is to be measured.
- A galvanometer can measure small electric currents.

6.Correct options: (B), (C)

The electric current is a flow of electric charges (called electrons) in a conductor such as a metal wire. The magnitude of electric current in a conductor is the amount of electric charge passing through a given point of the conductor in one second. If a charge of Q coulombs flows through a conductor in time t seconds, then the magnitude I of the electric current flowing through it is given by:

Current, $I = \frac{Q}{t}$

The SI unit of electric current is ampere which is denoted by the letter A.

If we put charge Q = 1 coulomb and time = 1 second in the above formula,

then current I becomes 1 ampere. This gives us the following definition of ampere: When 1 coulomb of charge flows through any cross-section of a conductor in 1 second, the electric current flowing through it is said to be 1 ampere. That is,

1 ampere = $\frac{1\ \text{coulomb}}{1\text{sec}}$

Hence, units of electric current are Ampere and coulomb/sec

7.Correct options: (A), (B), (C)

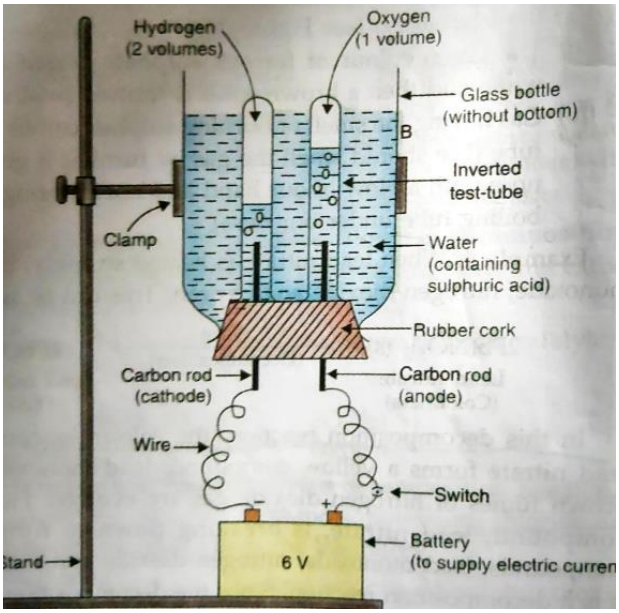
The most familiar uses of electroplating are:

- Used in ornamentation and decoration purposes
- Purification of metals
- Repairing worn out parts of machinery.
- Preventing corrosion/ rusting of metals

8.Correct options: (B), (C)

Electrolysis of water is using electricity to split water into oxygen gas and hydrogen gas by electrolysis.

9.Correct options: (B), (C), (D)



• When salt is dissolved in distilled water we get salt solution. Thus, distilled water becomes a good conductor of electricity on dissolving a little of salt in it.

• $CuSO_4$ dissociate completely into ions so it is strong electrolyte which conducts electricity.

• Lemon juice has charged particles in it. Hence, it act as a electrolyte.

10.Correct options: (A), (B)

Those substances through which electric charges cannot flow, are called insulators. In other words: Those substances through which electricity cannot flow are called insulators. Glass, ebonite, rubber, most plastics, paper, dry wood, cotton, mica, bakelite, porcelain, and dry air, are all insulators because they do not allow electric charges (or electricity) to flow through them.

11.Correct option: (A), (B), (C)

Liquids that are solutions of acids, bases and salts are good conductors. Distilled water is a poor conductor.

12.Correct options: (A), (B)

Rubbed area of non conducting body/ insulator gets charged, and this charge does not move to other parts of the body. The charge is static on rubbed portion only that is outer surface.

13.Correct option: (B)

Those substances through which electric charges cannot flow, are called insulators. In other words: Those substances through which electricity cannot flow are called insulators. Glass, ebonite, rubber, most plastics, paper, dry wood, cotton, mica, bakelite, porcelain, and dry air, are all insulators because they do not allow electric charges (or electricity) to flow through them**.**

14.Correct option: (C)

Those substances through which electric charges can flow, are called conductors. But the flow of electric charges is called electricity, so we can also say that: Those substances through which electricity can flow are called conductors. All the metals like silver, copper and aluminum, etc., are conductors

15.Correct option: (A)

All the conductors (like metals) have some electrons which are loosely held by the nuclei of their atoms. These electrons are called "free electrons" and can move from one atom to another atom throughout the conductor. The presence of "free electrons" in a substance makes it a conductor (of electricity).

16. Correct option: (D)

Electroplating a thin layer of chromium onto a metal item is known as chrome plating. It is done

- To make objects heavy
- To make the objects strongest
- To protect the objects from rusting

17. Correct option: (A)

The process in which a layer of one metal is coated with another metal by electrolysis is called electroplating.

18.Correct option: (A)

Electroplating is based on chemical effect of electric current. In electroplating, electric current is used to reduce metal cations which are dissolved in solvent to form thin metal coating on other electrode.

19. Correct Matching is: a-R; b-T; c-P; d-Q;

(a) The SI unit of electric charge is coulomb which is denoted by the letter C.

An electron possesses a charge of -1.6 x 10^{-19}C.

(b) The SI unit of electric potential is volt which is denoted by the letter V.

(c) The SI unit of electric current is ampere which is denoted by the letter A.

(d) The SI unit of electric charge is coulomb which is denoted by the letter C.

20. Correct Matching is: a-R; b-T; c-QS; d-P;

(a)Anode is the positive part of electrolyte where oxidation takes place

(b)cathode is the negative part of the cell where reduction takes place.

(c) The process in which a layer of one metal is coated with another metal by electrolysis is called electroplating.

Electroplating is based on chemical effect of electric current. In electroplating, electric current is used to reduce metal cations which are dissolved in solvent to form thin metal coating on other electrode.

(d) An electrolyte is a substance that produces an electrically conducting solution when dissolved in a polar solvent such as water.

Electric Charge and Current

LEVEL – I

1.Correct option: (A)

Electric cell is an instrument for the electricity generation where the electricity will generate after some chemical reaction takes place inside it. The generation is due to chemical reactions therefore the answer is chemical energy to electrical energy.

2.Correct option: (B)

The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell. Due to the chemical reactions going on inside the cell, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell is connected. A cell is a small source of electric current.

3. Correct option: (A)

When we rub two insulating substances against each other, we provide energy to overcome friction between them. This energy is used in removing electrons from one substance and transferring them to the other. The transfer takes place from the material in which electrons are held less tightly (with lower work function) to the material in which electrons are held more tightly (with higher work function), i.e., electrons are transferred from the material

whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

For example, when we rub an ebonite rod with cat's skin, electrons are transferred from cat's skin to the ebonite rod. The cat's skin becomes positively charged and ebonite rod acquires an equal negative charge.

4. Correct option: (A)

When we rub two insulating substances against each other, we provide energy to overcome friction between them. This energy is used in removing electrons from one substance and transferring them to the other. The transfer takes place from the material in which electrons are held less tightly (with lower work function) to the material in which electrons are held more tightly (with higher work function), i.e., electrons are transferred from the material whose work function is lower to the material whose work Junction is higher Consequently, the material which loses electrons acquires a positive charge and material which gains electrons acquires an equal negative charge.

For example, when we rub a glass rod with silk, electrons are transferred from glass rod to silk. The glass rod becomes positively charged and silk acquires an equal negative charge.

5.Correct option: (B)

The SI unit of electric charge is coulomb which is denoted by the letter C.

An electron possesses a negative charge of 1.6×10^{-19}C.

6.Correct option: (C)

The SI unit of electric charge is coulomb which is denoted by the letter C.

7. Correct option: (D)

The SI unit of electric charge is coulomb which is denoted by the letter C.

An electron possesses a negative charge of 1.6×10^{-19}coulombs.

8.Correct option: (D)

The vast amount of charge in an object is usually hidden as the object contains equal amounts of positive charge and negative charge. With such an equality or balance of charge, the object is said to be electrically neutral, ie, it contains no net charge.

9.Correct option: (C)

The electric current is a flow of electric charges (called electrons) in a conductor such as a metal wire. The magnitude of electric current in a conductor is the amount of electric charge passing through a given point of the conductor in one second. If a charge of Q coulombs flows through a conductor in time t seconds, then the magnitude I of the electric current flowing through it is given by:

Current, $I = \frac{Q}{t}$

The SI unit of electric current is ampere which is denoted by the letter A.

10.Correct option: (B)

The electric current is a flow of electric charges (called electrons) in a conductor such as a metal wire. The magnitude of electric current in a conductor is the amount of electric charge passing through a given point of the conductor in one second. If a charge of Q coulombs flows through a conductor in time t seconds, then the magnitude I of the electric current flowing through it is given by:

Current, $I = \frac{Q}{t}$

The SI unit of electric current is ampere which is denoted by the letter A.

11. Correct option: (D)

The electric current is a flow of electric charges (called electrons) in a conductor such as a metal wire. The magnitude of electric current in a conductor is the amount of electric charge

passing through a given point of the conductor in one second. If a charge of Q coulombs flows through a conductor in time t seconds, then the magnitude I of the electric current flowing through it is given by:

Current, $I = \frac{Q}{t}$

The SI unit of electric current is ampere which is denoted by the letter A.

12.Correct option: (A)

The electric current is a flow of electric charges (called electrons) in a conductor such as a metal wire. The magnitude of electric current in a conductor is the amount of electric charge passing through a given point of the conductor in one second. If a charge of Q coulombs flows through a conductor in time t seconds, then the magnitude I of the electric current flowing through it is given by:

Current, $I = \frac{Q}{t}$

The SI unit of electric current is ampere which is denoted by the letter A.

If we put charge Q = 1 coulomb and time = 1 second in the above formula,

then current I becomes 1 ampere. This gives us the following definition of ampere: When 1 coulomb of charge flows through any cross-section of a conductor in 1 second, the electric current flowing through it is said to be 1 ampere. That is,

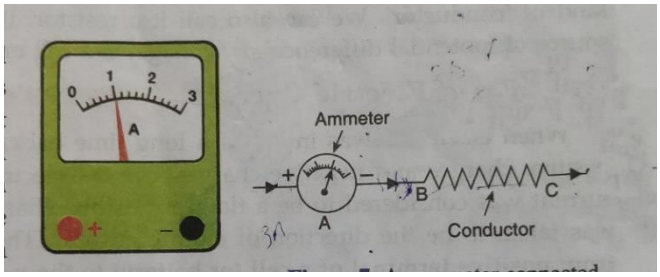
1 ampere = $\frac{1 \text{ coulomb}}{1\text{sec}}$

13.Correct option: (B)

Sensitive/ small magnitude currents are measured by galvanometer. It is connected in series with the circuit. When no current is flowing through a galvanometer, its pointer is at zero mark. When an electric current passes through the galvanometer, then the pointer deflects /moves either to the left side of zero mark or to the right side of the zero mark, depending on the direction of the current.

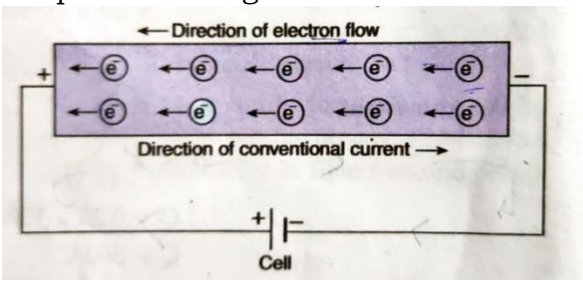
14.Correct option: (C)

Current is measured by an instrument called ammeter. The ammeter is always connected in series with the circuit in which the current is to be measured.



15.Correct option: (B)

When electricity was invented a long time back, it was known that there are two types of charges: positive charges and negative charges, but the electron had not been discovered at that time. So, electric current was considered to be a flow of positive charges and the direction of flow of the positive charges was taken to be the direction of electric current.



Thus, the conventional direction of electric current is from positive terminal of a cell (or battery) to the negative terminal, through the outer circuit.

16.Correct option: (D)

The difference in electric potential between two points is known as potential difference. The potential difference between two points in an electric circuit is defined as the amount of work done in moving a unit charge from one point to the other point. That is:

Potential difference=$\frac{\text{Work done}}{\text{Quantity of charge moved}}$

If W joules of work has to be done to move q coulombs of charge from one point to the other point, then the potential difference V between the two points is given by the formula:

$V = \frac{W}{q}$

where W =work done

and q= quantity of charge moved

17.Correct option: (B)

The difference in electric potential between two points is known as potential difference. The potential difference between two points in an electric circuit is defined as the amount of work done in moving a unit charge from one point to the other point. That is:

Potential difference=$\frac{\text{Work done}}{\text{Quantity of charge moved}}$

18.Correct option: (C)

The electric potential (or potential) at a point in an electric field is defined as the work done in moving a unit positive charge from infinity to that point.

Electric Potential =$\frac{\text{Work done}}{\text{Quantity of charge moved}}$

Electric Potential is denoted by the symbol V and its SI unit is volt.

19.Correct option: (A)

Potential difference=$\frac{\text{Work done}}{\text{Quantity of charge moved}}$

If W joules of work has to be done to move q coulombs of charge from one point to the other point, then the potential difference V between the two points is given by the formula:

$V = \frac{W}{q}$

where W =work done

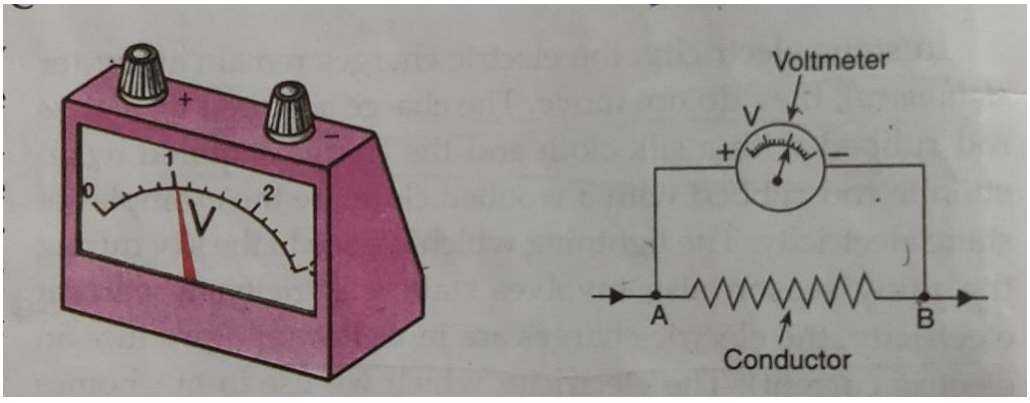
and q= quantity of charge moved

The SI unit of potential difference is volt which is denoted by the letter V. The potential difference between two points is said to be 1 volt if 1 joule of work is done in moving 1 coulomb of electric charge from one point to the other.

Thus, 1 volt= $\frac{1 \text{ joule}}{1 \text{ coulomb}}$

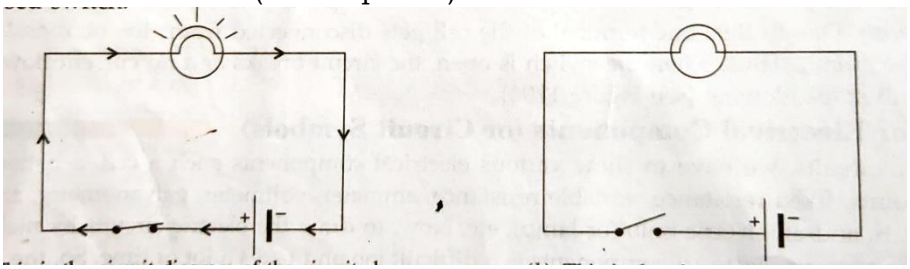
20. Correct option: (D)

The potential difference is measured by means of an instrument called voltmeter. The voltmeter is always connected in parallel across the two points where the potential difference is to be measured.



21.Correct option: (B)

The switch (or plug key) is a device for 'making or breaking an electric circuit. When the switch is open, then the circuit breaks' and no current flows in it. But when the switch is closed, then the circuit is 'made' (or completed) and current flows in it.



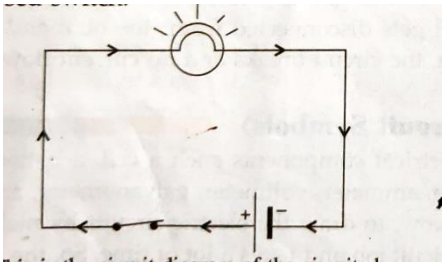
22.Correct option: (B), (C)

The switch (or plug key) is a device for 'making or breaking an electric circuit. When the switch/key is open, then the circuit breaks' and no current flows in it. Circuit is open or incomplete circuit.

23.Correct option: (A)

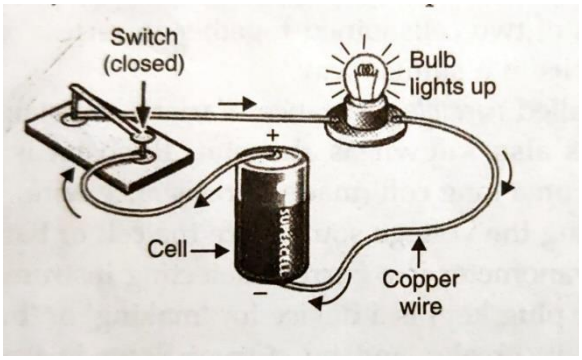
A diagram which indicates how different components in a circuit have been connected by using the electrical symbols for the components, is called a circuit diagram.

In the circuit diagram shown in Figure below, a bulb has been connected to the two terminals of a cell by copper wires through a closed switch.



24.Correct option: (D)

A continuous conducting path consisting of wires and other resistances (like electric bulb, etc.) and a switch, between the two terminals of a cell or a battery along which an electric current flow, is called a circuit.



25.Correct option: (C)

The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell. Due to the chemical reactions going on inside the cell, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell is connected.

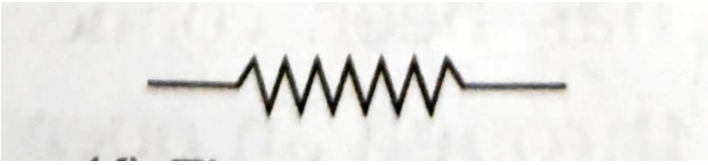
Electric cell is an instrument for the electricity generation where the electricity will generate after some chemical reaction takes place inside it. The generation is due to chemical reactions therefore the answer is chemical energy to electrical energy.

26.Correct option: (A)

The electric current is a flow of electrons through a conductor. When the electrons move from one part of the conductor to the other part, they collide with other electrons and with the atoms and ions present in the body of the conductor. Due to these collisions, there is some obstruction or opposition to the flow of electron current through the conductor. The property of a conductor due to which it opposes the flow of current through it is called resistance.

27.Correct option: (C)

Fixed resistance is shown as below.



28.Correct option: (B)

The property of a conductor due to which it opposes the flow of current through it is called resistance.

Resistance $= \frac{\text{Potential difference}}{\text{Current}}$

The SI unit of resistance is ohm which is denoted by the symbol omega Ω

29.Correct option: (A)

1 mega ohm (M Ω) = $10^6 \Omega$

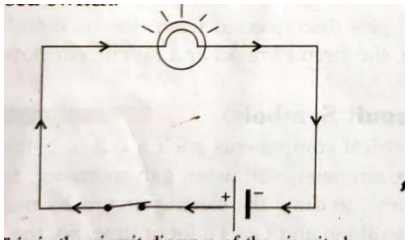
30.Correct option: (B)

The potential difference between two points in an electric circuit is defined as the amount of work done in moving a unit charge from one point to the other point.

The SI unit of potential difference is volt which is denoted by the letter V.

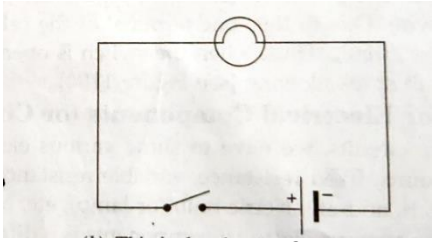
31.Correct option: (B)

(A)when the switch/key is closed, then the circuit is 'made' (or completed) and current flows in it. It is called a closed circuit.



(B) The potential difference is measured by means of an instrument called voltmeter. The voltmeter is always connected in parallel across the two points where the potential difference is to be measured.

32.Correct option: (B)



$$P = \frac{V^2}{R}$$

Where P= power

V = voltage

R = resistance

If we connect one bulb to two cells, Total Voltage of the circuit is received by the single bulb. As power/brightness is directly proportional to the voltage.
Hence, power consumption/brightness is high in case of single bulb connection to two cells.
(A) let the voltage of the cell = V
resistance of each cell = R
then
$P = \frac{V^2}{R}$
(B) voltage of two cells = 2V
resistance of each cell = R
then
$P = \frac{4V^2}{R}$
(C) voltage of 1 cells = V
resistance of each cell = R
total resistance = R + R = 2R
then
$P = \frac{V^2}{2R}$
Hence, power/brightness is maximum in case of option B

33.Correct option: (B)
Dilute sulphuric acid is used as electrolyte in voltaic cell.

34.Correct option: (C)
E.M.F. of a simple voltaic cell is 1.0 V.

35.Correct option: (A)
Dry cell is an improvement of wet leclanche cell.

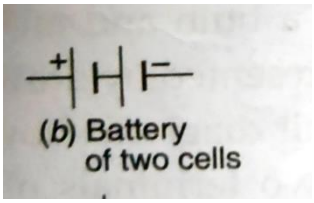
36.Correct option: (B)
The symbol of a single cell consists of two parallel vertical lines, one thin and long and the other thick and short (having horizontal lines on the sides). The longer vertical line represents the positive terminal of the cell (so it is marked plus, +), whereas the shorter vertical line represents the negative terminal of the cell(so it is marked minus,-)

37.Correct option: (D)
The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell. Due to the chemical reactions going on inside the cell, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell is connected.
Electric cell is an instrument for the electricity generation where the electricity will generate after some chemical reaction takes place inside it. The generation is due to chemical reactions therefore the answer is chemical energy to electrical energy.

38. Correct option: (D)
The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell. Due to the chemical reactions going on inside the cell, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell is connected.
Electric cell is an instrument for the electricity generation where the electricity will generate after some chemical reaction takes place inside it. The generation is due to chemical reactions therefore the answer is chemical energy to electrical energy.

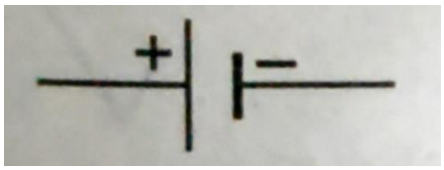
39. Correct option: (C)

When two (or more) cells are connected end to end consecutively, they are said to be connected in series. Figure below shows two cells which are connected in series.



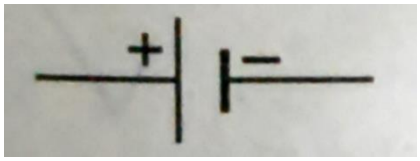
40. Correct option: (A)

The symbol of a single cell consists of two parallel vertical lines, one thin and long and the other thick and short (having horizontal lines on the sides). The longer vertical line represents the positive terminal of the cell (so it is marked plus, +), whereas the shorter vertical line represents the negative terminal of the cell(so it is marked minus,-)



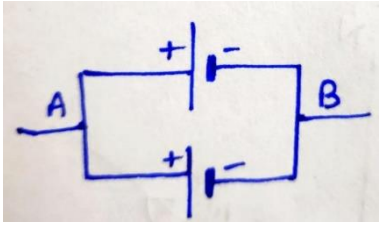
41. Correct option: (B)

The symbol of a single cell consists of two parallel vertical lines, one thin and long and the other thick and short (having horizontal lines on the sides). The longer vertical line represents the positive terminal of the cell (so it is marked plus, +), whereas the shorter vertical line represents the negative terminal of the cell(so it is marked minus,-)

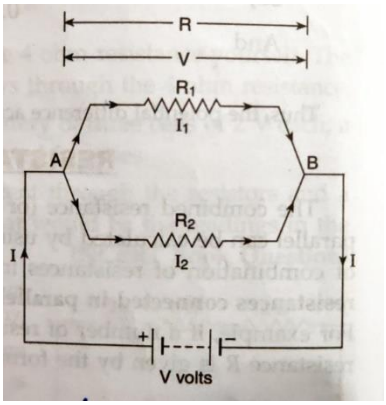


42.Correct option: (C)

When two (or more) cells are connected between the same two points, they are said to be connected in parallel (because they become parallel to one another). In Figure below, the two cells are connected in parallel arrangement between the same two points A and B.



43.Correct option: (B)



When a number of resistances connected in parallel are joined to the two terminals of a battery, then different amounts of current flow through each resistance (which depends on

the value of resistance). But the current flowing through all the individual parallel resistances, taken together, is equal to the current flowing in the circuit as a whole. Thus, when a number of resistances are connected in parallel, then the sum of the currents flowing through all the resistances is equal to the total current flowing in the circuit.

44.Correct option: (B)

When number of cells in an electric circuit are increased, the current will get increased.

45.Correct option: (B)

$$P = \frac{V^2}{R}$$

Where P= power

V = voltage

R = resistance

If we connect three bulbs of same power in series to one cell, the current flowing through each bulb is same and the voltage is equally divided among each of them.

When on bulb is connected to the same cell, then the voltage across the bulb increases.

Voltage received by each bulb in connection of three bulbs is less as compared to single bulb.

Hence, power consumption/brightness will be more in case of single bulb connection.

46.Correct option: (C)

In series circuit, if one electrical appliance stops working due to some defect, then all other appliances also stop working (because the whole circuit is broken). For example, if a number of bulbs are connected in series and just one bulb gets fused (or blows off), then all other bulbs will also stop glowing

47.Correct option: (A)

In parallel circuits, if one electrical appliance stops working due to some defect, then all other appliances keep working normally. For example, if a number of bulbs are connected in parallel circuits and one bulb gets fused (or blows out), then all the remaining bulbs will keep glowing.

48.Correct option: (C)

If the current passing through wires exceeds the maximum value, the copper wires get over-heated and may even cause fire leading to short-circuit.

49. Correct option: (A)

The heating effect of electric current is utilised in electric fuse for protecting household wiring and electrical appliances. A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing). This prevents the fire in house (due to over-heating of wiring) and also prevents damage to various electrical appliances in the house due to excessive current flowing through them.

50. Correct option: (D)

The full form of MCB is **Miniature Circuit Breaker**. It is a kind of electronic switch that functions automatically. It can detect a problem occurring inside a circuit by identifying the extreme amount of current flow in the circuit. If there is a large amount of power available in the loop, it can break the loop at any moment.

51.Correct option: (B)

The parallel electric circuit is better for connecting bulbs (and other electrical appliances) in a house because then we can have separate switches for each bulb (or electrical appliance) and hence operate it separately. In addition to having ease of operation, parallel

domestic circuits (or household circuits) have many other advantages over the series circuits.

52.Correct option: (B)

Commercial unit (or trade unit) of electrical energy called kilowatt-hour.

53.Correct option: (C)

Electrical energy = Power x Time

One kilowatt- hour is the amount of electrical energy consumed when an electrical appliance having a power rating of 1 kilowatt is used for 1 hour.

54.Correct option: (D)

kilowatt- hour is the 'unit' of electrical energy for which we pay the electrical supply department of our city.

55.Correct option: (C)

SI unit power is called Watt (W).

So, Wattage mentioned on a bulb indicates the power of the bulb.

56.Correct option: (D)

When electric current passes through a very thin, high resistance tungsten filament of an electric bulb, the filament becomes white-hot and emits light. The same current flowing through the tungsten filament of an electric bulb produces enormous heat but almost negligible heat is produced in the connecting wires of copper. This is because of the fact that the fine **tungsten filament has very high resistance** whereas copper connecting wires have very low resistance.

57.Correct option: (B)

Tungsten metal is used for making the filaments of electric bulbs.

58.Correct option: (B)

The heating effect of electric current is utilised in electric bulbs (electric lamps) for producing light. When electric current passes through a very thin, high resistance tungsten filament of an electric bulb, the filament becomes white-hot and emits light.

59.Correct option: (C)

A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing).

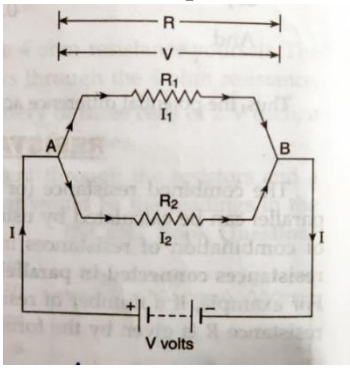
60.Correct option: (C)

Almost negligible heat is produced in the connecting wires of copper. This is because of the fact that copper connecting wires have very low resistance.

Hence, copper is used in household wiring of the circuits.

61.Correct option: (C)

When a number of resistances are connected in parallel, then the potential difference across each resistance is the same which is equal to the voltage of the battery applied.

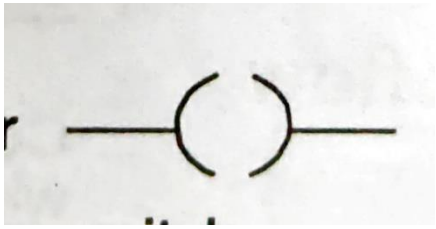


62.Correct option: (A)

According to the law of combination of resistances in series: The combined resistance of any number of resistances connected in series is equal to the sum of the individual resistances. Hence, the resistance is maximum in series connection of the bulbs.

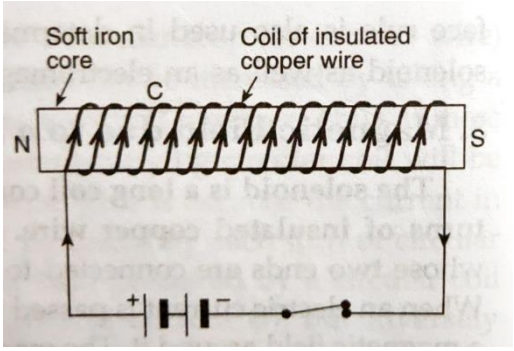
63.Correct option: (D)

The switch (or plug key) is a device for 'making or breaking an electric circuit. When the switch is open, then the circuit breaks' and no current flows in it. But when the switch is closed, then the circuit is 'made' (or completed) and current flows in it.



64. Correct option: (A)

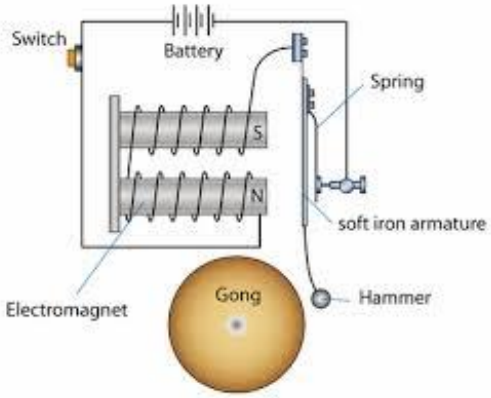
An electromagnet is a type of magnet in which the magnetic field is produced by an electric current. Electromagnets usually consist of wire wound into a coil. A current through the wire creates a magnetic field which is concentrated in the hole in the center of the coil. The magnetic field disappears when the current is turned off.



65. Correct option: (D)

An electric bell is a mechanical or electronic bell that functions by means of an electromagnet. When an electric current flows through the coil, the electromagnet creates a magnetic field which pulls the armature towards it, causing the hammer to strike the bell.

66.Correct option: (A)



The magnetic effect of current was discovered by Oersted in 1820. Oersted found that a wire carrying a current was able to deflect a compass needle. Now, the compass needle is a tiny magnet which can be deflected only by a magnetic field. Since a current carrying wire was able to deflect a compass needle, it was concluded that a current flowing in a wire always gives rise to a magnetic field around it.

LEVEL-II

1. Correct option: (B)

A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing).

2.Correct option: (A)

Battery is a combination of two (or more) cells connected in series. In order to obtain a battery, the negative terminal of the first cell is joined with the positive terminal of the second cell, and so on. The symbol for a battery is shown in Figure below. The battery shown in figure consists of two cells joined together in series.

3.Correct option: (B)

If we want to connect (or join) a large number of electric bulbs (say, hundreds or thousands of electric bulbs) for decorating buildings and trees as during festivals such as Diwali or marriage functions, then the series circuit is better because all the bulbs connected in series can be controlled with just one switch.

4. Correct option: (C)

If we connect three bulbs of same power in series, the current flowing through each bulb is same and the voltage is equally divided among each of them so, each bulb will get same amount of energy and hence, produce the same light. Hence, brightness of each bulb will be same.

5. Correct option: (C)

When numbers of cells in series combination are increased, When the battery (no of cells together) is connected then potential remains same. So increase in number of cells will increase the current.

6. Correct option: (B)

An electric current produces a magnetic field around it and this magnetic field around the current carrying coil lasts as long as current flows through it.

7.Correct option: (A)

Magnetic field produced by a straight current- carrying wire at a given point is directly proportional to the current passing in the wire. So, greater the current in the wire stronger will be the magnetic field produced.

8. Correct options: (A), (B)

The simplest way to maintain a potential difference between the two ends of a conductor so as to get a continuous flow of current is to connect the conductor between the terminals of a cell or a battery. Due to the chemical reactions going on inside the cell or battery, a potential difference is maintained between its terminals. And this potential difference drives the current in a circuit in which the cell or battery is connected.

9.Correct option: (A), (C)

- Large Current is measured by an instrument called ammeter. The ammeter is always connected in series with the circuit in which the current is to be measured.
- A galvanometer can measure small electric currents.

10.Correct options: (A), (D)

Potential difference=$\frac{\text{Work done}}{\text{Quantity of charge moved}}$

If W joules of work has to be done to move q coulombs of charge from one point to the other point, then the potential difference V between the two points is given by the formula:

$V = \frac{W}{q}$

Or w= v q

where W =work done

and q= quantity of charge moved

11. Correct options: (B), (C)

The electric current is a flow of electric charges (called electrons) in a conductor such as a metal wire. The magnitude of electric current in a conductor is the amount of electric charge passing through a given point of the conductor in one second. If a charge of Q coulombs flows through a conductor in time t seconds, then the magnitude I of the electric current flowing through it is given by:

Current, $I = \frac{Q}{t}$

The SI unit of electric current is ampere which is denoted by the letter A.

If we put charge Q = 1 coulomb and time = 1 second in the above formula,

then current I becomes 1 ampere. This gives us the following definition of ampere: When 1 coulomb of charge flows through any cross-section of a conductor in 1 second, the electric current flowing through it is said to be 1 ampere. That is,

1 ampere = $\frac{1 \text{ coulomb}}{1\text{sec}}$

Hence, units of electric current are Ampere and coulomb/sec

12.Correct options: (A), (D)

(A)The heating effect of electric current is utilised in electric fuse for protecting household wiring and electrical appliances. A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing). This prevents the fire in house (due to over-heating of wiring) and also prevents damage to various electrical appliances in the house due to excessive current flowing through them.

(D)The full form of MCB is **Miniature Circuit Breaker**. It is a kind of electronic switch that functions automatically. It can detect a problem occurring inside a circuit by identifying the extreme amount of current flow in the circuit. If there is a large amount of power available in the loop, it can break the loop at any moment.

13. Correct option: (C)

A simple Leclanche cell consists of conducting solution (electrolyte) of ammonium chloride.

14.Correct option: (C)

The positive terminal of simple Leclanche cell consists of carbon.

15.Correct option: (D)

Bichromate cell consists of

- Two carbon plates which act as positive plates and zinc plate act as the negative electrode.
- A solution of potassium bichromate and dilute sulphuric acid used a s an electrolyte.
- The emf of a bichromate cell is 2 volts.

16.Correct options: (A), (C)

The higher the wattage, the higher the Power, and the greater output of light.

Hence, if wattage of a bulb increases, power consumption of bulb increases and brightness of bulb also increases.

17.Correct options: (B), (C)

kilowatt- hour(KWh) is the 'unit' of electrical energy for which we pay the electrical supply department of our city.

18.Correct options: (C), (D)

$\text{power} = \frac{\text{work done}}{\text{time taken}}$

$P = \frac{W}{t}$

SI unit of work is joule (J)

SI unit of time is seconds (s)

SI unit of power is Joules/second (J/s)

SI unit power is called Watt (W)

$1W = \frac{1J}{1s}$--------(1)

1kWh = 1000 W x 3600 s (1Kw= 1000 W , 1h = 3600s)

From equation 1 $1W = \frac{1J}{1s}$

1KWh = 1000 x $\frac{1J}{1s}$ x 3600 s

1KWh = 36 x 10^5J

Or

1KWh = 3.6 x 10^6J

19.Correct options: (A), (B), (D)

Some conversion devices are sometimes called a "power source". An electrical device that supplies electric energy to an electrical load, by converting one form of electrical energy to another is called a power source.

An electric bulb converts electric energy into light energy.

A dry cell converts stored chemical energy into electrical energy.

A solar cell converts solar energy into electrical energy.

Hence, these are power sources.

20.Correct option: (B)

The electric potential (or potential) at a point in an electric field is defined as the work done in moving a unit positive charge from infinity to that point.

$\text{Electric Potential} = \frac{\text{Work done}}{\text{Quantity of charge moved}}$

Electric Potential is denoted by the symbol V and its SI unit is volt.

21. Correct option: (B)

The electric potential (or potential) at a point in an electric field is defined as the work done in moving a unit positive charge from infinity to that point.

$\text{Electric Potential} = \frac{\text{Work done}}{\text{Quantity of charge moved}}$

Electric Potential is denoted by the symbol V and its SI unit is volt.

22. Correct option: (D)

$\text{Potential difference} = \frac{\text{Work done}}{\text{Quantity of charge moved}}$

If W joules of work has to be done to move q coulombs of charge from one point to the other point, then the potential difference V between the two points is given by the formula:

$V = \frac{W}{q}$

where W =work done

and q= quantity of charge moved

The SI unit of potential difference is volt which is denoted by the letter V. The potential difference between two points is said to be 1 volt if 1 joule of work is done in moving 1 coulomb of electric charge from one point to the other.

Thus, 1 volt= $\frac{1 \text{ joule}}{1 \text{ coulomb}}$

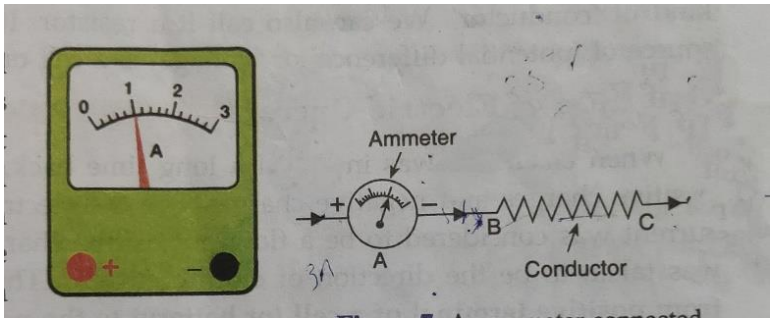
23.Correct option: (A)

When electricity was invented a long time back, it was known that there are two types of charges: positive charges and negative charges, but the electron had not been discovered at that time. So, electric current was considered to be a flow of positive charges and the direction of flow of the positive charges was taken to be the direction of electric current. Thus, the conventional direction of electric current is from positive terminal of a cell (or battery) to the negative terminal, through the outer circuit.

24. Correct option: (C)

Current is measured by an instrument called ammeter. The ammeter is always connected in series with the circuit in which the current is to be measured. For example, if we want to find out the current flowing through a conductor BC (such as a resistance wire), then we should connect the ammeter A in series with the conductor BC as shown in Figure below.

25. Correct option: (B)



Current is measured by an instrument called ammeter. The ammeter is always connected in series with the circuit in which the current is to be measured. For example, if we want to find out the current flowing through a conductor BC (such as a resistance wire), then we should connect the ammeter A in series with the conductor BC as shown in Figure below.

26. Correct option: (B)

Those substances through which electric charges cannot flow, are called insulators. In other words: Those substances through which electricity cannot flow are called insulators. Glass, ebonite, rubber, most plastics, paper, dry wood, cotton, mica, bakelite, porcelain, and dry air, are all insulators because they do not allow electric charges (or electricity) to flow through them**.**

27.Correct option: (C)

Those substances through which electric charges can flow, are called conductors. But the flow of electric charges is called electricity, so we can also say that: Those substances through which electricity can flow are called conductors. All the metals like silver, copper and aluminium, etc., are conductors.

28.Correct option: (A)

All the conductors (like metals) have some electrons which are loosely held by the nuclei of their atoms. These electrons are called "free electrons" and can move from one atom to another atom throughout the conductor. The presence of "free electrons" in a substance makes it a conductor (of electricity).

29.Correct option: (C)

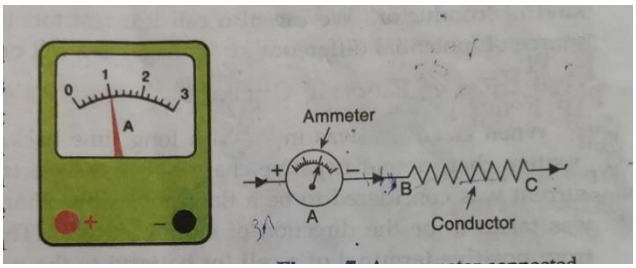
When work is done, an equal amount of energy is consumed. So, we can also say that **electric power** is the rate at which electrical energy is consumed. In other words, electric power is the electrical energy consumed per second. We can now write down another definition of watt based on electrical energy. When an electrical appliance consumes electrical energy at the rate of 1 joule per second, its power is said to be 1 watt.

$$\text{Electrical power(P)} = \frac{\text{electrical energy consumed (E)}}{\text{time (t)}}$$

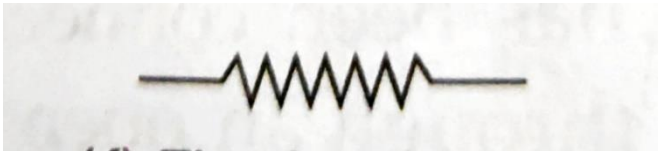
30.Correct option: (B)
Given: energy(E) = 10,000 J
Time(t) = 20 s
Electrical power(P) = $\frac{\text{electrical energy consumed (E)}}{\text{time (t)}}$
Putting the values in the above equation
Electrical power(P) = $\frac{10000}{20}$ = 500W
31.Correct option: (D)
Given: power (P) of bulb = 100 W
Time(t) = 10 hours
Electrical power(P) = $\frac{\text{electrical energy consumed (E)}}{\text{time (t)}}$
Putting the values in the above equation
100 W = $\frac{\text{E}}{10\,h}$
E = 100 x 10 Wh
E= 1000 Wh
Or electrical energy = 1 KWh or 1 Unit
electrical energy consumed in the month of December = 31 units
cost of 1 unit = Rs. 5
cost of 31 units = 31 x 5 = Rs.155
so, bill for December = Rs.155
32.Correct option: (B)
When an electric current is passed through a high resistance wire, like nichrome wire, the resistance wire becomes very hot and produces heat. This is called the heating effect of current. The heating effect of current is obtained by the transformation of electrical energy into heat energy. Just as mechanical energy used to overcome friction is converted into heat, in the same way, electrical energy is converted into heat energy when an electric current flows through a resistance wire. Thus, the role of resistance in electrical circuits is similar to the role of 'friction' in mechanics.
33.Correct option: (C)
The resistance (R) of a conductor is directly proportional to its length(l) and inversely proportional to its area of cross-section(A).
So, $R \propto \frac{l}{A}$
34.Correct option: (C)
When an electric current is passed through a high resistance wire, like nichrome wire, the resistance wire becomes very hot and produces heat. This is called the heating effect of current. The heating effect of current is obtained by the transformation of electrical energy into heat energy.
When electric current is passed through an electrical heating appliance such as an electric heater, electric kettle, hair dryer, immersion rod or a geyser, then most of the electrical energy is converted into heat. All the electrical heating appliances have a high resistance wire (like nichrome wire) which helps in converting most of the electrical energy into heat energy.
35. Correct Matching is: a-R; b-T; c-P; d-Q
(a) The SI unit of electric charge is coulomb which is denoted by the letter C.
An electron possesses a charge of -1.6 x 10^{-19}C.
(b) The SI unit of electric potential is volt which is denoted by the letter V.
(c) The SI unit of electric current is ampere which is denoted by the letter A.
(d) The SI unit of electric charge is coulomb which is denoted by the letter C.

36. Correct Matching is: a-S; b-T; c-P; d-Q

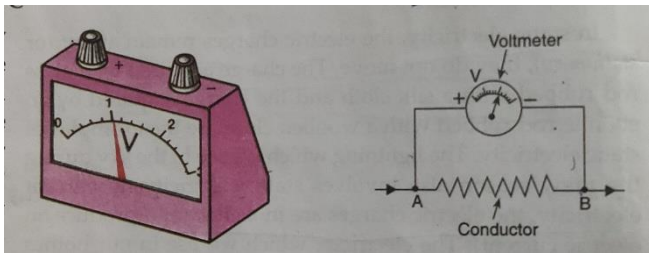
(a) Current is measured by an instrument called ammeter. The ammeter is always connected in series with the circuit in which the current is to be measured.



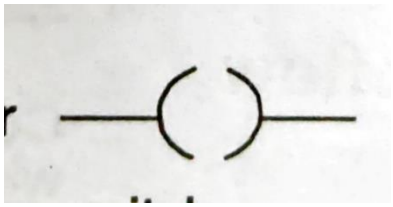
(b) symbol for resistance



(c) The potential difference is measured by means of an instrument called voltmeter. The voltmeter is always connected in parallel across the two points where the potential difference is to be measured.

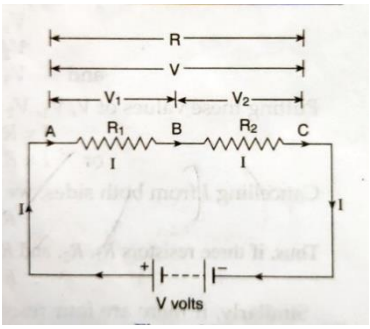


(d)symbol for key/switch in open



37. Correct Matching is: a-S; b-R; c-T; d-Q

(a) When a number of resistances are connected in series, then the same/constant current flows through each resistance (which is equal to the current flowing in the whole circuit).



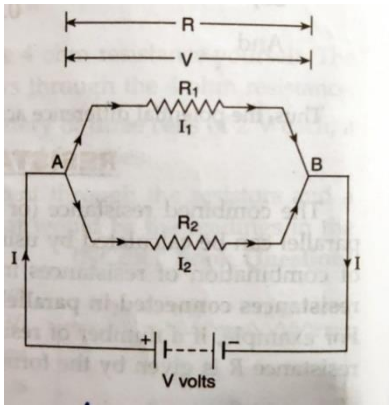
The heating effect of electric current is utilised in electric fuse for protecting household wiring and electrical appliances. A fuse is a short length of a thin tin- plated copper wire having low melting point. The thin fuse wire has a higher resistance than the rest of the electric wiring in a house. So, when the current in a household electric circuit rises too much due to some reason, then the fuse wire gets heated too much, melts and breaks the circuit (due to which the current stops flowing). This prevents the fire in house (due to over-

heating of wiring) and also prevents damage to various electrical appliances in the house due to excessive current flowing through them.

• The full form of MCB is **Miniature Circuit Breaker**. It is a kind of electronic switch that functions automatically. It can detect a problem occurring inside a circuit by identifying the extreme amount of current flow in the circuit. If there is a large amount of power available in the loop, it can break the loop at any moment.

(c) The parallel electric circuit is better for connecting bulbs (and other electrical appliances) in a house because then we can have separate switches for each bulb (or electrical appliance) and hence operate it separately. In addition to having ease of operation, parallel domestic circuits (or household circuits) have many other advantages over the series circuits.

(d) When a number of resistances are connected in parallel, then the potential difference across each resistance is the same which is equal to the voltage of the battery applied.



38. Correct Matching is: a-T; b-S; c-R; d-P

(a) $1W = \frac{1J}{1s}$--------(1)

1kWh = 1000 W x 3600 s (1Kw= 1000 W , 1h = 3600s)

From equation 1 $1W = \frac{1J}{1s}$

1KWh = 1000 x $\frac{1J}{1s}$ x 3600 s

1KWh = 36 x 10^5J

Or 1KWh = 3.6 x 10^6J

Or 1KWh = 3.6 MJ

(b) When electric current is passed through an electrical heating appliance such as an electric heater, electric kettle, hair dryer, immersion rod or a geyser, then most of the electrical energy is converted into heat. All the electrical heating appliances have a high resistance wire (like nichrome wire) which helps in converting most of the electrical energy into heat energy.

(c) Those substances through which electric charges cannot flow, are called insulators. In other words: Those substances through which electricity cannot flow are called insulators. Glass, ebonite, rubber, most plastics, paper, dry wood, cotton, mica, bakelite, porcelain, and dry air, are all insulators because they do not allow electric charges (or electricity) to flow through them**.**

(d) When work is done, an equal amount of energy is consumed. So, we can also say that **electric power** is the rate at which electrical energy is consumed. In other words, electric power is the electrical energy consumed per second. We can now write down another definition of watt based on electrical energy. When an electrical appliance consumes electrical energy at the rate of 1 joule per second, its power is said to be 1 watt.

Electrical power(P) = $\frac{\text{electrical energy consumed (E)}}{\text{time (t)}}$